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(54) **COMPOUNDS AND METHODS FOR MODULATING ANGIOTENSINOGEN EXPRESSION**

(71) Applicant: **Ionis Pharmaceuticals, Inc.**, Carlsbad, CA (US)

(72) Inventors: **Adam Mullick**, Carlsbad, CA (US); **Mark J. Graham**, San Clemente, CA (US); **Punit P. Seth**, Carlsbad, CA (US); **Susan M. Freier**, San Diego, CA (US)

(73) Assignee: **Ionis Pharmaceuticals, Inc.**, Carlsbad, CA (US)

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(57) **ABSTRACT**

Disclosed herein are compositions and compounds comprising modified oligonucleotides for modulating AGT and modulating a RAAS pathway related disease, disorder and/or condition in an individual in need thereof. A RAAS pathway related disease, disorder and/or condition in an individual such as hypertension can be treated, ameliorated, delayed or prevented with the administration of antisense compounds targeted to AGT.

Specification includes a Sequence Listing.

COMPOUNDS AND METHODS FOR MODULATING ANGIOTENSINOGEN EXPRESSION

SEQUENCE LISTING

[0001] The present application is being filed along with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled BIOL0270WOSEQ_ST25.txt created Oct. 3, 2016, which is 456 kb in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety

FIELD OF THE INVENTION

[0002] The present invention provides compounds, compositions and methods for modulating angiotensinogen (AGT) expression for the purpose of modulating a RAAS pathway related disease, disorder or condition in an animal. The present invention also provides compounds, compositions and methods for reducing hypertension and organ damage by administering an AGT inhibitor to an animal

BACKGROUND OF THE INVENTION

[0003] Angiotensinogen (AGT), also known as SERPINA8 or ANHU, is a member of the serpin family and is a component of the renin-angiotensin-aldosterone system (RAAS). It is primarily produced in the liver and is released into the circulation where renin converts it into angiotensin I. Angiotensin I is subsequently converted into angiotensin II by angiotension converting enzyme (ACE). Angiotensin II is a peptide hormone which causes vasoconstriction which, in turn, can increase blood pressure. Angiotensin II also stimulates secretion of the hormone aldosterone from the adrenal cortex. Aldosterone causes the kidneys to increase reabsorption of sodium and water leading to an increase of the fluid volume in a body which, in turn, can increase blood pressure. Over stimulation or activity of the RAAS pathway can lead to high blood pressure. Chronic high blood pressure is known as hypertension. The high blood pressure in a hypertensive subject requires the heart to work harder to circulate blood through the blood vessels.

[0004] The World Health Organization (WHO) has identified hypertension as a leading cause of cardiovascular morbidity. Hypertension is a major risk factor for various disease, disorders and conditions such as shortened life expectancy, chronic kidney disease, stroke, myocardial infarction, heart failure, aneurysms of the blood vessels (e.g. aortic aneurysm), peripheral artery disease, heart damage (e.g., heart enlargement or hypertrophy) and other cardiovascular related diseases, disorders and/or conditions.

[0005] The prevelance of resistant hypertension (RHTN), hypertension resistant to drug treatment, has steadily increased in number likely due to an ageing population and an ever increasing incidence of obesity. The current projection of approximately 10 million RHTN adults in the United States is expected to continue to rise.

[0006] Anti-hypertensive drugs, renal denervation, baroreceptor activation therapy, diet changes and lifestyle changes may reduce hypertension and reduce the diseases, disorders and/or conditions associated with hypertension (Paulis et al., Nat Rev Cardiol, 2012, 9:276-285). However, there are limitations to the therapies currently approved for treating hypertension as a significant subset of all hypertensive patients do not achieve adequate blood pressure control. For example, drugs such as ACE inhibitors and angiotensin receptor blockers (ARBs) that target parts of the renin-angiotensin system (RAS) pathway are limited in their ability to inhibit the RAAS pathway (Nobakht et al., Nat Rev Nephrol, 2011, 7:356-359). Additionally, certain anti-hypertensive drugs such as ACE inhibitors are contra-indicated in hypertensive patients with renal disease due to their potential to compromise renal function in patients.

[0007] Accordingly, there is a need to find alternative treatments to inhibit the RAAS pathway and treat hypertension. Antisense technology is emerging as an effective means for reducing the expression of certain gene products. However, early antisense oligonucleotides targeting AGT provided limited benefit (WO 1997/33623) or targeted non-human AGT (WO 2014/018930). The compounds and compositions herein provide novel, highly potent and tolerable compounds to inhibit human AGT and are suitable for use in human subjects. Additionally, compounds disclosed herein, by using a conjugate strategy that delivers antisense compounds to the liver and limits their renal distribution and activity, are predicted to mitigate the tolerability issues of traditional RAS blockers in patients at risk for hyperkalemia and/or renal disease.

[0008] All documents, or portions of documents, cited in this application, including, but not limited to, patents, patent applications, articles, books, and treatises, are hereby expressly incorporated-by-reference for the portions of the document discussed herein, as well as in their entirety.

SUMMARY OF THE INVENTION

[0009] Provided herein are compositions, compounds and methods for lowering the levels of AGT mRNA and/or protein in an animal.

[0010] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide targeting a nucleic acid sequence encoding AGT. In certain embodiments, the compound targets an AGT sequence as shown in the nucleobase sequences of any of SEQ ID NOs: 1-6.

[0011] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases complementary to an equal length portion of nucleobases 2250 to 2337 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

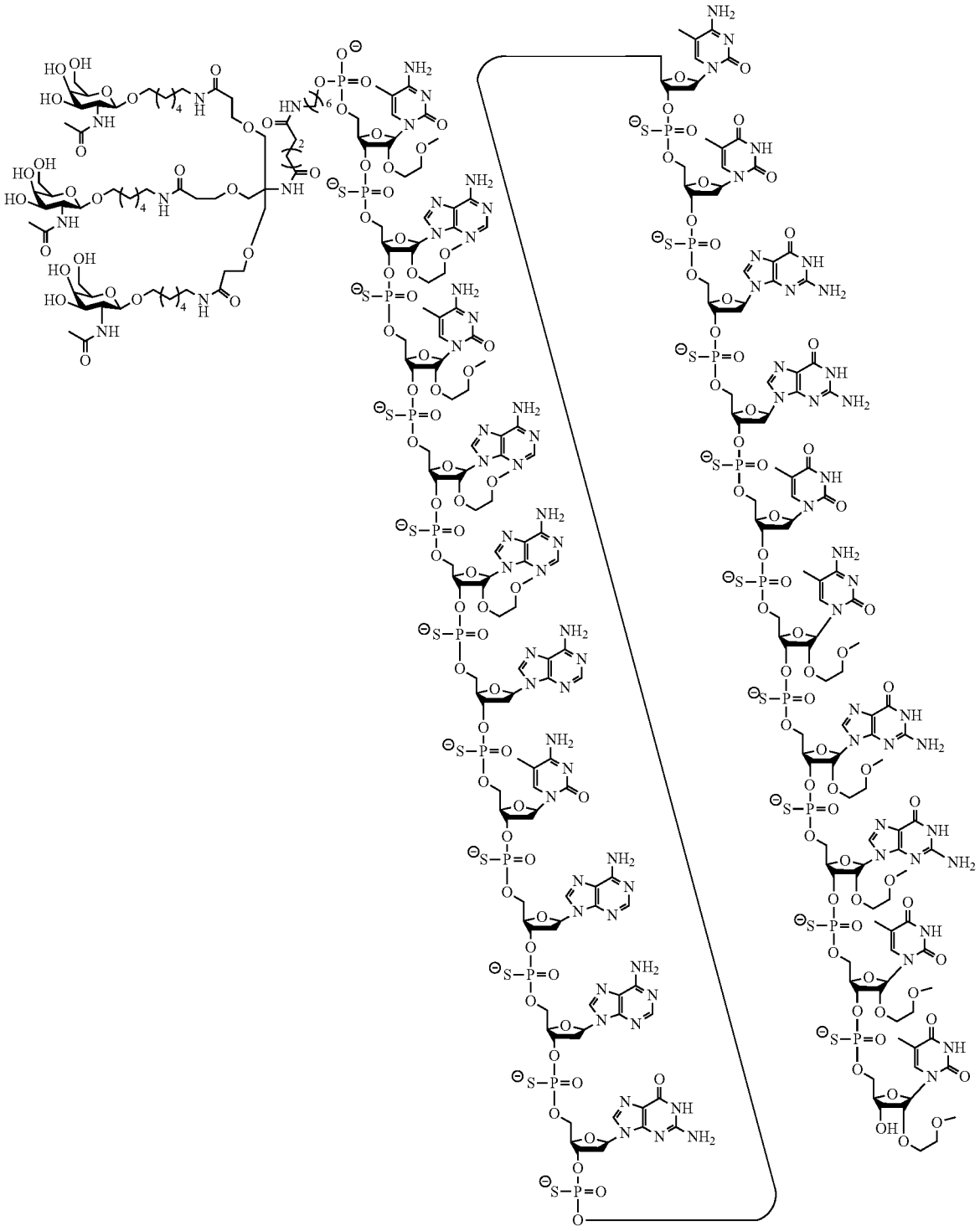
[0012] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases complementary to an equal length portion of nucleobases 2281 to 2300 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0013] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 46, 53-54, 61, 68, 76, 83, 85, 93, 96-97, 109, 127, 129-130, 132, 134-15, 137-39, 142, 163-172, 180-184, 186, 189, 234, 236, 238-239, 267, 313, 411, 452, 463-470, 475-478, 480, 500-503, 512, 517-518, 524-526, 654, 689, 702, 725-726, 728, 738, 779, 786-787, 800, 808, 810-811, 825, 865, 868, 889, 894, 903, 905, 909, 954, 966, 1011, 1015, 1021, 1024, 1080, 1085, 1258-1259, 1261-1262, 1293-1294, 1299, 1325, 1470, 1472-1473, 1522, 1542, 1604, 1623-1624, 1667, 1670, 1682-1683, 1687, 1700, 1703-1704, 1708, 1714, 1716, 1719-1720, 1724-1726, 1729-1730, 1827, 1936, 1843-1844, 1846, 1886, 1893-1894, 1914, 1923, 1925, 1932, 1979, 1986, 1988, 1990, 2003, 2015, 2018, 2020, 2027-2028, 2035, 2037, 2039, 2044.

[0014] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide according to the following formula: mCes Aes mCes Aes Aes Ads mCds Ads Ads Gds mCds Tds Gds Gds Tds mCes Ges Ges Tes Te (SEQ ID NO: 1914); wherein, A is an adenine, mC

is a 5'-methylcytosine, G is a guanine, T is a thymine, e is a 2'-O-methoxyethyl modified nucleoside, d is a 2'-deoxynucleoside, and s is a phosphorothioate internucleoside linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0015] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide with the following formula:



DETAILED DESCRIPTION OF THE INVENTION

[0016] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed. Herein, the use of the singular includes the plural unless specifically stated otherwise. As used herein, the use of "or" means "and/or" unless stated otherwise. Furthermore, the use of the term "including" as well as other forms, such as "includes" and "included", is not limiting. Also, terms such as "element" or "component" encompass both elements and components comprising one unit and elements and components that comprise more than one subunit, unless specifically stated otherwise.

[0017] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described. All documents, or portions of documents, cited in this application, including, but not limited to, patents, patent applications, articles, books, and treatises, are hereby expressly incorporated by reference for the portions of the document discussed herein, as well as in their entirety.

Definitions

[0018] Unless specific definitions are provided, the nomenclature utilized in connection with, and the procedures and techniques of, analytical chemistry, synthetic organic chemistry, and medicinal and pharmaceutical chemistry described herein are those well known and commonly used in the art. Standard techniques may be used for chemical synthesis, and chemical analysis. Where permitted, all patents, applications, published applications and other publications, GENBANK Accession Numbers and associated sequence information obtainable through databases such as National Center for Biotechnology Information (NCBI) and other data referred to throughout the disclosure herein are incorporated by reference for the portions of the document discussed herein, as well as in their entirety.

[0019] Unless otherwise indicated, the following terms have the following meanings:

[0020] "2'-O-methoxyethyl" (also 2'-MOE and 2'-O(CH_2)$_2$—OCH_3) refers to an O-methoxy-ethyl modification of the 2' position of a furosyl ring. A 2'-O-methoxyethyl modified sugar is a modified sugar.

[0021] "2'-O-methoxyethyl nucleotide" means a nucleotide comprising a 2'-O-methoxyethyl modified sugar moiety.

[0022] "5-methylcytosine" means a cytosine modified with a methyl group attached to the 5' position. A 5-methylcytosine is a modified nucleobase.

[0023] "About" means within +10% of a value. For example, if it is stated, "a marker may be increased by about 50%", it is implied that the marker may be increased between 45%-55%.

[0024] "ACE escape", also known as angiotensin II reactivation, refers to the inability of currently available ACE inhibitor treatment to reliably suppress plasma angiotensin II levels. The increase in plasma angiotensin II levels during ACE inhibition occurs via other enzymes converting angiotensin I to angiotensin. This incomplete blockage of angiotensin II levels prevents the ACE inhibitors from effectively treating some hypertensive subjects. Angiotensin Receptor Blockers (ARBs) may also be susceptible to ACE escape as other receptors besides the AT1 receptor engage angiotensin metabolites.

[0025] "Active pharmaceutical agent" or "Pharmaceutical agent" means the substance or substances in a pharmaceutical composition that provide a therapeutic benefit when administered to an individual. For example, in certain embodiments, an antisense oligonucleotide targeted to AGT is an active pharmaceutical agent.

[0026] "Active target region" or "target region" means a region to which one or more active antisense compounds is targeted.

[0027] "Active antisense compounds" means antisense compounds that reduce target nucleic acid levels or protein levels.

[0028] "Administered concomitantly" refers to the co-administration of two agents in any manner in which the pharmacological effects of both are manifest in the patient time. Concomitant administration does not require that both agents be administered in a single pharmaceutical composition, in the same dosage form, or by the same route of administration. The effects of both agents need not manifest themselves at the same time. The effects need only be overlapping for a period of time and need not be coextensive.

[0029] "Administering" means providing a pharmaceutical agent to an individual, and includes, but is not limited to administering by a medical professional and self-administering.

[0030] "Aldosterone escape" or "aldosterone breakthrough" refers to the inability of currently available ACE inhibitor Angiotensin Receptor Blocker (ARB) and/or Direct Renin Inhibitor (DRI) treatment to reliably suppress aldosterone release in some treated subjects. This incomplete blockage of aldosterone prevents the ACE inhibitors, DRIs and ARBs from effectively treating some hypertensive subjects.

[0031] "Agent" means an active substance that can provide a therapeutic benefit when administered to an animal. "First Agent" means a therapeutic compound provided herein. For example, a first agent is an antisense oligonucleotide targeting AGT. "Second agent" means a second therapeutic compound described herein. For example, a second agent can be a second antisense oligonucleotide targeting AGT or a non-AGT target. Alternatively, a second agent can be a compound other than an antisense oligonucleotide.

[0032] "Amelioration" or "ameliorate" refers to a lessening of at least one indicator, marker, sign, or symptom of an associated disease, disorder and/or condition. In certain embodiments, amelioration includes a delay or slowing in the progression of one or more indicators of a condition, disorder and/or disease. The severity of indicators may be determined by subjective or objective measures, which are known to those skilled in the art.

[0033] "Angiotensinogen" and "AGT" is used interchangeably herein. Angiotensinogen is also known as SERPINA8 and ANHU.

[0034] "Angiotensinogen nucleic acid" or "AGT nucleic acid" means any nucleic acid encoding AGT. For example, in certain embodiments, an AGT nucleic acid includes a DNA sequence encoding AGT, an RNA sequence transcribed from DNA encoding AGT (including genomic DNA

comprising introns and exons), and an mRNA sequence encoding AGT. "AGT mRNA" means an mRNA encoding an AGT protein.

[0035] "AGT specific inhibitor" refers to any agent capable of specifically inhibiting the expression of AGT mRNA and/or AGT protein. For example, AGT specific inhibitors include nucleic acids (including antisense compounds such as RNasH, siRNA and blockmer antisense compounds), peptides, antibodies, small molecules, and other agents capable of specifically inhibiting the expression of AGT mRNA and/or AGT protein. In certain embodiments, by specifically modulating AGT mRNA level and/or AGT protein expression, AGT specific inhibitors can affect components of the renin-angiotensin-aldosterone system (RAAS) pathway. In certain embodiments, by specifically modulating AGT mRNA level and/or AGT protein expression, AGT specific inhibitors can affect RAAS pathway related diseases, disorders and/or conditions such as blood pressure. Similarly, in certain embodiments, AGT specific inhibitors can affect other molecular processes in an animal.

[0036] "Animal" refers to a human or non-human animal, including, but not limited to, mice, rats, rabbits, dogs, cats, pigs, and non-human primates, including, but not limited to, monkeys and chimpanzees.

[0037] "Anti-hypertensive drug" refers to a drug capable of lowering blood pressure. Examples of such drugs include, but are not limited to, RAAS inhibitors, diuretics, calcium channel blockers, adrenergic receptor antagonists, adrenergic agonists and vasodilators. In one example, the anti-hypertensive drug captopril can be used in combination with the AGT compound described herein to treat an animal having or at risk of having a RAAS pathway related disease, disorder and/or condition.

[0038] "Anti-hypertensive procedure" refers to a medical procedure performed on a subject to reduce hypertension. Examples of such procedures include renal denervation and baroreceptor activation therapy.

[0039] "Antibody" refers to a molecule characterized by reacting specifically with an antigen in some way, where the antibody and the antigen are each defined in terms of the other. Antibody may refer to a complete antibody molecule or any fragment or region thereof, such as the heavy chain, the light chain, Fab region, and Fc region.

[0040] "Antisense activity" means any detectable or measurable activity attributable to the hybridization of an antisense compound to its target nucleic acid. In certain embodiments, antisense activity is a decrease in the amount or expression of a target nucleic acid or protein encoded by such target nucleic acid.

[0041] "Antisense compound" means an oligomeric compound that is capable of undergoing hybridization to a target nucleic acid through hydrogen bonding.

[0042] "Antisense inhibition" means reduction of target nucleic acid levels or target protein levels in the presence of an antisense compound complementary to a target nucleic acid compared to target nucleic acid levels or target protein levels in the absence of the antisense compound.

[0043] "Antisense oligonucleotide" means a single-stranded oligonucleotide having a nucleobase sequence that permits hybridization to a corresponding region or segment of a target nucleic acid.

[0044] "Bicyclic sugar" means a furosyl ring modified by the bridging of two non-geminal ring atoms. A bicyclic sugar is a modified sugar.

[0045] "Bicyclic nucleic acid" or "BNA" refers to a nucleoside or nucleotide wherein the furanose portion of the nucleoside or nucleotide includes a bridge connecting two carbon atoms on the furanose ring, thereby forming a bicyclic ring system.

[0046] "Blood pressure" refers to the pressure of the blood in the circulatory system against the walls of the blood vessel. The blood pressure is due mainly to the beating of the heart in an animal. During each heartbeat, the blood pressure varies between a maximum (systolic) blood pressure (SBP) and minimum (diastolic) blood pressure (DBP). The mean arterial pressure (MAP) is the average arterial pressure during a heartbeat cycle. Blood pressure can be measure by a blood pressure meter (i.e., a sphygmomanometer). Normal blood pressure at rest is within the range of 100-140 mmHg systolic and 60-90 mmHg diastolic and is commonly expressed as the systolic pressure (top reading)/diastolic pressure (bottom reading) mmHg.

[0047] "Cap structure" or "terminal cap moiety" means chemical modifications, which have been incorporated at either terminus of an antisense compound.

[0048] "cEt" or "constrained ethyl" means a bicyclic sugar moiety comprising a bridge connecting the 4'-carbon and the 2'-carbon, wherein the bridge has the formula: 4'-CH(CH_3)—O-2'.

[0049] "Constrained ethyl nucleoside" (also cEt nucleoside) means a nucleoside comprising a bicyclic sugar moiety comprising a 4'-CH(CH_3)—O-2' bridge.

[0050] "Chemically distinct region" refers to a region of an antisense compound that is in some way chemically different than another region of the same antisense compound. For example, a region having 2'-O-methoxyethyl nucleotides is chemically distinct from a region having nucleotides without 2'-O-methoxyethyl modifications.

[0051] "Chimeric antisense compound" means an antisense compound that has at least two chemically distinct regions.

[0052] "Co-administration" means administration of two or more pharmaceutical agents to an individual. The two or more pharmaceutical agents may be in a single pharmaceutical composition, or may be in separate pharmaceutical compositions. Each of the two or more pharmaceutical agents may be administered through the same or different routes of administration. Co-administration encompasses concomitant, parallel or sequential administration.

[0053] "Complementarity" means the capacity for pairing between nucleobases of a first nucleic acid and a second nucleic acid. In certain embodiments, the first nucleic acid is an antisense compound and the second nucleic acid is a target nucleic acid.

[0054] "Contiguous nucleobases" means nucleobases immediately adjacent to each other.

[0055] "Deoxyribonucleotide" means a nucleotide having a hydrogen at the 2' position of the sugar portion of the nucleotide. Deoxyribonucleotides may be modified with any of a variety of substituents.

[0056] "Diluent" means an ingredient in a composition that lacks pharmacological activity, but is pharmaceutically necessary or desirable. For example, the diluent in an injected composition may be a liquid, e.g. phosphate buffered saline (PBS) or water.

[0057] "Dosage unit" means a form in which a pharmaceutical agent is provided, e.g. pill, tablet, or other dosage unit known in the art. In certain embodiments, a dosage unit

is a vial containing lyophilized antisense oligonucleotide. In certain embodiments, a dosage unit is a vial containing reconstituted antisense oligonucleotide.

[0058] "Dose" means a specified quantity of a pharmaceutical agent provided in a single administration, or in a specified time period. In certain embodiments, a dose may be administered in one, two, or more boluses, tablets, or injections. For example, in certain embodiments where subcutaneous administration is desired, the desired dose requires a volume not easily accommodated by a single injection, therefore, two or more injections may be used to achieve the desired dose. In certain embodiments, the pharmaceutical agent is administered by infusion over an extended period of time or continuously. Doses may be stated as the amount of pharmaceutical agent per hour, day, week, or month.

[0059] "Effective amount" or "therapeutically effective amount" means the amount of active pharmaceutical agent sufficient to effectuate a desired physiological outcome in an individual in need of the agent. The effective amount can vary among individuals depending on the health and physical condition of the individual to be treated, the taxonomic group of the individuals to be treated, the formulation of the composition, assessment of the individual's medical condition, and other relevant factors. In an example, an effective amount of an AGT antisense oligonucleotide decreases blood pressure and/or ameliorates organ damage due to hypertension.

[0060] "Fully complementary" or "100% complementary" means that each nucleobase of a nucleobase sequence of a first nucleic acid has a complementary nucleobase in a second nucleobase sequence of a second nucleic acid. In certain embodiments, the first nucleic acid is an antisense compound and the second nucleic acid is a target nucleic acid.

[0061] "Gapmer" means a chimeric antisense compound in which an internal region having a plurality of nucleosides that support RNase H cleavage is positioned between external regions having one or more nucleosides, wherein the nucleosides comprising the internal region are chemically distinct from the nucleoside or nucleosides comprising the external regions. The internal region may be referred to as a "gap segment" and the external regions may be referred to as "wing segments."

[0062] "Gap-widened" means a chimeric antisense compound having a gap segment of 12 or more contiguous 2'-deoxynucleosides positioned between and immediately adjacent to 5' and 3' wing segments having from one to six nucleosides.

[0063] "Hybridization" means the annealing of complementary nucleic acid molecules. In certain embodiments, complementary nucleic acid molecules include an antisense compound and a target nucleic acid.

[0064] "Hypertension" or "HTN" refers to a chronic medical condition where the blood pressure in an animal is elevated. The elevated blood pressure requires the heart to work harder to circulate blood through the blood vessels. High blood pressure is said to be present if it is persistently at or above 140/90 mmHg. Hypertension is classified as primary (essential) or secondary. Primary hypertension has no clear cause and is thought to be linked to genetics, diet, lack of exercise and obesity. Secondary hypertension is caused by another medical condition. Hypertension is a major risk factor for shortened life expectancy, chronic kidney disease, stroke, myocardial infarction, heart failure, aneurysms of the blood vessels (e.g. aortic aneurysm), peripheral artery disease, organ damage (e.g., heart enlargement or hypertrophy) and other cardiovascular diseases, disorders and/or conditions or symptoms thereof. Antihypertensive drugs, diet changes and lifestyle changes may reduce hypertension and reduce the diseases, disorders and/or conditions associated with hypertension. Hypertension can be nonresistant to drug intervention (i.e., controllable by commercially available drug therapies) or resistant to drug intervention.

[0065] "Identifying an animal having, or at risk for, a RAAS related disease, disorder and/or condition" means identifying an animal having been diagnosed with a RAAS related disease, disorder and/or condition or identifying an animal predisposed to develop a RAAS related disease, disorder and/or condition. Individuals predisposed to develop a RAAS related disease, disorder and/or condition include, for example, individuals with a familial history a RAAS related disease such as hypertension. Such identification may be accomplished by any method including evaluating an individual's medical history and standard clinical tests or assessments.

[0066] "Immediately adjacent" means that there are no intervening elements between the immediately adjacent elements.

[0067] "Individual" or "subject" or "animal" means a human or non-human animal selected for treatment or therapy.

[0068] "Inhibiting the expression or activity" refers to a reduction or blockade of the expression or activity of a RNA or protein and does not necessarily indicate a total elimination of expression or activity.

[0069] "Intemucleoside linkage" refers to the chemical bond between nucleosides.

[0070] "Intravenous administration" means administration into a vein.

[0071] "Linked nucleosides" means adjacent nucleosides which are bonded together.

[0072] "Marker" or "biomarker" is any measurable and quantifiable biological parameter that serves as an index for health- or physiology-related assessments. For example, an increase in blood pressure, or a decrease in organ damage (e.g., fibrosis) can be considered markers of an RAAS related disease, disorder and/or condition.

[0073] "Mismatch" or "non-complementary nucleobase" or "MM" refers to the case when a nucleobase of a first nucleic acid is not capable of pairing with the corresponding nucleobase of a second or target nucleic acid.

[0074] "Modified internucleoside linkage" refers to a substitution or any change from a naturally occurring internucleoside bond (i.e. a phosphodiester internucleoside bond).

[0075] "Modified nucleobase" refers to any nucleobase other than adenine, cytosine, guanine, thymidine, or uracil. For example, a modified nucleobase can be 5'-methylcytosine. An "unmodified nucleobase" means the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C), and uracil (U).

[0076] "Modified nucleoside" means a nucleoside having, independently, a modified sugar moiety and/or modified nucleobase.

[0077] "Modified nucleotide" means a nucleotide having, independently, a modified sugar moiety, modified internucleoside linkage, and/or modified nucleobase.

[0078] "Modified oligonucleotide" means an oligonucleotide comprising a modified internucleoside linkage, a modified sugar, and/or a modified nucleobase.

[0079] "Modified sugar" refers to a substitution or change from a natural sugar. For example, a modified sugar can be 2'-MOE.

[0080] "Modulating" refers to changing or adjusting a feature in a cell, tissue, organ or organism. For example, modulating AGT mRNA can mean to increase or decrease the level of AGT mRNA and/or AGT protein in a cell, tissue, organ or organism. Modulating AGT mRNA and/or protein can lead to an increase or decrease in a RAAS related disease, disorder and/or condition in a cell, tissue, organ or organism. A "modulator" effects the change in the cell, tissue, organ or organism. For example, an AGT antisense compound can be a modulator that increases or decreases the amount of AGT mRNA and/or AGT protein in a cell, tissue, organ or organism.

[0081] "Monomer" refers to a single unit of an oligomer. Monomers include, but are not limited to, nucleosides and nucleotides, whether naturally occuring or modified.

[0082] "Motif" means the pattern of chemically distinct regions in an antisense compound.

[0083] "Naturally occurring internucleoside linkage" means a 3' to 5' phosphodiester linkage.

[0084] "Natural sugar moiety" means a sugar found in DNA (2'-H) or RNA (2'-OH).

[0085] "Nonresistant hypertension", "nonrefractory hypertension" or "controlled hypertension" is defined as hypertension that responds to treatment resulting in, for example, blood pressure <140 mmHg SBP or <90 mmHg DBP with concurrent use of up to 3 anti-hypertensive agents.

[0086] "Nucleic acid" refers to molecules composed of monomeric nucleotides. A nucleic acid includes ribonucleic acids (RNA), deoxyribonucleic acids (DNA), single-stranded nucleic acids, double-stranded nucleic acids, small interfering ribonucleic acids (siRNA), and microRNAs (miRNA).

[0087] "Nucleobase" means a heterocyclic moiety capable of pairing with a base of another nucleic acid.

[0088] "Nucleobase sequence" means the order of contiguous nucleobases independent of any sugar, linkage, or nucleobase modification.

[0089] "Nucleoside" means a nucleobase linked to a sugar.

[0090] "Nucleoside mimetic" includes those structures used to replace the sugar or the sugar and the base and not necessarily the linkage at one or more positions of an oligomeric compound; such as, for example, nucleoside mimetics having morpholino, cyclohexenyl, cyclohexyl, tetrahydropyranyl, bicyclo or tricyclo sugar mimetics e.g. non furanose sugar units.

[0091] "Nucleotide" means a nucleoside having a phosphate group covalently linked to the sugar portion of the nucleoside.

[0092] "Nucleotide mimetic" includes those structures used to replace the nucleoside and the linkage at one or more positions of an oligomeric compound; such as, for example, peptide nucleic acids or morpholinos (morpholinos linked by —N(H)—C(═O)—O— or other non-phosphodiester linkage).

[0093] "Organ damage" or "end organ damage" refers to damage occurring in major organs fed by the circulatory system such as the heart (e.g., heart muscle hypertrophy, reduced heart function and/or heart failure), kidney (e.g., albuminurea, proteinurea, reduced renal function and/or renal failure), eyes (e.g., hypertensive retinopathy), brain (e.g., stroke) and the like. The organs can be damaged by hypertension in an animal. In certain embodiments, the heart damage is fibrosis, heart cell and/or muscle hypertrophy leading to heart enlargement.

[0094] "Oligomeric compound" or "oligomer" refers to a polymeric structure comprising two or more sub-structures (monomers) and capable of hybridizing to a region of a nucleic acid molecule. In certain embodiments, oligomeric compounds are oligonucleosides. In certain embodiments, oligomeric compounds are oligonucleotides. In certain embodiments, oligomeric compounds are antisense compounds. In certain embodiments, oligomeric compounds are antisense oligonucleotides. In certain embodiments, oligomeric compounds are chimeric oligonucleotides.

[0095] "Oligonucleotide" means a polymer of linked nucleosides each of which can be modified or unmodified, independent one from another.

[0096] "Parenteral administration" means administration through injection or infusion. Parenteral administration includes subcutaneous administration, intravenous administration, intramuscular administration, intra-arterial administration, intraperitoneal administration, or intracranial administration, e.g., intrathecal or intracerebroventricular administration. Administration can be continuous, or chronic, or short or intermittent.

[0097] "Peptide" refers to a molecule formed by linking at least two amino acids by amide bonds. Peptide refers to polypeptides and proteins.

[0098] "Pharmaceutical composition" means a mixture of substances suitable for administering to an individual. For example, a pharmaceutical composition may comprise one or more active pharmaceutical agents and a sterile aqueous solution.

[0099] "Pharmaceutically acceptable carrier" means a medium or diluent that does not interfere with the structure of the oligonucleotide. Certain of such carriers enable pharmaceutical compositions to be formulated as, for example, tablets, pills, dragees, capsules, liquids, gels, syrups, slurries, suspension and lozenges for the oral ingestion by a subject. For example, a pharmaceutically acceptable carrier can be a sterile aqueous solution, such as sterile water or PBS.

[0100] "Pharmaceutically acceptable derivative" encompasses pharmaceutically acceptable salts, conjugates, prodrugs or isomers of the compounds described herein.

[0101] "Pharmaceutically acceptable salts" means physiologically and pharmaceutically acceptable salts of antisense compounds, i.e., salts that retain the desired biological activity of the parent oligonucleotide and do not impart undesired toxicological effects thereto.

[0102] "Phosphorothioate linkage" means a linkage between nucleosides where the phosphodiester bond is modified by replacing one of the non-bridging oxygen atoms with a sulfur atom. A phosphorothioate linkage is a modified internucleoside linkage.

[0103] "Portion" means a defined number of contiguous (i.e. linked) nucleobases of a nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucle-

obases of a target nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucleobases of an antisense compound.

[0104] "Prevent" refers to delaying or forestalling the onset, development, or progression of a disease, disorder, or condition for a period of time from minutes to indefinitely. Prevent also means reducing risk of developing a disease, disorder, or condition.

[0105] "Prodrug" means a therapeutic agent that is prepared in an inactive form that is converted to an active form within the body or cells thereof by the action of endogenous enzymes or other chemicals or conditions.

[0106] "Renin-angiotensin-aldosterone system", "Renin-angiotensin-aldosterone system pathway", "RAAS pathway" or "RAAS" refer to a multi-component enzymatic pathway where a precursor component (angiotensinogen) is converted by various enzymes such as renin and enzyme angiotensin-converting-enzyme (ACE) into downstream components such as angiotensin I and angiotensin II. Angiotensin I stimulates secretion of the steroid aldosterone in the pathway. The RAAS pathway regulates blood pressure and fluid balance in a body.

[0107] "Renin-angiotensin System", or "RAS" or "RAS pathway" refer to a portion of the RAAS pathway. Various components of this pathway have been targeted by agonists or antagonists to block the production of the components. For example renin inhibitors, ACE inhibitors, angiotensin-receptor blockers (ARBs) and the like have been developed to inhibit or block the RAS pathway. However, commercially available therapies targeting various RAS pathway components have been ineffective in completely inhibiting or blocking the RAS pathway due to various mechanisms (Nobakht et al., Nat Rev Nephrol, 2011, 7:356-359).

[0108] "RAAS related disease, disorder and/or condition" or "RAAS pathway related disease, disorder and/or condition" refers to any disease, disorder or condition related to RAAS in an animal. Examples of RAAS related diseases, disorders and/or conditions include shortened life expectancy, hypertension (e.g. nonresistant hypertension, resistant hypertension), kidney disease (e.g., chronic kidney disease, polycystic kidney disease), stroke, heart disease (e.g., myocardial infarction, heart failure, valvular heart disease), aneurysms of the blood vessels (e.g. aortic aneurysm), peripheral artery disease, organ damage (e.g., heart damage or hypertrophy), tissue fibrosis and other cardiovascular diseases, disorders and/or conditions or symptoms thereof. In certain embodiments, RAAS related disease, disorder and/or condition does not include hypertension.

[0109] "Resistant hypertension" or "RHTN" is defined as (1) blood pressure >140 mmHg SBP or >90 mmHg DBP despite concurrent use of 3 anti-hypertensive agents from different drug classes or (2) use of >4 anti-hypertensive drugs regardless of blood pressure.

[0110] "Side effects" means physiological disease and/or conditions attributable to a treatment other than the desired effects. In certain embodiments, side effects include injection site reactions, liver function test abnormalities, renal function abnormalities, liver toxicity, renal toxicity, central nervous system abnormalities, myopathies, and malaise. For example, increased aminotransferase levels in serum may indicate liver toxicity or liver function abnormality. For example, increased bilirubin may indicate liver toxicity or liver function abnormality.

[0111] "Single-stranded oligonucleotide" means an oligonucleotide which is not hybridized to a complementary strand.

[0112] "Specifically hybridizable" refers to an antisense compound having a sufficient degree of complementarity between an antisense oligonucleotide and a target nucleic acid to induce a desired effect, while exhibiting minimal or no effects on non-target nucleic acids under conditions in which specific binding is desired, e.g., under physiological conditions in the case of in vivo assays and therapeutic treatments. In an example, an antisense compound is specifically hybridizable to a target when binding of the compound to the target nucleic acid interferes with the normal function of the target nucleic acid to cause a loss of activity, and there is a sufficient degree of complementarity to avoid non-specific binding of the antisense compound to non-target nucleic acid sequences under conditions in which specific binding is desired.

[0113] "Subcutaneous administration" means administration just below the skin.

[0114] "Targeting" or "targeted" means the process of design and selection of an antisense compound that will specifically hybridize to a target nucleic acid and induce a desired effect.

[0115] "Target nucleic acid," "target RNA," and "target RNA transcript" all refer to a nucleic acid capable of being targeted by antisense compounds.

[0116] "Target segment" means the sequence of nucleotides of a target nucleic acid to which an antisense compound is targeted. "5' target site" refers to the 5'-most nucleotide of a target segment. "3' target site" refers to the 3'-most nucleotide of a target segment.

[0117] "Therapeutically effective amount" means an amount of a pharmaceutical agent that provides a therapeutic benefit to an animal.

[0118] "Treat" refers to administering a pharmaceutical composition to an animal in order to effect an alteration or improvement of a disease, disorder, or condition in the animal. In certain embodiments, one or more pharmaceutical compositions can be administered to the animal.

[0119] "Unmodified nucleotide" means a nucleotide composed of naturally occuring nucleobases, sugar moieties, and internucleoside linkages. In certain embodiments, an unmodified nucleotide is an RNA nucleotide (i.e. (3-D-ribonucleotide) or a DNA nucleotide (i.e. (3-D-deoxyribonucleotide).

CERTAIN EMBODIMENTS

[0120] Certain embodiments provide compounds specifically modulating AGT. In certain embodiments, the AGT specific modulators are AGT specific inhibitors, for use in treating, preventing, or ameliorating a RAAS related disease, disorder and/or condition. In certain embodiments, AGT specific inhibitors are nucleic acid compounds capable of inhibiting the expression of AGT mRNA and/or AGT protein. In certain embodiments, the nucleic acid compounds are oligomeric compounds. In certain embodiments, the oligomeric compounds are antisense oligonucleotides. In certain embodiments, the antisense oligonucleotides are modified antisense oligonucleotides. In certain embodiments, the modified antisense oligonucleotides are chimeric antisense oligonucleotides.

[0121] In certain embodiments, the compounds target an AGT nucleic acid. In certain embodiments, the AGT nucleic

acid is any of the human sequences set forth in GENBANK Accession No. NM_000029.3 (incorporated herein as SEQ ID NO: 1), the complement of the nucleotides 24354000 to 24370100 of GENBANK Accession No. NT_167186.1 (incorporated herein as SEQ ID NO: 2), GENBANK Accession No. AK307978.1 (incorporated herein as SEQ ID NO: 3), GENBANK Accession No. AK303755.1 (incorporated herein as SEQ ID NO: 4), GENBANK Accession No. AK293507.1 (incorporated herein as SEQ ID NO: 5), and GENBANK Accession No. CR606672.1 (incorporated herein as SEQ ID NO: 6).

[0122] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide targeting a nucleic acid sequence encoding AGT. In certain embodiments, the compound targets an AGT sequence as shown in the nucleobase sequences of any of SEQ ID NOs: 1-6.

[0123] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising at least 8, least 9, least 10, least 11, at least 12, least 13, at least 14, at least 15, at least 16, least 17, least 18, least 19, or 20 contiguous nucleobases complementary to an equal length portion of SEQ ID NOs: 1-6.

[0124] In certain embodiments, the nucleobase sequence of the modified oligonucleotide is at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% complementary to an equal length portion of any of SEQ ID NOs: 1-6. In certain embodiments, the modified oligonucleotide comprises a nucleobase sequence 100% complementary to an equal length portion of any of SEQ ID NOs: 1-6.

[0125] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases complementary to an equal length portion of nucleobases 2027-2068 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0126] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 contiguous nucleobases complementary to an equal length portion of nucleobases 2027 to 2068 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0127] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases complementary to an equal length portion of nucleobases 2250 to 2337 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0128] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 contiguous nucleobases complementary to an equal length portion of nucleobases 2250 to 2337 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0129] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases complementary to an equal length portion of nucleobases 2266 to 2337 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0130] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 contiguous nucleobases complementary to an equal length portion of nucleobases 2266 to 2337 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0131] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases complementary to an equal length portion of nucleobases 2281 to 2300 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0132] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 contiguous nucleobases complementary to an equal length portion of nucleobases 2281 to 2300 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0133] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases complementary to an equal length portion of nucleobases 2324 to 2346 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0134] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising a portion of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 contiguous nucleobases complementary to an equal length portion of nucleobases 2324 to 2346 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

[0135] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or 20 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 14-2051.

[0136] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or 20 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 40, 42, 46, 47, 49, 53 to 55, 61, 62, 68, 71, 76, 82, 84, 85, 89, 93, 96 to 98, 102, 109, 114, 119, 127, 129, 130 to 135, 137 to 140, 142, 143, 160, 162 to 207, 209, 210, 223, 225 to 227, 230 to 243, 252 to 254, 257, 258, 262 to 273, 276, 278, 279, 281, 284, 452, 463, 464, 466, 467, 470, 477, 480, 500, 502, 512, 517, 525, 526, 726, 728, 868, 905, 906, 954, 961, 962, 963, 965, 966, 971, 973, 986, 987, 989, 990, 991, 994, 997, 998, 1000, 1001, 1011, 1015, 1021, 1024, 1035, 1080, 1085, 1150, 1258, 1259 to 1262, 1293, 1294, 1299, 1325, 1326, 1354, 1355 to 1357, 1370, 1384, 1391, 1393 to 1395, 1406 to 1408, 1431, 1467, 1468, 1470, 1472 to 1474, 1476, 1488, 1489, 1500, 1503, 1504, 1522, 1524, 1526, 1528, 1535, 1536, 1539, 1542, 1543, 1545, 1585, 1592, 1594, 1595, 1599, 1604, 1610 to 1612, 1615, 1618, 1619 to 1624, 1626, 1628, 1629, 1631, 1632, 1635 to 1637, 1640, 1658, 1662, 1665 to 1671, 1673, 1676 to 1679, 1681 to 1683, 1686, 1687, 1699 to 1710, 1712, 1714 to 1721, 1724 to 1726, 1728 to 1731, 1735, 1736, 1739 to 1741, 1751, 1755, 1771, 1778, 1781 to 1783, 1827, 1834, 1836, 1843 to 1846, 1872, 1874, 1875 to 1888, 1890 to 1895, 1897, 1898, 1900, 1904 to 1927, 1931 to 1933, 1937, 1939, 1940, 1943, 1950, 1951, 1953, 1955 to 1959, 1962, 1964 to 1967, 1969 to 1971, 1973, 1977 to 1981, 1984 to 1991, 1993 to 1996, 2000 to 2005, 2007 to 2012, 2014 to 2025, 2027, 2028, 2030, 2032 to 2037, 2039-2045, 2047, 2051.

[0137] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or 20 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 46, 53, 54, 68, 76, 85, 96, 97, 114, 127, 129 to 132, 134, 135, 137 to 139, 142, 162 to 207, 225, 226, 230 to 243, 252, 264, 266 to 270, 284, 464, 467, 962, 963, 965, 966, 973, 990, 991, 997, 1000, 1001, 1011, 1261, 1299, 1355, 1356, 1470, 1472, 1473, 1503, 1504, 1522, 1526, 1535, 1536, 1542, 1543, 1545, 1595, 1599, 1604, 1620, 1623, 1624, 1626, 1640, 1662, 1666, 1667, 1669, 1670, 1673, 1682, 1683, 1687, 1699 to 1706, 1708, 1712, 1714 to 1716, 1719 to 1721, 1724 to 1726, 1729, 1730, 1736, 1778, 1783, 1836, 1843, 1875 to 1888, 1893 to 1895, 1897, 1900, 1904 to 1908, 1911, 1914 to 1918, 1920, 1922, 1923, 1925, 1926, 1931 to 1933, 1937, 1939, 1955, 1958, 1959, 1962, 1966, 1967, 1970, 1971, 1973, 1977, 1978 to 1981, 1985, 1986, 1987, 1988, 1990, 1991, 1994, 1996, 2000, 2002 to 2005, 2010, 2011, 2014 to 2025, 2027, 2028, 2035 to 2037, 2039, 2041 to 2045.

[0138] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or 20 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 96, 127, 129 to 132, 139, 162 to 169, 171 to 189, 191 to 193, 195, 196, 198 to 206, 234, 236, 238 to 240, 267 to 270, 966, 1000, 1522, 1542, 1623, 1624, 1667, 1682, 1683, 1700, 1703, 1704, 1708, 1714, 1719, 1720, 1724 to 1726, 1729, 1875, 1876, 1878, 1884 to 1886, 1893, 1894, 1906, 1908, 1914, 1917, 1918, 1922, 1923, 1925, 1926, 1932, 1933, 1967, 1970, 1978 to 1981, 1985, 1986, 1988, 1990, 1991, 2003, 2010, 2015, 2016, 2018, 2020, 2021, 2024, 2025, 2027, 2028, 2035, 2037, 2039, 2044.

[0139] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or 20 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 129, 130, 132, 163 to 168, 171, 172, 175 to 186, 188, 189, 192, 193, 195, 198 to 206, 238, 239, 966, 1703, 1720, 1726, 1923, 1925, 2003, 2015.

[0140] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or 20 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 46, 53-54, 61, 68, 76, 83, 85, 93, 96-97, 109, 127, 129-130, 132, 134-15, 137-39, 142, 163-172, 180-184, 186, 189, 234, 236, 238-239, 267, 313, 411, 452, 463-470, 475-478, 480, 500-503, 512, 517-518, 524-526, 654, 689, 702, 725-726, 728, 738, 779, 786-787, 800, 808, 810-811, 825, 865, 868, 889, 894, 903, 905, 909, 954, 966, 1011, 1015, 1021, 1024, 1080, 1085, 1258-1259, 1261-1262, 1293-1294, 1299, 1325, 1470, 1472-1473, 1522, 1542, 1604, 1623-1624, 1667, 1670, 1682-1683, 1687, 1700, 1703-1704, 1708, 1714, 1716, 1719-1720, 1724-1726, 1729-1730, 1827, 1936, 1843-1844, 1846, 1886, 1893-1894, 1914, 1923, 1925, 1932, 1979, 1986, 1988, 1990, 2003, 2015, 2018, 2020, 2027-2028, 2035, 2037, 2039, 2044. Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or 20 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 238, 1714, 1719, 1893-1894, 1914, 1923, 1925, 2003. In certain embodiments, the compound comprises a modified oligonucleotide consisting of 8 to 80, 20 to 80, 10 to 50, 20 to 35, 10 to 30, 12 to 30, 15 to 30, 16 to 30, 20 to 30, 20 to 29, 20 to 28, 20 to 27, 20 to 26, 20 to 25, 20 to 24, 20 to 23, 20 to 22, 20 to 21, 15 to 25, 16 to 25, 15 to 24, 16 to 24, 17 to 24, 18 to 24, 19 to 24, 19 to 22, 16 to 21, 18 to 21 or 16 to 20 linked nucleobases. In certain embodiments, the compound comprises a modified oligonucleotide consisting of 16 linked nucleosides. In certain embodiments, the compound comprises a modified oligonucleotide consisting of 20 linked nucleosides.

[0141] In certain embodiments, the compound comprises a modified oligonucleotide consisting of 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, or 80 linked nucleobases in length, or a range defined by any two of the above values.

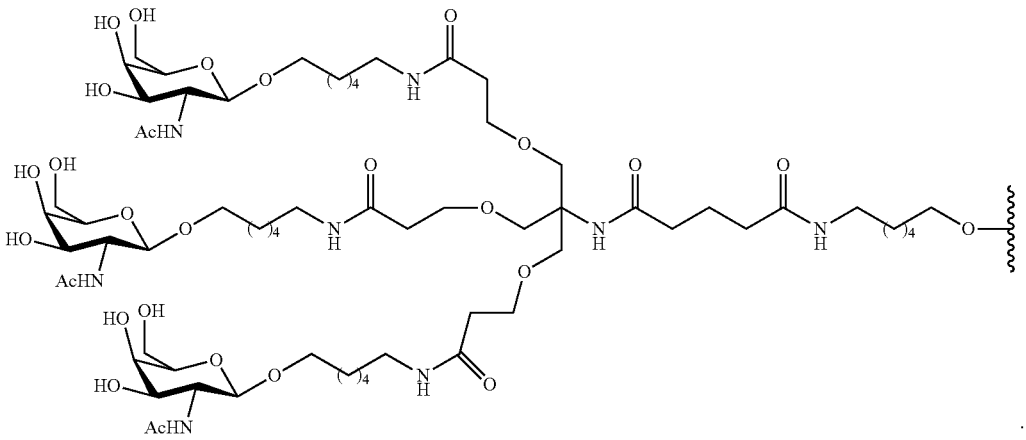
[0142] In certain embodiments, the modified oligonucleotide is single-stranded.

[0143] In certain embodiments, the modified oligonucleotide comprises at least one modified internucleoside linkage. In certain embodiments, the modified internucleoside linkage is a phosphorothioate internucleoside linkage. In certain embodiments, at least one modified internucleoside linkage is a phosphorothioate internucleoside linkage. In certain embodiments, each modified internucleoside linkage is a phosphorothioate internucleoside linkage.

[0144] In certain embodiments, the modified oligonucleotide comprises at least one nucleoside comprising a modified sugar. In certain embodiments, at least one modified sugar comprises a bicyclic sugar. In certain embodiments, at least one modified sugar comprises a 2'-O-methoxyethyl, a constrained ethyl, a 3'-fluoro-HNA or a 4'-$(CH_2)_n$—O-2' bridge, wherein n is 1 or 2.

[0145] In certain embodiments, the modified oligonucleotide comprises at least one nucleoside comprising a modified nucleobase. In certain embodiments, the modified nucleobase is a 5-methylcytosine.

[0146] In certain embodiments, the modified oligonucleotide comprises a conjugate group. In certain embodiments, the conjugate is a carbohydrate moiety. In certain embodiments, the conjugate is a GalNAc moiety. In certain embodiments, the GalNAc is 5'-Trishexylamino-(THA)-C6 $GalNAc_3$. In certain embodiments, the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate has the formula



[0147] In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0148] In certain embodiments, the compound comprises a modified oligonucleotide consisting of 12 to 30 linked nucleosides and targeted to or complementary to an equal length portion of region 2250 to 2337 of SEQ ID NO: 1, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. In certain embodiments, the modified oligonucleotide further comprises at least one phosphorothioate internucleoside linkage. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0149] In certain embodiments, the compound comprises a modified oligonucleotide consisting of 12 to 30 linked nucleosides and targeted to or complementary to an equal length portion of region 2266 to 2337 of SEQ ID NO: 1, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. In certain embodiments, the modified oligonucleotide further comprises at least one phosphorothioate internucleoside linkage. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0150] In certain embodiments, the compound comprises a modified oligonucleotide consisting of 12 to 30 linked nucleosides and targeted to or complementary to an equal length portion of region 2281 to 2300 of SEQ ID NO: 1, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. In certain embodiments, the modified oligonucleotide further comprises at least one phosphorothioate internucleoside linkage. In certain embodiments, each internucleoside

linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0151] In certain embodiments, the compound comprises a modified oligonucleotide consisting of 20 linked nucleosides and targeted to or complementary to an equal length portion of region 2027 to 2068 of SEQ ID NO: 1, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a 2'-O-methoxyethyl sugar, wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine residue is a 5-methylcytosine. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0152] In certain embodiments, the compound comprising a modified oligonucleotide consisting of 16 to 20 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of SEQ ID NOs: 14-2051, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a modified sugar, wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine residue is a 5-methylcytosine. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group. In certain embodiments, the compound comprising a modified oligonucleotide consisting of 16 to 20 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of SEQ ID NOs: 40, 42, 46, 47, 49, 53 to 55, 61, 62, 68, 71, 76, 82, 84, 85, 89, 93, 96 to 98, 102, 109, 114, 119, 127, 129, 130 to 135, 137 to 140, 142, 143, 160, 162 to 207, 209, 210, 223, 225 to 227, 230 to 243, 252 to 254, 257, 258, 262 to 273, 276, 278, 279, 281, 284, 452, 463, 464, 466, 467, 470, 477, 480, 500, 502, 512, 517, 525, 526, 726, 728, 868, 905, 906, 954, 961, 962, 963, 965, 966, 971, 973, 986, 987, 989, 990, 991, 994, 997, 998, 1000, 1001, 1011, 1015, 1021, 1024, 1035, 1080, 1085, 1150, 1258, 1259 to 1262, 1293, 1294, 1299, 1325, 1326, 1354, 1355 to 1357, 1370, 1384, 1391, 1393 to 1395, 1406 to 1408, 1431, 1467, 1468, 1470, 1472 to 1474, 1476, 1488, 1489, 1500, 1503, 1504, 1522, 1524, 1526, 1528, 1535, 1536, 1539, 1542, 1543, 1545, 1585, 1592, 1594, 1595, 1599, 1604, 1610 to 1612, 1615, 1618, 1619 to 1624, 1626, 1628, 1629, 1631, 1632, 1635 to 1637, 1640, 1658, 1662, 1665 to 1671, 1673, 1676 to 1679, 1681 to 1683, 1686, 1687, 1699 to 1710, 1712, 1714 to 1721, 1724 to 1726, 1728 to 1731, 1735, 1736, 1739 to 1741, 1751, 1755, 1771, 1778, 1781 to 1783, 1827, 1834, 1836, 1843 to 1846, 1872, 1874, 1875 to 1888, 1890 to 1895, 1897, 1898, 1900, 1904 to 1927, 1931 to 1933, 1937, 1939, 1940, 1943, 1950, 1951, 1953, 1955 to 1959, 1962, 1964 to 1967, 1969 to 1971, 1973, 1977 to 1981, 1984 to 1991, 1993 to 1996, 2000 to 2005, 2007 to 2012, 2014 to 2025, 2027, 2028, 2030, 2032 to 2037, 2039-2045, 2047, 2051, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a modified sugar, wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine residue is a 5-methylcytosine. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0153] In certain embodiments, the compound comprising a modified oligonucleotide consisting of 16 to 20 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of SEQ ID NOs: 46, 53, 54, 68, 76, 85, 96, 97, 114, 127, 129 to 132, 134, 135, 137 to 139, 142, 162 to 207, 225, 226, 230 to 243, 252, 264, 266 to 270, 284, 464, 467, 962, 963, 965, 966, 973, 990, 991, 997, 1000, 1001, 1011, 1261, 1299, 1355, 1356, 1470, 1472, 1473, 1503, 1504, 1522, 1526, 1535, 1536, 1542, 1543, 1545, 1595, 1599, 1604, 1620, 1623, 1624, 1626, 1640, 1662, 1666, 1667, 1669, 1670, 1673, 1682, 1683, 1687, 1699 to 1706, 1708, 1712, 1714 to 1716, 1719 to 1721, 1724 to 1726, 1729, 1730, 1736, 1778, 1783, 1836, 1843, 1875 to 1888, 1893 to 1895, 1897, 1900, 1904 to 1908, 1911, 1914 to 1918, 1920, 1922, 1923, 1925, 1926, 1931 to 1933, 1937, 1939, 1955, 1958, 1959, 1962, 1966, 1967, 1970, 1971, 1973, 1977, 1978 to 1981, 1985, 1986, 1987, 1988, 1990, 1991, 1994, 1996, 2000, 2002 to 2005, 2010, 2011, 2014 to 2025, 2027, 2028, 2035 to 2037, 2039, 2041 to 2045, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a modified sugar, wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine residue is a 5-methylcytosine. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0154] In certain embodiments, the compound comprising a modified oligonucleotide consisting of 16 to 20 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of SEQ ID NOs: 96, 127, 129 to 132, 139, 162 to 169, 171 to 189, 191 to 193, 195, 196,

198 to 206, 234, 236, 238 to 240, 267 to 270, 966, 1000, 1522, 1542, 1623, 1624, 1667, 1682, 1683, 1700, 1703, 1704, 1708, 1714, 1719, 1720, 1724 to 1726, 1729, 1875, 1876, 1878, 1884 to 1886, 1893, 1894, 1906, 1908, 1914, 1917, 1918, 1922, 1923, 1925, 1926, 1932, 1933, 1967, 1970, 1978 to 1981, 1985, 1986, 1988, 1990, 1991, 2003, 2010, 2015, 2016, 2018, 2020, 2021, 2024, 2025, 2027, 2028, 2035, 2037, 2039, 2044, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a modified sugar, wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine residue is a 5-methylcytosine. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0155] In certain embodiments, the compound comprising a modified oligonucleotide consisting of 16 to 20 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of SEQ ID NOs: 129, 130, 132, 163 to 168, 171, 172, 175 to 186, 188, 189, 192, 193, 195, 198 to 206, 238, 239, 966, 1703, 1720, 1726, 1923, 1925, 2003, 2015, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a modified sugar, wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine residue is a 5-methylcytosine. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0156] In certain embodiments, the compound comprising a modified oligonucleotide consisting of 16 to 20 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of SEQ ID NOs: 46, 53-54, 61, 68, 76, 83, 85, 93, 96-97, 109, 127, 129-130, 132, 134-15, 137-39, 142, 163-172, 180-184, 186, 189, 234, 236, 238-239, 267, 313, 411, 452, 463-470, 475-478, 480, 500-503, 512, 517-518, 524-526, 654, 689, 702, 725-726, 728, 738, 779, 786-787, 800, 808, 810-811, 825, 865, 868, 889, 894, 903, 905, 909, 954, 966, 1011, 1015, 1021, 1024, 1080, 1085, 1258-1259, 1261-1262, 1293-1294, 1299, 1325, 1470, 1472-1473, 1522, 1542, 1604, 1623-1624, 1667, 1670, 1682-1683, 1687, 1700, 1703-1704, 1708, 1714, 1716, 1719-1720, 1724-1726, 1729-1730, 1827, 1936, 1843-1844, 1846, 1886, 1893-1894, 1914, 1923, 1925, 1932, 1979, 1986, 1988, 1990, 2003, 2015, 2018, 2020, 2027-2028, 2035, 2037, 2039, 2044, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a modified sugar, wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine residue is a 5-methylcytosine. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

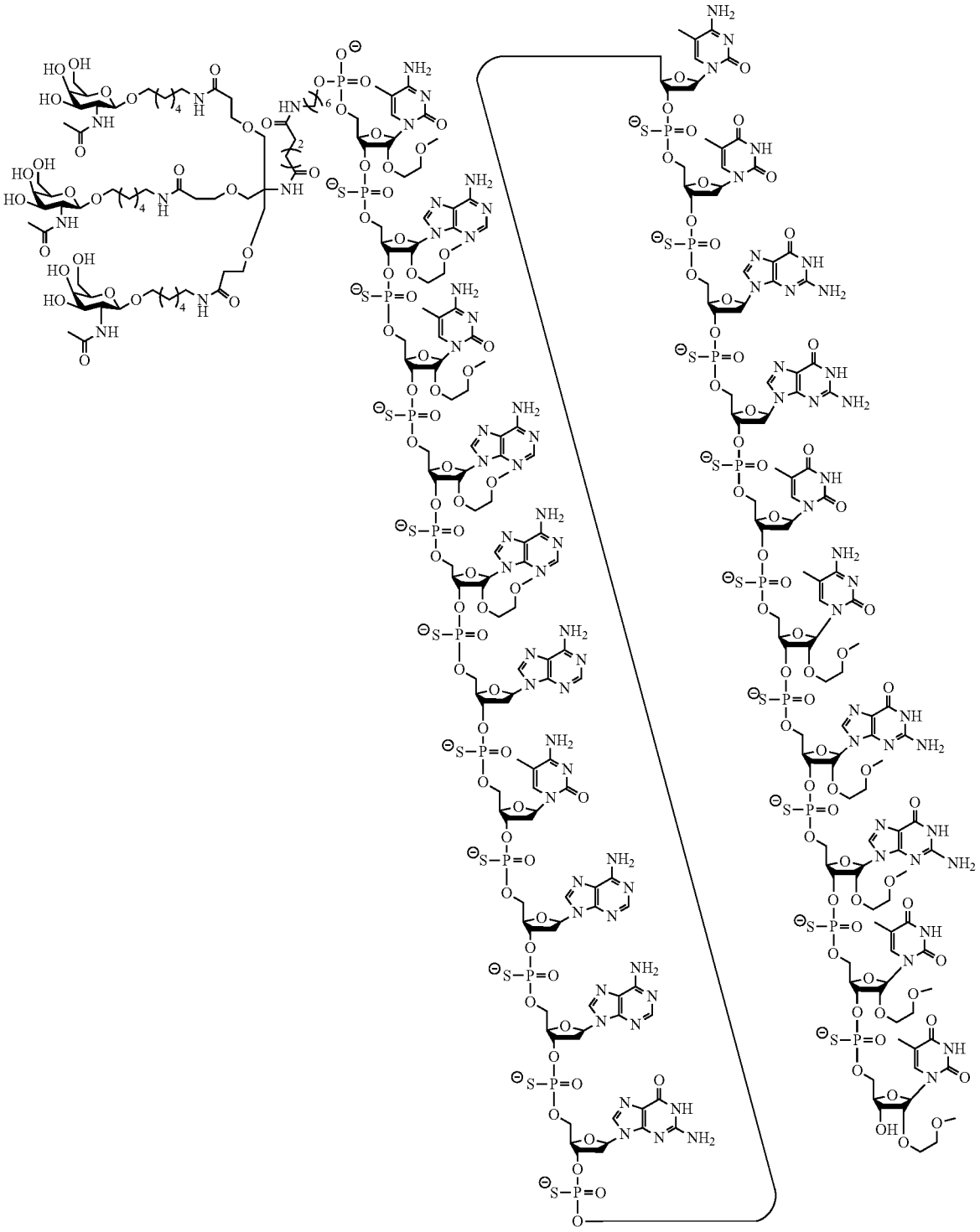
[0157] In certain embodiments, the compound comprising a modified oligonucleotide consisting of 16 to 20 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of SEQ ID NOs: 238, 1714, 1719, 1893-1894, 1914, 1923, 1925, 2003, wherein the modified oligonucleotide comprises: (a) a gap segment consisting of linked deoxynucleosides; (b) a 5' wing segment consisting of linked nucleosides; and (c) a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a modified sugar, wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine residue is a 5-methylcytosine. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0158] In certain embodiments, the compound comprising a modified oligonucleotide consisting of 20 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of SEQ ID NO: 1914, wherein the modified oligonucleotide comprises: (a) a gap segment consisting often linked deoxynucleosides; (b) a 5' wing segment consisting of five linked nucleosides; and (c) a 3' wing segment consisting of five linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a 2'-O-methoxyethyl sugar, wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine residue is a 5-methylcytosine. In certain embodiments, each internucleoside linkage is a phosphorothioate linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0159] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide according to the following formula: mCes Aes mCes Aes Aes Ads mCds Ads Ads Gds mCds Tds Gds Gds Tds mCes Ges Ges Tes Te (SEQ ID NO: 1914); wherein, A is an adenine, mC is a 5'-methylcytosine, G is a guanine, T is a thymine, e is a 2'-O-methoxyethyl modified nucleoside, d is a 2'-deoxy-

nucleoside, and s is a phosphorothioate internucleoside linkage. In certain embodiments, the modified oligonucleotide further comprises a GalNAc conjugate. In certain embodiments, the conjugate is a 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate. In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

[0160] Certain embodiments disclosed herein provide a compound comprising a modified oligonucleotide with the following formula:



[0161] In certain embodiments, the compounds or compositions disclosed herein comprise a salt of the modified oligonucleotide.

[0162] In certain embodiments, the compounds or compositions disclosed herein further comprise a pharmaceutically acceptable carrier or diluent.

[0163] In certain embodiments, the animal is a human.

[0164] Certain embodiments provide a composition or compound comprising a modified oligonucleotide as described herein, wherein the viscosity level is less than 40 cP. In certain embodiments, the composition has a viscosity level less than 15 cP. In certain embodiments, the composition has a viscosity level less than 12 cP. In certain embodiments, the composition has a viscosity level less than 10 cP.

[0165] Certain embodiments disclosed herein provide compounds and compositions comprising a modified oligonucleotide targeting AGT for use in reducing AGT in a cell, tissue, organ or animal. In certain embodiments, reducing AGT treats, prevents, slows the progression, delays the onset of, and/or reduces a RAAS pathway related disease, disorder and/or condition, or symptom thereof. In certain embodiments, reducing AGT decreases hypertension. In certain embodiments, reducing AGT decreases or prevents fibrosis. In certain embodiments, reducing AGT modulates a symptom or marker of a RAAS pathway related disease, disorder and/or condition. In certain embodiments, the marker can be selected from one or more of shortened life expectancy, hypertension, chronic kidney disease, stroke, myocardial infarction, heart failure, valvular heart disease, aneurysms of the blood vessels, peripheral artery disease, organ damage and other cardiovascular diseases, disorders and/or conditions or symptoms thereof.

[0166] In certain embodiments, provided are compounds and compositions comprising a modified oligonucleotide targeting AGT for use in therapy. In certain embodiments, the compounds and compositions comprising a modified oligonucleotide targeting AGT are administered to an animal in a therapeutically effective amount.

[0167] In certain embodiments, provided are compounds and compositions comprising a modified oligonucleotide targeting AGT for use in the preparation of a medicament. In certain embodiments, the medicament is used for treating, preventing, slowing the progression, delaying the onset of, and/or reducing a RAAS pathway related disease, disorder and/or condition, or symptom thereof.

[0168] In certain embodiments, provided is a kit for treating, preventing, or ameliorating a RAAS pathway related disease and/or condition, disease, disorder or condition, wherein the kit comprises: (i) an AGT specific inhibitor as described herein; and optionally (ii) an additional agent or therapy as described herein. A kit of the present invention may further include instructions for using the kit to treat, prevent, or ameliorate a RAAS pathway related disease, disorder or condition as described herein.

[0169] In certain embodiments, the RAAS pathway related disease, disorder or condition is shortened life expectancy, hypertension, kidney disease (e.g., chronic kidney disease), stroke, cardiac disease (e.g., myocardial infarction, heart failure, valvular heart disease), aneurysms of the blood vessels, peripheral artery disease, organ damage and other RAAS related diseases, disorders and/or conditions or symptoms thereof. In certain embodiments, the hypertension is nonresistant hypertension or resistant hypertension. In certain embodiments, the aneurysm of the blood vessels is aortic aneurysm. In certain embodiments, the organ damage is heart muscle hypertrophy or fibrosis in an organ or tissue. In certain embodiments, the organ is heart, liver or kidney and the tissue is derived from the heart, liver or kidney.

[0170] The compound can be used in combination therapy with one or more additional agent or therapy as described herein. Agents or therapies can be administered concomitantly or sequentially to an animal. In certain embodiments, the composition or compound comprising a modified oligonucleotide targeting AGT is co-administered with one or more second agent(s). In certain embodiments the second agent includes procedures to reduce hypertension, diet changes, lifestyle changes, anti-fibrotic drugs and anti-hypertensive drugs such as RAS or RAAS inhibitors, diuretics, calcium channel blockers, adrenergic receptor antagonists, adrenergic agonists and vasodilators. In certain embodiments, the second agent is a second antisense compound. In further embodiments, the second antisense compound targets AGT. In other embodiments, the second antisense compound targets a non-AGT compound.

Antisense Compounds

[0171] Oligomeric compounds include, but are not limited to, oligonucleotides, oligonucleosides, oligonucleotide analogs, oligonucleotide mimetics, antisense compounds, antisense oligonucleotides, and siRNAs. An oligomeric compound can be "antisense" to a target nucleic acid, meaning that it is capable of undergoing hybridization to a target nucleic acid through hydrogen bonding.

[0172] In certain embodiments, an antisense compound has a nucleobase sequence that, when written in the 5' to 3' direction, comprises the reverse complement of the target segment of a target nucleic acid to which it is targeted. In certain such embodiments, an antisense oligonucleotide has a nucleobase sequence that, when written in the 5' to 3' direction, comprises the reverse complement of the target segment of a target nucleic acid to which it is targeted.

[0173] In certain embodiments, an antisense compound targeted to AGT nucleic acid is 10 to 30 nucleotides in length. In other words, antisense compounds are from 10 to 30 linked nucleobases. In other embodiments, the antisense compound comprises a modified oligonucleotide consisting of 8 to 80, 10 to 80, 12 to 50, 15 to 30, 18 to 24, 19 to 22, or 20 linked nucleobases. In certain such embodiments, the antisense compound comprises a modified oligonucleotide consisting of 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, or 80 linked nucleobases in length, or a range defined by any two of the above values. In some embodiments, the antisense compound is an antisense oligonucleotide.

[0174] In certain embodiments, the antisense compound comprises a shortened or truncated modified oligonucleotide. The shortened or truncated modified oligonucleotide can have a single nucleoside deleted from the 5' end (5' truncation), the central portion or alternatively from the 3' end (3' truncation). A shortened or truncated oligonucleotide can have two or more nucleosides deleted from the 5' end, two or more nucleosides deleted from the central portion or alternatively can have two or more nucleosides deleted from the 3' end. Alternatively, the deleted nucleosides can be

dispersed throughout the modified oligonucleotide, for example, in an antisense compound having one or more nucleoside deleted from the 5' end, one or more nucleoside deleted from the central portion and/or one or more nucleoside deleted from the 3' end.

[0175] When a single additional nucleoside is present in a lengthened oligonucleotide, the additional nucleoside can be located at the 5' end, 3' end or central portion of the oligonucleotide. When two or more additional nucleosides are present, the added nucleosides can be adjacent to each other, for example, in an oligonucleotide having two nucleosides added to the 5' end (5' addition), to the 3' end (3' addition) or the central portion, of the oligonucleotide. Alternatively, the added nucleoside can be dispersed throughout the antisense compound, for example, in an oligonucleotide having one or more nucleoside added to the 5' end, one or more nucleoside added to the 3' end, and/or one or more nucleoside added to the central portion.

[0176] It is possible to increase or decrease the length of an antisense compound, such as an antisense oligonucleotide, and/or introduce mismatch bases without eliminating activity. For example, in Woolf et al. (Proc. Natl. Acad. Sci. USA 89:7305-7309, 1992), a series of antisense oligonucleotides 13-25 nucleobases in length were tested for their ability to induce cleavage of a target RNA in an oocyte injection model. Antisense oligonucleotides 25 nucleobases in length with 8 or 11 mismatch bases near the ends of the antisense oligonucleotides were able to direct specific cleavage of the target mRNA, albeit to a lesser extent than the antisense oligonucleotides that contained no mismatches. Similarly, target specific cleavage was achieved using 13 nucleobase antisense oligonucleotides, including those with 1 or 3 mismatches.

[0177] Gautschi et al (J. Natl. Cancer Inst. 93:463-471, March 2001) demonstrated the ability of an oligonucleotide having 100% complementarity to the bcl-2 mRNA and having 3 mismatches to the bcl-xL mRNA to reduce the expression of both bcl-2 and bcl-xL in vitro and in vivo. Furthermore, this oligonucleotide demonstrated potent antitumor activity in vivo.

[0178] Maher and Dolnick (Nuc. Acid. Res. 16:3341-3358, 1988) tested a series of tandem 14 nucleobase antisense oligonucleotides, and a 28 and 42 nucleobase antisense oligonucleotides comprised of the sequence of two or three of the tandem antisense oligonucleotides, respectively, for their ability to arrest translation of human DHFR in a rabbit reticulocyte assay. Each of the three 14 nucleobase antisense oligonucleotides alone was able to inhibit translation, albeit at a more modest level than the 28 or 42 nucleobase antisense oligonucleotides.

Certain Antisense Compound Motifs and Mechanisms

[0179] In certain embodiments, antisense compounds have chemically modified subunits arranged in patterns, or motifs, to confer to the antisense compounds properties such as enhanced inhibitory activity, increased binding affinity for a target nucleic acid, or resistance to degradation by in vivo nucleases.

[0180] Chimeric antisense compounds typically contain at least one region modified so as to confer increased resistance to nuclease degradation, increased cellular uptake, increased binding affinity for the target nucleic acid, and/or increased inhibitory activity. A second region of a chimeric antisense compound may confer another desired property e.g., serve as a substrate for the cellular endonuclease RNase H, which cleaves the RNA strand of an RNA:DNA duplex.

[0181] Antisense activity may result from any mechanism involving the hybridization of the antisense compound (e.g., oligonucleotide) with a target nucleic acid, wherein the hybridization ultimately results in a biological effect. In certain embodiments, the amount and/or activity of the target nucleic acid is modulated. In certain embodiments, the amount and/or activity of the target nucleic acid is reduced. In certain embodiments, hybridization of the antisense compound to the target nucleic acid ultimately results in target nucleic acid degradation. In certain embodiments, hybridization of the antisense compound to the target nucleic acid does not result in target nucleic acid degradation. In certain such embodiments, the presence of the antisense compound hybridized with the target nucleic acid (occupancy) results in a modulation of antisense activity. In certain embodiments, antisense compounds having a particular chemical motif or pattern of chemical modifications are particularly suited to exploit one or more mechanisms. In certain embodiments, antisense compounds function through more than one mechanism and/or through mechanisms that have not been elucidated. Accordingly, the antisense compounds described herein are not limited by particular mechanism.

[0182] Antisense mechanisms include, without limitation, RNase H mediated antisense; RNAi mechanisms, which utilize the RISC pathway and include, without limitation, siRNA, ssRNA and microRNA mechanisms; and occupancy based mechanisms. Certain antisense compounds may act through more than one such mechanism and/or through additional mechanisms.

RNase H-Mediated Antisense

[0183] In certain embodiments, antisense activity results at least in part from degradation of target RNA by RNase H. RNase H is a cellular endonuclease that cleaves the RNA strand of an RNA:DNA duplex. It is known in the art that single-stranded antisense compounds which are "DNA-like" elicit RNase H activity in mammalian cells. Accordingly, antisense compounds comprising at least a portion of DNA or DNA-like nucleosides may activate RNase H, resulting in cleavage of the target nucleic acid. In certain embodiments, antisense compounds that utilize RNase H comprise one or more modified nucleosides. In certain embodiments, such antisense compounds comprise at least one block of 1-8 modified nucleosides. In certain such embodiments, the modified nucleosides do not support RNase H activity. In certain embodiments, such antisense compounds are gapmers, as described herein. In certain such embodiments, the gap of the gapmer comprises DNA nucleosides. In certain such embodiments, the gap of the gapmer comprises DNA-like nucleosides. In certain such embodiments, the gap of the gapmer comprises DNA nucleosides and DNA-like nucleosides.

[0184] Certain antisense compounds having a gapmer motif are considered chimeric antisense compounds. In a gapmer an internal region having a plurality of nucleotides that supports RNaseH cleavage is positioned between external regions having a plurality of nucleotides that are chemically distinct from the nucleosides of the internal region. In the case of an antisense oligonucleotide having a gapmer motif, the gap segment generally serves as the substrate for endonuclease cleavage, while the wing segments comprise modified nucleosides. In certain embodiments, the regions

of a gapmer are differentiated by the types of sugar moieties comprising each distinct region. The types of sugar moieties that are used to differentiate the regions of a gapmer may in some embodiments include β-D-ribonucleosides, β-D-deoxyribonucleosides, 2'-modified nucleosides (such 2'-modified nucleosides may include 2'-MOE and 2'-O—CH_3, among others), and bicyclic sugar modified nucleosides (such bicyclic sugar modified nucleosides may include those having a constrained ethyl). In certain embodiments, nucleosides in the wings may include several modified sugar moieties, including, for example 2'-MOE and bicyclic sugar moieties such as constrained ethyl (cEt) or LNA. In certain embodiments, wings may include several modified and unmodified sugar moieties. In certain embodiments, wings may include various combinations of 2'-MOE nucleosides, bicyclic sugar moieties such as constrained ethyl nucleosides or LNA nucleosides, and 2'-deoxynucleosides.

[0185] Each distinct region may comprise uniform sugar moieties, variant, or alternating sugar moieties. The wing-gap-wing motif is frequently described as "X-Y-Z", where "X" represents the length of the 5'-wing, "Y" represents the length of the gap, and "Z" represents the length of the 3'-wing. "X" and "Z" may comprise uniform, variant, or alternating sugar moieties. In certain embodiments, "X" and "Y" may include one or more 2'-deoxynucleosides. "Y" may comprise 2'-deoxynucleosides. As used herein, a gapmer described as "X-Y-Z" has a configuration such that the gap is positioned immediately adjacent to each of the 5'-wing and the 3' wing. Thus, no intervening nucleotides exist between the 5'-wing and gap, or the gap and the 3'-wing. Any of the antisense compounds described herein can have a gapmer motif. In certain embodiments, "X" and "Z" are the same; in other embodiments they are different. In certain embodiments, "Y" is between 8 and 15 nucleosides. X, Y, or Z can be any of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30 or more nucleosides.

[0186] In certain embodiments, the antisense compound targeted to an AGT nucleic acid has a gapmer motif in which the gap consists of 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or 16 linked nucleosides.

[0187] In certain embodiments, the antisense oligonucleotide has a sugar motif described by Formula A as follows: $(J)_m(B)_n(J)_p$-$(B)_r$-$(A)_t$-$(D)_g$-$(A)_v$-$(B)_w$-$(J)_x$-$(B)_y$-$(J)_z$

[0188] wherein:

[0189] each A is independently a 2'-substituted nucleoside;

[0190] each B is independently a bicyclic nucleoside;

[0191] each J is independently either a 2'-substituted nucleoside or a 2'-deoxynucleoside;

[0192] each D is a 2'-deoxynucleoside;

[0193] m is 0-4; n is 0-2; p is 0-2; r is 0-2; t is 0-2; v is 0-2; w is 0-4; x is 0-2; y is 0-2; z is 0-4; g is 6-14; provided that:

[0194] at least one of m, n, and r is other than 0;

[0195] at least one of w and y is other than 0;

[0196] the sum of m, n, p, r, and t is from 2 to 5; and

[0197] the sum of v, w, x, y, and z is from 2 to 5.

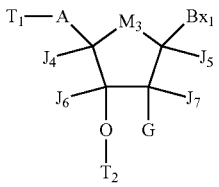
[0198] RNAi Compounds

[0199] In certain embodiments, antisense compounds are interfering RNA compounds (RNAi), which include double-stranded RNA compounds (also referred to as short-interfering RNA or siRNA) and single-stranded RNAi compounds (or ssRNA). Such compounds work at least in part through the RISC pathway to degrade and/or sequester a target nucleic acid (thus, include microRNA/microRNA-mimic compounds). In certain embodiments, antisense compounds comprise modifications that make them particularly suited for such mechanisms.

[0200] i. ssRNA Compounds

[0201] In certain embodiments, antisense compounds including those particularly suited for use as single-stranded RNAi compounds (ssRNA) comprise a modified 5'-terminal end. In certain such embodiments, the 5'-terminal end comprises a modified phosphate moiety. In certain embodiments, such modified phosphate is stabilized (e.g., resistant to degradation/cleavage compared to unmodified 5'-phosphate). In certain embodiments, such 5'-terminal nucleosides stabilize the 5'-phosphorous moiety. Certain modified 5'-terminal nucleosides may be found in the art, for example in WO 2011/139702.

[0202] In certain embodiments, the 5'-nucleoside of an ssRNA compound has Formula IIc:



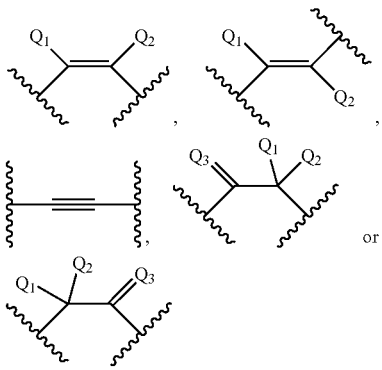
IIc

wherein:

[0203] T_1 is an optionally protected phosphorus moiety;

[0204] T_2 is an internucleoside linking group linking the compound of Formula IIc to the oligomeric compound;

[0205] A has one of the formulas:



[0206] Q_1 and Q_2 are each, independently, H, halogen, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_1-C_6 alkoxy, substituted C_1-C_6 alkoxy, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl, substituted C_2-C_6 alkynyl or $N(R_3)(R_4)$;

[0207] Q_3 is O, S, $N(R_5)$ or $C(R_6)(R_7)$;

[0208] each R_3, R_4 R_5, R_6 and R_7 is, independently, H, C_1-C_6 alkyl, substituted C_1-C_6 alkyl or C_1-C_6 alkoxy;

[0209] M_3 is O, S, NR_{14}, $C(R_{15})(R_{16})$, $C(R_{15})(R_{16})C(R_7)(R_{15})$, $C(R_{15})=C(R_{17})$, $OC(R_{15})(R_{16})$ or $OC(R_{15})(Bx_2)$;

[0210] R_{14} is H, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_1-C_6 alkoxy, substituted C_1-C_6 alkoxy, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl;

[0211] R_{15}, R_{16}, R_{17} and R_{15} are each, independently, H, halogen, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_1-C_6 alkoxy, substituted C_1-C_6 alkoxy, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl;

[0212] Bx_1 is a heterocyclic base moiety;

[0213] or if Bx_2 is present then Bx_2 is a heterocyclic base moiety and Bx_1 is H, halogen, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_1-C_6 alkoxy, substituted C_1-C_6 alkoxy, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl;

[0214] J_4, J_5, J_6 and J_7 are each, independently, H, halogen, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_1-C_6 alkoxy, substituted C_1-C_6 alkoxy, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl;

[0215] or J_4 forms a bridge with one of J_5 or J_7 wherein said bridge comprises from 1 to 3 linked biradical groups selected from O, S, NR_{19}, $C(R_{20})(R_{21})$, $C(R_{20})=C(R_{21})$, $C[=C(R_{20})(R_{21})]$ and C(=O) and the other two of J_5, J_6 and J_7 are each, independently, H, halogen, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_1-C_6 alkoxy, substituted C_1-C_6 alkoxy, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl;

[0216] each R_{19}, R_{20} and R_{21} is, independently, H, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_1-C_6 alkoxy, substituted C_1-C_6 alkoxy, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl;

[0217] G is H, OH, halogen or $O—[C(R_8)(R_9)]_n—[(C=O)_m—X_1]_j—Z$;

[0218] each R_8 and R_9 is, independently, H, halogen, C_1-C_6 alkyl or substituted C_1-C_6 alkyl;

[0219] X_1 is O, S or $N(E_1)$;

[0220] Z is H, halogen, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl, substituted C_2-C_6 alkynyl or $N(E_2)(E_3)$;

[0221] E_1, E_2 and E_3 are each, independently, H, C_1-C_6 alkyl or substituted C_1-C_6 alkyl;

[0222] n is from 1 to about 6;

[0223] m is 0 or 1;

[0224] j is 0 or 1;

[0225] each substituted group comprises one or more optionally protected substituent groups independently selected from halogen, OJ_1, $N(J_1)(J_2)$, $=NJ_1$, SJ_1, N_3, CN, $OC(=X_2)J_1$, $OC(=X_2)N(J_1)(J_2)$ and $C(=X_2)N(J_1)(J_2)$;

[0226] X_2 is O, S or NJ_3;

[0227] each J_1, J_2 and J_3 is, independently, H or C_1-C_6 alkyl;

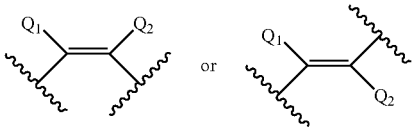
[0228] when j is 1 then Z is other than halogen or $N(E_2)(E_3)$; and

[0229] wherein said oligomeric compound comprises from 8 to 40 monomeric subunits and is hybridizable to at least a portion of a target nucleic acid.

[0230] In certain embodiments, M_3 is O, CH=CH, OCH_2 or $OC(H)(Bx_2)$. In certain embodiments, M_3 is O.

[0231] In certain embodiments, J_4, J_5, J_6 and J_7 are each H. In certain embodiments, J_4 forms a bridge with one of J_5 or J_7.

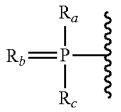
[0232] In certain embodiments, A has one of the formulas:



wherein:

[0233] Q_1 and Q_2 are each, independently, H, halogen, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_1-C_6 alkoxy or substituted C_1-C_6 alkoxy. In certain embodiments, Q_1 and Q_2 are each H. In certain embodiments, Q_1 and Q_2 are each, independently, H or halogen. In certain embodiments, Q_1 and Q_2 is H and the other of Q_1 and Q_2 is F, CH_3 or OCH_3.

[0234] In certain embodiments, T_1 has the formula:



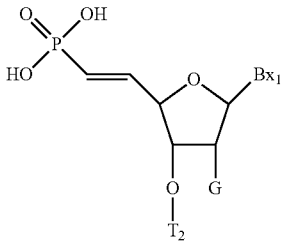
wherein:

[0235] R_a and R_c are each, independently, protected hydroxyl, protected thiol, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_1-C_6 alkoxy, substituted C_1-C_6 alkoxy, protected amino or substituted amino; and

[0236] R_b is O or S. In certain embodiments, R_b is O and R_a and R_c are each, independently, OCH_3, OCH_2CH_3 or $CH(CH_3)_2$.

[0237] In certain embodiments, G is halogen, OCH_3, OCH_2F, $OCHF_2$, OCF_3, OCH_2CH_3, $O(CH_2)_2F$, OCH_2CHF_2, OCH_2CF_3, $OCH_2—CH=CH_2$, $O(CH_2)_2—OCH_3$, $O(CH_2)_2—SCH_3$, $O(CH_2)_2—OCF_3$, $O(CH_2)_3—N(R_{10})(R_{11})$, $O(CH_2)_2—ON(R_{10})(R_{11})$, $O(CH_2)_2—O(CH_2)_2—N(R_{10})(R_{11})$, $OCH_2C(=O)—N(R_{10})(R_{11})$, $OCH_2C(=O)—N(R_{12})—(CH_2)_2—N(R_{10})(R_{11})$ or $O(CH_2)_2—N(R_{12})—C(=NR_{13})[N(R_{10})(R_{11})]$ wherein R_{10}, R_{11}, R_{12} and R_{13} are each, independently, H or C_1-C_6 alkyl. In certain embodiments, G is halogen, OCH_3, OCF_3, OCH_2CH_3, OCH_2CF_3, $OCH_2—CH=CH_2$, $O(CH_2)_2—OCH_3$, $O(CH_2)_2—O(CH_2)_2—N(CH_3)_2$, $OCH_2C(=O)—N(H)CH_3$, $OCH_2C(=O)—N(H)—(CH_2)_2—N(CH_3)_2$ or $OCH_2—N(H)—C(=NH)NH_2$. In certain embodiments, G is F, OCH_3 or $O(CH_2)_2—OCH_3$. In certain embodiments, G is $O(CH_2)_2—OCH_3$.

[0238] In certain embodiments, the 5'-terminal nucleoside has Formula IIe:



IIe

[0239] In certain embodiments, antisense compounds, including those particularly suitable for ssRNA comprise one or more type of modified sugar moieties and/or naturally occurring sugar moieties arranged along an oligonucleotide or region thereof in a defined pattern or sugar modification motif. Such motifs may include any of the sugar modifications discussed herein and/or other known sugar modifications.

[0240] In certain embodiments, the oligonucleotides comprise or consist of a region having uniform sugar modifications. In certain such embodiments, each nucleoside of the region comprises the same RNA-like sugar modification. In certain embodiments, each nucleoside of the region is a 2'-F

nucleoside. In certain embodiments, each nucleoside of the region is a 2'-OMe nucleoside. In certain embodiments, each nucleoside of the region is a 2'-MOE nucleoside. In certain embodiments, each nucleoside of the region is a cEt nucleoside. In certain embodiments, each nucleoside of the region is an LNA nucleoside. In certain embodiments, the uniform region constitutes all or essentially all of the oligonucleotide. In certain embodiments, the region constitutes the entire oligonucleotide except for 1-4 terminal nucleosides.

[0241] In certain embodiments, oligonucleotides comprise one or more regions of alternating sugar modifications, wherein the nucleosides alternate between nucleotides having a sugar modification of a first type and nucleotides having a sugar modification of a second type. In certain embodiments, nucleosides of both types are RNA-like nucleosides. In certain embodiments the alternating nucleosides are selected from: 2'-OMe, 2'-F, 2'-MOE, LNA, and cEt. In certain embodiments, the alternating modificatios are 2'-F and 2'-OMe. Such regions may be contiguous or may be interrupted by differently modified nucleosides or conjugated nucleosides.

[0242] In certain embodiments, the alternating region of alternating modifications each consist of a single nucleoside (i.e., the pattern is $(AB)_xA_y$ wherein A is a nucleoside having a sugar modification of a first type and B is a nucleoside having a sugar modification of a second type; x is 1-20 and y is 0 or 1). In certain embodiments, one or more alternating regions in an alternating motif includes more than a single nucleoside of a type. For example, oligonucleotides may include one or more regions of any of the following nucleoside motifs:

[0243] AABBAA;

[0244] ABBABB;

[0245] AABAAB;

[0246] ABBABAABB;

[0247] ABABAA;

[0248] AABABAB;

[0249] ABABAA;

[0250] ABBAABBABABAA;

[0251] BABBAABBABABAA; or

[0252] ABABBAABBABABAA;

[0253] wherein A is a nucleoside of a first type and B is a nucleoside of a second type. In certain embodiments, A and B are each selected from 2'-F, 2'-OMe, BNA, and MOE.

[0254] In certain embodiments, oligonucleotides having such an alternating motif also comprise a modified 5' terminal nucleoside, such as those of formula IIc or IIe.

[0255] In certain embodiments, oligonucleotides comprise a region having a 2-2-3 motif. Such regions comprises the following motif:

$$-(A)_2-(B)_x-(A)_2-(C)_y-(A)_3-$$

[0256] wherein: A is a first type of modified nucleoside;

[0257] B and C, are nucleosides that are differently modified than A, however, B and C may have the same or different modifications as one another;

[0258] x and y are from 1 to 15.

[0259] In certain embodiments, A is a 2'-OMe modified nucleoside. In certain embodiments, B and C are both 2'-F modified nucleosides. In certain embodiments, A is a 2'-OMe modified nucleoside and B and C are both 2'-F modified nucleosides.

[0260] In certain embodiments, oligonucleosides have the following sugar motif:

$$5'-(Q)-(AB)_xA_y-(D)_z$$

wherein:

[0261] Q is a nucleoside comprising a stabilized phosphate moiety. In certain embodiments, Q is a nucleoside having Formula IIc or IIe;

[0262] A is a first type of modified nucleoside;

[0263] B is a second type of modified nucleoside;

[0264] D is a modified nucleoside comprising a modification different from the nucleoside adjacent to it. Thus, if y is 0, then D must be differently modified than B and if y is 1, then D must be differently modified than A. In certain embodiments, D differs from both A and B.

[0265] X is 5-15;

[0266] Y is 0 or 1;

[0267] Z is 0-4.

[0268] In certain embodiments, oligonucleosides have the following sugar motif:

$$5'-(Q)-(A)_x-(D)_z$$

wherein:

[0269] Q is a nucleoside comprising a stabilized phosphate moiety. In certain embodiments, Q is a nucleoside having Formula IIc or IIe;

[0270] A is a first type of modified nucleoside;

[0271] D is a modified nucleoside comprising a modification different from A.

[0272] X is 11-30;

[0273] Z is 0-4.

[0274] In certain embodiments A, B, C, and D in the above motifs are selected from: 2'-OMe, 2'-F, 2'-MOE, LNA, and cEt. In certain embodiments, D represents terminal nucleosides. In certain embodiments, such terminal nucleosides are not designed to hybridize to the target nucleic acid (though one or more might hybridize by chance). In certain embodiments, the nucleobase of each D nucleoside is adenine, regardless of the identity of the nucleobase at the corresponding position of the target nucleic acid. In certain embodiments the nucleobase of each D nucleoside is thymine.

[0275] In certain embodiments, antisense compounds, including those particularly suited for use as ssRNA comprise modified internucleoside linkages arranged along the oligonucleotide or region thereof in a defined pattern or modified internucleoside linkage motif. In certain embodiments, oligonucleotides comprise a region having an alternating internucleoside linkage motif. In certain embodiments, oligonucleotides comprise a region of uniformly modified internucleoside linkages. In certain such embodiments, the oligonucleotide comprises a region that is uniformly linked by phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide is uniformly linked by phosphorothioate internucleoside linkages. In certain embodiments, each internucleoside linkage of the oligonucleotide is selected from phosphodiester and phosphorothioate. In certain embodiments, each internucleoside linkage of the oligonucleotide is selected from phosphodiester and phosphorothioate and at least one internucleoside linkage is phosphorothioate.

[0276] In certain embodiments, the oligonucleotide comprises at least 6 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least 8 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least 10 phosphorothioate internucleoside linkages. In certain

embodiments, the oligonucleotide comprises at least one block of at least 6 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 8 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 10 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least one 12 consecutive phosphorothioate internucleoside linkages. In certain such embodiments, at least one such block is located at the 3' end of the oligonucleotide. In certain such embodiments, at least one such block is located within 3 nucleosides of the 3' end of the oligonucleotide.

[0277] Oligonucleotides having any of the various sugar motifs described herein, may have any linkage motif. For example, the oligonucleotides, including but not limited to those described above, may have a linkage motif selected from non-limiting the table below:

5' most linkage	Central region	3'-region
PS	Alternating PO/PS	6 PS
PS	Alternating PO/PS	7 PS
PS	Alternating PO/PS	8 PS

ii. siRNA Compounds

[0278] In certain embodiments, antisense compounds are double-stranded RNAi compounds (siRNA). In such embodiments, one or both strands may comprise any modification motif described above for ssRNA. In certain embodiments, ssRNA compounds may be unmodified RNA. In certain embodiments, siRNA compounds may comprise unmodified RNA nucleosides, but modified internucleoside linkages.

[0279] Several embodiments relate to double-stranded compositions wherein each strand comprises a motif defined by the location of one or more modified or unmodified nucleosides. In certain embodiments, compositions are provided comprising a first and a second oligomeric compound that are fully or at least partially hybridized to form a duplex region and further comprising a region that is complementary to and hybridizes to a nucleic acid target. It is suitable that such a composition comprise a first oligomeric compound that is an antisense strand having full or partial complementarity to a nucleic acid target and a second oligomeric compound that is a sense strand having one or more regions of complementarity to and forming at least one duplex region with the first oligomeric compound.

[0280] The compositions of several embodiments modulate gene expression by hybridizing to a nucleic acid target resulting in loss of its normal function. In some embodiments, the target nucleic acid is AGT. In certain embodiment, the degradation of the targeted AGT is facilitated by an activated RISC complex that is formed with compositions of the invention.

[0281] Several embodiments are directed to double-stranded compositions wherein one of the strands is useful in, for example, influencing the preferential loading of the opposite strand into the RISC (or cleavage) complex. The compositions are useful for targeting selected nucleic acid molecules and modulating the expression of one or more genes. In some embodiments, the compositions of the present invention hybridize to a portion of a target RNA resulting in loss of normal function of the target RNA.

[0282] Certain embodiments are drawn to double-stranded compositions wherein both the strands comprises a hemimer motif, a fully modified motif, a positionally modified motif or an alternating motif. Each strand of the compositions of the present invention can be modified to fulfil a particular role in for example the siRNA pathway. Using a different motif in each strand or the same motif with different chemical modifications in each strand permits targeting the antisense strand for the RISC complex while inhibiting the incorporation of the sense strand. Within this model, each strand can be independently modified such that it is enhanced for its particular role. The antisense strand can be modified at the 5'-end to enhance its role in one region of the RISC while the 3'-end can be modified differentially to enhance its role in a different region of the RISC.

[0283] The double-stranded oligonucleotide molecules can be a double-stranded polynucleotide molecule comprising self-complementary sense and antisense regions, wherein the antisense region comprises nucleotide sequence that is complementary to nucleotide sequence in a target nucleic acid molecule or a portion thereof and the sense region having nucleotide sequence corresponding to the target nucleic acid sequence or a portion thereof. The double-stranded oligonucleotide molecules can be assembled from two separate oligonucleotides, where one strand is the sense strand and the other is the antisense strand, wherein the antisense and sense strands are self-complementary (i.e. each strand comprises nucleotide sequence that is complementary to nucleotide sequence in the other strand; such as where the antisense strand and sense strand form a duplex or double-stranded structure, for example wherein the double-stranded region is about 15 to about 30, e.g., about 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 base pairs; the antisense strand comprises nucleotide sequence that is complementary to nucleotide sequence in a target nucleic acid molecule or a portion thereof and the sense strand comprises nucleotide sequence corresponding to the target nucleic acid sequence or a portion thereof (e.g., about 15 to about 25 or more nucleotides of the double-stranded oligonucleotide molecule are complementary to the target nucleic acid or a portion thereof). Alternatively, the double-stranded oligonucleotide is assembled from a single oligonucleotide, where the self-complementary sense and antisense regions of the siRNA are linked by means of a nucleic acid based or non-nucleic acid-based linker(s).

[0284] The double-stranded oligonucleotide can be a polynucleotide with a duplex, asymmetric duplex, hairpin or asymmetric hairpin secondary structure, having self-complementary sense and antisense regions, wherein the antisense region comprises nucleotide sequence that is complementary to nucleotide sequence in a separate target nucleic acid molecule or a portion thereof and the sense region having nucleotide sequence corresponding to the target nucleic acid sequence or a portion thereof. The double-stranded oligonucleotide can be a circular single-stranded polynucleotide having two or more loop structures and a stem comprising self-complementary sense and antisense regions, wherein the antisense region comprises nucleotide sequence that is complementary to nucleotide sequence in a target nucleic acid molecule or a portion thereof and the sense region having nucleotide sequence

corresponding to the target nucleic acid sequence or a portion thereof, and wherein the circular polynucleotide can be processed either in vivo or in vitro to generate an active siRNA molecule capable of mediating RNAi.

[0285] In certain embodiments, the double-stranded oligonucleotide comprises separate sense and antisense sequences or regions, wherein the sense and antisense regions are covalently linked by nucleotide or non-nucleotide linkers molecules as is known in the art, or are alternately non-covalently linked by ionic interactions, hydrogen bonding, van der waals interactions, hydrophobic interactions, and/or stacking interactions. In certain embodiments, the double-stranded oligonucleotide comprises nucleotide sequence that is complementary to nucleotide sequence of a target gene. In another embodiment, the double-stranded oligonucleotide interacts with nucleotide sequence of a target gene in a manner that causes inhibition of expression of the target gene.

[0286] As used herein, double-stranded oligonucleotides need not be limited to those molecules containing only RNA, but further encompasses chemically modified nucleotides and non-nucleotides. In certain embodiments, the short interfering nucleic acid molecules lack 2'-hydroxy (2'-OH) containing nucleotides. In certain embodiments short interfering nucleic acids optionally do not include any ribonucleotides (e.g., nucleotides having a 2'—OH group). Such double-stranded oligonucleotides that do not require the presence of ribonucleotides within the molecule to support RNAi can however have an attached linker or linkers or other attached or associated groups, moieties, or chains containing one or more nucleotides with 2'—OH groups. Optionally, double-stranded oligonucleotides can comprise ribonucleotides at about 5, 10, 20, 30, 40, or 50% of the nucleotide positions. As used herein, the term siRNA is meant to be equivalent to other terms used to describe nucleic acid molecules that are capable of mediating sequence specific RNAi, for example short interfering RNA (siRNA), double-stranded RNA (dsRNA), micro-RNA (miRNA), short hairpin RNA (shRNA), short interfering oligonucleotide, short interfering nucleic acid, short interfering modified oligonucleotide, chemically modified siRNA, post-transcriptional gene silencing RNA (ptgsRNA), and others. In addition, as used herein, the term RNAi is meant to be equivalent to other terms used to describe sequence specific RNA interference, such as post transcriptional gene silencing, translational inhibition, or epigenetics. For example, double-stranded oligonucleotides can be used to epigenetically silence genes at both the post-transcriptional level and the pre-transcriptional level. In a non-limiting example, epigenetic regulation of gene expression by siRNA molecules of the invention can result from siRNA mediated modification of chromatin structure or methylation pattern to alter gene expression (see, for example, Verdel et al., 2004, Science, 303, 672-676; Pal-Bhadra et al., 2004, Science, 303, 669-672; Allshire, 2002, Science, 297, 1818-1819; Volpe et al., 2002, Science, 297, 1833-1837; Jenuwein, 2002, Science, 297, 2215-2218; and Hall et al., 2002, Science, 297, 2232-2237).

[0287] It is contemplated that compounds and compositions of several embodiments provided herein can target AGT by a dsRNA-mediated gene silencing or RNAi mechanism, including, e.g., "hairpin" or stem-loop double-stranded RNA effector molecules in which a single RNA strand with self-complementary sequences is capable of assuming a double-stranded conformation, or duplex dsRNA effector molecules comprising two separate strands of RNA. In various embodiments, the dsRNA consists entirely of ribonucleotides or consists of a mixture of ribonucleotides and deoxynucleotides, such as the RNA/DNA hybrids disclosed, for example, by WO 00/63364, filed Apr. 19, 2000, or U.S. Ser. No. 60/130,377, filed Apr. 21, 1999. The dsRNA or dsRNA effector molecule may be a single molecule with a region of self-complementarity such that nucleotides in one segment of the molecule base pair with nucleotides in another segment of the molecule. In various embodiments, a dsRNA that consists of a single molecule consists entirely of ribonucleotides or includes a region of ribonucleotides that is complementary to a region of deoxyribonucleotides. Alternatively, the dsRNA may include two different strands that have a region of complementarity to each other.

[0288] In various embodiments, both strands consist entirely of ribonucleotides, one strand consists entirely of ribonucleotides and one strand consists entirely of deoxyribonucleotides, or one or both strands contain a mixture of ribonucleotides and deoxyribonucleotides. In certain embodiments, the regions of complementarity are at least 70, 80, 90, 95, 98, or 100% complementary to each other and to a target nucleic acid sequence. In certain embodiments, the region of the dsRNA that is present in a double-stranded conformation includes at least 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 50, 75, 100, 200, 500, 1000, 2000 or 5000 nucleotides or includes all of the nucleotides in a cDNA or other target nucleic acid sequence being represented in the dsRNA. In some embodiments, the dsRNA does not contain any single stranded regions, such as single stranded ends, or the dsRNA is a hairpin. In other embodiments, the dsRNA has one or more single stranded regions or overhangs. In certain embodiments, RNA/DNA hybrids include a DNA strand or region that is an antisense strand or region (e.g, has at least 70, 80, 90, 95, 98, or 100% complementarity to a target nucleic acid) and an RNA strand or region that is a sense strand or region (e.g, has at least 70, 80, 90, 95, 98, or 100% identity to a target nucleic acid), and vice versa.

[0289] In various embodiments, the RNA/DNA hybrid is made in vitro using enzymatic or chemical synthetic methods such as those described herein or those described in WO 00/63364, filed Apr. 19, 2000, or U.S. Ser. No. 60/130,377, filed Apr. 21, 1999. In other embodiments, a DNA strand synthesized in vitro is complexed with an RNA strand made in vivo or in vitro before, after, or concurrent with the transformation of the DNA strand into the cell. In yet other embodiments, the dsRNA is a single circular nucleic acid containing a sense and an antisense region, or the dsRNA includes a circular nucleic acid and either a second circular nucleic acid or a linear nucleic acid (see, for example, WO 00/63364, filed Apr. 19, 2000, or U.S. Ser. No. 60/130,377, filed Apr. 21, 1999.) Exemplary circular nucleic acids include lariat structures in which the free 5' phosphoryl group of a nucleotide becomes linked to the 2' hydroxyl group of another nucleotide in a loop back fashion.

[0290] In other embodiments, the dsRNA includes one or more modified nucleotides in which the 2' position in the sugar contains a halogen (such as fluorine group) or contains an alkoxy group (such as a methoxy group) which increases the half-life of the dsRNA in vitro or in vivo compared to the corresponding dsRNA in which the corresponding 2' position contains a hydrogen or an hydroxyl group. In yet other

embodiments, the dsRNA includes one or more linkages between adjacent nucleotides other than a naturally-occurring phosphodiester linkage. Examples of such linkages include phosphoramide, phosphorothioate, and phosphorodithioate linkages. The dsRNAs may also be chemically modified nucleic acid molecules as taught in U.S. Pat. No. 6,673,661. In other embodiments, the dsRNA contains one or two capped strands, as disclosed, for example, by WO 00/63364, filed Apr. 19, 2000, or U.S. Ser. No. 60/130,377, filed Apr. 21, 1999.

[0291] In other embodiments, the dsRNA can be any of the at least partially dsRNA molecules disclosed in WO 00/63364, as well as any of the dsRNA molecules described in U.S. Provisional Application 60/399,998; and U.S. Provisional Application 60/419,532, and PCT/US2003/033466, the teaching of which is hereby incorporated by reference. Any of the dsRNAs may be expressed in vitro or in vivo using the methods described herein or standard methods, such as those described in WO 00/63364.

Occupancy

[0292] In certain embodiments, antisense compounds are not expected to result in cleavage or the target nucleic acid via RNase H or to result in cleavage or sequestration through the RISC pathway. In certain such embodiments, antisense activity may result from occupancy, wherein the presence of the hybridized antisense compound disrupts the activity of the target nucleic acid. In certain such embodiments, the antisense compound may be uniformly modified or may comprise a mix of modifications and/or modified and unmodified nucleosides.

Target Nucleic Acids, Target Regions and Nucleotide Sequences

[0293] Nucleotide sequences that encode AGT include, without limitation, the following: GENBANK Accession No. NM_000029.3 (incorporated herein as SEQ ID NO: 1), the complement of the nucleotides 24354000 to 24370100 of GENBANK Accession No. NT_167186.1 (incorporated herein as SEQ ID NO: 2), GENBANK Accession No. AK307978.1 (incorporated herein as SEQ ID NO: 3), GENBANK Accession No. AK303755.1 (incorporated herein as SEQ ID NO: 4), GENBANK Accession No. AK293507.1 (incorporated herein as SEQ ID NO: 5), and GENBANK Accession No. CR606672.1 (incorporated herein as SEQ ID NO: 6). In certain embodiments, an antisense compound described herein targets a nucleic acid sequence encoding AGT. In certain embodiments, an antisense compound described herein targets the sequence of any of SEQ ID NOs: 1-6.

[0294] It is understood that the sequence set forth in each SEQ ID NO in the examples contained herein is independent of any modification to a sugar moiety, an internucleoside linkage, or a nucleobase. As such, antisense compounds defined by a SEQ ID NO may comprise, independently, one or more modifications to a sugar moiety, an internucleoside linkage, or a nucleobase. Antisense compounds described by Isis Number (Isis No) indicate a combination of nucleobase sequence and motif.

[0295] In certain embodiments, a target region is a structurally defined region of the target nucleic acid. For example, a target region may encompass a 3' UTR, a 5' UTR, an exon, an intron, an exon/intron junction, a coding region, a translation initiation region, translation termination region, or other defined nucleic acid region. The structurally defined regions for AGT can be obtained by accession number from sequence databases such as NCBI and such information is incorporated herein by reference. In certain embodiments, a target region may encompass the sequence from a 5' target site of one target segment within the target region to a 3' target site of another target segment within the target region. In certain embodiments, a target region may encompass at least 8 consecutive nucleobases selected from within an antisense compound at least 8 consecutive nucleobases from the 5'-terminus of the antisense compound (the remaining nucleobases being a consecutive stretch the beginning immediately upstream of the 5'-terminus of the antisense compound which is specifically hybridizable to the target nucleic acid and continuing until the region contains about 8 to about 80 nucleobases). In certain embodiments, a target region may encompass at least 8 consecutive nucleobases selected from within an antisense compound at least 8 consecutive nucleobases from the 3'-terminus of the antisense compound (the remaining nucleobases being a consecutive stretch beginning immediately downstream of the 3'-terminus of the antisense compound which is specifically hybridizable to the target nucleic acid and continuing until the region contains about 8 to about 80 nucleobases). In certain embodiments, the target region comprises at least 8 consecutive nucleobases selected from any of SEQ ID NOs: 14-2051 and continues up to 80 nucleobases 5' or 3' of the 8 consecutive nucleobase sequence.

[0296] In certain embodiments, a "target segment" is a smaller, sub-portion of a target region within a nucleic acid. For example, a target segment can be the sequence of nucleotides of a target nucleic acid to which one or more antisense compound is targeted. "5' target site" refers to the 5'-most nucleotide of a target segment. "3' target site" refers to the 3'-most nucleotide of a target segment.

[0297] Targeting includes determination of at least one target segment to which an antisense compound hybridizes, such that a desired effect occurs. In certain embodiments, the desired effect is a reduction in mRNA target nucleic acid levels. In certain embodiments, the desired effect is reduction of levels of protein encoded by the target nucleic acid or a phenotypic change associated with the target nucleic acid.

[0298] A target region may contain one or more target segments. Multiple target segments within a target region may be overlapping. Alternatively, they may be non-overlapping. In certain embodiments, target segments within a target region are separated by no more than about 300 nucleotides. In certain embodiments, target segments within a target region are separated by a number of nucleotides that is, is about, is no more than, is no more than about, 250, 200, 150, 100, 90, 80, 70, 60, 50, 40, 30, 20, or 10 nucleotides on the target nucleic acid, or is a range defined by any two of the preceeding values. In certain embodiments, target segments within a target region are separated by no more than, or no more than about, 5 nucleotides on the target nucleic acid. In certain embodiments, target segments are contiguous. Contemplated are target regions defined by a range having a starting nucleic acid that is any of the 5' target sites or 3' target sites listed herein.

[0299] Suitable target segments may be found within a 5' UTR, a coding region, a 3' UTR, an intron, an exon, or an exon/intron junction. Target segments containing a start

codon or a stop codon are also suitable target segments. A suitable target segment may specifically exclude a certain structurally defined region such as the start codon or stop codon.

[0300] The determination of suitable target segments may include a comparison of the sequence of a target nucleic acid to other sequences throughout the genome. For example, the BLAST algorithm may be used to identify regions of similarity amongst different nucleic acids. This comparison can prevent the selection of antisense compound sequences that may hybridize in a non-specific manner to sequences other than a selected target nucleic acid (i.e., non-target or off-target sequences).

[0301] There may be variation in activity (e.g., as defined by percent reduction of target nucleic acid levels) of the antisense compounds within an active target region. In certain embodiments, reductions in AGT mRNA levels are indicative of inhibition of AGT expression. Reductions in levels of an AGT protein are also indicative of inhibition of AGT expression. Further, phenotypic changes are indicative of inhibition of AGT expression. For example, a decrease in fibrosis in tissues can be indicative of inhibition of AGT expression. In another example, an decrease in hypertension can be indicative of inhibition of AGT expression.

Hybridization

[0302] In some embodiments, hybridization occurs between an antisense compound disclosed herein and an AGT nucleic acid. The most common mechanism of hybridization involves hydrogen bonding (e.g., Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding) between complementary nucleobases of the nucleic acid molecules.

[0303] Hybridization can occur under varying conditions. Stringent conditions are sequence-dependent and are determined by the nature and composition of the nucleic acid molecules to be hybridized.

[0304] Methods of determining whether a sequence is specifically hybridizable to a target nucleic acid are well known in the art (Sambrook and Russell, Molecular Cloning: A Laboratory Manual, 3rd Ed., 2001). In certain embodiments, the antisense compounds provided herein are specifically hybridizable with an AGT nucleic acid.

Complementarity

[0305] An antisense compound and a target nucleic acid are complementary to each other when a sufficient number of nucleobases of the antisense compound can hydrogen bond with the corresponding nucleobases of the target nucleic acid, such that a desired effect will occur (e.g., antisense inhibition of a target nucleic acid, such as an AGT nucleic acid).

[0306] Non-complementary nucleobases between an antisense compound and an AGT nucleic acid may be tolerated provided that the antisense compound remains able to specifically hybridize to the AGT nucleic acid. Moreover, an antisense compound may hybridize over one or more segments of an AGT nucleic acid such that intervening or adjacent segments are not involved in the hybridization event (e.g., a loop structure, mismatch or hairpin structure).

[0307] In certain embodiments, the antisense compounds provided herein, or a specified portion thereof, are, or are at least 70%, at least 80%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% complementary to an AGT nucleic acid, a target region, target segment, or specified portion thereof. Percent complementarity of an antisense compound with a target nucleic acid can be determined using routine methods. For example, an antisense compound in which 18 of 20 nucleobases of the antisense compound are complementary to a target region, and would therefore specifically hybridize, would represent 90 percent complementarity. In this example, the remaining noncomplementary nucleobases may be clustered or interspersed with complementary nucleobases and need not be contiguous to each other or to complementary nucleobases. As such, an antisense compound which is 18 nucleobases in length having 4 (four) noncomplementary nucleobases which are flanked by two regions of complete complementarity with the target nucleic acid would have 77.8% overall complementarity with the target nucleic acid and would thus fall within the scope of the present invention.

[0308] Percent complementarity of an antisense compound with a region of a target nucleic acid can be determined routinely using BLAST programs (basic local alignment search tools) and PowerBLAST programs known in the art (Altschul et al., J. Mol. Biol., 1990, 215, 403 410; Zhang and Madden, Genome Res., 1997, 7, 649 656). Percent homology, sequence identity or complementarity, can be determined by, for example, the Gap program (Wisconsin Sequence Analysis Package, Version 8 for Unix, Genetics Computer Group, University Research Park, Madison Wis.), using default settings, which uses the algorithm of Smith and Waterman (Adv. Appl. Math., 1981, 2, 482 489).

[0309] In certain embodiments, the antisense compounds provided herein, or specified portions thereof, are fully complementary (i.e. 100% complementary) to a target nucleic acid, or specified portion thereof. For example, antisense compound may be fully complementary to an AGT nucleic acid, or a target region, or a target segment or target sequence thereof. As used herein, "fully complementary" means each nucleobase of an antisense compound is capable of precise base pairing with the corresponding nucleobases of a target nucleic acid. For example, a 20 nucleobase antisense compound is fully complementary to a target sequence that is 400 nucleobases long, so long as there is a corresponding 20 nucleobase portion of the target nucleic acid that is fully complementary to the antisense compound. Fully complementary can also be used in reference to a specified portion of the first and/or the second nucleic acid. For example, a 20 nucleobase portion of a 30 nucleobase antisense compound can be "fully complementary" to a target sequence that is 400 nucleobases long. The 20 nucleobase portion of the 30 nucleobase oligonucleotide is fully complementary to the target sequence if the target sequence has a corresponding 20 nucleobase portion wherein each nucleobase is complementary to the 20 nucleobase portion of the antisense compound. At the same time, the entire 30 nucleobase antisense compound may or may not be fully complementary to the target sequence, depending on whether the remaining 10 nucleobases of the antisense compound are also complementary to the target sequence.

[0310] The location of a non-complementary nucleobase may be at the 5' end or 3' end of the antisense compound. Alternatively, the non-complementary nucleobase or nucleobases may be at an internal position of the antisense

compound. When two or more non-complementary nucleobases are present, they may be contiguous (i.e. linked) or non-contiguous. In one embodiment, a non-complementary nucleobase is located in the wing segment of a gapmer antisense oligonucleotide.

[0311] In certain embodiments, antisense compounds that are, or are up to, 12, 13, 14, 15, 16, 17, 18, 19, or 20 nucleobases in length comprise no more than 4, no more than 3, no more than 2, or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as an AGT nucleic acid, or specified portion thereof.

[0312] In certain embodiments, antisense compounds that are, or are up to, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleobases in length comprise no more than 6, no more than 5, no more than 4, no more than 3, no more than 2, or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as an AGT nucleic acid, or specified portion thereof.

[0313] The antisense compounds provided herein also include those which are complementary to a portion of a target nucleic acid. As used herein, "portion" refers to a defined number of contiguous (i.e. linked) nucleobases within a region or segment of a target nucleic acid. A "portion" can also refer to a defined number of contiguous nucleobases of an antisense compound. In certain embodiments, the antisense compounds, are complementary to at least an 8 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least a 12 nucleobase portion of a target segment. In certain embodiments, the antisense compounds are complementary to at least a 15 nucleobase portion of a target segment. Also contemplated are antisense compounds that are complementary to at least a 9, at least a 10, at least an 11, at least a 12, at least a 13, at least a 14, at least a 15, at least a 16, at least a 17, at least an 18, at least a 19, at least a 20, or more nucleobase portion of a target segment, or a range defined by any two of these values.

Identity

[0314] The antisense compounds provided herein may also have a defined percent identity to a particular nucleotide sequence, SEQ ID NO, or compound represented by a specific Isis number, or portion thereof. As used herein, an antisense compound is identical to the sequence disclosed herein if it has the same nucleobase pairing ability. For example, a RNA which contains uracil in place of thymidine in a disclosed DNA sequence would be considered identical to the DNA sequence since both uracil and thymidine pair with adenine. Shortened and lengthened versions of the antisense compounds described herein as well as compounds having non-identical bases relative to the antisense compounds provided herein also are contemplated. The non-identical bases may be adjacent to each other or dispersed throughout the antisense compound. Percent identity of an antisense compound is calculated according to the number of bases that have identical base pairing relative to the sequence to which it is being compared.

[0315] In certain embodiments, the antisense compounds, or portions thereof, are at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100% identical to one or more of the antisense compounds or SEQ ID NOs, or a portion thereof, disclosed herein.

Modifications

[0316] A nucleoside is a base-sugar combination. The nucleobase (also known as base) portion of the nucleoside is normally a heterocyclic base moiety. Nucleotides are nucleosides that further include a phosphate group covalently linked to the sugar portion of the nucleoside. For those nucleosides that include a pentofuranosyl sugar, the phosphate group can be linked to the 2', 3' or 5' hydroxyl moiety of the sugar. Oligonucleotides are formed through the covalent linkage of adjacent nucleosides to one another, to form a linear polymeric oligonucleotide. Within the oligonucleotide structure, the phosphate groups are commonly referred to as forming the internucleoside linkages of the oligonucleotide.

[0317] Modifications to antisense compounds encompass substitutions or changes to internucleoside linkages, sugar moieties, or nucleobases. Modified antisense compounds are often preferred over native forms because of desirable properties such as, for example, enhanced cellular uptake, enhanced affinity for nucleic acid target, increased stability in the presence of nucleases, or increased inhibitory activity.

[0318] Chemically modified nucleosides may also be employed to increase the binding affinity of a shortened or truncated antisense oligonucleotide for its target nucleic acid. Consequently, comparable results can often be obtained with shorter antisense compounds that have such chemically modified nucleosides.

Modified Internucleoside Linkages

[0319] The naturally occuring internucleoside linkage of RNA and DNA is a 3' to 5' phosphodiester linkage. Antisense compounds having one or more modified, i.e. non-naturally occurring, internucleoside linkages are often selected over antisense compounds having naturally occurring internucleoside linkages because of desirable properties such as, for example, enhanced cellular uptake, enhanced affinity for target nucleic acids, and increased stability in the presence of nucleases.

[0320] Oligonucleotides having modified internucleoside linkages include internucleoside linkages that retain a phosphorus atom as well as internucleoside linkages that do not have a phosphorus atom. Representative phosphorus containing internucleoside linkages include, but are not limited to, phosphodiesters, phosphotriesters, methylphosphonates, phosphoramidate, and phosphorothioates. Methods of preparation of phosphorous-containing and non-phosphorous-containing linkages are well known.

[0321] In certain embodiments, antisense compounds targeted to an AGT nucleic acid comprise one or more modified internucleoside linkages. In certain embodiments, at least one of the modified internucleoside linkages are phosphorothioate linkages. In certain embodiments, each internucleoside linkage of an antisense compound is a phosphorothioate internucleoside linkage.

Modified Sugar Moieties

[0322] Antisense compounds of the invention can optionally contain one or more nucleosides wherein the sugar group has been modified. Such sugar modified nucleosides may impart enhanced nuclease stability, increased binding affinity, or some other beneficial biological property to the antisense compounds. In certain embodiments, nucleosides comprise chemically modified ribofuranose ring moieties.

Examples of chemically modified ribofuranose rings include without limitation, addition of substitutent groups (including 5' and 2' substituent groups, bridging of non-geminal ring atoms to form bicyclic nucleic acids (BNA), replacement of the ribosyl ring oxygen atom with S, N(R), or $C(R_1)(R_2)$ (R, R_1 and R_2 are each independently H, C_1-C_{12} alkyl or a protecting group) and combinations thereof. Examples of chemically modified sugars include 2'-F-5'-methyl substituted nucleoside (see PCT International Application WO 2008/101157 Published on Aug. 21, 2008 for other disclosed 5',2'-bis substituted nucleosides) or replacement of the ribosyl ring oxygen atom with S with further substitution at the 2'-position (see published U.S. Patent Application US2005-0130923, published on Jun. 16, 2005) or alternatively 5'-substitution of a BNA (see PCT International Application WO 2007/134181 Published on Nov. 22, 2007 wherein LNA is substituted with for example a 5'-methyl or a 5'-vinyl group).

[0323] Examples of nucleosides having modified sugar moieties include without limitation nucleosides comprising 5'-vinyl, 5'-methyl (R or S), 4'-S, 2'-F, 2'-OCH_3, 2'-OCH_2CH_3, 2'-OCH_2CH_2F and 2'-$O(CH_2)_2OCH_3$ substituent groups. The substituent at the 2' position can also be selected from allyl, amino, azido, thio, O-allyl, O—C_1-C_{10} alkyl, OCF_3, OCH_2F, $O(CH_2)_2SCH_3$, $O(CH_2)_2$—O—$N(R_m)(R_n)$, O—CH_2—C(=O)—$N(R_m)(R_n)$, and O—CH_2—C(=O)—$N(R_l)$—$(CH_2)_2$—$N(R_m)(R_n)$, where each R_l, R_m and R_n is, independently, H or substituted or unsubstituted C_1-C_{10} alkyl.

[0324] As used herein, "bicyclic nucleosides" refer to modified nucleosides comprising a bicyclic sugar moiety. Examples of bicyclic nucleic acids (BNAs) include without limitation nucleosides comprising a bridge between the 4' and the 2' ribosyl ring atoms. In certain embodiments, antisense compounds provided herein include one or more BNA nucleosides wherein the bridge comprises one of the formulas: 4'-(CH_2)—O-2' (LNA); 4'-(CH_2)—S-2'; 4'-$(CH_2)_2$—O-2' (ENA); 4'-$CH(CH_3)$—O-2' (cEt) and 4'-$CH(CH_2OCH_3)$—O-2' (and analogs thereof see U.S. Pat. No. 7,399,845, issued on Jul. 15, 2008); 4'-$C(CH_3)(CH_3)$—O-2' (and analogs thereof see PCT/US2008/068922 published as WO/2009/006478, published Jan. 8, 2009); 4'-CH_2—$N(OCH_3)$-2' (and analogs thereof see PCT/US2008/064591 published as WO/2008/150729, published Dec. 11, 2008); 4'-CH_2—O—$N(CH_3)$-2' (see published U.S. Patent Application US2004-0171570, published Sep. 2, 2004); 4'-CH_2—N(R)—O-2', wherein R is H, C_1-C_{12} alkyl, or a protecting group (see U.S. Pat. No. 7,427,672, issued on Sep. 23, 2008); 4'-CH_2—$C(H)(CH_3)$-2' (see Zhou et al., *J. Org. Chem.,* 2009, 74, 118-134); and 4'-CH_2—$C(=CH_2)$-2' (and analogs thereof see PCT/US2008/066154 published as WO 2008/154401, published on Dec. 8, 2008).

[0325] Further bicyclic nucleosides have been reported in published literature (see for example: Srivastava et al., *J Am. Chem. Soc.,* 2007, 129(26) 8362-8379; Frieden et al., *Nucleic Acids Research,* 2003, 21, 6365-6372; Elayadi et al., *Curr. Opinion Invens. Drugs,* 2001, 2, 558-561; Braasch et al., *Chem. Biol.,* 2001, 8, 1-7; Orum et al., *Curr. Opinion Mol. Ther.,* 2001, 3, 239-243; Wahlestedt et al., *Proc. Natl. Acad. Sci. U.S.A,* 2000, 97, 5633-5638; Singh et al., *Chem. Commun.,* 1998, 4, 455-456; Koshkin et al., *Tetrahedron,* 1998, 54, 3607-3630; Kumar et al., *Bioorg. Med. Chem. Lett.,* 1998, 8, 2219-2222; Singh et al., *J Org. Chem.,* 1998, 63, 10035-10039; U.S. Pat. Nos. 7,399,845; 7,053,207; 7,034,133; 6,794,499; 6,770,748; 6,670,461; 6,525,191; 6,268,490; U.S. Patent Publication Nos.: US2008-0039618; US2007-0287831; US2004-0171570; U.S. Patent Applications, Ser. Nos. 12/129,154; 61/099,844; 61/097,787; 61/086,231; 61/056,564; 61/026,998; 61/026,995; 60/989,574; International applications WO 2007/134181; WO 2005/021570; WO 2004/106356; WO 99/14226; and PCT International Applications Nos.: PCT/US2008/068922; PCT/US-2008/066154; and PCT/US2008/064591). Each of the foregoing bicyclic nucleosides can be prepared having one or more stereochemical sugar configurations including for example α-L-ribofuranose and β-D-ribofuranose (see PCT international application PCT/DK98/00393, published on Mar. 25, 1999 as WO 99/14226).

[0326] As used herein, "monocyclic nucleosides" refer to nucleosides comprising modified sugar moieties that are not bicyclic sugar moieties. In certain embodiments, the sugar moiety, or sugar moiety analogue, of a nucleoside may be modified or substituted at any position.

[0327] As used herein, "4'-2' bicyclic nucleoside" or "4' to 2' bicyclic nucleoside" refers to a bicyclic nucleoside comprising a furanose ring comprising a bridge connecting two carbon atoms of the furanose ring connects the 2' carbon atom and the 4' carbon atom of the sugar ring.

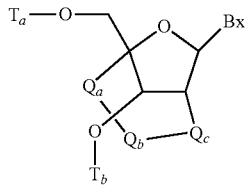
[0328] In certain embodiments, bicyclic sugar moieties of BNA nucleosides include, but are not limited to, compounds having at least one bridge between the 4' and the 2' carbon atoms of the pentofuranosyl sugar moiety including without limitation, bridges comprising 1 or from 1 to 4 linked groups independently selected from —$[C(R_a)(R_b)]_n$—, —$C(R_a)$=$C(R_b)$—, —$C(R_a)$=N—, —$C(=NR_a)$—, —C(=O)—, —C(=S)—, —O—, —$Si(R_a)_2$—, —$S(=O)_x$—, and —$N(R_a)$—; wherein: x is 0, 1, or 2; n is 1, 2, 3, or 4; each R_a and R_b is, independently, H, a protecting group, hydroxyl, C_1-C_{12} alkyl, substituted C_1-C_{12} alkyl, C_2-C_{12} alkenyl, substituted C_2-C_{12} alkenyl, C_2-C_{12} alkynyl, substituted C_2-C_{12} alkynyl, C_5-C_{20} aryl, substituted C_5-C_{20} aryl, heterocycle radical, substituted heterocycle radical, heteroaryl, substituted heteroaryl, C_5-C_7 alicyclic radical, substituted C_5-C_7 alicyclic radical, halogen, OJ_1, NJ_1J_2, SJ_1, N_3, $COOJ_1$, acyl (C(=O)—H), substituted acyl, CN, sulfonyl ($S(=O)_2$-J_1), or sulfoxyl (S(=O)-J_1); and each J_1 and J_2 is, independently, H, C_1-C_{12} alkyl, substituted C_1-C_{12} alkyl, C_2-C_{12} alkenyl, substituted C_2-C_{12} alkenyl, C_2-C_{12} alkynyl, substituted C_2-C_{12} alkynyl, C_5-C_{20} aryl, substituted C_5-C_{20} aryl, acyl (C(=O)—H), substituted acyl, a heterocycle radical, a substituted heterocycle radical, C_1-C_{12} aminoalkyl, substituted C_1-C_{12} aminoalkyl or a protecting group.

[0329] In certain embodiments, the bridge of a bicyclic sugar moiety is, —$[C(R_a)(R_b)]_n$—, —$[C(R_a)(R_b)]_n$—O—, —$C(R_aR_b)$—N(R)—O— or —$C(R_aR_b)$—O—N(R)—. In certain embodiments, the bridge is 4'-CH_2-2', 4'-$(CH_2)_2$-2', 4'-$(CH_2)_3$-2', 4'-CH_2—O-2', 4'-$(CH_2)_2$—O-2', 4'-CH_2—O—N(R)-2' and 4'-CH_2—N(R)—O-2'- wherein each R is, independently, H, a protecting group or C_1-C_{12} alkyl.

[0330] In certain embodiments, bicyclic nucleosides are further defined by isomeric configuration. For example, a nucleoside comprising a 4'-(CH_2)—O-2' bridge, may be in the α-L configuration or in the β-D configuration. Previously, α-L-methyleneoxy (4'-CH_2—O-2') BNA's have been incorporated into antisense oligonucleotides that showed antisense activity (Frieden et al., *Nucleic Acids Research,* 2003, 21, 6365-6372).

[0331] In certain embodiments, bicyclic nucleosides include those having a 4' to 2' bridge wherein such bridges include without limitation, α-L-4'-(CH_2)—O-2', β-D-4'-CH_2—O-2', 4'-(CH_2)$_2$—O-2', 4'-CH_2—O—N(R)-2', 4'-CH_2—N(R)—O-2', 4'-CH(CH_3)—O-2', 4'-CH_2—S-2', 4'-CH_2—N(R)-2', 4'-CH_2—CH(CH_3)-2', and 4'-(CH_2)$_3$-2', wherein R is H, a protecting group or C_1-C_{12} alkyl.

[0332] In certain embodiment, bicyclic nucleosides have the formula:



wherein:

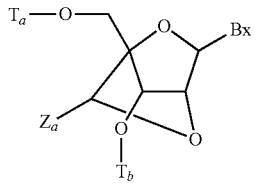
[0333] Bx is a heterocyclic base moiety;

[0334] -Q_a-Q_b-Q_c- is —CH_2—N(R_c)—CH_2—, —C(=O)—N(R_c)—CH_2—, —CH_2—O—N(R_c)—, —CH_2—N(R_c)—O— or —N(R_c)—O—CH_2;

[0335] R_c is C_1-C_{12} alkyl or an amino protecting group; and

[0336] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium.

[0337] In certain embodiments, bicyclic nucleosides have the formula:



wherein:

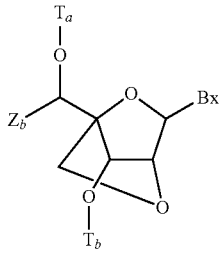
[0338] Bx is a heterocyclic base moiety;

[0339] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium;

[0340] Z_a is C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, substituted C_1-C_6 alkyl, substituted C_2-C_6 alkenyl, substituted C_2-C_6 alkynyl, acyl, substituted acyl, substituted amide, thiol or substituted thiol.

[0341] In one embodiment, each of the substituted groups, is, independently, mono or poly substituted with substituent groups independently selected from halogen, oxo, hydroxyl, OJ_c, NJ_eJ_d, SJ_c, N_3, $OC(=X)J_c$, and $NJ_eC(=X)NJ_cJ_d$, wherein each J_c, J_d and J_e is, independently, H, C_1-C_6 alkyl, or substituted C_1-C_6 alkyl and X is O or NJ_c.

[0342] In certain embodiments, bicyclic nucleosides have the formula:



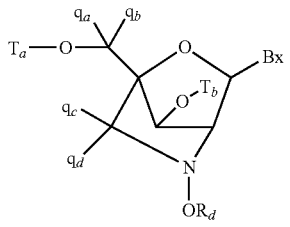
wherein:

[0343] Bx is a heterocyclic base moiety;

[0344] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium;

[0345] Z_b is C_1-C_6 alkyl, C_2-C_6 alkenyl, C_2-C_6 alkynyl, substituted C_1-C_6 alkyl, substituted C_2-C_6 alkenyl, 5 substituted C_2-C_6 alkynyl or substituted acyl (C(=O)—).

[0346] In certain embodiments, bicyclic nucleosides have the formula:



wherein:

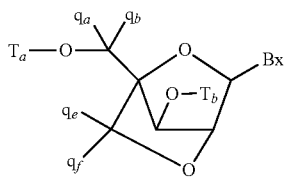
[0347] Bx is a heterocyclic base moiety;

[0348] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium;

[0349] R_d is C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl;

[0350] each q_a, q_b, q_c and q_d is, independently, H, halogen, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl, C_1-C_6 alkoxyl, substituted C_1-C_6 alkoxyl, acyl, substituted acyl, C_1-C_6 aminoalkyl or substituted C_1-C_6 aminoalkyl;

[0351] In certain embodiments, bicyclic nucleosides have the formula:



wherein:

[0352] Bx is a heterocyclic base moiety;

[0353] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium;

[0354] q_a, q_b, q_e and q_f are each, independently, hydrogen, halogen, C_1-C_{12} alkyl, substituted C_1-C_{12} alkyl, C_2-C_{12} alkenyl, substituted C_2-C_{12} alkenyl, C_2-C_{12} alkynyl, substituted C_2-C_{12} alkynyl, C_1-C_{12} alkoxy, substituted C_1-C_{12} alkoxy, OJ_j, SJ_j, SOJ_j, SO_2J_j, NJ_jJ_k, N_3, CN, $C(=O)OJ_j$, $C(=O)NJ_jJ_k$, $C(=O)J_j$, $O—C(=O)NJ_jJ_k$, $N(H)C(=NH)NJ_jJ_k$, $N(H)C(=O)NJ_jJ_k$ or $N(H)C(=S)NJ_jJ_k$;

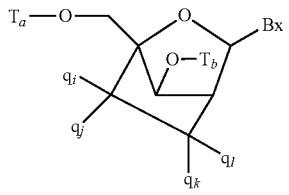
[0355] or q_e and q_f together are $=C(q_g)(q_h)$;

[0356] q_g and q_h are each, independently, H, halogen, C_1-C_{12} alkyl or substituted C_1-C_{12} alkyl.

[0357] The synthesis and preparation of adenine, cytosine, guanine, 5-methyl-cytosine, thymine and uracil bicyclic nucleosides having a 4'-CH_2—O-2' bridge, along with their oligomerization, and nucleic acid recognition properties have been described (Koshkin et al., *Tetrahedron,* 1998, 54, 3607-3630). The synthesis of bicyclic nucleosides has also been described in WO 98/39352 and WO 99/14226.

[0358] Analogs of various bicyclic nucleosides that have 4' to 2' bridging groups such as 4'-CH_2—O-2' and 4'-CH_2—S-2', have also been prepared (Kumar et al., *Bioorg. Med. Chem. Lett.,* 1998, 8, 2219-2222). Preparation of oligodeoxyribonucleotide duplexes comprising bicyclic nucleosides for use as substrates for nucleic acid polymerases has also been described (Wengel et al., WO 99/14226). Furthermore, synthesis of 2'-amino-BNA, a novel conformationally restricted high-affinity oligonucleotide analog has been described in the art (Singh et al., *J Org. Chem.,* 1998, 63, 10035-10039). In addition, 2'-amino- and 2'-methylamino-BNA's have been prepared and the thermal stability of their duplexes with complementary RNA and DNA strands has been previously reported.

[0359] In certain embodiments, bicyclic nucleosides have the formula:



wherein:

[0360] Bx is a heterocyclic base moiety;

[0361] T_a and T_b are each, independently H, a hydroxyl protecting group, a conjugate group, a reactive phosphorus group, a phosphorus moiety or a covalent attachment to a support medium;

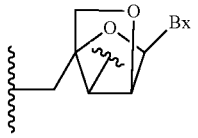
[0362] each q_i, q_j, q_k and q_l is, independently, H, halogen, C_1-C_{12} alkyl, substituted C_1-C_{12} alkyl, C_2-C_{12} alkenyl, substituted C_2-C_{12} alkenyl, C_2-C_{12} alkynyl, substituted C_2-C_{12} alkynyl, C_1-C_{12} alkoxyl, substituted C_1-C_{12} alkoxyl, OJ_j, SJ_j, SOJ_j, SO_2J_j, NJ_jJ_k, N_3, CN, $C(=O)OJ_j$, $C(=O)NJ_jJ_k$, $C(=O)J_j$, $O—C(=O)NJ_jJ_k$, $N(H)C(=NH)NJ_jJ_k$, $N(H)C(=O)NJ_jJ_k$ or $N(H)C(=S)NJ_jJ_k$; and

[0363] q_i and q_j or q_l and q_k together are $=C(q_g)(q_h)$, wherein q_g and q_h are each, independently, H, halogen, C_1-C_{12} alkyl or substituted C_1-C_{12} alkyl.

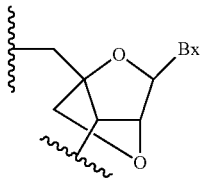
[0364] One carbocyclic bicyclic nucleoside having a 4'-$(CH_2)_3$-2' bridge and the alkenyl analog bridge 4'-CH=CH—CH_2-2' have been described (Frier et al., *Nucleic Acids Research,* 1997, 25(22), 4429-4443 and Albaek et al., *J Org. Chem.,* 2006, 71, 7731-7740). The synthesis and preparation of carbocyclic bicyclic nucleosides along with their oligomerization and biochemical studies have also been described (Srivastava et al., *J Am. Chem. Soc.* 2007, 129(26), 8362-8379).

[0365] In certain embodiments, bicyclic nucleosides include, but are not limited to, (A) α-L-methyleneoxy (4'-CH_2—O-2') BNA, (B) β-D-methyleneoxy (4'-CH_2—O-2') BNA, (C) ethyleneoxy (4'-$(CH_2)_2$—O-2') BNA, (D) aminooxy (4'-CH_2—O—N(R)-2') BNA, (E) oxyamino (4'-CH_2—N(R)—O-2') BNA, (F) methyl(methyleneoxy) (4'-$CH(CH_3)$—O-2') BNA (also referred to as constrained ethyl or cEt), (G) methylene-thio (4'-CH_2—S-2') BNA, (H) methylene-amino (4'-CH_2—N(R)-2') BNA, (I) methyl carbocyclic (4'-CH_2—$CH(CH_3)$-2') BNA, (J) propylene carbocyclic (4'-$(CH_2)_3$-2') BNA, and (K) vinyl BNA as depicted below.

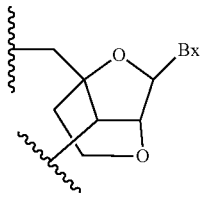
(A)



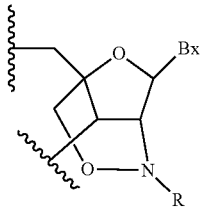
(B)



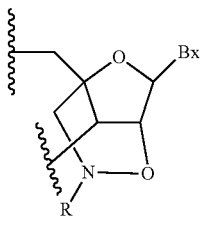
(C)



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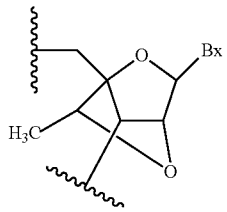


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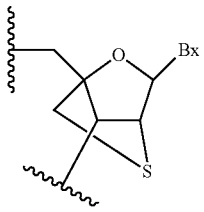


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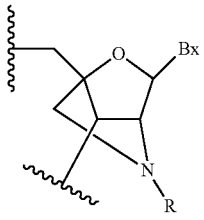
(F)



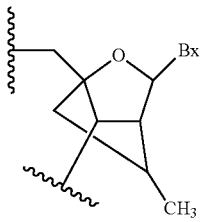
(G)



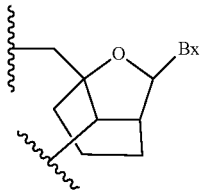
(H)



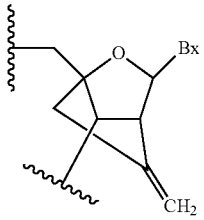
(I)



(J)

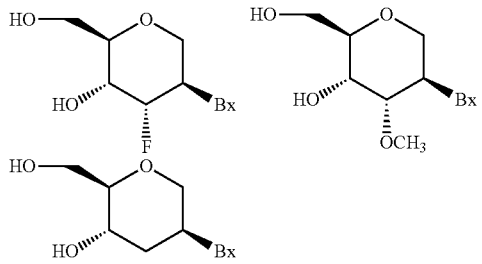


(K)

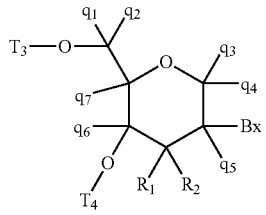


[0366] wherein Bx is the base moiety and R is, independently, H, a protecting group, C_1-C_6 alkyl or C_1-C_6 alkoxy.

[0367] As used herein, the term "modified tetrahydropyran nucleoside" or "modified THP nucleoside" means a nucleoside having a six-membered tetrahydropyran "sugar" substituted for the pentofuranosyl residue in normal nucleosides and can be referred to as a sugar surrogate. Modified THP nucleosides include, but are not limited to, what is referred to in the art as hexitol nucleic acid (HNA), anitol nucleic acid (ANA), manitol nucleic acid (MNA) (see Leumann, *Bioorg. Med. Chem.,* 2002, 10, 841-854) or fluoro HNA (F-HNA) having a tetrahydropyranyl ring system as illustrated below.



[0368] In certain embodiment, sugar surrogates are selected having the formula:



wherein:

[0369] Bx is a heterocyclic base moiety;

[0370] T_3 and T_4 are each, independently, an internucleoside linking group linking the tetrahydropyran nucleoside analog to the oligomeric compound or one of T_3 and T_4 is an internucleoside linking group linking the tetrahydropyran nucleoside analog to an oligomeric compound or oligonucleotide and the other of T_3 and T_4 is H, a hydroxyl protecting group, a linked conjugate group or a 5' or 3'-terminal group;

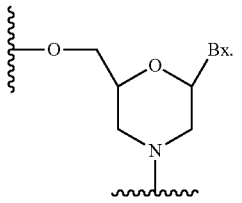
[0371] q_1, q_2, q_3, q_4, q_5, q_6 and q_7 are each independently, H, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl or substituted C_2-C_6 alkynyl; and

[0372] one of R_1 and R_2 is hydrogen and the other is selected from halogen, substituted or unsubstituted alkoxy, NJ_1J_2, SJ_1, N_3, $OC(=X)J_1$, $OC(=X)NJ_1J_2$, $NJ_3C(=X)NJ_1J_2$ and CN, wherein X is O, S or NJ_1 and each J_1, J_2 and J_3 is, independently, H or C_1-C_6 alkyl.

[0373] In certain embodiments, q_1, q_2, q_3, q_4, q_5, q_6 and q_7 are each H. In certain embodiments, at least one of q_1, q_2, q_3, q_4, q_5, q_6 and q_7 is other than H. In certain embodiments, at least one of q_1, q_2, q_3, q_4, q_5, q_6 and q_7 is methyl. In certain embodiments, THP nucleosides are provided wherein one of R_1 and R_2 is F. In certain embodiments, R_1 is fluoro and R_2 is H; R_1 is methoxy and R_2 is H, and R_1 is methoxyethoxy and R_2 is H.

[0374] In certain embodiments, sugar surrogates comprise rings having more than 5 atoms and more than one heteroatom. For example nucleosides comprising morpholino sugar moieties and their use in oligomeric compounds has been reported (see for example: Braasch et al., *Biochemistry,* 2002, 41, 4503-4510; and U.S. Pat. Nos. 5,698,685; 5,166,

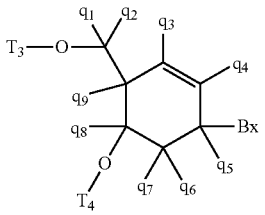
315; 5,185,444; and 5,034,506). As used here, the term "morpholino" means a sugar surrogate having the following formula:



In certain embodiments, morpholinos may be modified, for example by adding or altering various substituent groups from the above morpholino structure. Such sugar surrogates are referred to herein as "modified morpholinos."

[0375] Combinations of modifications are also provided without limitation, such as 2'-F-5'-methyl substituted nucleosides (see PCT International Application WO 2008/101157 published on Aug. 21, 2008 for other disclosed 5', 2'-bis substituted nucleosides) and replacement of the ribosyl ring oxygen atom with S and further substitution at the 2'-position (see published U.S. Patent Application US2005-0130923, published on Jun. 16, 2005) or alternatively 5'-substitution of a bicyclic nucleic acid (see PCT International Application WO 2007/134181, published on Nov. 22, 2007 wherein a 4'-CH_2—O-2' bicyclic nucleoside is further substituted at the 5' position with a 5'-methyl or a 5'-vinyl group). The synthesis and preparation of carbocyclic bicyclic nucleosides along with their oligomerization and biochemical studies have also been described (see, e.g., Srivastava et al., *J. Am. Chem. Soc.* 2007, 129(26), 8362-8379).

[0376] In certain embodiments, antisense compounds comprise one or more modified cyclohexenyl nucleosides, which is a nucleoside having a six-membered cyclohexenyl in place of the pentofuranosyl residue in naturally occurring nucleosides. Modified cyclohexenyl nucleosides include, but are not limited to those described in the art (see for example commonly owned, published PCT Application WO 2010/036696, published on Apr. 10, 2010, Robeyns et al., *J. Am. Chem. Soc.,* 2008, 130(6), 1979-1984; Horváth et al., *Tetrahedron Letters,* 2007, 48, 3621-3623; Nauwelaerts et al., *J Am. Chem. Soc.,* 2007, 129(30), 9340-9348; Gu et al., *Nucleosides, Nucleotides & Nucleic Acids,* 2005, 24(5-7), 993-998; Nauwelaerts et al., *Nucleic Acids Research,* 2005, 33(8), 2452-2463; Robeyns et al., *Acta Crystallographica, Section F: Structural Biology and Crystallization Communications,* 2005, F61(6), 585-586; Gu et al., *Tetrahedron,* 2004, 60(9), 2111-2123; Gu et al., *Oligonucleotides,* 2003, 13(6), 479-489; Wang et al., *J. Org. Chem.,* 2003, 68, 4499-4505; Verbeure et al., *Nucleic Acids Research,* 2001, 29(24), 4941-4947; Wang et al., *J. Org. Chem.,* 2001, 66, 8478-82; Wang et al., *Nucleosides, Nucleotides & Nucleic Acids,* 2001, 20(4-7), 785-788; Wang et al., *J Am. Chem.,* 2000, 122, 8595-8602; Published PCT application, WO 06/047842; and Published PCT Application WO 01/049687; the text of each is incorporated by reference herein, in their entirety). Certain modified cyclohexenyl nucleosides have Formula X.



X

[0377] wherein independently for each of said at least one cyclohexenyl nucleoside analog of Formula X:

[0378] Bx is a heterocyclic base moiety;

[0379] T_3 and T_4 are each, independently, an internucleoside linking group linking the cyclohexenyl nucleoside analog to an antisense compound or one of T_3 and T_4 is an internucleoside linking group linking the tetrahydropyran nucleoside analog to an antisense compound and the other of T_3 and T_4 is H, a hydroxyl protecting group, a linked conjugate group, or a 5'- or 3'-terminal group; and

[0380] q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8 and q_9 are each, independently, H, C_1-C_6 alkyl, substituted C_1-C_6 alkyl, C_2-C_6 alkenyl, substituted C_2-C_6 alkenyl, C_2-C_6 alkynyl, substituted C_2-C_6 alkynyl or other sugar substituent group.

[0381] Many other monocyclic, bicyclic and tricyclic ring systems are known in the art and are suitable as sugar surrogates that can be used to modify nucleosides for incorporation into oligomeric compounds as provided herein (see for example review article: Leumann, Christian J. *Bioorg. & Med. Chem.,* 2002, 10, 841-854). Such ring systems can undergo various additional substitutions to further enhance their activity.

[0382] As used herein, "2'-modified sugar" means a furanosyl sugar modified at the 2' position. In certain embodiments, such modifications include substituents selected from: a halide, including, but not limited to substituted and unsubstituted alkoxy, substituted and unsubstituted thioalkyl, substituted and unsubstituted amino alkyl, substituted and unsubstituted alkyl, substituted and unsubstituted allyl, and substituted and unsubstituted alkynyl. In certain embodiments, 2' modifications are selected from substituents including, but not limited to: $O[(CH_2)_nO]_mCH_3$, $O(CH_2)_nNH_2$, $O(CH_2)_nCH_3$, $O(CH_2)_nF$, $O(CH_2)_nONH_2$, $OCH_2C(=O)N(H)CH_3$, and $O(CH_2)_nON[(CH_2)_nCH_3]_2$, where n and m are from 1 to about 10. Other 2'-substituent groups can also be selected from: C_1-C_{12} alkyl, substituted alkyl, alkenyl, alkynyl, alkaryl, aralkyl, O-alkaryl or O-aralkyl, SH, SCH_3, OCN, Cl, Br, CN, F, CF_3, OCF_3, $SOCH_3$, SO_2CH_3, ONO_2, NO_2, N_3, NH_2, heterocycloalkyl, heterocycloalkaryl, aminoalkylamino, polyalkylamino, substituted silyl, an RNA cleaving group, a reporter group, an intercalator, a group for improving pharmacokinetic properties, or a group for improving the pharmacodynamic properties of an antisense compound, and other substituents having similar properties. In certain embodiments, modified nucleosides comprise a 2'-MOE side chain (Baker et al., *J Biol. Chem.,* 1997, 272, 11944-12000). Such 2'-MOE substitution have been described as having improved binding affinity compared to unmodified nucleosides and to other modified nucleosides, such as 2'-O-methyl, O-propyl, and O-aminopropyl. Oligonucleotides having the 2'-MOE substituent also have been shown to be antisense inhibitors of gene expression with promising features for in vivo use

(Martin, *Helv. Chim. Acta,* 1995, 78, 486-504; Altmann et al., *Chimia,* 1996, 50, 168-176; Altmann et al., *Biochem. Soc. Trans.,* 1996, 24, 630-637; and Altmann et al., *Nucleosides Nucleotides,* 1997, 16, 917-926).

[0383] As used herein, "2'-modified" or "2'-substituted" refers to a nucleoside comprising a sugar comprising a substituent at the 2' position other than H or OH. 2'-modified nucleosides, include, but are not limited to, nucleosides with non-bridging 2'substituents, such as allyl, amino, azido, thio, O-allyl, O—C_1-C_{10} alkyl, —OCF_3, O—$(CH_2)_2$—O—CH_3, 2'-O$(CH_2)_2SCH_3$, O—$(CH_2)_2$—O—$N(R_m)(R_n)$, or O—CH_2—C(=O)—$N(R_m)(R_n)$, where each R_m and R_n is, independently, H or substituted or unsubstituted C_1-C_{10} alkyl. 2'-modified nucleosides may further comprise other modifications, for example at other positions of the sugar and/or at the nucleobase.

[0384] As used herein, "2'-F" refers to a nucleoside comprising a sugar comprising a fluoro group at the 2' position of the sugar ring.

[0385] As used herein, "2'-OMe" or "2'-OCH_3", "2'-O-methyl" or "2'-methoxy" each refers to a nucleoside comprising a sugar comprising an —OCH_3 group at the 2' position of the sugar ring.

[0386] As used herein, "MOE" or "2'-MOE" or "2'-$OCH_2CH_2OCH_3$" or "2'-O-methoxyethyl" each refers to a nucleoside comprising a sugar comprising a —$OCH_2CH_2OCH_3$ group at the 2' position of the sugar ring.

[0387] Methods for the preparations of modified sugars are well known to those skilled in the art. Some representative U.S. patents that teach the preparation of such modified sugars include without limitation, U.S. Pat. Nos. 4,981,957; 5,118,800; 5,319,080; 5,359,044; 5,393,878; 5,446,137; 5,466,786; 5,514,785; 5,519,134; 5,567,811; 5,576,427; 5,591,722; 5,597,909; 5,610,300; 5,627,053; 5,639,873; 5,646,265; 5,670,633; 5,700,920; 5,792,847 and 6,600,032 and International Application PCT/US2005/019219, filed Jun. 2, 2005 and published as WO 2005/121371 on Dec. 22, 2005, and each of which is herein incorporated by reference in its entirety.

[0388] As used herein, "oligonucleotide" refers to a compound comprising a plurality of linked nucleosides. In certain embodiments, one or more of the plurality of nucleosides is modified. In certain embodiments, an oligonucleotide comprises one or more ribonucleosides (RNA) and/or deoxyribonucleosides (DNA).

[0389] In nucleotides having modified sugar moieties, the nucleobase moieties (natural, modified or a combination thereof) are maintained for hybridization with an appropriate nucleic acid target.

[0390] In certain embodiments, antisense compounds comprise one or more nucleosides having modified sugar moieties. In certain embodiments, the modified sugar moiety is 2'-MOE. In certain embodiments, the 2'-MOE modified nucleosides are arranged in a gapmer motif. In certain embodiments, the modified sugar moiety is a bicyclic nucleoside having a (4'-CH(CH_3)—O-2') bridging group. In certain embodiments, the (4'-CH(CH_3)—O-2') modified nucleosides are arranged throughout the wings of a gapmer motif.

Modified Nucleobases

[0391] Nucleobase (or base) modifications or substitutions are structurally distinguishable from, yet functionally interchangeable with, naturally occurring or synthetic unmodified nucleobases. Both natural and modified nucleobases are capable of participating in hydrogen bonding. Such nucleobase modifications may impart nuclease stability, binding affinity or some other beneficial biological property to antisense compounds. Modified nucleobases include synthetic and natural nucleobases such as, for example, 5-methylcytosine (5-me-C). Certain nucleobase substitutions, including 5-methylcytosine substitutions, are particularly useful for increasing the binding affinity of an antisense compound for a target nucleic acid. For example, 5-methylcytosine substitutions have been shown to increase nucleic acid duplex stability by 0.6-1.2° C. (Sanghvi, Y. S., Crooke, S. T. and Lebleu, B., eds., *Antisense Research and Applications*, CRC Press, Boca Raton, 1993, pp. 276-278).

[0392] Additional unmodified nucleobases include 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-halouracil and cytosine, 5-propynyl (—C≡C—CH_3) uracil and cytosine and other alkynyl derivatives of pyrimidine bases, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl and other 8-substituted adenines and guanines, 5-halo particularly 5-bromo, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine and 7-methyladenine, 2-F-adenine, 2-amino-adenine, 8-azaguanine and 8-azaadenine, 7-deazaguanine and 7-deazaadenine and 3-deazaguanine and 3-deazaadenine.

[0393] Heterocyclic base moieties may also include those in which the purine or pyrimidine base is replaced with other heterocycles, for example 7-deaza-adenine, 7-deazaguanosine, 2-aminopyridine and 2-pyridone. Nucleobases that are particularly useful for increasing the binding affinity of antisense compounds include 5-substituted pyrimidines, 6-azapyrimidines and N-2, N-6 and 0-6 substituted purines, including 2 aminopropyladenine, 5-propynyluracil and 5-propynylcytosine.

[0394] In certain embodiments, antisense compounds targeted to an AGT nucleic acid comprise one or more modified nucleobases. In certain embodiments, gap-widened antisense oligonucleotides targeted to an AGT nucleic acid comprise one or more modified nucleobases. In certain embodiments, at least one of the modified nucleobases is 5-methylcytosine. In certain embodiments, each cytosine is a 5-methylcytosine.

Compositions and Methods for Formulating Pharmaceutical Compositions

[0395] Antisense oligonucleotides may be admixed with pharmaceutically acceptable active or inert substance for the preparation of pharmaceutical compositions or formulations. Compositions and methods for the formulation of pharmaceutical compositions are dependent upon a number of criteria, including, but not limited to, route of administration, extent of disease, or dose to be administered.

[0396] Antisense compound targeted to an AGT nucleic acid can be utilized in pharmaceutical compositions by combining the antisense compound with a suitable pharmaceutically acceptable diluent or carrier. A pharmaceutically acceptable diluent includes water e.g., water-for-injection (WFI). A pharmaceutically acceptable diluent includes saline e.g., phosphate-buffered saline (PBS). Water or saline is a diluent suitable for use in compositions to be delivered

parenterally. Accordingly, in one embodiment, employed in the methods described herein is a pharmaceutical composition comprising an antisense compound targeted to an AGT nucleic acid and a pharmaceutically acceptable diluent. In certain embodiments, the pharmaceutically acceptable diluent is water or saline. In certain embodiments, the antisense compound is an antisense oligonucleotide.

[0397] Pharmaceutical compositions comprising antisense compounds encompass any pharmaceutically acceptable salts, esters, or salts of such esters, or any other oligonucleotide which, upon administration to an animal, including a human, is capable of providing (directly or indirectly) the biologically active metabolite or residue thereof. Accordingly, for example, the disclosure herein is also drawn to pharmaceutically acceptable salts of antisense compounds, prodrugs, pharmaceutically acceptable salts of such prodrugs, and other bioequivalents. Suitable pharmaceutically acceptable salts include, but are not limited to, sodium and potassium salts.

[0398] Pharmaceutically acceptable salts of the compounds described herein may be prepared by methods well-known in the art. For a review of pharmaceutically acceptable salts, see Stahl and Wermuth, Handbook of Pharmaceutical Salts: Properties, Selection and Use (Wiley-VCH, Weinheim, Germany, 2002). Sodium salts of antisense oligonucleotides are useful and are well accepted for therapeutic administration to humans. Accordingly, in one embodiment the compounds described herein are in the form of a sodium salt.

[0399] A prodrug can include the incorporation of additional nucleosides at one or both ends of an antisense compound which are cleaved by endogenous nucleases within the body, to form the active antisense compound.

Dosing

[0400] In certain embodiments, pharmaceutical compositions are administered according to a dosing regimen (e.g., dose, dose frequency, and duration) wherein the dosing regimen can be selected to achieve a desired effect. The desired effect can be, for example, reduction of AGT or the prevention, reduction, amelioration or slowing the progression of a disease, disorder or condition associated with AGT.

[0401] In certain embodiments, the variables of the dosing regimen are adjusted to result in a desired concentration of pharmaceutical composition in a subject. "Concentration of pharmaceutical composition" as used with regard to dose regimen can refer to the compound, oligonucleotide, or active ingredient of the pharmaceutical composition. For example, in certain embodiments, dose and dose frequency are adjusted to provide a tissue concentration or plasma concentration of a pharmaceutical composition at an amount sufficient to achieve a desired effect.

[0402] Dosing is dependent on severity and responsiveness of the disease state to be treated, with the course of treatment lasting from several days to several months, or until a cure is effected or a diminution of the disease state is achieved. Dosing is also dependent on drug potency and metabolism. In certain embodiments, dosage is from 0.01 μg to 100 mg per kg of body weight, or within a range of 0.001 mg to 1000 mg dosing, and may be given once or more daily, weekly, biweekly, monthly, quarterly, semi-annually or yearly, or even once every 2 to 20 years. Following successful treatment, it may be desirable to have the patient undergo maintenance therapy to prevent the recurrence of the disease state, wherein the oligonucleotide is administered in maintenance doses, ranging from 0.01 μg to 100 mg per kg of body weight, once or more daily, to once every 20 years or ranging from 0.001 mg to 1000 mg dosing.

Administration

[0403] The compounds or pharmaceutical compositions of the present invention can be administered in a number of ways depending upon whether local or systemic treatment is desired and upon the area to be treated. Administration can be inhaled (i.e., pulmonary), enteral (i.e., enteric), parenteral or topical.

[0404] In certain embodiments, the compounds and compositions as described herein are administered parenterally. Parenteral administration includes, but is not limited to, intravenous, intra-arterial, subcutaneous, intraperitoneal, intraocular, intramuscular, intracranial, intrathecal, intramedullary, intraventricular or intratumoral injection or infusion. Parenteral administration also includes intranasal administration.

[0405] In certain embodiments, parenteral administration is by infusion. Infusion can be chronic or continuous or short or intermittent. In certain embodiments, infused pharmaceutical agents are delivered with a pump.

[0406] In certain embodiments, parenteral administration is by injection. The injection can be delivered with a syringe or a pump. In certain embodiments, the injection is a bolus injection. In certain embodiments, the injection is administered directly to a tissue or organ.

[0407] In certain embodiments, formulations for parenteral administration can include sterile aqueous solutions which can also contain buffers, diluents and other suitable additives such as, but not limited to, penetration enhancers, carrier compounds and other pharmaceutically acceptable carriers or excipients.

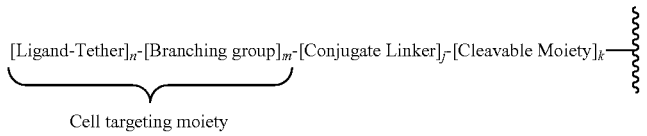
[0408] In certain embodiments, the compounds and compositions as described herein are administered enterally. Enteric administration includes, but is not limited to, oral, transmucosal, intestinal or rectal (e.g., suppository, enema). In certain embodiments, formulations for enteral administration of the compounds or compositions can include, but is not limited to, pharmaceutical carriers, excipients, powders or granules, microparticulates, nanoparticulates, suspensions or solutions in water or non-aqueous media, capsules, gel capsules, sachets, tablets or minitablets. Thickeners, flavoring agents, diluents, emulsifiers, dispersing aids or binders can be desirable. In certain embodiments, enteral formulations are those in which compounds provided herein are administered in conjunction with one or more penetration enhancers, surfactants and chelators.

[0409] In certain embodiments, administration includes pulmonary administration. In certain embodiments, pulmonary administration comprises delivery of aerosolized oligonucleotide to the lung of a subject by inhalation. Following inhalation by a subject of aerosolized oligonucleotide, oligonucleotide distributes to cells of both normal and inflamed lung tissue, including alveolar macrophages, eosinophils, epithelium, blood vessel endothelium, and bronchiolar epithelium. A suitable device for the delivery of a pharmaceutical composition comprising a modified oligonucleotide includes, but is not limited to, a standard nebulizer device. Additional suitable devices include dry powder inhalers or metered dose inhalers.

[0410] In certain embodiments, pharmaceutical compositions are administered to achieve local rather than systemic exposures. For example, pulmonary administration delivers a pharmaceutical composition to the lung, with minimal systemic exposure.

Conjugated Antisense Compounds

[0411] In certain embodiments, the oligonucleotides or oligomeric compounds as provided herein are modified by covalent attachment of one or more conjugate groups. In general, conjugate groups modify one or more properties of the attached oligonucleotide or oligomeric compound including but not limited to pharmacodynamics, pharmacokinetics, stability, binding, absorption, cellular distribution, cellular uptake, charge and clearance. As used herein, "conjugate group" means a radical group comprising a group of atoms that are attached to an oligonucleotide or oligomeric compound. In general, conjugate groups modify one or more properties of the compound to which they are attached, including, but not limited to pharmacodynamic, pharmacokinetic, binding, absorption, cellular distribution, cellular uptake, charge and/or clearance properties. Conjugate groups are routinely used in the chemical arts and can include a conjugate linker that covalently links the conjugate group to an oligonucleotide or oligomeric compound. In certain embodiments, conjugate groups include a cleavable moiety that covalently links the conjugate group to an oligonucleotide or oligomeric compound. In certain embodiments, conjugate groups include a conjugate linker and a cleavable moiety to covalently link the conjugate group to an oligonucleotide or oligomeric compound. In certain embodiments, a conjugate group has the general formula:



[0412] wherein n is from 1 to about 3, m is 0 when n is 1 or m is 1 when n is 2 or 3, j is 1 or 0, k is 1 or 0 and the sum of j and k is at least one.

[0413] In certain embodiments, n is 1, j is 1 and k is 0. In certain embodiments, n is 1, j is 0 and k is 1. In certain embodiments, n is 1, j is 1 and k is 1. In certain embodiments, n is 2, j is 1 and k is 0. In certain embodiments, n is 2, j is 0 and k is 1. In certain embodiments, n is 2, j is 1 and k is 1. In certain embodiments, n is 3, j is 1 and k is 0. In certain embodiments, n is 3, j is 0 and k is 1. In certain embodiments, n is 3, j is 1 and k is 1.

[0414] Conjugate groups are shown herein as radicals, providing a bond for forming covalent attachment to an oligomeric compound such as an oligonucleotide. In certain embodiments, the point of attachment on the oligomeric compound is at the 3'-terminal nucleoside or modified nucleoside. In certain embodiments, the point of attachment on the oligomeric compound is the 3'-oxygen atom of the 3'-hydroxyl group of the 3' terminal nucleoside or modified nucleoside. In certain embodiments, the point of attachment on the oligomeric compound is at the 5'-terminal nucleoside or modified nucleoside. In certain embodiments the point of attachment on the oligomeric compound is the 5'-oxygen atom of the 5'-hydroxyl group of the 5'-terminal nucleoside or modified nucleoside. In certain embodiments, the point of attachment on the oligomeric compound is at any reactive site on a nucleoside, a modified nucleoside or an internucleoside linkage.

[0415] As used herein, "cleavable moiety" and "cleavable bond" mean a cleavable bond or group of atoms that is capable of being split or cleaved under certain physiological conditions. In certain embodiments, a cleavable moiety is a cleavable bond. In certain embodiments, a cleavable moiety comprises a cleavable bond. In certain embodiments, a cleavable moiety is a group of atoms. In certain embodiments, a cleavable moiety is selectively cleaved inside a cell or sub-cellular compartment, such as a lysosome. In certain embodiments, a cleavable moiety is selectively cleaved by endogenous enzymes, such as nucleases. In certain embodiments, a cleavable moiety comprises a group of atoms having one, two, three, four, or more than four cleavable bonds.

[0416] In certain embodiments, conjugate groups comprise a cleavable moiety. In certain such embodiments, the cleavable moiety covalently attaches the oligomeric compound to the conjugate linker. In certain such embodiments, the cleavable moiety covalently attaches the oligomeric compound to the cell-targeting moiety.

[0417] In certain embodiments, a cleavable bond is selected from among: an amide, a polyamide, an ester, an ether, one or both esters of a phosphodiester, a phosphate ester, a carbamate, a di-sulfide, or a peptide. In certain embodiments, a cleavable bond is one of the esters of a phosphodiester. In certain embodiments, a cleavable bond is one or both esters of a phosphodiester. In certain embodiments, the cleavable moiety is a phosphodiester linkage between an oligomeric compound and the remainder of the conjugate group. In certain embodiments, the cleavable moiety comprises a phosphodiester linkage that is located between an oligomeric compound and the remainder of the conjugate group. In certain embodiments, the cleavable moiety comprises a phosphate or phosphodiester. In certain embodiments, the cleavable moiety is attached to the conjugate linker by either a phosphodiester or a phosphorothioate linkage. In certain embodiments, the cleavable moiety is attached to the conjugate linker by a phosphodiester linkage. In certain embodiments, the conjugate group does not include a cleavable moiety.

[0418] In certain embodiments, the cleavable moiety is a cleavable nucleoside or a modified nucleoside. In certain embodiments, the nucleoside or modified nucleoside comprises an optionally protected heterocyclic base selected from a purine, substituted purine, pyrimidine or substituted pyrimidine. In certain embodiments, the cleavable moiety is a nucleoside selected from uracil, thymine, cytosine, 4-N-benzoylcytosine, 5-methylcytosine, 4-N-benzoyl-5-methylcytosine, adenine, 6-N-benzoyladenine, guanine and 2-N-isobutyrylguanine.

[0419] In certain embodiments, the cleavable moiety is 2'-deoxy nucleoside that is attached to either the 3' or 5'-terminal nucleoside of an oligomeric compound by a phosphodiester linkage and covalently attached to the remainder of the conjugate group by a phosphodiester or phosphorothioate linkage. In certain embodiments, the cleavable moiety is 2'-deoxy adenosine that is attached to either the 3' or 5'-terminal nucleoside of an oligomeric compound by a phosphodiester linkage and covalently attached to the remainder of the conjugate group by a phosphodiester or phosphorothioate linkage. In certain embodiments, the cleavable moiety is 2'-deoxy adenosine that is attached to the 3'-oxygen atom of the 3'-hydroxyl group of the 3'-terminal nucleoside or modified nucleoside by a phosphodiester linkage. In certain embodiments, the cleavable moiety is 2'-deoxy adenosine that is attached to the 5'-oxygen atom of the 5'-hydroxyl group of the 5'-terminal nucleoside or modified nucleoside by a phosphodiester linkage. In certain embodiments, the cleavable moiety is attached to a 2'-position of a nucleoside or modified nucleoside of an oligomeric compound.

[0420] As used herein, "conjugate linker" in the context of a conjugate group means a portion of a conjugate group comprising any atom or group of atoms that covalently link the cell-targeting moiety to the oligomeric compound either directly or through the cleavable moiety. In certain embodiments, the conjugate linker comprises groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether (—S—) and hydroxylamino (—O—N(H)—). In certain embodiments, the conjugate linker comprises groups selected from alkyl, amino, oxo, amide and ether groups. In certain embodiments, the conjugate linker comprises groups selected from alkyl and amide groups. In certain embodiments, the conjugate linker comprises groups selected from alkyl and ether groups. In certain embodiments, the conjugate linker comprises at least one phosphorus linking group. In certain embodiments, the conjugate linker comprises at least one phosphodiester group. In certain embodiments, the conjugate linker includes at least one neutral linking group.

[0421] In certain embodiments, the conjugate linker is covalently attached to the oligomeric compound. In certain embodiments, the conjugate linker is covalently attached to the oligomeric compound and the branching group. In certain embodiments, the conjugate linker is covalently attached to the oligomeric compound and a tethered ligand. In certain embodiments, the conjugate linker is covalently attached to the cleavable moiety. In certain embodiments, the conjugate linker is covalently attached to the cleavable moiety and the branching group. In certain embodiments, the conjugate linker is covalently attached to the cleavable moiety and a tethered ligand. In certain embodiments, the conjugate linker includes one or more cleavable bonds. In certain embodiments, the conjugate group does not include a conjugate linker.

[0422] As used herein, "branching group" means a group of atoms having at least 3 positions that are capable of forming covalent linkages to two or more tether-ligands and the remainder of the conjugate group. In general a branching group provides a plurality of reactive sites for connecting tethered ligands to the oligomeric compound through the conjugate linker and/or the cleavable moiety. In certain embodiments, the branching group comprises groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether and hydroxylamino groups. In certain embodiments, the branching group comprises a branched aliphatic group comprising groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether and hydroxylamino groups. In certain such embodiments, the branched aliphatic group comprises groups selected from alkyl, amino, oxo, amide and ether groups. In certain such embodiments, the branched aliphatic group comprises groups selected from alkyl, amino and ether groups. In certain such embodiments, the branched aliphatic group comprises groups selected from alkyl and ether groups. In certain embodiments, the branching group comprises a mono or polycyclic ring system.

[0423] In certain embodiments, the branching group is covalently attached to the conjugate linker. In certain embodiments, the branching group is covalently attached to the cleavable moiety. In certain embodiments, the branching group is covalently attached to the conjugate linker and each of the tethered ligands. In certain embodiments, the branching group comprises one or more cleavable bond. In certain embodiments, the conjugate group does not include a branching group.

[0424] In certain embodiments, conjugate groups as provided herein include a cell-targeting moiety that has at least one tethered ligand. In certain embodiments, the cell-targeting moiety comprises two tethered ligands covalently attached to a branching group. In certain embodiments, the cell-targeting moiety comprises three tethered ligands covalently attached to a branching group.

[0425] As used herein, "tether" means a group of atoms that connect a ligand to the remainder of the conjugate group. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, substituted alkyl, ether, thioether, disulfide, amino, oxo, amide, phosphodiester and polyethylene glycol groups in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, ether, thioether, disulfide, amino, oxo, amide and polyethylene glycol groups in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, substituted alkyl, phosphodiester, ether and amino, oxo, amide groups in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, ether and amino, oxo, amide groups in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, amino and oxo groups in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl and oxo groups in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl and phosphodiester in any combination. In certain embodiments, each tether comprises at least one phosphorus linking group or neutral linking group.

[0426] In certain embodiments, tethers include one or more cleavable bond. In certain embodiments, each tethered ligand is attached to a branching group. In certain embodiments, each tethered ligand is attached to a branching group through an amide group. In certain embodiments, each tethered ligand is attached to a branching group through an ether group. In certain embodiments, each tethered ligand is attached to a branching group through a phosphorus linking

group or neutral linking group. In certain embodiments, each tethered ligand is attached to a branching group through a phosphodiester group. In certain embodiments, each tether is attached to a ligand through either an amide or an ether group. In certain embodiments, each tether is attached to a ligand through an ether group.

[0427] In certain embodiments, each tether comprises from about 8 to about 20 atoms in chain length between the ligand and the branching group. In certain embodiments, each tether comprises from about 10 to about 18 atoms in chain length between the ligand and the branching group. In certain embodiments, each tether comprises about 13 atoms in chain length.

[0428] In certain embodiments, the present disclosure provides ligands wherein each ligand is covalently attached to the remainder of the conjugate group through a tether. In certain embodiments, each ligand is selected to have an affinity for at least one type of receptor on a target cell. In certain embodiments, ligands are selected that have an affinity for at least one type of receptor on the surface of a mammalian liver cell. In certain embodiments, ligands are selected that have an affinity for the hepatic asialoglycoprotein receptor (ASGP-R). In certain embodiments, each ligand is a carbohydrate. In certain embodiments, each ligand is, independently selected from galactose, N-acetyl galactoseamine, mannose, glucose, glucosamone and fucose. In certain embodiments, each ligand is N-acetyl galactoseamine (GalNAc). In certain embodiments, the targeting moiety comprises 1 to 3 ligands. In certain embodiments, the targeting moiety comprises 3 ligands. In certain embodiments, the targeting moiety comprises 2 ligands. In certain embodiments, the targeting moiety comprises 1 ligand. In certain embodiments, the targeting moiety comprises 3 N-acetyl galactoseamine ligands. In certain embodiments, the targeting moiety comprises 2 N-acetyl galactoseamine ligands. In certain embodiments, the targeting moiety comprises 1 N-acetyl galactoseamine ligand.

[0429] In certain embodiments, each ligand is a carbohydrate, carbohydrate derivative, modified carbohydrate, multivalent carbohydrate cluster, polysaccharide, modified polysaccharide, or polysaccharide derivative. In certain embodiments, each ligand is an amino sugar or a thio sugar. For example, amino sugars may be selected from any number of compounds known in the art, for example glucosamine, sialic acid, α-D-galactosamine, N-Acetylgalactosamine, 2-acetamido-2-deoxy-D-galactopyranose (GalNAc), 2-Amino-3-O—[(R)-1-carboxyethyl]-2-deoxy-β-D-glucopyranose (β-muramic acid), 2-Deoxy-2-methylamino-L-glucopyranose, 4,6-Dideoxy-4-formamido-2,3-di-O-methyl-D-mannopyranose, 2-Deoxy-2-sulfoamino-D-glucopyranose and N-sulfo-D-glucosamine, and N-Glycoloyl-α-neuraminic acid. For example, thio sugars may be selected from the group consisting of 5-Thio-β-D-glucopyranose, Methyl 2,3,4-tri-O-acetyl-1-thio-6-O-trityl-α-D-glucopyranoside, 4-Thio-β-D-galactopyranose, and ethyl 3,4,6,7-tetra-O-acetyl-2-deoxy-1,5-dithio-α-D-gluco-heptopyranoside.

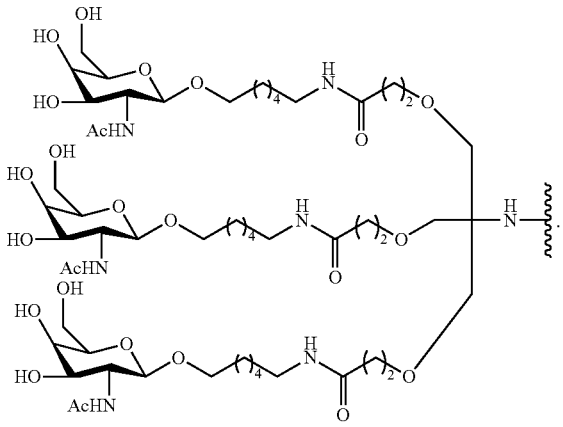
[0430] In certain embodiments, conjugate groups as provided herein comprise a carbohydrate cluster. As used herein, "carbohydrate cluster" means a portion of a conjugate group wherein two or more carbohydrate residues are attached to a branching group through tether groups. (see, e.g., Maier et al., "Synthesis of Antisense Oligonucleotides Conjugated to a Multivalent Carbohydrate Cluster for Cellular Targeting," *Bioconjugate Chemistry,* 2003, (14): 18-29, which is incorporated herein by reference in its entirety, or Rensen et al., "Design and Synthesis of Novel N-Acetylgalactosamine-Terminated Glycolipids for Targeting of Lipoproteins to the Hepatic Asiaglycoprotein Receptor," *J. Med. Chem.* 2004, (47): 5798-5808, for examples of carbohydrate conjugate clusters).

[0431] As used herein, "modified carbohydrate" means any carbohydrate having one or more chemical modifications relative to naturally occurring carbohydrates.

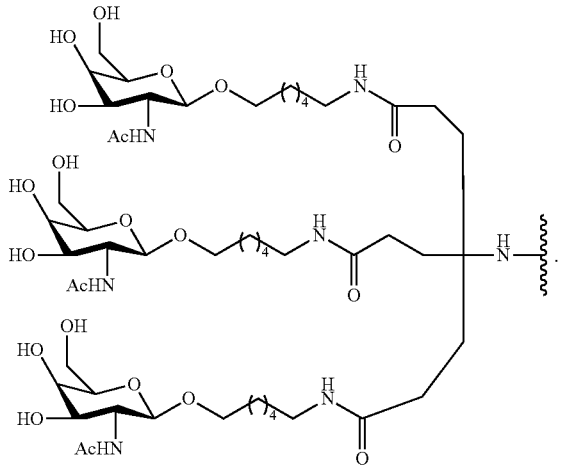
[0432] As used herein, "carbohydrate derivative" means any compound which may be synthesized using a carbohydrate as a starting material or intermediate.

[0433] As used herein, "carbohydrate" means a naturally occurring carbohydrate, a modified carbohydrate, or a carbohydrate derivative.

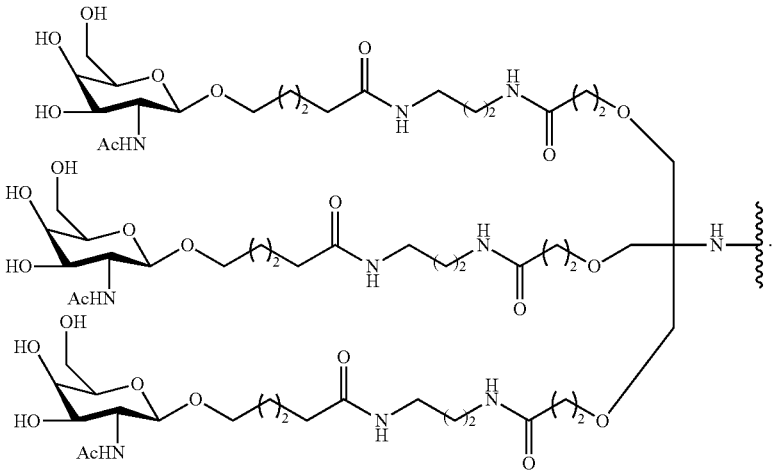
[0434] In certain embodiments, conjugate groups are provided wherein the cell-targeting moiety has the formula:



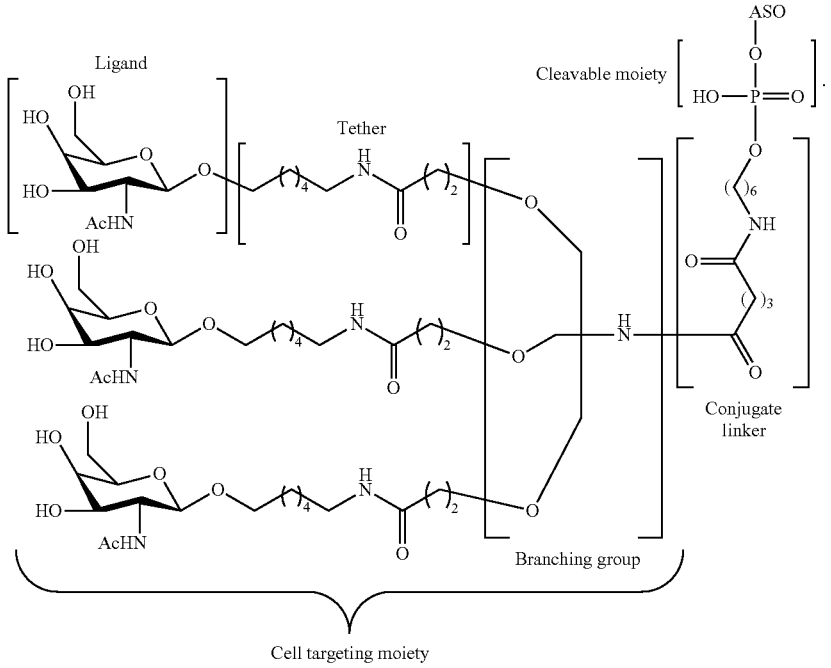
[0435] In certain embodiments, conjugate groups are provided wherein the cell-targeting moiety has the formula:



[0436] In certain embodiments, conjugate groups are provided wherein the cell-targeting moiety has the formula:



[0437] In certain embodiments, conjugate groups have the formula:



[0438] In certain embodiments, an antisense oligonucleotide linked to the conjugate group shown in the formula above has the nucleobase sequence of SEQ ID NO: 1914.

[0439] Representative United States patents, United States patent application publications, and international patent application publications that teach the preparation of certain of the above noted conjugate groups, conjugated oligomeric compounds such as antisense compounds comprising a conjugate group, tethers, conjugate linkers, branching groups, ligands, cleavable moieties as well as other modifications include without limitation, U.S. Pat. Nos. 5,994,517, 6,300,319, 6,660,720, 6,906,182, 7,262,177, 7,491,805, 8,106,022, 7,723,509, US 2006/0148740, US 2011/0123520, WO 2013/033230, WO 2014/179620 and WO 2012/037254, each of which is incorporated by reference herein in its entirety.

[0440] Representative publications that teach the preparation of certain of the above noted conjugate groups, conjugated oligomeric compounds such as antisense compounds comprising a conjugate group, tethers, conjugate linkers, branching groups, ligands, cleavable moieties as well as other modifications include without limitation, BIESSEN et al., "The Cholesterol Derivative of a Triantennary Galactoside with High Affinity for the Hepatic Asialoglycoprotein Receptor: a Potent Cholesterol Lowering Agent" J. Med. Chem. (1995) 38:1846-1852, BIESSEN et al., "Synthesis of Cluster Galactosides with High Affinity for the Hepatic Asialoglycoprotein Receptor" J. Med. Chem. (1995) 38:1538-1546, LEE et al., "New and more efficient multivalent glyco-ligands for asialoglycoprotein receptor of mammalian hepatocytes" Bioorganic & Medicinal Chemistry (2011) 19:2494-2500, RENSEN et al., "Determination of the Upper Size Limit for Uptake and Processing of Ligands by the Asialoglycoprotein Receptor on Hepatocytes in Vitro and in Vivo" J. Biol. Chem. (2001) 276(40):37577-37584,

RENSEN et al., "Design and Synthesis of Novel N-Acetylgalactosamine-Terminated Glycolipids for Targeting of Lipoproteins to the Hepatic Asialoglycoprotein Receptor" J. Med. Chem. (2004) 47:5798-5808, SLIEDREGT et al., "Design and Synthesis of Novel Amphiphilic Dendritic Galactosides for Selective Targeting of Liposomes to the Hepatic Asialoglycoprotein Receptor" J. Med. Chem. (1999) 42:609-618, and Valentijn et al., "Solid-phase synthesis of lysine-based cluster galactosides with high affinity for the Asialoglycoprotein Receptor" *Tetrahedron,* 1997, 53(2), 759-770, each of which is incorporated by reference herein in its entirety.

[0441] In certain embodiments, conjugate groups include without limitation, intercalators, reporter molecules, polyamines, polyamides, polyethylene glycols, thioethers, polyethers, cholesterols, thiocholesterols, cholic acid moieties, folate, lipids, phospholipids, biotin, phenazine, phenanthridine, anthraquinone, adamantane, acridine, fluoresceins, rhodamines, coumarins and dyes. Certain conjugate groups have been described previously, for example: cholesterol moiety (Letsinger et al., Proc. Natl. Acad. Sci. USA, 1989, 86, 6553-6556), cholic acid (Manoharan et al., Bioorg. Med. Chem. Let., 1994, 4, 1053-1060), a thioether, e.g., hexyl-S-tritylthiol (Manoharan et al., Ann. N.Y. Acad. Sci., 1992, 660, 306-309; Manoharan et al., Bioorg. Med. Chem. Let., 1993, 3, 2765-2770), a thiocholesterol (Oberhauser et al., Nucl. Acids Res., 1992, 20, 533-538), an aliphatic chain, e.g., do-decan-diol or undecyl residues (Saison-Behmoaras et al., EMBO J., 1991, 10, 1111-1118; Kabanov et al., FEBS Lett., 1990, 259, 327-330; Svinarchuk et al., Biochimie, 1993, 75, 49-54), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethyl-ammonium 1,2-di-O-hexadecyl-rac-glycero-3-H-phosphonate (Manoharan et al., Tetrahedron Lett., 1995, 36, 3651-3654; Shea et al., Nucl. Acids Res., 1990, 18, 3777-3783), a polyamine or a polyethylene glycol chain (Manoharan et al., Nucleosides & Nucleotides, 1995, 14, 969-973), or adamantane acetic acid (Manoharan et al., Tetrahedron Lett., 1995, 36, 3651-3654), a palmityl moiety (Mishra et al., Biochim. Biophys. Acta, 1995, 1264, 229-237), or an octadecylamine or hexylamino-carbonyl-oxycholesterol moiety (Crooke et al., J. Pharmacol. Exp. Ther., 1996, 277, 923-937).

[0442] In certain embodiments, a conjugate group comprises an active drug substance, for example, aspirin, warfarin, phenylbutazone, ibuprofen, suprofen, fen-bufen, ketoprofen, (S)-(+)-pranoprofen, carprofen, dansylsarcosine, 2,3,5-triiodobenzoic acid, flufenamic acid, folinic acid, a benzothiadiazide, chlorothiazide, a diazepine, indo-methicin, a barbiturate, a cephalosporin, a sulfa drug, an antidiabetic, an antibacterial or an antibiotic.

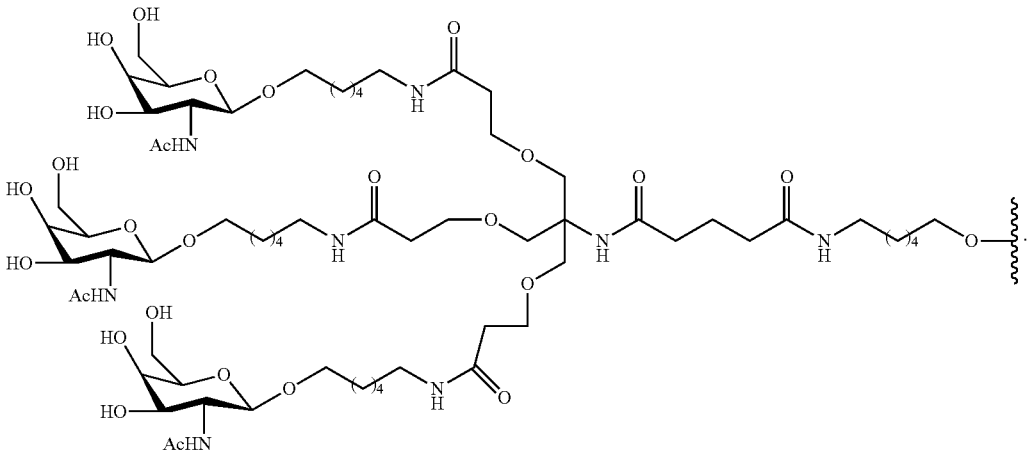
[0443] Some nonlimiting examples of conjugate linkers include pyrrolidine, 8-amino-3,6-dioxaoctanoic acid (ADO), succinimidyl 4-(N-maleimidomethyl) cyclohexane-1-carboxylate (SMCC) and 6-aminohexanoic acid (AHEX or AHA). Other conjugate linkers include, but are not limited to, substituted C_1-C_{10} alkyl, substituted or unsubstituted C_2-C_{10} alkenyl or substituted or unsubstituted C_2-C_{10} alkynyl, wherein a nonlimiting list of preferred substituent groups includes hydroxyl, amino, alkoxy, carboxy, benzyl, phenyl, nitro, thiol, thioalkoxy, halogen, alkyl, aryl, alkenyl and alkynyl.

[0444] Conjugate groups may be attached to either or both ends of an oligonucleotide (terminal conjugate groups) and/or at any internal position.

[0445] In certain embodiments, conjugate groups are at the 3'-end of an oligonucleotide of an oligomeric compound. In certain embodiments, conjugate groups are near the 3'-end. In certain embodiments, conjugates are attached at the 3'end of an oligomeric compound, but before one or more terminal group nucleosides. In certain embodiments, conjugate groups are placed within a terminal group.

[0446] In certain embodiments, conjugate groups are at the 5'-end of an oligonucleotide of an oligomeric compound. In certain embodiments, conjugate groups are near the 5'-end.

[0447] In certain embodiments, a modified oligonucleotide targeting AGT described herein further comprises a GalNAc conjugate group. In certain embodiments, the GalNAc conjugate group is 5'-Trishexylamino-(THA)-C6 Gal-NAc_3. In certain embodiments, the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate has the formula



[0448] In certain embodiments, the modified oligonucleotide is linked to the 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ conjugate by a cleavable moiety. In certain embodiments, the cleavable moiety is a phosphate group.

Cell Culture and Antisense Compounds Treatment

[0449] The effects of antisense compounds on the level, activity or expression of AGT nucleic acids can be tested in vitro in a variety of cell types. Cell types used for such analyses are available from commercial vendors (e.g., American Type Culture Collection, Manassas, Va.; Zen-Bio, Inc., Research Triangle Park, N.C.; Clonetics Corporation, Walkersville, Md.) and cells are cultured according to the

vendor's instructions using commercially available reagents (e.g., Invitrogen Life Technologies, Carlsbad, Calif.). Illustrative cell types include, but are not limited to, HepG2 cells, Hep3B cells, Huh7 (hepatocellular carcinoma) cells, primary hepatocytes, A549 cells, GM04281 fibroblasts and LLC-MK2 cells.

In Vitro Testing of Antisense Oligonucleotides

[0450] Described herein are methods for treatment of cells with antisense oligonucleotides, which can be modified appropriately for treatment with other antisense compounds.

[0451] In general, cells are treated with antisense oligonucleotides when the cells reach approximately 60-80% confluence in culture.

[0452] One reagent commonly used to introduce antisense oligonucleotides into cultured cells includes the cationic lipid transfection reagent LIPOFECTIN® (Invitrogen, Carlsbad, Calif.). Antisense oligonucleotides are mixed with LIPOFECTIN® in OPTI-MEM® 1 (Invitrogen, Carlsbad, Calif.) to achieve the desired final concentration of antisense oligonucleotide and a LIPOFECTIN® concentration that typically ranges 2 to 12 ug/mL per 100 nM antisense oligonucleotide.

[0453] Another reagent used to introduce antisense oligonucleotides into cultured cells includes LIPOFECTAMINE 2000® (Invitrogen, Carlsbad, Calif.). Antisense oligonucleotide is mixed with LIPOFECTAMINE 2000® in OPTI-MEM® 1 reduced serum medium (Invitrogen, Carlsbad, Calif.) to achieve the desired concentration of antisense oligonucleotide and a LIPOFECTAMINE® concentration that typically ranges 2 to 12 ug/mL per 100 nM antisense oligonucleotide.

[0454] Another reagent used to introduce antisense oligonucleotides into cultured cells includes Cytofectin® (Invitrogen, Carlsbad, Calif.). Antisense oligonucleotide is mixed with Cytofectin® in OPTI-MEM® 1 reduced serum medium (Invitrogen, Carlsbad, Calif.) to achieve the desired concentration of antisense oligonucleotide and a Cytofectin® concentration that typically ranges 2 to 12 ug/mL per 100 nM antisense oligonucleotide.

[0455] Another reagent used to introduce antisense oligonucleotides into cultured cells includes Oligofectamine™ (Invitrogen Life Technologies, Carlsbad, Calif.). Antisense oligonucleotide is mixed with Oligofectamine™ in Opti-MEM™-1 reduced serum medium (Invitrogen Life Technologies, Carlsbad, Calif.) to achieve the desired concentration of oligonucleotide with an Oligofectamine™ to oligonucleotide ratio of approximately 0.2 to 0.8 μL per 100 nM.

[0456] Another reagent used to introduce antisense oligonucleotides into cultured cells includes FuGENE 6 (Roche Diagnostics Corp., Indianapolis, Ind.). Antisense oligomeric compound was mixed with FuGENE 6 in 1 mL of serum-free RPMI to achieve the desired concentration of oligonucleotide with a FuGENE 6 to oligomeric compound ratio of 1 to 4 μL of FuGENE 6 per 100 nM.

[0457] Another technique used to introduce antisense oligonucleotides into cultured cells includes electroporation (Sambrook and Russell in *Molecular Cloning. A Laboratory Manual*. Third Edition. Cold Spring Harbor laboratory Press, Cold Spring Harbor, N.Y. 2001).

[0458] Cells are treated with antisense oligonucleotides by routine methods. Cells are typically harvested 16-24 hours after antisense oligonucleotide treatment, at which time RNA or protein levels of target nucleic acids are measured by methods known in the art and described herein (Sambrook and Russell in *Molecular Cloning. A Laboratory Manual*. Third Edition. Cold Spring Harbor laboratory Press, Cold Spring Harbor, N.Y. 2001). In general, when treatments are performed in multiple replicates, the data are presented as the average of the replicate treatments.

[0459] The concentration of antisense oligonucleotide used varies from cell line to cell line. Methods to determine the optimal antisense oligonucleotide concentration for a particular cell line are well known in the art (Sambrook and Russell in *Molecular Cloning. A Laboratory Manual*. Third Edition. Cold Spring Harbor laboratory Press, Cold Spring Harbor, N.Y. 2001). Antisense oligonucleotides are typically used at concentrations ranging from 1 nM to 300 nM when transfected with LIPOFECTAMINE2000®, Lipofectin or Cytofectin. Antisense oligonucleotides are used at higher concentrations ranging from 625 to 20,000 nM when transfected using electroporation.

RNA Isolation

[0460] RNA analysis can be performed on total cellular RNA or poly(A)+ mRNA. Methods of RNA isolation are well known in the art (Sambrook and Russell, Molecular Cloning: A Laboratory Manual, 3rd Ed., 2001). RNA is prepared using methods well known in the art, for example, using the TRIZOL® Reagent (Invitrogen, Carlsbad, Calif.) according to the manufacturer's recommended protocols.

Analysis of Inhibition of Target Levels or Expression

[0461] Inhibition of levels or expression of an AGT nucleic acid can be assayed in a variety of ways known in the art (Sambrook and Russell, Molecular Cloning: A Laboratory Manual, 3rd Ed., 2001). For example, target nucleic acid levels can be quantitated by, e.g., Northern blot analysis, competitive polymerase chain reaction (PCR), or quantitaive real-time PCR. RNA analysis can be performed on total cellular RNA or poly(A)+ mRNA. Methods of RNA isolation are well known in the art. Northern blot analysis is also routine in the art. Quantitative real-time PCR can be conveniently accomplished using the commercially available ABI PRISM® 7600, 7700, or 7900 Sequence Detection System, available from PE-Applied Biosystems, Foster City, Calif. and used according to manufacturer's instructions.

Quantitative Real-Time PCR Analysis of Target RNA Levels

[0462] Quantitation of target RNA levels may be accomplished by quantitative real-time PCR using the ABI PRISM® 7600, 7700, or 7900 Sequence Detection System (PE-Applied Biosystems, Foster City, Calif.) according to manufacturer's instructions. Methods of quantitative real-time PCR are well known in the art.

[0463] Prior to real-time PCR, the isolated RNA is subjected to a reverse transcriptase (RT) reaction, which produces complementary DNA (cDNA) that is then used as the substrate for the real-time PCR amplification. The RT and real-time PCR reactions are performed sequentially in the same sample well. RT and real-time PCR reagents are obtained from Invitrogen (Carlsbad, Calif.). RT, real-time-PCR reactions are carried out by methods well known to those skilled in the art.

[0464] Gene (or RNA) target quantities obtained by real time PCR are normalized using either the expression level of

a gene whose expression is constant, such as cyclophilin A, or by quantifying total RNA using RIBOGREEN® (Invitrogen, Inc. Carlsbad, Calif.). Cyclophilin A expression is quantified by real time PCR, by being run simultaneously with the target, multiplexing, or separately. Total RNA is quantified using RIBOGREEN® RNA quantification reagent (Invitrogen, Inc. Eugene, Oreg.). Methods of RNA quantification by RIBOGREEN® are taught in Jones, L. J., et al, (Analytical Biochemistry, 1998, 265, 368-374). A CYTOFLUOR® 4000 instrument (PE Applied Biosystems) is used to measure RIBOGREEN® fluorescence.

[0465] Probes and primers are designed to hybridize to an AGT nucleic acid. Methods for designing real-time PCR probes and primers are well known in the art, and may include the use of software such as PRIMER EXPRESS® Software (Applied Biosystems, Foster City, Calif.).

Analysis of Protein Levels

[0466] Antisense inhibition of AGT nucleic acids can be assessed by measuring AGT protein levels. Protein levels of AGT can be evaluated or quantitated in a variety of ways well known in the art, such as immunoprecipitation, Western blot analysis (immunoblotting), enzyme-linked immunosorbent assay (ELISA), quantitative protein assays, protein activity assays (for example, caspase activity assays), immunohistochemistry, immunocytochemistry or fluorescence-activated cell sorting (FACS) (Sambrook and Russell, Molecular Cloning: A Laboratory Manual, 3rd Ed., 2001). Antibodies directed to a target can be identified and obtained from a variety of commercially available sources, or can be prepared via conventional monoclonal or polyclonal antibody generation methods well known in the art.

In Vivo Testing of Antisense Compounds

[0467] Antisense compounds, for example, antisense oligonucleotides, are tested in animals to assess their ability to inhibit expression of AGT and produce phenotypic changes, such as, reduced hypertension in the body. Testing can be performed in normal animals, or in experimental disease models. For administration to animals, antisense oligonucleotides are formulated in a pharmaceutically acceptable diluent, such as sterile water-for-injection or phosphate-buffered saline. Administration includes parenteral routes of administration, such as intraperitoneal, intravenous, and subcutaneous. Calculation of antisense oligonucleotide dosage and dosing frequency depends upon factors such as route of administration and animal body weight. In one embodiment, following a period of treatment with antisense oligonucleotides, RNA is isolated from liver tissue and changes in AGT nucleic acid expression are measured. Changes in AGT protein levels can be directly measured. Changes in AGT expression can also be measured by determining the level of inhibition of the RAAS pathway. RAAS pathway related diseases, disorders and/or conditions may be used as markers for determining the level of AGT inhibition.

Certain Indications

[0468] Certain embodiments of the invention provide compounds, compositions and methods of using the compounds and compositions to reduce AGT levels. In certain embodiments, the invention provides compounds, compositions and methods of using the compounds and compositions to treat a subject comprising administering a therapeutically effective amount of the compounds or compositions to the subject. In certain embodiments, the subject has, or is at risk for, a RAAS pathway related disease, disorder or condition. In certain embodiments, the compound or composition comprises and antisense compound.

[0469] In certain embodiments, administration of a therapeutically effective amount of an antisense compound targeted to an AGT nucleic acid is accompanied by monitoring of AGT levels in the serum or tissue of a subject to determine a subject's response to the antisense compound. A subject's response to administration of the antisense compound is used by a physician to determine the amount and duration of therapeutic intervention.

[0470] In certain embodiments, administration of an antisense compound targeted to an AGT nucleic acid results in reduction of AGT expression by at least 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 99% or 100% or a range defined by any two of these values. In certain embodiments, administration of an antisense compound targeted to an AGT nucleic acid results in inhibition of the RAAS pathway by at least 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 99% or 100% or a range defined by any two of these values. In certain embodiments, administration of an antisense compound targeted to an AGT nucleic acid results in a change the RAAS pathway related disease, disorder, condition, symptom or marker (e.g., hypertension or organ damage). In certain embodiments, administration of an AGT antisense compound increases or decreases the RAAS related disease, disorder, condition, symptom or marker by at least 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 99% or 100% or a range defined by any two of these values.

[0471] In certain embodiments, pharmaceutical compositions comprising an antisense compound targeted to AGT are used in the preparation of a medicament for reducing AGT levels. In certain embodiments, pharmaceutical compositions comprising an antisense compound targeted to AGT are used in the preparation of a medicament for treating a subject suffering from, or susceptible to, a RAAS related disease, disorder or condition.

[0472] In certain embodiments, reducing AGT levels in a subject treats, ameliorates, prevents, slows the progression, or delays the onset of a disease, condition or disorder. In certain embodiments, the disease, condition or disorder is shortened life expectancy, hypertension, hypertensive emergency (i.e. malignant hypertension), kidney disease (e.g., chronic kidney disease, polycystic kidney disease), pre-eclampsia, Marfan Syndrome, stroke, cardiac disease (e.g., myocardial infarction, heart failure, congestive heart failure, valvular heart disease), aneurysms of the blood vessels, abdominal aneurysm, peripheral artery disease, organ damage, pulmonary arterial hypertension, obesity, metabolic syndrome, non-alcoholic steatohepatitis (NASH), non-alcoholic fatty liver disease (NAFLD) and RAAS related diseases, disorders and/or conditions or symptoms thereof. In certain embodiments, the hypertension is nonresistant hypertension or resistant hypertension. In certain embodiments, the aneurysm of the blood vessels is aortic aneurysm. In certain embodiments, the organ damage is heart muscle hypertrophy or fibrosis in an organ or tissue. In certain embodiments, the organ is heart, liver or kidney and the tissue is derived from the heart, liver or kidney.

[0473] In certain embodiments, reducing AGT levels in a subject treats, ameliorates, prevents, slows the progression, or delays the onset of a RAAS pathway related disease, disorder or condition. In certain embodiments, the RAAS pathway related disease, disorder or condition is shortened life expectancy, hypertension, hypertensive emergency (i.e. malignant hypertension), kidney disease (e.g., chronic kidney disease, polycystic kidney disease), pre-eclampsia, Marfan Syndrome, stroke, cardiac disease (e.g., myocardial infarction, heart failure, congestive heart failure, valvular heart disease), aneurysms of the blood vessels, abdominal aneurysm, peripheral artery disease, organ damage, pulmonary arterial hypertension, obesity, metabolic syndrome, NASH, NAFLD and other RAAS related diseases, disorders and/or conditions or symptoms thereof. In certain embodiments, the hypertension is nonresistant hypertension or resistant hypertension. In certain embodiments, the aneurysm of the blood vessels is aortic aneurysm. In certain embodiments, the organ damage is heart muscle hypertrophy or fibrosis in an organ or tissue. In certain embodiments, the organ is heart, liver or kidney and the tissue is derived from the heart, liver or kidney.

[0474] In certain embodiments, provided are compounds, compositions and methods for modulating a symptom or marker of a disease, disorder and/or condition. In certain embodiments, the marker can be selected from one or more of shortened life expectancy, hypertension, hypertensive emergency (i.e. malignant hypertension), kidney disease (e.g., chronic kidney disease, polycystic kidney disease), pre-eclampsia, Marfan Syndrome, stroke, cardiac disease (e.g., myocardial infarction, heart failure, congestive heart failure, valvular heart disease), aneurysms of the blood vessels, abdominal aneurysm, peripheral artery disease, organ damage and other RAAS related diseases, disorders and/or conditions or symptoms thereof.

Certain Combination Therapies

[0475] In certain embodiments, a first agent comprising an antisense compound provided herein is co-administered with one or more secondary agents. In certain embodiments, the antisense compound is an antisense oligonucleotide. In certain embodiments, the antisense oligonucleotide is a modified oligonucleotide.

[0476] In certain embodiments, such second agents are designed to treat the same RAAS pathway related disease, disorder or condition as the first agent described herein. In certain embodiments, such second agents are designed to treat a different disease, disorder, or condition as the first agent described herein. In certain embodiments, such second agents are designed to treat an undesired side effect of one or more pharmaceutical compositions as described herein. In certain embodiments, such first agents are designed to treat an undesired side effect of a second agent. In certain embodiments, second agents are co-administered with the first agent to treat an undesired effect of the first agent. In certain embodiments, second agents are co-administered with the first agent to produce a combinational or additive effect. In certain embodiments, second agents are co-administered with the first agent to produce a synergistic effect.

[0477] In certain embodiments, the co-administration of the first and second agents permits use of lower dosages than would be required to achieve a therapeutic or prophylactic effect if the agents were administered as independent therapy. In certain embodiments the dose of a co-administered second agent is the same as the dose that would be administered if the second agent was administered alone. In certain embodiments the dose of a co-administered second agent is greater than the dose that would be administered if the second agent was administered alone.

[0478] In certain embodiments, a first agent and one or more second agents are administered at the same time. In certain embodiments, the first agent and one or more second agents are administered at different times. In certain embodiments, the first agent and one or more second agents are prepared together in a single pharmaceutical formulation. In certain embodiments, the first agent and one or more second agents are prepared separately.

[0479] In certain embodiments, second agents include, but are not limited to, certain procedures to reduce hypertension, diet changes, lifestyle changes, anti-fibrotic drugs and anti-hypertensive drugs such as RAAS inhibitors, endothelin receptor antagonists, neprilysin inhibitors, diuretics, calcium channel blockers, adrenergic receptor antagonists, adrenergic agonists and vasodilators.

[0480] Examples of procedures that can reduce hypertension include, but are not limited to, renal denervation and baroreceptor activation therapy.

[0481] Examples of RAS or RAAS inhibitors include, but are not limited to ACE inhibitors (e.g., captopril, enalapril, fosinopril, lisinopril, perindopril, quinapril, ramipril, trandolapril and benazepril), angiotensin II receptor antagonists (e.g., candesartan, eprosartan, irbesartan, losartan, olmesartan, telmisartan and valsartan), renin inhibitors (e.g., aliskiren), aldosterone receptor antagonists (e.g., eplerenone, spironolactone and finerenone).

[0482] Examples of endothelin receptor antagonists include ambrisentan, sitaxentan, atrasentan, BQ-123, zibotentan, bosentan, macitentan and tezosentan.

[0483] Examples of neprilysin inhibitors include sacubitril and omapatrilat.

[0484] Examples of diuretics include loop diuretics (e.g., bumetanide, ethacrynic acid, furosemide, torsemide), thiazide diuretics (e.g., epitizide, hydrochlorothiazide, chlorothiazide and bendroflumethiazide), thiazide-like diuretics (e.g., indapamide, chlorthalidone and metolazone) and potassium-sparing diuretics (e.g., amiloride, triamterene and spironolactone).

[0485] Examples of calcium channel blockers include dihydropyridines (e.g., amlodipine, felodipine, isradipine, lercanidipine, nicardipine, nifedipine, nimodipine and nitrendipine) and non-dihydropyridines (e.g., diltiazem and verapamil).

[0486] Examples of adrenergic receptor antagonists include Beta blockers (e.g., atenolol, metoprolol, nadolol, oxprenolol, pindolol, propranolol and timolol), Alpha blockers (e.g., doxazosin, phentolamine, indoramin, phenoxybenzamine, prazosin, terazosin and tolazoline) and mixed Alpha+Beta blockers (e.g., bucindolol, carvedilol and labetalol).

[0487] Examples of vasodilators include sodium nitroprusside and hydralazine and its derivatives.

[0488] Examples of adrenergic agonists include alpha-2 agonists (e.g., clonidine, guanabenz, methyldopa and moxonidine).

[0489] Additional examples of anti-hypertensive drugs include guanethidine, reserpine and the like.

[0490] The second agents can be used in combination with the therapeutic compounds described herein to decrease a disease, disorder and/or condition such as hypertension, organ damage and the like.

Certain Compounds

[0491] Preferred antisense compounds with beneficial properties that enhance their use as therapeutic treatments in humans are demonstrated in the examples herein. For brevity, only the studies that contributed to the selection of the preferred antisense compounds are described. A non-exhaustive summary of the examples is provided below for ease of reference.

[0492] Over 2000 antisense compounds with a MOE containing and/or a cEt containing gapmer motif targeting human AGT were designed. Example 1 shows representative single dose inhibition data for the over 2000 potent antisense compounds tested in HepG2 cells for their effect on human AGT mRNA.

[0493] Of the over 2000 antisense compounds tested with a single dose in vitro, over 160 antisense compounds were chosen for testing in dose-dependent inhibition studies to determine their half maximal inhibitory concentration (IC_{50}) in HepG2 cells (Example 2).

[0494] Base on the in vitro dose response studies, over 50 antisense compounds were selected for single dose potency and tolerability testing in human AGT transgenic (huAGT tg) mice as described in the exemplary studies in Example 3. Of the over 50 antisense compounds, about 14 antisense compounds were further selected for dose response and tolerability studies in huAGT tg mice (Example 4).

[0495] Nine antisense compounds exhibiting significant potency and tolerability in huAGT mice were chosen for further studies: in a viscosity assay (Example 5); in CD1 mice (Example 6) and Sprague-Dawlay rats (Example 7) to assess tolerability of the antisense compounds; in monkey hepatocytes to test cross-species potency in inhibiting monkey AGT (Example 8); and in cynomolgus monkeys to assess potency and tolerability (Example 9). Although the antisense compounds in the studies described in Example 9 were tested in cynomolgus monkeys, the cynomolgus monkey AGT sequence was not available for comparison to the sequences of the antisense compounds, therefore the sequences of the antisense compounds were compared to that of the closely related rhesus monkey (Example 8).

[0496] Based on the extensive characterization of the 9 antisense compounds, the sequence of antisense compound ISIS 654472 (parent compound) was selected for further study (Example 10). Six antisense compounds were designed with the sequence of parent compound ISIS 654472 but with different chemical modifications and a GalNAc conjugate. The 6 newly designed compounds were administered to CD1 mice (Example 10) and Sprague-Dawley rats (Example 11) to test their tolerability in these animal models. Of the 6 GalNAc conjugated antisense compounds, compound ISIS 757456 was selected to test in huAGT mice compared to the parent antisense compound ISIS 654472. ISIS 757456 showed an 8× improvement in potency compared to unconjugated compound ISIS 654472.

[0497] Accordingly, provided herein are antisense compounds with any one or more characteristics that are beneficial for their use as a therapeutic agent. In certain embodiments, provided herein are antisense compounds comprising a modified oligonucleotide as described herein targeted to, or specifically hybridizable with, a region of nucleotides selected from any of SEQ ID NOs: 1-6.

[0498] In certain embodiments, certain antisense compounds as described herein are efficacious by virtue of their potency in inhibiting AGT expression. In certain embodiments, the compounds or compositions inhibit AGT by at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% or at least 95%.

[0499] In certain embodiments, certain antisense compounds as described herein are efficacious by virtue of an in vitro IC_{50} of less than 20 μM, less than 10 μM, less than 8 μM, less than 5 μM, less than 2 μM, less than 1 μM, less than 0.9 μM, less than 0.8 μM, less than 0.7 μM, less than 0.6 μM, or less than 0.5 μM when tested in human cells, for example, in the Hep3B cell line (as described in Example 2).

[0500] In certain embodiments, certain antisense compounds as described herein are efficacious by virtue of a median effective dose (ED_{50}) of ≤10 mpk/wk, ≤9 mpk/wk, ≤8 mpk/wk, ≤7 mpk/wk, ≤6 mpk/wk, ≤5 mpk/wk, ≤4 mpk/wk, ≤3 mpk/wk, ≤2 mpk/wk, or ≤1 mpk/wk in vivo as shown in Example 4. In certain embodiments, a preferred antisense compound such as antisense compound ISIS 757456 has an ED_{50}≤3 mpk/wk as shown in Example 12.

[0501] In certain embodiments, certain antisense compounds as described herein are efficacious by virtue of having a viscosity of less than 40 cP, less than 35 cP, less than 30 cP, less than 25 cP, less than 20 cP, less than 15 cP, or less than 12 cP as described in Example 5. Oligonucleotides having a viscosity greater than 40 cP would have less than optimal viscosity.

[0502] In certain embodiments, certain antisense compounds as described herein are highly tolerable, as demonstrated by the in vivo tolerability measurements described in the examples. In certain embodiments, the certain antisense compounds as described herein are highly tolerable, as demonstrated by having an increase in ALT and/or AST value of no more than 3 fold, 2 fold or 1.5 fold over saline treated animals.

[0503] In certain embodiments, certain antisense compounds as described herein are efficacious by virtue of having one or more of an inhibition potency of greater than 50%, an ED_{50}≤5 mpk/wk, a viscosity of less than 40 cP, and no more than a 3 fold increase in ALT and/or AST in transgenic mice.

[0504] In certain embodiments, ISIS 757456 (SEQ ID NO: 1914) is preferred. This compound was found to be a potent inhibitor in AGT transgenic mice and a very tolerable antisense compound in CD-1 mice. In mice it had less than a 3 fold increase in ALT and/or AST levels over saline treated animals. It had an ED_{50}≤3 mpk/wk in huAGT transgenic mice.

EXAMPLES

Non-Limiting Disclosure and Incorporation by Reference

[0505] While certain compounds, compositions and methods described herein have been described with specificity in accordance with certain embodiments, the following examples serve only to illustrate the compounds described herein and are not intended to limit the same. Each of the references recited in the present application is incorporated herein by reference in its entirety.

Example 1: Antisense Inhibition of Human Angiotensinogen (AGT) in HepG2 Cells

[0506] Over 2000 antisense oligonucleotides were designed targeting human AGT nucleic acid and were tested for their effects on AGT mRNA in vitro in a series of experiments that had similar culture conditions. The results for representative antisense oligonucleotides are presented in tables shown below.

[0507] The newly designed chimeric antisense oligonucleotides in the Tables below were designed as MOE and/or cEt containing gapmers. The MOE containing oligonucleotides have a central gap segment comprising 2'-deoxynucleosides which is flanked by wing segments on the 5' direction and the 3' direction. At least one nucleoside in the 5' wing segment and/or one nucleoside in the 3' wing segment has a 2'-MOE sugar modification. The cEt containing oligonucleotides have a central gap segment comprising 2'-deoxynucleosides which is flanked by wing segments on the 5' direction and the 3' direction. At least one nucleoside in the 5' wing segment and/or one nucleoside in the 3' wing segment has a cEt sugar modification.

[0508] In some instances oligonucleotides were designed to contain both a MOE and a cEt. The MOE and cEt containing oligonucleotides have a central gap segment comprising 2'-deoxynucleosides which is flanked by wing segments on the 5' direction and the 3' direction. At least one nucleoside in the 5' wing segment and/or one nucleoside in the 3' wing segment has a MOE and/or cEt sugar modification.

[0509] The "Chemistry" column describes the sugar modifications of each oligonucleotide. "k" indicates an cEt sugar modification; "d" indicates deoxyribose; and "e" indicates a MOE modification. The internucleoside linkages throughout each gapmer are phosphorothioate (P═S) linkages. All cytosine residues throughout each gapmer are 5-methylcytosines.

[0510] "Start site" indicates the 5'-most nucleoside to which the gapmer is targeted in the human gene sequence. "Stop site" indicates the 3'-most nucleoside to which the gapmer is targeted human gene sequence. Each gapmer listed in the Tables below is targeted to either the human AGT mRNA, designated herein as SEQ ID NO: 1 (GENBANK Accession NM_000029.3) and/or the human AGT genomic sequence, designated herein as SEQ ID NO: 2 (GENBANK Accession NT_167186.1 truncated from nucleotides 24354000 to 24370100).

[0511] Table 1 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 4500 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 (forward sequence CCCTGATGGGAGCCAGTGT, designated herein as SEQ ID NO: 8; reverse sequence AGCAGGGAGAAGCCCTTCA, designated herein as SEQ ID NO: 9; and probe sequence CCCTGGCTTTCAACACCTACGTCCACTX, where X is a fluorescent label, designated herein as SEQ ID NO: 10) was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 1

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568518	1	16	TGCCCGCTCATGGGAT	eekddddddddddkke	26	1986	2001	14
568519	20	35	GGGCCACTTCTGACCC	eekddddddddddkke	34	2005	2020	15
568520	35	50	GCTTAGGCAACACGGG	eekddddddddddkke	20	2020	2035	16
568521	45	60	GGAGAGTCTTGCTTAG	eekddddddddddkke	26	2030	2045	17
568522	80	95	CATGCAGGCCGGAGGT	eekddddddddddkke	25	2065	2080	18
568523	90	105	GCCACAGGGACATGCA	eekddddddddddkke	30	2075	2090	19
568524	122	137	TGACCCAGCCCCGGGA	eekddddddddddkke	40	2107	2122	20
568525	155	170	TGTGACAGCCTGAGGC	eekddddddddddkke	25	2140	2155	21
568526	165	180	TCCCTAGGTGTGTGAC	eekddddddddddkke	34	2150	2165	22
568527	179	194	GAAACGGGAGCATCTC	eekddddddddddkke	9	2164	2179	23
568528	189	204	AAGGTTCCCAGAAACG	eekddddddddddkke	19	2174	2189	24
568529	209	224	AGTTTGCAGGAGTCGG	eekddddddddddkke	26	2194	2209	25
568530	229	244	TCGAGTTACACATTTA	eekddddddddddkke	24	2214	2229	26
568531	248	263	AGAGTGAGCCGGTGCA	eekddddddddddkke	21	2233	2248	27
568532	258	273	ACTGCTGAACAGAGTG	eekddddddddddkke	19	2243	2258	28

TABLE 1-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568533	268	283	GCAGAGTTTCACTGCT	eekddddddddddkke	16	2253	2268	29
568534	278	293	AGTGATCGATGCAGAG	eekddddddddddkke	32	2263	2278	30
568535	288	303	AGGAAGTCTTAGTGAT	eekddddddddddkke	15	2273	2288	31
568536	301	316	TGGGACCTCTTCCAGG	eekddddddddddkke	31	2286	2301	32
568537	353	368	GGCCAGACCACAGGCT	eekddddddddddkke	16	2338	2353	33
568538	363	378	TACATCACTTGGCCAG	eekddddddddddkke	30	2348	2363	34
568539	373	388	AGAGGAGGGTTACATC	eekddddddddddkke	24	2358	2373	35
568540	386	401	GTGCACAGGCTGGAGA	eekddddddddddkke	43	2371	2386	36
568541	431	446	TATTTATAGCTGAGGG	eekddddddddddkke	29	2416	2431	37
568542	441	456	CACGATGCCCTATTTA	eekddddddddddkke	34	2426	2441	38
568543	478	493	TACCCAGAACAACGGC	eekddddddddddkke	28	2463	2478	39
568544	525	540	GCCATCTCAGACTGGG	eekddddddddddkke	69	5742	5757	40
568545	535	550	ACCGGCAGGAGCCATC	eekddddddddddkke	42	5752	5767	41
568546	545	560	TCAGGCTCACACCGGC	eekddddddddddkke	67	5762	5777	42
568547	555	570	ATGGTGGCCCTCAGGC	eekddddddddddkke	39	5772	5787	43
568548	596	611	GGTCACCTGCAGCCAG	eekddddddddddkke	59	5813	5828	44
568549	606	621	ATGTACACCCGGTCAC	eekddddddddddkke	25	5823	5838	45
568550	643	658	GGTACTCTCATTGTGG	eekddddddddddkke	76	5860	5875	46
568551	654	669	AGCTGCTCACAGGTAC	eekddddddddddkke	60	5871	5886	47
568552	676	691	CTTCCCGGCATTGGCC	eekddddddddddkke	50	5893	5908	48
568553	703	718	AGCAGGTATGAAGGTG	eekddddddddddkke	62	5920	5935	49
568554	713	728	CCTGAATTGGAGCAGG	eekddddddddddkke	43	5930	5945	50
568555	723	738	GATGTCTTGGCCTGAA	eekddddddddddkke	21	5940	5955	51
568556	739	754	CTTTTCATCCACAGGG	eekddddddddddkke	39	5956	5971	52
568557	762	777	AGCACCAGCTGGTCCT	eekddddddddddkke	71	5979	5994	53
568558	772	787	TGCAGCGACTAGCACC	eekddddddddddkke	71	5989	6004	54
568559	782	797	TGTCAAGTTTTGCAGC	eekddddddddddkke	61	5999	6014	55
568560	803	818	CGGCCCTCAACTTGTC	eekddddddddddkke	45	6020	6035	56
568561	815	830	TCCCGACCATTGCGGC	eekddddddddddkke	25	6032	6047	57
568562	825	840	TTGGCCAGCATCCCGA	eekddddddddddkke	51	6042	6057	58
568563	835	850	GCCCAAGAAGTTGGCC	eekddddddddddkke	13	6052	6067	59
568564	845	860	ATATACGGAAGCCCAA	eekddddddddddkke	52	6062	6077	60
568565	855	870	TGCATGCCATATATAC	eekddddddddddkke	64	6072	6087	61
568566	871	886	GCCCCATAGCTCACTG	eekddddddddddkke	64	6088	6103	62

TABLE 1-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568567	886	901	GGCCCCATGGACCACG	eekddddddddddkke	38	6103	6118	63
568568	913	928	AAAGACAGCCGTTGGG	eekddddddddddkke	58	6130	6145	64
568569	923	938	CCAGGGTGCCAAAGAC	eekddddddddddkke	36	6140	6155	65
568570	937	952	CAGATAGAGAGAGGCC	eekddddddddddkke	59	6154	6169	66
568571	954	969	GTGTGGTCCAAGGCTC	eekddddddddddkke	37	6171	6186	67
568572	983	998	CACCCAGGATTGCCTG	eekddddddddddkke	72	6200	6215	68
568573	993	1008	TTCCAAGGAACACCCA	eekddddddddddkke	35	6210	6225	69
568574	1017	1032	AGCCGGGAGGTGCAGT	eekddddddddddkke	53	6234	6249	70
568575	1020	1035	TCCAGCCGGGAGGTGC	eekddddddddddkke	62	6237	6252	71
568576	1053	1068	ACAGCCTGCAGGGCAG	eekddddddddddkke	47	6270	6285	72
568577	1070	1085	CCACTAGCAGGCCCTG	eekddddddddddkke	33	6287	6302	73
568578	1088	1103	TATCAGCCCTGCCCTG	eekddddddddddkke	37	6305	6320	74
568579	1098	1113	TGGGCCTGGCTATCAG	eekddddddddddkke	42	6315	6330	75
568580	1114	1129	CGTGGACAGCAGCAGC	eekddddddddddkke	70	6331	6346	76
568581	1131	1146	GTGAACACGCCCACCA	eekddddddddddkke	48	6348	6363	77
568582	1151	1166	TCAGGTGCAGGCCTGG	eekddddddddddkke	36	6368	6383	78
568583	1171	1186	GCCCTGCACAAACGGC	eekddddddddddkke	16	6388	6403	79
568584	1182	1197	TAGAGAGCCAGGCCCT	eekddddddddddkke	52	6399	6414	80
568585	1203	1218	CGTGGGAGGACCACAG	eekddddddddddkke	47	6420	6435	81
568586	1217	1232	TGAAGTCCAGAGAGCG	eekddddddddddkke	60	6434	6449	82
568587	1233	1248	GCAACATCCAGTTCTG	eekddddddddddkke	50	6450	6465	83
568588	1244	1259	TCTTCTCAGCAGCAAC	eekddddddddddkke	54	6461	6476	84
568589	1272	1287	CCTGTCACAGCCTGCA	eekddddddddddkke	77	6489	6504	85
568590	1278	1293	TTCCATCCTGTCACAG	eekddddddddddkke	51	6495	6510	86
568595	1403	1418	TGTCCACCCAGAACTC	eekddddddddddkke	33	10414	10429	87
568596	1406	1421	TGTTGTCCACCCAGAA	eekddddddddddkke	59	10417	10432	88
568597	1409	1424	TGCTGTTGTCCACCCA	eekddddddddddkke	60	10420	10435	89
568598	1412	1427	AGGTGCTGTTGTCCAC	eekddddddddddkke	57	10423	10438	90
568599	1415	1430	CTGAGGTGCTGTTGTC	eekddddddddddkke	56	10426	10441	91
568600	1418	1433	ACACTGAGGTGCTGTT	eekddddddddddkke	28	10429	10444	92
568601	1421	1436	CAGACACTGAGGTGCT	eekddddddddddkke	67	10432	10447	93
568602	1431	1446	AGCATGGGAACAGACA	eekddddddddddkke	27	10442	10457	94
568603	1443	1458	CCCATGCCAGAGAGCA	eekddddddddddkke	30	10454	10469	95
568604	1462	1477	ACTCCAGTGCTGGAAG	eekddddddddddkke	41	10473	10488	96
568605	1465	1480	GTCACTCCAGTGCTGG	eekddddddddddkke	73	10476	10491	97

TABLE 1-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568606	1474	1489	GTCCTGGATGTCACTC	eekddddddddddkke	68	10485	10500	98
568607	1484	1499	CCGAGAAGTTGTCCTG	eekddddddddddkke	47	10495	10510	99
568608	1494	1509	ACTTGAGTCACCGAGA	eekddddddddddkke	39	10505	10520	100
568609	1504	1519	AGTGAAGGGCACTTGA	eekddddddddddkke	28	10515	10530	101
568610	1531	1546	CTGGATCAGCAGCAGG	eekddddddddddkke	69	10542	10557	102
568611	1550	1565	GGTCAGAGGCATAGTG	eekddddddddddkke	43	10561	10576	103
568612	1578	1593	TGGAAAGTGAGACCCT	eekddddddddddkke	45	10589	10604	104
568613	1588	1603	GGAGTTTTGCTGGAAA	eekddddddddddkke	53	10599	10614	105
568614	1598	1613	TCCAGTTGAGGGAGTT	eekddddddddddkke	38	10609	10624	106
568615	1614	1629	GGAGATAGTTTCTTCA	eekddddddddddkke	24	10625	10640	107
568616	1631	1646	TCAGGTGGATGGTCCG	eekddddddddddkke	34	N/A	N/A	108
568617	1653	1668	TGCAGCACCAGTTGGG	eekddddddddddkke	65	12259	12274	109
568618	1663	1678	ATAAGATCCTTGCAGC	eekddddddddddkke	21	12269	12284	110
568619	1680	1695	AGCAGGTCCTGCAGGT	eekddddddddddkke	50	12286	12301	111
568620	1700	1715	CGGGCAGCTCAGCCTG	eekddddddddddkke	39	12306	12321	112
568621	1710	1725	TGCAGAATGGCGGGCA	eekddddddddddkke	57	12316	12331	113
568622	1720	1735	CAGCTCGGTGTGCAGA	eekddddddddddkke	70	12326	12341	114
568623	1730	1745	TTTGCAGGTTCAGCTC	eekddddddddddkke	44	12336	12351	115
568624	1745	1760	GGTCATTGCTCAATTT	eekddddddddddkke	45	12351	12366	116
568625	1755	1770	ACCCTGATGCGGTCAT	eekddddddddddkke	43	12361	12376	117
568626	1794	1809	GCTTCAAGCTCAAAAA	eekddddddddddkke	56	13263	13278	118
568627	1827	1842	TGTTGGGTAGACTCTG	eekddddddddddkke	61	13296	13311	119
568628	1841	1856	CAGGCTTGTTAAGCTG	eekddddddddddkke	53	13310	13325	120
568629	1851	1866	TCCAAGACCTCAGGCT	eekddddddddddkke	46	13320	13335	121
568630	1875	1890	AGGAATGGGCGGTTCA	eekddddddddddkke	58	13344	13359	122
568631	1923	1938	CGGCCCAGGAAGTGCA	eekddddddddddkke	30	13392	13407	123
568632	1933	1948	GTTGGCCACGCGGCCC	eekddddddddddkke	11	13402	13417	124
568633	1943	1958	TGCTCAGCGGGTTGGC	eekddddddddddkke	49	13412	13427	125
568634	1961	1976	GGCCCTGGCCTCATGC	eekddddddddddkke	44	13430	13445	126
568635	1986	2001	GGCCTTGCCAGGCACT	eekddddddddddkke	86	13455	13470	127
568636	2007	2022	GCCTCAAAGGCCAGGG	eekddddddddddkke	49	13476	13491	128
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	92	13515	13530	129
568638	2056	2071	GGTGACACATCGCTGA	eekddddddddddkke	86	13525	13540	130
568639	2075	2090	GAAAAGGTGGGAGACT	eekddddddddddkke	39	13544	13559	131

TABLE 1-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568640	2088	2103	CGACTCATTAGAAGAA	eekddddddddddkke	87	13557	13572	132
568641	2111	2126	ACGGCTGCTTTCCAGC	eekddddddddddkke	64	13580	13595	133
568642	2121	2136	CCAAGGAGAAACGGCT	eekddddddddddkke	79	13590	13605	134
568643	2131	2146	CACACTTAGACCAAGG	eekddddddddddkke	78	13600	13615	135
568644	2166	2181	TGCCGCTGCAGGCTTC	eekddddddddddkke	57	13635	13650	136
568645	2176	2191	GGTGCATTTGTGCCGC	eekddddddddddkke	75	13645	13660	137
568646	2274	2289	TGGTCGGTTGGAATTC	eekddddddddddkke	77	13743	13758	138
568647	2284	2299	ACAAACAAGCTGGTCG	eekddddddddddkke	84	13753	13768	139
568648	2311	2326	CTTGAAAAGGGAACAC	eekddddddddddkke	62	13780	13795	140
568649	2331	2346	AACCCAATTTTTGTTC	eekddddddddddkke	56	13800	13815	141
568650	2362	2377	GGCAATGCAAAAATGT	eekddddddddddkke	78	13831	13846	142
568651	2391	2406	TACATTCAAGACACTA	eekddddddddddkke	60	13860	13875	143
568652	2402	2417	GGTCATGTTCTTACAT	eekddddddddddkke	55	13871	13886	144
568653	2412	2427	ACTACACGGAGGTCAT	eekddddddddddkke	55	13881	13896	145
568654	2422	2437	TATTACAGACACTACA	eekddddddddddkke	35	13891	13906	146
568655	2482	2497	GGTGCTTGCATCTTTC	eekddddddddddkke	58	13951	13966	147
568656	2492	2507	CAGAAATTCAGGTGCT	eekddddddddddkke	47	13961	13976	148
568657	2503	2518	CCGCATTCAAACAGAA	eekddddddddddkke	38	13972	13987	149
568658	2513	2528	AGCTATGGTTCCGCAT	eekddddddddddkke	55	13982	13997	150
568659	2537	2552	TACTAACACAAGGGAG	eekddddddddddkke	37	14006	14021	151
568660	2558	2573	TTATTGTGGCAAGACG	eekddddddddddkke	48	14027	14042	152
568661	N/A	N/A	TTACTAATACAGCCCA	eekddddddddddkke	31	3322	3337	153
568662	N/A	N/A	GGTTTCCCTGATGCAG	eekddddddddddkke	34	3516	3531	154
568663	N/A	N/A	TGATAGTTGGATTCCT	eekddddddddddkke	21	4783	4798	155
568664	N/A	N/A	TGTGGTCCCAACATGC	eekddddddddddkke	41	4944	4959	156
568665	N/A	N/A	TTGAAGTCCTCAACCC	eekddddddddddkke	26	5460	5475	157
568670	N/A	N/A	CTCTTGGATGTCACAG	eekddddddddddkke	56	10997	11012	158
568671	N/A	N/A	GATGGCAAATTTTGTT	eekddddddddddkke	23	11321	11336	159
568672	N/A	N/A	TGTGTTACTTGGGTAA	eekddddddddddkke	68	11933	11948	160
568673	N/A	N/A	GCCACACAGTGAGGGC	eekddddddddddkke	22	12189	12204	161

[0512] Table 2 shows the percent inhibition of AGT mRNA by additional gapmer oligonucleotides. Cultured HepG2 cells at a density of about 20,000 cells per well were transfected using electroporation with 4,000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 2

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	92	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	91	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	97	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	94	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	92	13515	13530	129
568638	2056	2071	GGTGACACATCGCTGA	eekddddddddddkke	82	13525	13540	130
594621	2022	2037	CTGCTGCTGGCCTTTG	kkkddddddddddkkk	87	13491	13506	162
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	97	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	99	13496	13511	163
594623	2032	2047	GGGTTGTTATCTGCTG	kkkddddddddddkkk	90	13501	13516	164
594624	2046	2061	CGCTGATTTGTCCGGG	kkkddddddddddkkk	94	13515	13530	129
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	91	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	97	13516	13531	165
594626	2049	2064	CATCGCTGATTTGTCC	kkkddddddddddkkk	0	13518	13533	166
594627	2053	2068	GACACATCGCTGATTT	kkkddddddddddkkk	92	13522	13537	167
594628	2073	2088	AAAGGTGGGAGACTGG	kkkddddddddddkkk	81	13542	13557	168
594629	2082	2097	ATTAGAAGAAAAGGTG	kkkddddddddddkkk	84	13551	13566	169
594630	2090	2105	GTCGACTCATTAGAAG	kkkddddddddddkkk	79	13559	13574	170
594631	2095	2110	TCAAAGTCGACTCATT	kkkddddddddddkkk	91	13564	13579	171
594632	2099	2114	CAGCTCAAAGTCGACT	kkkddddddddddkkk	96	13568	13583	172
594641	2022	2037	CTGCTGCTGGCCTTTG	eekddddddddddkke	61	13491	13506	162
594642	2027	2042	GTTATCTGCTGCTGGC	eekddddddddddkke	91	13496	13511	163
594643	2032	2047	GGGTTGTTATCTGCTG	eekddddddddddkke	91	13501	13516	164
594644	2047	2062	TCGCTGATTTGTCCGG	eekddddddddddkke	87	13516	13531	165
594645	2049	2064	CATCGCTGATTTGTCC	eekddddddddddkke	79	13518	13533	166
594646	2053	2068	GACACATCGCTGATTT	eekddddddddddkke	80	13522	13537	167
594647	2073	2088	AAAGGTGGGAGACTGG	eekddddddddddkke	62	13542	13557	168
609078	2020	2035	GCTGCTGGCCTTTGCC	kkkddddddddddkkk	73	13489	13504	173
609079	2021	2036	TGCTGCTGGCCTTTGC	kkkddddddddddkkk	69	13490	13505	174
609080	2023	2038	TCTGCTGCTGGCCTTT	kkkddddddddddkkk	91	13492	13507	175
609081	2024	2039	ATCTGCTGCTGGCCTT	kkkddddddddddkkk	90	13493	13508	176
609082	2025	2040	TATCTGCTGCTGGCCT	kkkddddddddddkkk	84	13494	13509	177
609083	2026	2041	TTATCTGCTGCTGGCC	kkkddddddddddkkk	91	13495	13510	178
609084	2028	2043	TGTTATCTGCTGCTGG	kkkddddddddddkkk	89	13497	13512	179
609085	2029	2044	TTGTTATCTGCTGCTG	kkkddddddddddkkk	91	13498	13513	180

TABLE 2-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
609086	2030	2045	GTTGTTATCTGCTGCT	kkkddddddddddkkk	98	13499	13514	181
609087	2031	2046	GGTTGTTATCTGCTGC	kkkddddddddddkkk	97	13500	13515	182
609088	2048	2063	ATCGCTGATTTGTCCG	kkkddddddddddkkk	98	13517	13532	183
609089	2050	2065	ACATCGCTGATTTGTC	kkkddddddddddkkk	92	13519	13534	184
609090	2051	2066	CACATCGCTGATTTGT	kkkddddddddddkkk	91	13520	13535	185
609091	2052	2067	ACACATCGCTGATTTG	kkkddddddddddkkk	96	13521	13536	186
609092	2054	2069	TGACACATCGCTGATT	kkkddddddddddkkk	34	13523	13538	187
609093	2055	2070	GTGACACATCGCTGAT	kkkddddddddddkkk	78	13524	13539	188
609094	2056	2071	GGTGACACATCGCTGA	kkkddddddddddkkk	93	13525	13540	130
609095	2057	2072	GGGTGACACATCGCTG	kkkddddddddddkkk	96	13526	13541	189
609096	2074	2089	AAAAGGTGGGAGACTG	kkkddddddddddkkk	70	13543	13558	190
609097	2075	2090	GAAAAGGTGGGAGACT	kkkddddddddddkkk	80	13544	13559	131
609098	2076	2091	AGAAAAGGTGGGAGAC	kkkddddddddddkkk	85	13545	13560	191
609099	2080	2095	TAGAAGAAAAGGTGGG	kkkddddddddddkkk	90	13549	13564	192
609100	2081	2096	TTAGAAGAAAAGGTGG	kkkddddddddddkkk	95	13550	13565	193
609101	2083	2098	CATTAGAAGAAAAGGT	kkkddddddddddkkk	76	13552	13567	194
609102	2084	2099	TCATTAGAAGAAAAGG	kkkddddddddddkkk	97	13553	13568	195
609103	2085	2100	CTCATTAGAAGAAAAG	kkkddddddddddkkk	87	13554	13569	196
609104	2086	2101	ACTCATTAGAAGAAAA	kkkddddddddddkkk	70	13555	13570	197
609105	2087	2102	GACTCATTAGAAGAAA	kkkddddddddddkkk	93	13556	13571	198
609106	2088	2103	CGACTCATTAGAAGAA	kkkddddddddddkkk	98	13557	13572	132
609107	2089	2104	TCGACTCATTAGAAGA	kkkddddddddddkkk	97	13558	13573	199
609108	2091	2106	AGTCGACTCATTAGAA	kkkddddddddddkkk	97	13560	13575	200
609109	2092	2107	AAGTCGACTCATTAGA	kkkddddddddddkkk	96	13561	13576	201
609110	2093	2108	AAAGTCGACTCATTAG	kkkddddddddddkkk	96	13562	13577	202
609111	2094	2109	CAAAGTCGACTCATTA	kkkddddddddddkkk	92	13563	13578	203
609112	2096	2111	CTCAAAGTCGACTCAT	kkkddddddddddkkk	93	13565	13580	204
609113	2097	2112	GCTCAAAGTCGACTCA	kkkddddddddddkkk	97	13566	13581	205
609114	2098	2113	AGCTCAAAGTCGACTC	kkkddddddddddkkk	95	13567	13582	206
609115	2020	2035	GCTGCTGGCCTTTGCC	eekddddddddddkke	71	13489	13504	173
609116	2021	2036	TGCTGCTGGCCTTTGC	eekddddddddddkke	47	13490	13505	174
609117	2023	2038	TCTGCTGCTGGCCTTT	eekddddddddddkke	74	13492	13507	175
609118	2024	2039	ATCTGCTGCTGGCCTT	eekddddddddddkke	81	13493	13508	176
609119	2025	2040	TATCTGCTGCTGGCCT	eekddddddddddkke	76	13494	13509	177

TABLE 2-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
609120	2026	2041	TTATCTGCTGCTGGCC	eekddddddddddkke	56	13495	13510	178
609121	2028	2043	TGTTATCTGCTGCTGG	eekddddddddddkke	73	13497	13512	179
609122	2029	2044	TTGTTATCTGCTGCTG	eekddddddddddkke	87	13498	13513	180
609123	2030	2045	GTTGTTATCTGCTGCT	eekddddddddddkke	92	13499	13514	181
609124	2031	2046	GGTTGTTATCTGCTGC	eekddddddddddkke	90	13500	13515	182
609125	2048	2063	ATCGCTGATTTGTCCG	eekddddddddddkke	91	13517	13532	183
609126	2050	2065	ACATCGCTGATTTGTC	eekddddddddddkke	66	13519	13534	184
609127	2051	2066	CACATCGCTGATTTGT	eekddddddddddkke	79	13520	13535	185
609128	2052	2067	ACACATCGCTGATTTG	eekddddddddddkke	72	13521	13536	186
609129	2054	2069	TGACACATCGCTGATT	eekddddddddddkke	60	13523	13538	187
609130	2055	2070	GTGACACATCGCTGAT	eekddddddddddkke	77	13524	13539	188
609131	2057	2072	GGGTGACACATCGCTG	eekddddddddddkke	85	13526	13541	189
609132	2020	2035	GCTGCTGGCCTTTGCC	eekkddddddddkeee	47	13489	13504	173
609133	2021	2036	TGCTGCTGGCCTTTGC	eekkddddddddkeee	44	13490	13505	174
609134	2022	2037	CTGCTGCTGGCCTTTG	eekkddddddddkeee	62	13491	13506	162
609135	2023	2038	TCTGCTGCTGGCCTTT	eekkddddddddkeee	59	13492	13507	175
609136	2024	2039	ATCTGCTGCTGGCCTT	eekkddddddddkeee	70	13493	13508	176
609137	2025	2040	TATCTGCTGCTGGCCT	eekkddddddddkeee	59	13494	13509	177
609138	2026	2041	TTATCTGCTGCTGGCC	eekkddddddddkeee	78	13495	13510	178
609139	2027	2042	GTTATCTGCTGCTGGC	eekkddddddddkeee	79	13496	13511	163
609140	2028	2043	TGTTATCTGCTGCTGG	eekkddddddddkeee	83	13497	13512	179
609141	2029	2044	TTGTTATCTGCTGCTG	eekkddddddddkeee	67	13498	13513	180
609142	2030	2045	GTTGTTATCTGCTGCT	eekkddddddddkeee	68	13499	13514	181
609143	2031	2046	GGTTGTTATCTGCTGC	eekkddddddddkeee	81	13500	13515	182
609144	2032	2047	GGGTTGTTATCTGCTG	eekkddddddddkeee	81	13501	13516	164
609145	2046	2061	CGCTGATTTGTCCGGG	eekkddddddddkeee	53	13515	13530	129
609146	2047	2062	TCGCTGATTTGTCCGG	eekkddddddddkeee	80	13516	13531	165
609147	2048	2063	ATCGCTGATTTGTCCG	eekkddddddddkeee	88	13517	13532	183
609148	2049	2064	CATCGCTGATTTGTCC	eekkddddddddkeee	75	13518	13533	166
609149	2050	2065	ACATCGCTGATTTGTC	eekkddddddddkeee	64	13519	13534	184
609150	2051	2066	CACATCGCTGATTTGT	eekkddddddddkeee	77	13520	13535	185
609151	2052	2067	ACACATCGCTGATTTG	eekkddddddddkeee	57	13521	13536	186
609152	2053	2068	GACACATCGCTGATTT	eekkddddddddkeee	52	13522	13537	167
609153	2054	2069	TGACACATCGCTGATT	eekkddddddddkeee	37	13523	13538	187
609154	2055	2070	GTGACACATCGCTGAT	eekkddddddddkeee	50	13524	13539	188

TABLE 2-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
609155	2056	2071	GGTGACACATCGCTGA	eekkddddddddkkeee	60	13525	13540	130
609156	2057	2072	GGGTGACACATCGCTG	eekkddddddddkkeee	54	13526	13541	189
609157	2073	2088	AAAGGTGGGAGACTGG	eekkddddddddkkeee	40	13542	13557	168
609158	2020	2035	GCTGCTGGCCTTTGCC	eekkddddddddkkee	77	13489	13504	173
609159	2021	2036	TGCTGCTGGCCTTTGC	eekkddddddddkkee	85	13490	13505	174
609160	2022	2037	CTGCTGCTGGCCTTTG	eekkddddddddkkee	81	13491	13506	162
609161	2023	2038	TCTGCTGCTGGCCTTT	eekkddddddddkkee	91	13492	13507	175
609162	2024	2039	ATCTGCTGCTGGCCTT	eekkddddddddkkee	92	13493	13508	176
609163	2025	2040	TATCTGCTGCTGGCCT	eekkddddddddkkee	83	13494	13509	177
609164	2026	2041	TTATCTGCTGCTGGCC	eekkddddddddkkee	93	13495	13510	178
609165	2027	2042	GTTATCTGCTGCTGGC	eekkddddddddkkee	93	13496	13511	163
609166	2028	2043	TGTTATCTGCTGCTGG	eekkddddddddkkee	98	13497	13512	179
609167	2029	2044	TTGTTATCTGCTGCTG	eekkddddddddkkee	95	13498	13513	180
609168	2030	2045	GTTGTTATCTGCTGCT	eekkddddddddkkee	95	13499	13514	181
609169	2031	2046	GGTTGTTATCTGCTGC	eekkddddddddkkee	95	13500	13515	182
609170	2032	2047	GGGTTGTTATCTGCTG	eekkddddddddkkee	96	13501	13516	164
609171	2046	2061	CGCTGATTTGTCCGGG	eekkddddddddkkee	90	13515	13530	129
609172	2047	2062	TCGCTGATTTGTCCGG	eekkddddddddkkee	92	13516	13531	165
609173	2048	2063	ATCGCTGATTTGTCCG	eekkddddddddkkee	94	13517	13532	183
609174	2049	2064	CATCGCTGATTTGTCC	eekkddddddddkkee	96	13518	13533	166
609175	2050	2065	ACATCGCTGATTTGTC	eekkddddddddkkee	91	13519	13534	184
609176	2051	2066	CACATCGCTGATTTGT	eekkddddddddkkee	94	13520	13535	185
609177	2052	2067	ACACATCGCTGATTTG	eekkddddddddkkee	96	13521	13536	186
609178	2053	2068	GACACATCGCTGATTT	eekkddddddddkkee	88	13522	13537	167
609179	2054	2069	TGACACATCGCTGATT	eekkddddddddkkee	84	13523	13538	187
609180	2055	2070	GTGACACATCGCTGAT	eekkddddddddkkee	83	13524	13539	188
609181	2056	2071	GGTGACACATCGCTGA	eekkddddddddkkee	87	13525	13540	130
609182	2057	2072	GGGTGACACATCGCTG	eekkddddddddkkee	90	13526	13541	189
609183	2073	2088	AAAGGTGGGAGACTGG	eekkddddddddkkee	82	13542	13557	168
609184	2020	2035	GCTGCTGGCCTTTGCC	ekkddddddddkkee	84	13489	13504	173
609185	2021	2036	TGCTGCTGGCCTTTGC	ekkddddddddkkee	88	13490	13505	174
609186	2022	2037	CTGCTGCTGGCCTTTG	ekkddddddddkkee	88	13491	13506	162
609187	2023	2038	TCTGCTGCTGGCCTTT	ekkddddddddkkee	74	13492	13507	175
609188	2024	2039	ATCTGCTGCTGGCCTT	ekkddddddddkkee	90	13493	13508	176

TABLE 2-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
609189	2025	2040	TATCTGCTGCTGGCCT	ekkddddddddddkkee	91	13494	13509	177
609190	2026	2041	TTATCTGCTGCTGGCC	ekkddddddddddkkee	87	13495	13510	178
609191	2027	2042	GTTATCTGCTGCTGGC	ekkddddddddddkkee	97	13496	13511	163
609192	2028	2043	TGTTATCTGCTGCTGG	ekkddddddddddkkee	95	13497	13512	179
609193	2029	2044	TTGTTATCTGCTGCTG	ekkddddddddddkkee	96	13498	13513	180
609194	2030	2045	GTTGTTATCTGCTGCT	ekkddddddddddkkee	97	13499	13514	181
609195	2031	2046	GGTTGTTATCTGCTGC	ekkddddddddddkkee	97	13500	13515	182
609196	2032	2047	GGGTTGTTATCTGCTG	ekkddddddddddkkee	98	13501	13516	164
609197	2046	2061	CGCTGATTTGTCCGGG	ekkddddddddddkkee	96	13515	13530	129
609198	2047	2062	TCGCTGATTTGTCCGG	ekkddddddddddkkee	95	13516	13531	165
609199	2048	2063	ATCGCTGATTTGTCCG	ekkddddddddddkkee	96	13517	13532	183
609200	2049	2064	CATCGCTGATTTGTCC	ekkddddddddddkkee	94	13518	13533	166
609201	2050	2065	ACATCGCTGATTTGTC	ekkddddddddddkkee	94	13519	13534	184
609202	2051	2066	CACATCGCTGATTTGT	ekkddddddddddkkee	94	13520	13535	185
609203	2052	2067	ACACATCGCTGATTTG	ekkddddddddddkkee	91	13521	13536	186
609204	2053	2068	GACACATCGCTGATTT	ekkddddddddddkkee	94	13522	13537	167
609205	2054	2069	TGACACATCGCTGATT	ekkddddddddddkkee	87	13523	13538	187
609206	2055	2070	GTGACACATCGCTGAT	ekkddddddddddkkee	91	13524	13539	188
609207	2056	2071	GGTGACACATCGCTGA	ekkddddddddddkkee	93	13525	13540	130
609208	2057	2072	GGGTGACACATCGCTG	ekkddddddddddkkee	97	13526	13541	189
609209	2073	2088	AAAGGTGGGAGACTGG	ekkddddddddddkkee	95	13542	13557	168
609983	1983	2002	AGGCCTTGCCAGGCACTGTG	eeeeeddddddddddeeeee	75	13452	13471	207
609984	1984	2003	GAGGCCTTGCCAGGCACTGT	eeeeeddddddddddeeeee	54	13453	13472	208
609985	1985	2004	AGAGGCCTTGCCAGGCACTG	eeeeeddddddddddeeeee	63	13454	13473	209
609986	1986	2005	CAGAGGCCTTGCCAGGCACT	eeeeeddddddddddeeeee	63	13455	13474	210
609987	1987	2006	GCAGAGGCCTTGCCAGGCAC	eeeeeddddddddddeeeee	36	13456	13475	211
609988	1988	2007	GGCAGAGGCCTTGCCAGGCA	eeeeeddddddddddeeeee	48	13457	13476	212
609989	1989	2008	GGGCAGAGGCCTTGCCAGGC	eeeeeddddddddddeeeee	55	13458	13477	213
609990	2007	2026	CTTTGCCTCAAAGGCCAGGG	eeeeeddddddddddeeeee	38	13476	13495	214
609991	2008	2027	CCTTTGCCTCAAAGGCCAGG	eeeeeddddddddddeeeee	12	13477	13496	215
609992	2009	2028	GCCTTTGCCTCAAAGGCCAG	eeeeeddddddddddeeeee	11	13478	13497	216
609993	2010	2029	GGCCTTTGCCTCAAAGGCCA	eeeeeddddddddddeeeee	16	13479	13498	217
609994	2011	2030	TGGCCTTTGCCTCAAAGGCC	eeeeeddddddddddeeeee	13	13480	13499	218
609995	2012	2031	CTGGCCTTTGCCTCAAAGGC	eeeeeddddddddddeeeee	13	13481	13500	219
609996	2013	2032	GCTGGCCTTTGCCTCAAAGG	eeeeeddddddddddeeeee	35	13482	13501	220

TABLE 2-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
609997	2014	2033	TGCTGGCCTTTGCCTCAAAG	eeeeeddddddddddeeeee	20	13483	13502	221
609998	2015	2034	CTGCTGGCCTTTGCCTCAAA	eeeeeddddddddddeeeee	33	13484	13503	222
609999	2016	2035	GCTGCTGGCCTTTGCCTCAA	eeeeeddddddddddeeeee	69	13485	13504	223
610000	2017	2036	TGCTGCTGGCCTTTGCCTCA	eeeeeddddddddddeeeee	55	13486	13505	224
610001	2018	2037	CTGCTGCTGGCCTTTGCCTC	eeeeeddddddddddeeeee	73	13487	13506	225
610002	2019	2038	TCTGCTGCTGGCCTTTGCCT	eeeeeddddddddddeeeee	72	13488	13507	226
610003	2020	2039	ATCTGCTGCTGGCCTTTGCC	eeeeeddddddddddeeeee	69	13489	13508	227
610004	2021	2040	TATCTGCTGCTGGCCTTTGC	eeeeeddddddddddeeeee	56	13490	13509	228
610005	2022	2041	TTATCTGCTGCTGGCCTTTG	eeeeeddddddddddeeeee	29	13491	13510	229
610006	2023	2042	GTTATCTGCTGCTGGCCTTT	eeeeeddddddddddeeeee	74	13492	13511	230
610007	2024	2043	TGTTATCTGCTGCTGGCCTT	eeeeeddddddddddeeeee	74	13493	13512	231
610008	2025	2044	TTGTTATCTGCTGCTGGCCT	eeeeeddddddddddeeeee	72	13494	13513	232
610009	2026	2045	GTTGTTATCTGCTGCTGGCC	eeeeeddddddddddeeeee	73	13495	13514	233
610010	2027	2046	GGTTGTTATCTGCTGCTGGC	eeeeeddddddddddeeeee	83	13496	13515	234
610011	2028	2047	GGGTTGTTATCTGCTGCTGG	eeeeeddddddddddeeeee	76	13497	13516	235
610012	2046	2065	ACATCGCTGATTTGTCCGGG	eeeeeddddddddddeeeee	79	13515	13534	236
610013	2047	2066	CACATCGCTGATTTGTCCGG	eeeeeddddddddddeeeee	79	13516	13535	237
610014	2048	2067	ACACATCGCTGATTTGTCCG	eeeeeddddddddddeeeee	77	13517	13536	238
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	89	13518	13537	239
610016	2050	2069	TGACACATCGCTGATTTGTC	eeeeeddddddddddeeeee	83	13519	13538	240
610017	2051	2070	GTGACACATCGCTGATTTGT	eeeeeddddddddddeeeee	74	13520	13539	241
610018	2052	2071	GGTGACACATCGCTGATTTG	eeeeeddddddddddeeeee	74	13521	13540	242
610019	2053	2072	GGGTGACACATCGCTGATTT	eeeeeddddddddddeeeee	76	13522	13541	243
610020	2073	2092	AAGAAAAGGTGGGAGACTGG	eeeeeddddddddddeeeee	24	13542	13561	244
610021	2074	2093	GAAGAAAAGGTGGGAGACTG	eeeeeddddddddddeeeee	23	13543	13562	245
610022	2075	2094	AGAAGAAAAGGTGGGAGACT	eeeeeddddddddddeeeee	26	13544	13563	246
610023	2076	2095	TAGAAGAAAAGGTGGGAGAC	eeeeeddddddddddeeeee	24	13545	13564	247
610024	2077	2096	TTAGAAGAAAAGGTGGGAGA	eeeeeddddddddddeeeee	19	13546	13565	248
610025	2078	2097	ATTAGAAGAAAAGGTGGGAG	eeeeeddddddddddeeeee	30	13547	13566	249
610026	2079	2098	CATTAGAAGAAAAGGTGGGA	eeeeeddddddddddeeeee	40	13548	13567	250
610027	2080	2099	TCATTAGAAGAAAAGGTGGG	eeeeeddddddddddeeeee	56	13549	13568	251
610028	2081	2100	CTCATTAGAAGAAAAGGTGG	eeeeeddddddddddeeeee	74	13550	13569	252
610029	2082	2101	ACTCATTAGAAGAAAAGGTG	eeeeeddddddddddeeeee	62	13551	13570	253
610030	2083	2102	GACTCATTAGAAGAAAAGGT	eeeeeddddddddddeeeee	69	13552	13571	254

TABLE 2-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
610031	2084	2103	CGACTCATTAGAAGAAAAGG	eeeeeddddddddddeeeee	59	13553	13572	255
610032	2085	2104	TCGACTCATTAGAAGAAAAG	eeeeeddddddddddeeeee	50	13554	13573	256
610033	2086	2105	GTCGACTCATTAGAAGAAAA	eeeeeddddddddddeeeee	67	13555	13574	257
610034	2087	2106	AGTCGACTCATTAGAAGAAA	eeeeeddddddddddeeeee	62	13556	13575	258
610035	2088	2107	AAGTCGACTCATTAGAAGAA	eeeeeddddddddddeeeee	45	13557	13576	259
610036	2089	2108	AAAGTCGACTCATTAGAAGA	eeeeeddddddddddeeeee	43	13558	13577	260
610037	2090	2109	CAAAGTCGACTCATTAGAAG	eeeeeddddddddddeeeee	46	13559	13578	261
610038	2091	2110	TCAAAGTCGACTCATTAGAA	eeeeeddddddddddeeeee	67	13560	13579	262
610039	2092	2111	CTCAAAGTCGACTCATTAGA	eeeeeddddddddddeeeee	65	13561	13580	263
610040	2093	2112	GCTCAAAGTCGACTCATTAG	eeeeeddddddddddeeeee	74	13562	13581	264
610041	2094	2113	AGCTCAAAGTCGACTCATTA	eeeeeddddddddddeeeee	61	13563	13582	265
610042	2095	2114	CAGCTCAAAGTCGACTCATT	eeeeeddddddddddeeeee	71	13564	13583	266
610043	2096	2115	CCAGCTCAAAGTCGACTCAT	eeeeeddddddddddeeeee	77	13565	13584	267
610044	2097	2116	TCCAGCTCAAAGTCGACTCA	eeeeeddddddddddeeeee	82	13566	13585	268
610045	2098	2117	TTCCAGCTCAAAGTCGACTC	eeeeeddddddddddeeeee	80	13567	13586	269
610046	2099	2118	TTTCCAGCTCAAAGTCGACT	eeeeeddddddddddeeeee	84	13568	13587	270
610047	2100	2119	CTTTCCAGCTCAAAGTCGAC	eeeeeddddddddddeeeee	65	13569	13588	271
610048	2101	2120	GCTTTCCAGCTCAAAGTCGA	eeeeeddddddddddeeeee	61	13570	13589	272
610049	2102	2121	TGCTTTCCAGCTCAAAGTCG	eeeeeddddddddddeeeee	69	13571	13590	273
610050	2103	2122	CTGCTTTCCAGCTCAAAGTC	eeeeeddddddddddeeeee	54	13572	13591	274
610051	2104	2123	GCTGCTTTCCAGCTCAAAGT	eeeeeddddddddddeeeee	57	13573	13592	275
610052	2105	2124	GGCTGCTTTCCAGCTCAAAG	eeeeeddddddddddeeeee	63	13574	13593	276
610053	2106	2125	CGGCTGCTTTCCAGCTCAAA	eeeeeddddddddddeeeee	40	13575	13594	277
610054	2107	2126	ACGGCTGCTTTCCAGCTCAA	eeeeeddddddddddeeeee	62	13576	13595	278
610055	2108	2127	AACGGCTGCTTTCCAGCTCA	eeeeeddddddddddeeeee	69	13577	13596	279
610056	2109	2128	AAACGGCTGCTTTCCAGCTC	eeeeeddddddddddeeeee	54	13578	13597	280
610057	2110	2129	GAAACGGCTGCTTTCCAGCT	eeeeeddddddddddeeeee	64	13579	13598	281
610058	2111	2130	AGAAACGGCTGCTTTCCAGC	eeeeeddddddddddeeeee	57	13580	13599	282
610059	2112	2131	GAGAAACGGCTGCTTTCCAG	eeeeeddddddddddeeeee	56	13581	13600	283
610060	2113	2132	GGAGAAACGGCTGCTTTCCA	eeeeeddddddddddeeeee	73	13582	13601	284

[0513] Table 3 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 500 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 3

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	65	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	75	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	0	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	72	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	65	13515	13530	129
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	80	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	81	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	81	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	76	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	88	13496	13511	163
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	68	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	75	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	62	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	54	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	81	13516	13531	165
611901	1	16	TGCCCGCTCATGGGAT	ekkddddddddddkke	13	1986	2001	14
611902	6	21	CCTGCTGCCCGCTCAT	ekkddddddddddkke	19	1991	2006	285
611903	11	26	CTGACCCTGCTGCCCG	ekkddddddddddkke	19	1996	2011	286
611904	16	31	CACTTCTGACCCTGCT	ekkddddddddddkke	0	2001	2016	287
611905	35	50	GCTTAGGCAACACGGG	ekkddddddddddkke	0	2020	2035	16
611906	40	55	GTCTTGCTTAGGCAAC	ekkddddddddddkke	27	2025	2040	288
611907	45	60	GGAGAGTCTTGCTTAG	ekkddddddddddkke	0	2030	2045	17
611908	67	82	GGTGCAGAGGGCAGAG	ekkddddddddddkke	0	2052	2067	289
611909	72	87	CCGGAGGTGCAGAGGG	ekkddddddddddkke	17	2057	2072	290
611910	77	92	GCAGGCCGGAGGTGCA	ekkddddddddddkke	10	2062	2077	291
611911	82	97	GACATGCAGGCCGGAG	ekkddddddddddkke	16	2067	2082	292
611912	87	102	ACAGGGACATGCAGGC	ekkddddddddddkke	20	2072	2087	293
611913	92	107	AGGCCACAGGGACATG	ekkddddddddddkke	5	2077	2092	294
611914	97	112	CCAAGAGGCCACAGGG	ekkddddddddddkke	0	2082	2097	295
611915	102	117	TACCCCCAAGAGGCCA	ekkddddddddddkke	0	2087	2102	296
611916	107	122	AGATGTACCCCCAAGA	ekkddddddddddkke	8	2092	2107	297
611917	112	127	CCGGGAGATGTACCCC	ekkddddddddddkke	16	2097	2112	298
611918	117	132	CAGCCCCGGGAGATGT	ekkddddddddddkke	15	2102	2117	299
611919	122	137	TGACCCAGCCCCGGGA	ekkddddddddddkke	4	2107	2122	20
611920	127	142	CCTTCTGACCCAGCCC	ekkddddddddddkke	23	2112	2127	300

TABLE 3-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
611921	132	147	CCAGGCCTTCTGACCC	ekkddddddddddkke	20	2117	2132	301
611922	137	152	ACCACCCAGGCCTTCT	ekkddddddddddkke	15	2122	2137	302
611923	142	157	GGCCAACCACCCAGGC	ekkddddddddddkke	11	2127	2142	303
611924	147	162	CCTGAGGCCAACCACC	ekkddddddddddkke	0	2132	2147	304
611925	152	167	GACAGCCTGAGGCCAA	ekkddddddddddkke	18	2137	2152	305
611926	157	172	TGTGTGACAGCCTGAG	ekkddddddddddkke	12	2142	2157	306
611927	162	177	CTAGGTGTGTGACAGC	ekkddddddddddkke	23	2147	2162	307
611928	167	182	TCTCCCTAGGTGTGTG	ekkddddddddddkke	9	2152	2167	308
611929	172	187	GAGCATCTCCCTAGGT	ekkddddddddddkke	12	2157	2172	309
611930	177	192	AACGGGAGCATCTCCC	ekkddddddddddkke	8	2162	2177	310
611931	182	197	CCAGAAACGGGAGCAT	ekkddddddddddkke	9	2167	2182	311
611932	187	202	GGTTCCCAGAAACGGG	ekkddddddddddkke	13	2172	2187	312
611933	192	207	GCCAAGGTTCCCAGAA	ekkddddddddddkke	33	2177	2192	313
611934	208	223	GTTTGCAGGAGTCGGG	ekkddddddddddkke	17	2193	2208	314
611935	213	228	CCGAAGTTTGCAGGAG	ekkddddddddddkke	27	2198	2213	315
611936	218	233	ATTTACCGAAGTTTGC	ekkddddddddddkke	7	2203	2218	316
611937	223	238	TACACATTTACCGAAG	ekkddddddddddkke	14	2208	2223	317
611938	228	243	CGAGTTACACATTTAC	ekkddddddddddkke	12	2213	2228	318
611939	233	248	AGGGTCGAGTTACACA	ekkddddddddddkke	9	2218	2233	319
611940	238	253	GGTGCAGGGTCGAGTT	ekkddddddddddkke	28	2223	2238	320
611941	243	258	GAGCCGGTGCAGGGTC	ekkddddddddddkke	26	2228	2243	321
611942	248	263	AGAGTGAGCCGGTGCA	ekkddddddddddkke	8	2233	2248	27
611943	253	268	TGAACAGAGTGAGCCG	ekkddddddddddkke	16	2238	2253	322
611944	258	273	ACTGCTGAACAGAGTG	ekkddddddddddkke	17	2243	2258	28
611945	263	278	GTTTCACTGCTGAACA	ekkddddddddddkke	17	2248	2263	323
611946	268	283	GCAGAGTTTCACTGCT	ekkddddddddddkke	0	2253	2268	29
611947	273	288	TCGATGCAGAGTTTCA	ekkddddddddddkke	2	2258	2273	324
611948	278	293	AGTGATCGATGCAGAG	ekkddddddddddkke	12	2263	2278	30
611949	283	298	GTCTTAGTGATCGATG	ekkddddddddddkke	0	2268	2283	325
611950	288	303	AGGAAGTCTTAGTGAT	ekkddddddddddkke	3	2273	2288	31
611951	293	308	CTTCCAGGAAGTCTTA	ekkddddddddddkke	10	2278	2293	326
611952	299	314	GGACCTCTTCCAGGAA	ekkddddddddddkke	21	2284	2299	327
611953	304	319	CGCTGGGACCTCTTCC	ekkddddddddddkke	20	2289	2304	328
611954	309	324	ACTCACGCTGGGACCT	ekkddddddddddkke	0	2294	2309	329

TABLE 3-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
611955	314	329	GCGACACTCACGCTGG	ekkddddddddddkke	21	2299	2314	330
611956	319	334	CAGAAGCGACACTCAC	ekkddddddddddkke	18	2304	2319	331
611957	324	339	GATGCCAGAAGCGACA	ekkddddddddddkke	1	2309	2324	332
611958	329	344	GGACAGATGCCAGAAG	ekkddddddddddkke	16	2314	2329	333
611959	334	349	CAGAAGGACAGATGCC	ekkddddddddddkke	0	2319	2334	334
611960	339	354	CTGGCCAGAAGGACAG	ekkddddddddddkke	13	2324	2339	335
611961	344	359	ACAGGCTGGCCAGAAG	ekkddddddddddkke	15	2329	2344	336
611962	349	364	AGACCACAGGCTGGCC	ekkddddddddddkke	17	2334	2349	337
611963	354	369	TGGCCAGACCACAGGC	ekkddddddddddkke	21	2339	2354	338
611964	359	374	TCACTTGGCCAGACCA	ekkddddddddddkke	7	2344	2359	339
611965	364	379	TTACATCACTTGGCCA	ekkddddddddddkke	21	2349	2364	340
611966	369	384	GAGGGTTACATCACTT	ekkddddddddddkke	20	2354	2369	341
611967	374	389	GAGAGGAGGGTTACAT	ekkddddddddddkke	18	2359	2374	342
611968	386	401	GTGCACAGGCTGGAGA	ekkddddddddddkke	4	2371	2386	36
611969	391	406	TGCCTGTGCACAGGCT	ekkddddddddddkke	10	2376	2391	343
611970	396	411	CAGGCTGCCTGTGCAC	ekkddddddddddkke	26	2381	2396	344
611971	401	416	GTTCCCAGGCTGCCTG	ekkddddddddddkke	30	2386	2401	345
611972	406	421	GAGCTGTTCCCAGGCT	ekkddddddddddkke	15	2391	2406	346
611973	411	426	GGATGGAGCTGTTCCC	ekkddddddddddkke	19	2396	2411	347
611974	431	446	TATTTATAGCTGAGGG	ekkddddddddddkke	11	2416	2431	37
611975	436	451	TGCCCTATTTATAGCT	ekkddddddddddkke	20	2421	2436	348
611976	441	456	CACGATGCCCTATTTA	ekkddddddddddkke	11	2426	2441	38
612381	1852	1867	CTCCAAGACCTCAGGC	ekkddddddddddkke	4	13321	13336	349
612382	1855	1870	CACCTCCAAGACCTCA	ekkddddddddddkke	18	13324	13339	350
612383	1858	1873	GGTCACCTCCAAGACC	ekkddddddddddkke	0	13327	13342	351
612384	1861	1876	CAGGGTCACCTCCAAG	ekkddddddddddkke	16	13330	13345	352
612385	1864	1879	GTTCAGGGTCACCTCC	ekkddddddddddkke	28	13333	13348	353
612386	1867	1882	GCGGTTCAGGGTCACC	ekkddddddddddkke	18	13336	13351	354
612387	1873	1888	GAATGGGCGGTTCAGG	ekkddddddddddkke	6	13342	13357	355
612388	1876	1891	CAGGAATGGGCGGTTC	ekkddddddddddkke	13	13345	13360	356
612389	1879	1894	AAACAGGAATGGGCGG	ekkddddddddddkke	16	13348	13363	357
612390	1883	1898	CAGCAAACAGGAATGG	ekkddddddddddkke	11	13352	13367	358
612391	1887	1902	TACACAGCAAACAGGA	ekkddddddddddkke	8	13356	13371	359
612392	1892	1907	GATCATACACAGCAAA	ekkddddddddddkke	6	13361	13376	360
612393	1895	1910	TTTGATCATACACAGC	ekkddddddddddkke	15	13364	13379	361

TABLE 3-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612394	1898	1913	CGCTTTGATCATACAC	ekkddddddddddkke	16	13367	13382	362
612395	1916	1931	GGAAGTGCAGGGCAGT	ekkddddddddddkke	8	13385	13400	363
612396	1923	1938	CGGCCCAGGAAGTGCA	ekkddddddddddkke	0	13392	13407	123
612397	1926	1941	ACGCGGCCCAGGAAGT	ekkddddddddddkke	1	13395	13410	364
612398	1929	1944	GCCACGCGGCCCAGGA	ekkddddddddddkke	6	13398	13413	365
612399	1932	1947	TTGGCCACGCGGCCCA	ekkddddddddddkke	7	13401	13416	366
612400	1935	1950	GGGTTGGCCACGCGGC	ekkddddddddddkke	29	13404	13419	367
612401	1938	1953	AGCGGGTTGGCCACGC	ekkddddddddddkke	13	13407	13422	368
612402	1941	1956	CTCAGCGGGTTGGCCA	ekkddddddddddkke	0	13410	13425	369
612403	1944	1959	GTGCTCAGCGGGTTGG	ekkddddddddddkke	13	13413	13428	370
612404	1947	1962	GCTGTGCTCAGCGGGT	ekkddddddddddkke	39	13416	13431	371
612405	1949	1964	ATGCTGTGCTCAGCGG	ekkddddddddddkke	13	13418	13433	372
612406	1950	1965	CATGCTGTGCTCAGCG	ekkddddddddddkke	20	13419	13434	373
612407	1951	1966	TCATGCTGTGCTCAGC	ekkddddddddddkke	23	13420	13435	374
612408	1952	1967	CTCATGCTGTGCTCAG	ekkddddddddddkke	29	13421	13436	375
612409	1954	1969	GCCTCATGCTGTGCTC	ekkddddddddddkke	36	13423	13438	376
612410	1956	1971	TGGCCTCATGCTGTGC	ekkddddddddddkke	0	13425	13440	377
612411	1957	1972	CTGGCCTCATGCTGTG	ekkddddddddddkke	2	13426	13441	378
612412	1959	1974	CCCTGGCCTCATGCTG	ekkddddddddddkke	5	13428	13443	379
612413	1960	1975	GCCCTGGCCTCATGCT	ekkddddddddddkke	6	13429	13444	380
612414	1961	1976	GGCCCTGGCCTCATGC	ekkddddddddddkke	0	13430	13445	126
612415	1976	1991	GGCACTGTGTTCTGGG	ekkddddddddddkke	45	13445	13460	381
612416	1987	2002	AGGCCTTGCCAGGCAC	ekkddddddddddkke	35	13456	13471	382
612417	1992	2007	GGCAGAGGCCTTGCCA	ekkddddddddddkke	14	13461	13476	383
612418	2007	2022	GCCTCAAAGGCCAGGG	ekkddddddddddkke	0	13476	13491	128
612419	2008	2023	TGCCTCAAAGGCCAGG	ekkddddddddddkke	10	13477	13492	384
612420	2009	2024	TTGCCTCAAAGGCCAG	ekkddddddddddkke	7	13478	13493	385
612421	2010	2025	TTTGCCTCAAAGGCCA	ekkddddddddddkke	13	13479	13494	386
612422	2011	2026	CTTTGCCTCAAAGGCC	ekkddddddddddkke	0	13480	13495	387
612423	2012	2027	CCTTTGCCTCAAAGGC	ekkddddddddddkke	14	13481	13496	388
612424	2013	2028	GCCTTTGCCTCAAAGG	ekkddddddddddkke	3	13482	13497	389
612425	2014	2029	GGCCTTTGCCTCAAAG	ekkddddddddddkke	15	13483	13498	390
612426	2015	2030	TGGCCTTTGCCTCAAA	ekkddddddddddkke	0	13484	13499	391
612427	2016	2031	CTGGCCTTTGCCTCAA	ekkddddddddddkke	0	13485	13500	392

TABLE 3-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612428	2017	2032	GCTGGCCTTTGCCTCA	ekkddddddddddkke	5	13486	13501	393
612429	2100	2115	CCAGCTCAAAGTCGAC	ekkddddddddddkke	46	13569	13584	394
612430	2101	2116	TCCAGCTCAAAGTCGA	ekkddddddddddkke	34	13570	13585	395
612431	2102	2117	TTCCAGCTCAAAGTCG	ekkddddddddddkke	16	13571	13586	396
612432	2103	2118	TTTCCAGCTCAAAGTC	ekkddddddddddkke	5	13572	13587	397
612433	2105	2120	GCTTTCCAGCTCAAAG	ekkddddddddddkke	9	13574	13589	398
612434	2110	2125	CGGCTGCTTTCCAGCT	ekkddddddddddkke	0	13579	13594	399
612435	2111	2126	ACGGCTGCTTTCCAGC	ekkddddddddddkke	24	13580	13595	133
612436	2112	2127	AACGGCTGCTTTCCAG	ekkddddddddddkke	14	13581	13596	400
612437	2113	2128	AAACGGCTGCTTTCCA	ekkddddddddddkke	14	13582	13597	401
612438	2114	2129	GAAACGGCTGCTTTCC	ekkddddddddddkke	13	13583	13598	402
612439	2115	2130	AGAAACGGCTGCTTTC	ekkddddddddddkke	15	13584	13599	403
612440	2116	2131	GAGAAACGGCTGCTTT	ekkddddddddddkke	33	13585	13600	404
612441	2117	2132	GGAGAAACGGCTGCTT	ekkddddddddddkke	26	13586	13601	405
612442	2118	2133	AGGAGAAACGGCTGCT	ekkddddddddddkke	50	13587	13602	406
612443	2119	2134	AAGGAGAAACGGCTGC	ekkddddddddddkke	21	13588	13603	407
612444	2120	2135	CAAGGAGAAACGGCTG	ekkddddddddddkke	30	13589	13604	408
612445	2121	2136	CCAAGGAGAAACGGCT	ekkddddddddddkke	43	13590	13605	134
612446	2122	2137	ACCAAGGAGAAACGGC	ekkddddddddddkke	32	13591	13606	409
612447	2123	2138	GACCAAGGAGAAACGG	ekkddddddddddkke	33	13592	13607	410
612448	2124	2139	AGACCAAGGAGAAACG	ekkddddddddddkke	55	13593	13608	411
612449	2125	2140	TAGACCAAGGAGAAAC	ekkddddddddddkke	15	13594	13609	412
612450	2126	2141	TTAGACCAAGGAGAAA	ekkddddddddddkke	17	13595	13610	413
612451	2128	2143	ACTTAGACCAAGGAGA	ekkddddddddddkke	32	13597	13612	414
612452	2129	2144	CACTTAGACCAAGGAG	ekkddddddddddkke	38	13598	13613	415
612453	2130	2145	ACACTTAGACCAAGGA	ekkddddddddddkke	48	13599	13614	416
612454	2133	2148	AGCACACTTAGACCAA	ekkddddddddddkke	29	13602	13617	417
612455	2134	2149	CAGCACACTTAGACCA	ekkddddddddddkke	31	13603	13618	418
612456	2135	2150	GCAGCACACTTAGACC	ekkddddddddddkke	13	13604	13619	419
612457	2136	2151	TGCAGCACACTTAGAC	ekkddddddddddkke	18	13605	13620	420
612458	2137	2152	ATGCAGCACACTTAGA	ekkddddddddddkke	0	13606	13621	421
612459	2138	2153	CATGCAGCACACTTAG	ekkddddddddddkke	0	13607	13622	422
612460	2139	2154	CCATGCAGCACACTTA	ekkddddddddddkke	0	13608	13623	423
612461	2140	2155	TCCATGCAGCACACTT	ekkddddddddddkke	24	13609	13624	424
612462	2141	2156	CTCCATGCAGCACACT	ekkddddddddddkke	40	13610	13625	425

TABLE 3-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612463	2142	2157	ACTCCATGCAGCACAC	ekkddddddddddkke	0	13611	13626	426
612464	2143	2158	CACTCCATGCAGCACA	ekkddddddddddkke	18	13612	13627	427
612465	2144	2159	TCACTCCATGCAGCAC	ekkddddddddddkke	14	13613	13628	428
612466	2162	2177	GCTGCAGGCTTCTACT	ekkddddddddddkke	12	13631	13646	429
612467	2163	2178	CGCTGCAGGCTTCTAC	ekkddddddddddkke	2	13632	13647	430
612468	2164	2179	CCGCTGCAGGCTTCTA	ekkddddddddddkke	2	13633	13648	431
612469	2165	2180	GCCGCTGCAGGCTTCT	ekkddddddddddkke	12	13634	13649	432
612470	2166	2181	TGCCGCTGCAGGCTTC	ekkddddddddddkke	1	13635	13650	136
612471	2167	2182	GTGCCGCTGCAGGCTT	ekkddddddddddkke	12	13636	13651	433
612472	2168	2183	TGTGCCGCTGCAGGCT	ekkddddddddddkke	31	13637	13652	434
612473	2169	2184	TTGTGCCGCTGCAGGC	ekkddddddddddkke	20	13638	13653	435
612474	2170	2185	TTTGTGCCGCTGCAGG	ekkddddddddddkke	27	13639	13654	436
612475	2171	2186	ATTTGTGCCGCTGCAG	ekkddddddddddkke	29	13640	13655	437
612476	2172	2187	CATTTGTGCCGCTGCA	ekkddddddddddkke	33	13641	13656	438
612477	2173	2188	GCATTTGTGCCGCTGC	ekkddddddddddkke	48	13642	13657	439
612478	2174	2189	TGCATTTGTGCCGCTG	ekkddddddddddkke	13	13643	13658	440
612479	2175	2190	GTGCATTTGTGCCGCT	ekkddddddddddkke	49	13644	13659	441
612480	2176	2191	GGTGCATTTGTGCCGC	ekkddddddddddkke	32	13645	13660	137
612481	2177	2192	AGGTGCATTTGTGCCG	ekkddddddddddkke	40	13646	13661	442
612482	2178	2193	GAGGTGCATTTGTGCC	ekkddddddddddkke	48	13647	13662	443
612483	2179	2194	GGAGGTGCATTTGTGC	ekkddddddddddkke	17	13648	13663	444
612484	2180	2195	GGGAGGTGCATTTGTG	ekkddddddddddkke	15	13649	13664	445
612485	2181	2196	TGGGAGGTGCATTTGT	ekkddddddddddkke	25	13650	13665	446
612486	2182	2197	CTGGGAGGTGCATTTG	ekkddddddddddkke	25	13651	13666	447
612487	2183	2198	ACTGGGAGGTGCATTT	ekkddddddddddkke	19	13652	13667	448
612488	2184	2199	AACTGGGAGGTGCATT	ekkddddddddddkke	0	13653	13668	449
612489	2185	2200	AAACTGGGAGGTGCAT	ekkddddddddddkke	14	13654	13669	450
612490	2186	2201	CAAACTGGGAGGTGCA	ekkddddddddddkke	53	13655	13670	451
612491	2187	2202	GCAAACTGGGAGGTGC	ekkddddddddddkke	63	13656	13671	452
612492	2188	2203	AGCAAACTGGGAGGTG	ekkddddddddddkke	26	13657	13672	453
612493	2192	2207	ACCCAGCAAACTGGGA	ekkddddddddddkke	0	13661	13676	454
612494	2193	2208	AACCCAGCAAACTGGG	ekkddddddddddkke	0	13662	13677	455
612495	2195	2210	TAAACCCAGCAAACTG	ekkddddddddddkke	8	13664	13679	456
612496	2196	2211	ATAAACCCAGCAAACT	ekkddddddddddkke	4	13665	13680	457

TABLE 3-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612497	2210	2225	CCCCATTCTCTAAAAT	ekkddddddddddkke	24	13679	13694	458
612498	2211	2226	CCCCCATTCTCTAAAA	ekkddddddddddkke	0	13680	13695	459
612499	2212	2227	ACCCCCATTCTCTAAA	ekkddddddddddkke	0	13681	13696	460
612500	2213	2228	CACCCCCATTCTCTAA	ekkddddddddddkke	6	13682	13697	461
612501	2214	2229	CCACCCCCATTCTCTA	ekkddddddddddkke	39	13683	13698	462
612502	2226	2241	GTTCTTGCCTCCCCAC	ekkddddddddddkke	61	13695	13710	463
612503	2227	2242	GGTTCTTGCCTCCCCA	ekkddddddddddkke	76	13696	13711	464
612504	2228	2243	TGGTTCTTGCCTCCCC	ekkddddddddddkke	59	13697	13712	465
612505	2229	2244	CTGGTTCTTGCCTCCC	ekkddddddddddkke	66	13698	13713	466
612506	2230	2245	ACTGGTTCTTGCCTCC	ekkddddddddddkke	70	13699	13714	467
612507	2231	2246	CACTGGTTCTTGCCTC	ekkddddddddddkke	57	13700	13715	468
612508	2232	2247	ACACTGGTTCTTGCCT	ekkddddddddddkke	45	13701	13716	469
612509	2233	2248	AACACTGGTTCTTGCC	ekkddddddddddkke	66	13702	13717	470
612510	2234	2249	AAACACTGGTTCTTGC	ekkddddddddddkke	52	13703	13718	471
612511	2235	2250	TAAACACTGGTTCTTG	ekkddddddddddkke	17	13704	13719	472
612512	2236	2251	CTAAACACTGGTTCTT	ekkddddddddddkke	35	13705	13720	473
612513	2237	2252	GCTAAACACTGGTTCT	ekkddddddddddkke	53	13706	13721	474
612514	2238	2253	CGCTAAACACTGGTTC	ekkddddddddddkke	56	13707	13722	475
612515	2239	2254	GCGCTAAACACTGGTT	ekkddddddddddkke	59	13708	13723	476
612516	2240	2255	CGCGCTAAACACTGGT	ekkddddddddddkke	66	13709	13724	477
612517	2241	2256	CCGCGCTAAACACTGG	ekkddddddddddkke	57	13710	13725	478
612518	2242	2257	CCCGCGCTAAACACTG	ekkddddddddddkke	35	13711	13726	479
612519	2243	2258	TCCCGCGCTAAACACT	ekkddddddddddkke	60	13712	13727	480
612520	2244	2259	GTCCCGCGCTAAACAC	ekkddddddddddkke	38	13713	13728	481
612521	2245	2260	AGTCCCGCGCTAAACA	ekkddddddddddkke	35	13714	13729	482
612522	2246	2261	TAGTCCCGCGCTAAAC	ekkddddddddddkke	1	13715	13730	483
612524	2248	2263	AGTAGTCCCGCGCTAA	ekkddddddddddkke	47	13717	13732	484
612525	2249	2264	CAGTAGTCCCGCGCTA	ekkddddddddddkke	13	13718	13733	485
612526	2250	2265	ACAGTAGTCCCGCGCT	ekkddddddddddkke	32	13719	13734	486
612527	2251	2266	AACAGTAGTCCCGCGC	ekkddddddddddkke	46	13720	13735	487
612528	2252	2267	GAACAGTAGTCCCGCG	ekkddddddddddkke	27	13721	13736	488
612529	2253	2268	GGAACAGTAGTCCCGC	ekkddddddddddkke	46	13722	13737	489
612530	2254	2269	TGGAACAGTAGTCCCG	ekkddddddddddkke	17	13723	13738	490
612531	2255	2270	TTGGAACAGTAGTCCC	ekkddddddddddkke	42	13724	13739	491
612532	2256	2271	TTTGGAACAGTAGTCC	ekkddddddddddkke	14	13725	13740	492

TABLE 3-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612533	2257	2272	TTTTGGAACAGTAGTC	ekkddddddddddkke	7	13726	13741	493
612534	2258	2273	TTTTTGGAACAGTAGT	ekkddddddddddkke	4	13727	13742	494
612535	2259	2274	CTTTTTGGAACAGTAG	ekkddddddddddkke	31	13728	13743	495
612536	2264	2279	GAATTCTTTTTGGAAC	ekkddddddddddkke	6	13733	13748	496
612537	2265	2280	GGAATTCTTTTTGGAA	ekkddddddddddkke	45	13734	13749	497
612538	2266	2281	TGGAATTCTTTTTGGA	ekkddddddddddkke	42	13735	13750	498
612539	2267	2282	TTGGAATTCTTTTTGG	ekkddddddddddkke	26	13736	13751	499
612540	2270	2285	CGGTTGGAATTCTTTT	ekkddddddddddkke	61	13739	13754	500
612541	2271	2286	TCGGTTGGAATTCTTT	ekkddddddddddkke	58	13740	13755	501
612542	2272	2287	GTCGGTTGGAATTCTT	ekkddddddddddkke	60	13741	13756	502
612543	2273	2288	GGTCGGTTGGAATTCT	ekkddddddddddkke	58	13742	13757	503
612544	2274	2289	TGGTCGGTTGGAATTC	ekkddddddddddkke	46	13743	13758	138
612545	2275	2290	CTGGTCGGTTGGAATT	ekkddddddddddkke	0	13744	13759	504
612546	2276	2291	GCTGGTCGGTTGGAAT	ekkddddddddddkke	27	13745	13760	505
612547	2277	2292	AGCTGGTCGGTTGGAA	ekkddddddddddkke	33	13746	13761	506
612548	2278	2293	AAGCTGGTCGGTTGGA	ekkddddddddddkke	51	13747	13762	507
612549	2279	2294	CAAGCTGGTCGGTTGG	ekkddddddddddkke	32	13748	13763	508
612550	2280	2295	ACAAGCTGGTCGGTTG	ekkddddddddddkke	19	13749	13764	509
612551	2281	2296	AACAAGCTGGTCGGTT	ekkddddddddddkke	39	13750	13765	510
612552	2282	2297	AAACAAGCTGGTCGGT	ekkddddddddddkke	49	13751	13766	511
612553	2283	2298	CAAACAAGCTGGTCGG	ekkddddddddddkke	63	13752	13767	512
612554	2284	2299	ACAAACAAGCTGGTCG	ekkddddddddddkke	48	13753	13768	139
612555	2285	2300	CACAAACAAGCTGGTC	ekkddddddddddkke	37	13754	13769	513
612556	2286	2301	TCACAAACAAGCTGGT	ekkddddddddddkke	28	13755	13770	514
612557	2287	2302	TTCACAAACAAGCTGG	ekkddddddddddkke	52	13756	13771	515
612558	2288	2303	TTTCACAAACAAGCTG	ekkddddddddddkke	14	13757	13772	516
612559	2289	2304	GTTTCACAAACAAGCT	ekkddddddddddkke	65	13758	13773	517
612560	2290	2305	TGTTTCACAAACAAGC	ekkddddddddddkke	58	13759	13774	518
612561	2291	2306	TTGTTTCACAAACAAG	ekkddddddddddkke	8	13760	13775	519
612562	2304	2319	AGGGAACACTTTTTTG	ekkddddddddddkke	26	13773	13788	520
612563	2311	2326	CTTGAAAAGGGAACAC	ekkddddddddddkke	29	13780	13795	140
612564	2312	2327	ACTTGAAAAGGGAACA	ekkddddddddddkke	19	13781	13796	521
612565	2313	2328	AACTTGAAAAGGGAAC	ekkddddddddddkke	2	13782	13797	522
612566	2316	2331	CTCAACTTGAAAAGGG	ekkddddddddddkke	49	13785	13800	523

TABLE 3-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612567	2321	2336	TTGTTCTCAACTTGAA	ekkddddddddddkke	58	13790	13805	524
612568	2322	2337	TTTGTTCTCAACTTGA	ekkddddddddddkke	63	13791	13806	525
612569	2329	2344	CCCAATTTTTGTTCTC	ekkddddddddddkke	65	13798	13813	526
612570	2330	2345	ACCCAATTTTTGTTCT	ekkddddddddddkke	37	13799	13814	527
612571	2331	2346	AACCCAATTTTTGTTC	ekkddddddddddkke	30	13800	13815	141
612572	2362	2377	GGCAATGCAAAAATGT	ekkddddddddddkke	53	13831	13846	142
612573	2366	2381	CGAAGGCAATGCAAAA	ekkddddddddddkke	7	13835	13850	528
612574	2367	2382	CCGAAGGCAATGCAAA	ekkddddddddddkke	25	13836	13851	529
612575	2368	2383	ACCGAAGGCAATGCAA	ekkddddddddddkke	36	13837	13852	530
612576	2369	2384	AACCGAAGGCAATGCA	ekkddddddddddkke	36	13838	13853	531
612577	2370	2385	AAACCGAAGGCAATGC	ekkddddddddddkke	29	13839	13854	532
612578	2371	2386	CAAACCGAAGGCAATG	ekkddddddddddkke	6	13840	13855	533
612579	2372	2387	ACAAACCGAAGGCAAT	ekkddddddddddkke	0	13841	13856	534
612580	2373	2388	TACAAACCGAAGGCAA	ekkddddddddddkke	27	13842	13857	535
612581	2374	2389	ATACAAACCGAAGGCA	ekkddddddddddkke	13	13843	13858	536
612582	2375	2390	AATACAAACCGAAGGC	ekkddddddddddkke	0	13844	13859	537
612583	2376	2391	AAATACAAACCGAAGG	ekkddddddddddkke	0	13845	13860	538
612584	2377	2392	TAAATACAAACCGAAG	ekkddddddddddkke	25	13846	13861	539
612585	2378	2393	CTAAATACAAACCGAA	ekkddddddddddkke	0	13847	13862	540
612586	2379	2394	ACTAAATACAAACCGA	ekkddddddddddkke	19	13848	13863	541
612587	2380	2395	CACTAAATACAAACCG	ekkddddddddddkke	15	13849	13864	542
612588	2382	2397	GACACTAAATACAAAC	ekkddddddddddkke	0	13851	13866	543
612589	2385	2400	CAAGACACTAAATACA	ekkddddddddddkke	9	13854	13869	544
612590	2386	2401	TCAAGACACTAAATAC	ekkddddddddddkke	19	13855	13870	545
612591	2387	2402	TTCAAGACACTAAATA	ekkddddddddddkke	0	13856	13871	546
612592	2388	2403	ATTCAAGACACTAAAT	ekkddddddddddkke	2	13857	13872	547
612593	2389	2404	CATTCAAGACACTAAA	ekkddddddddddkke	0	13858	13873	548
612594	2390	2405	ACATTCAAGACACTAA	ekkddddddddddkke	8	13859	13874	549
612595	2391	2406	TACATTCAAGACACTA	ekkddddddddddkke	1	13860	13875	143
612596	2392	2407	TTACATTCAAGACACT	ekkddddddddddkke	3	13861	13876	550
612597	2393	2408	CTTACATTCAAGACAC	ekkddddddddddkke	0	13862	13877	551
612598	2394	2409	TCTTACATTCAAGACA	ekkddddddddddkke	0	13863	13878	552
612599	2395	2410	TTCTTACATTCAAGAC	ekkddddddddddkke	0	13864	13879	553
612600	2398	2413	ATGTTCTTACATTCAA	ekkddddddddddkke	10	13867	13882	554
612601	2401	2416	GTCATGTTCTTACATT	ekkddddddddddkke	0	13870	13885	555

TABLE 3-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612602	2402	2417	GGTCATGTTCTTACAT	ekkddddddddddkke	34	13871	13886	144
612603	2403	2418	AGGTCATGTTCTTACA	ekkddddddddddkke	35	13872	13887	556
612604	2404	2419	GAGGTCATGTTCTTAC	ekkddddddddddkke	37	13873	13888	557
612605	2405	2420	GGAGGTCATGTTCTTA	ekkddddddddddkke	25	13874	13889	558
612606	2406	2421	CGGAGGTCATGTTCTT	ekkddddddddddkke	31	13875	13890	559
612607	2407	2422	ACGGAGGTCATGTTCT	ekkddddddddddkke	23	13876	13891	560
612608	2408	2423	CACGGAGGTCATGTTC	ekkddddddddddkke	24	13877	13892	561
612685	2565	2580	TGGAGGCTTATTGTGG	ekkddddddddddkke	25	14034	14049	562
612686	2566	2581	TTGGAGGCTTATTGTG	ekkddddddddddkke	30	14035	14050	563
612687	2567	2582	TTTGGAGGCTTATTGT	ekkddddddddddkke	20	14036	14051	564
612688	N/A	N/A	CGGCTTACCTTCTGCT	ekkddddddddddkke	30	2483	2498	565
612689	N/A	N/A	CCTCCCGGCCTTTTCC	ekkddddddddddkke	23	2562	2577	566
612690	N/A	N/A	TAGGGTGACCACTCTG	ekkddddddddddkke	26	2897	2912	567
612691	N/A	N/A	AGCAAATCGAGGTTCA	ekkddddddddddkke	25	2970	2985	568
612692	N/A	N/A	TATTAGTTCTCTTCAG	ekkddddddddddkke	9	3047	3062	569
612693	N/A	N/A	CCTTTTAGCTTATCCC	ekkddddddddddkke	24	3089	3104	570
612694	N/A	N/A	AATCTGCCTTTTAGCT	ekkddddddddddkke	20	3095	3110	571
612695	N/A	N/A	CAATCTACGCTGCCCT	ekkddddddddddkke	27	3124	3139	572
612696	N/A	N/A	AGCACCAATCTACGCT	ekkddddddddddkke	16	3129	3144	573
612697	N/A	N/A	CATCCTGGAGAAGTAG	ekkddddddddddkke	9	3276	3291	574
612698	N/A	N/A	GCATCCTGGAGAAGTA	ekkddddddddddkke	13	3277	3292	575
612699	N/A	N/A	ATACAGCCCACATTCC	ekkddddddddddkke	17	3316	3331	576
612700	N/A	N/A	CTGTACCATGTAGTTA	ekkddddddddddkke	32	3418	3433	577
612701	N/A	N/A	CCACACCGGGCACTCT	ekkddddddddddkke	12	3476	3491	578
612702	N/A	N/A	CCCACCACACCGGGCA	ekkddddddddddkke	22	3480	3495	579
612703	N/A	N/A	TTCCCCACCACACCGG	ekkddddddddddkke	19	3483	3498	580
612704	N/A	N/A	TTCACCCTGCAGCTTT	ekkddddddddddkke	13	3497	3512	581
612705	N/A	N/A	CATAGTCCTCACCTTC	ekkddddddddddkke	16	3537	3552	582
612706	N/A	N/A	GTGAAGATGACGGCTC	ekkddddddddddkke	24	3615	3630	583
612707	N/A	N/A	TATGTCTCCCTACTTC	ekkddddddddddkke	25	3651	3666	584

TABLE 3-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612708	N/A	N/A	GGGAGTAATGGTGCTC	ekkddddddddddkke	33	3755	3770	585
612709	N/A	N/A	GTCCTGGGAGTAATGG	ekkddddddddddkke	24	3760	3775	586
612710	N/A	N/A	GGGAACCGACTGCTGG	ekkddddddddddkke	24	3977	3992	587
612711	N/A	N/A	CCTGTGGGAACCGACT	ekkddddddddddkke	14	3982	3997	588
612712	N/A	N/A	CCTAATCTAGACAGTC	ekkddddddddddkke	5	4024	4039	589
612713	N/A	N/A	CATCCGCTGTTCTCAG	ekkddddddddddkke	2	4133	4148	590
612714	N/A	N/A	CTCCATCCGCTGTTCT	ekkddddddddddkke	28	4136	4151	591
612715	N/A	N/A	GACTCCATCCGCTGTT	ekkddddddddddkke	30	4138	4153	592
612716	N/A	N/A	TGACTCCATCCGCTGT	ekkddddddddddkke	25	4139	4154	593
612717	N/A	N/A	GCTGAAGTACCTGGTG	ekkddddddddddkke	34	4230	4245	594
612718	N/A	N/A	GCCCTCAACACGGTGC	ekkddddddddddkke	25	4250	4265	595
612719	N/A	N/A	TGCCCTCAACACGGTG	ekkddddddddddkke	20	4251	4266	596
612720	N/A	N/A	GTCATTCTTCTTACAT	ekkddddddddddkke	14	4307	4322	597
612721	N/A	N/A	GCTTCCTTGGAGCTGT	ekkddddddddddkke	5	4390	4405	598
612722	N/A	N/A	GTGTACTGCAATATCG	ekkddddddddddkke	39	4446	4461	599
612723	N/A	N/A	CACTCATTTCTTGTGG	ekkddddddddddkke	8	4468	4483	600
612724	N/A	N/A	TTGTACCACATCTCAC	ekkddddddddddkke	21	4481	4496	601
612725	N/A	N/A	GTTCTCTCAAAGGCCT	ekkddddddddddkke	32	4651	4666	602
612726	N/A	N/A	GCAGGGTTTAGAACCC	ekkddddddddddkke	18	4694	4709	603
612727	N/A	N/A	TATGTAAGCAGGGTTT	ekkddddddddddkke	11	4701	4716	604
612728	N/A	N/A	AAACCAGCTCTCAACC	ekkddddddddddkke	5	4864	4879	605
612729	N/A	N/A	TAAGACATGCTCCTGC	ekkddddddddddkke	12	5094	5109	606
612730	N/A	N/A	ACTTATGGCAGCCCAA	ekkddddddddddkke	20	5116	5131	607
612731	N/A	N/A	TACTTATGGCAGCCCA	ekkddddddddddkke	12	5117	5132	608
612732	N/A	N/A	CCATTATTTGGAGACA	ekkddddddddddkke	9	5426	5441	609
612733	N/A	N/A	TGCCATCTAACCAGAT	ekkddddddddddkke	15	5655	5670	610
612745	N/A	N/A	GTTTTCAGTAATGCCC	ekkddddddddddkke	21	7085	7100	611

[0514] Table 4 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 1000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 4

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	43	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	36	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	20	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	51	13515	13530	129
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	92	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	91	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	92	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	90	13496	13511	163
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	79	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	88	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	80	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	73	13516	13531	165
611977	446	461	CGGGTCACGATGCCCT	ekkddddddddddkke	13	2431	2446	612
611978	451	466	CCGGCCGGGTCACGAT	ekkddddddddddkke	20	2436	2451	613
611979	454	469	CCCCCGGCCGGGTCAC	ekkddddddddddkke	26	2439	2454	614
611980	457	472	CTTCCCCCGGCCGGGT	ekkddddddddddkke	20	2442	2457	615
611981	460	475	CTTCTTCCCCCGGCCG	ekkddddddddddkke	24	2445	2460	616
611982	463	478	CAGCTTCTTCCCCCGG	ekkddddddddddkke	41	2448	2463	617
611983	466	481	CGGCAGCTTCTTCCCC	ekkddddddddddkke	15	2451	2466	618
611984	469	484	CAACGGCAGCTTCTTC	ekkddddddddddkke	20	2454	2469	619
611985	472	487	GAACAACGGCAGCTTC	ekkddddddddddkke	27	2457	2472	620
611986	475	490	CCAGAACAACGGCAGC	ekkddddddddddkke	23	2460	2475	621
611987	478	493	TACCCAGAACAACGGC	ekkddddddddddkke	40	2463	2478	39
611988	481	496	TAGTACCCAGAACAAC	ekkddddddddddkke	10	2466	2481	622
611989	484	499	CTGTAGTACCCAGAAC	ekkddddddddddkke	21	2469	2484	623
611990	487	502	CTGCTGTAGTACCCAG	ekkddddddddddkke	28	2472	2487	624
611991	490	505	CTTCTGCTGTAGTACC	ekkddddddddddkke	33	2475	2490	625
611992	493	508	ACCCTTCTGCTGTAGT	ekkddddddddddkke	39	N/A	N/A	626
611993	496	511	CATACCCTTCTGCTGT	ekkddddddddddkke	19	N/A	N/A	627
611994	499	514	CCGCATACCCTTCTGC	ekkddddddddddkke	11	N/A	N/A	628
611995	502	517	CTTCCGCATACCCTTC	ekkddddddddddkke	21	N/A	N/A	629
611996	505	520	TCGCTTCCGCATACCC	ekkddddddddddkke	53	5722	5737	630
611997	508	523	TGCTCGCTTCCGCATA	ekkddddddddddkke	6	5725	5740	631
611998	511	526	GGGTGCTCGCTTCCGC	ekkddddddddddkke	38	5728	5743	632
611999	525	540	GCCATCTCAGACTGGG	ekkddddddddddkke	31	5742	5757	40

TABLE 4-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612000	533	548	CGGCAGGAGCCATCTC	ekkddddddddddkke	31	5750	5765	633
612001	536	551	CACCGGCAGGAGCCAT	ekkddddddddddkke	19	5753	5768	634
612002	539	554	TCACACCGGCAGGAGC	ekkddddddddddkke	19	5756	5771	635
612003	542	557	GGCTCACACCGGCAGG	ekkddddddddddkke	41	5759	5774	636
612004	545	560	TCAGGCTCACACCGGC	ekkddddddddddkke	46	5762	5777	42
612005	549	564	GCCCTCAGGCTCACAC	ekkddddddddddkke	18	5766	5781	637
612006	552	567	GTGGCCCTCAGGCTCA	ekkddddddddddkke	29	5769	5784	638
612007	555	570	ATGGTGGCCCTCAGGC	ekkddddddddddkke	32	5772	5787	43
612008	561	576	CAGAGGATGGTGGCCC	ekkddddddddddkke	33	5778	5793	639
612009	596	611	GGTCACCTGCAGCCAG	ekkddddddddddkke	38	5813	5828	44
612010	599	614	CCCGGTCACCTGCAGC	ekkddddddddddkke	47	5816	5831	640
612011	602	617	ACACCCGGTCACCTGC	ekkddddddddddkke	29	5819	5834	641
612012	605	620	TGTACACCCGGTCACC	ekkddddddddddkke	22	5822	5837	642
612013	608	623	GTATGTACACCCGGTC	ekkddddddddddkke	5	5825	5840	643
612014	611	626	GGTGTATGTACACCCG	ekkddddddddddkke	0	5828	5843	644
612015	626	641	TGACGAGGTGGAAGGG	ekkddddddddddkke	21	5843	5858	645
612016	629	644	GGATGACGAGGTGGAA	ekkddddddddddkke	32	5846	5861	646
612017	632	647	TGTGGATGACGAGGTG	ekkddddddddddkke	48	5849	5864	647
612018	635	650	CATTGTGGATGACGAG	ekkddddddddddkke	28	5852	5867	648
612019	638	653	TCTCATTGTGGATGAC	ekkddddddddddkke	34	5855	5870	649
612020	639	654	CTCTCATTGTGGATGA	ekkddddddddddkke	38	5856	5871	650
612021	640	655	ACTCTCATTGTGGATG	ekkddddddddddkke	45	5857	5872	651
612022	641	656	TACTCTCATTGTGGAT	ekkddddddddddkke	29	5858	5873	652
612023	642	657	GTACTCTCATTGTGGA	ekkddddddddddkke	46	5859	5874	653
612024	643	658	GGTACTCTCATTGTGG	ekkddddddddddkke	58	5860	5875	46
612025	645	660	CAGGTACTCTCATTGT	ekkddddddddddkke	59	5862	5877	654
612026	646	661	ACAGGTACTCTCATTG	ekkddddddddddkke	50	5863	5878	655
612027	647	662	CACAGGTACTCTCATT	ekkddddddddddkke	37	5864	5879	656
612028	648	663	TCACAGGTACTCTCAT	ekkddddddddddkke	31	5865	5880	657
612029	649	664	CTCACAGGTACTCTCA	ekkddddddddddkke	22	5866	5881	658
612030	652	667	CTGCTCACAGGTACTC	ekkddddddddddkke	4	5869	5884	659
612031	659	674	TTGCCAGCTGCTCACA	ekkddddddddddkke	39	5876	5891	660
612032	662	677	CCTTTGCCAGCTGCTC	ekkddddddddddkke	45	5879	5894	661
612033	665	680	TGGCCTTTGCCAGCTG	ekkddddddddddkke	30	5882	5897	662

TABLE 4-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612034	668	683	CATTGGCCTTTGCCAG	ekkddddddddddkke	18	5885	5900	663
612035	671	686	CGGCATTGGCCTTTGC	ekkddddddddddkke	18	5888	5903	664
612036	674	689	TCCCGGCATTGGCCTT	ekkddddddddddkke	27	5891	5906	665
612037	677	692	GCTTCCCGGCATTGGC	ekkddddddddddkke	15	5894	5909	666
612038	680	695	TGGGCTTCCCGGCATT	ekkddddddddddkke	2	5897	5912	667
612039	683	698	CTTTGGGCTTCCCGGC	ekkddddddddddkke	44	5900	5915	668
612040	686	701	GGTCTTTGGGCTTCCC	ekkddddddddddkke	36	5903	5918	669
612041	701	716	CAGGTATGAAGGTGGG	ekkddddddddddkke	42	5918	5933	670
612042	704	719	GAGCAGGTATGAAGGT	ekkddddddddddkke	39	5921	5936	671
612043	707	722	TTGGAGCAGGTATGAA	ekkddddddddddkke	28	5924	5939	672
612044	710	725	GAATTGGAGCAGGTAT	ekkddddddddddkke	20	5927	5942	673
612045	713	728	CCTGAATTGGAGCAGG	ekkddddddddddkke	7	5930	5945	50
612046	716	731	TGGCCTGAATTGGAGC	ekkddddddddddkke	23	5933	5948	674
612047	719	734	TCTTGGCCTGAATTGG	ekkddddddddddkke	29	5936	5951	675
612048	722	737	ATGTCTTGGCCTGAAT	ekkddddddddddkke	22	5939	5954	676
612049	725	740	GGGATGTCTTGGCCTG	ekkddddddddddkke	35	5942	5957	677
612050	739	754	CTTTTCATCCACAGGG	ekkddddddddddkke	21	5956	5971	52
612051	742	757	GGCCTTTTCATCCACA	ekkddddddddddkke	3	5959	5974	678
612052	745	760	TAGGGCCTTTTCATCC	ekkddddddddddkke	10	5962	5977	679
612053	748	763	CTGTAGGGCCTTTTCA	ekkddddddddddkke	5	5965	5980	680
612054	751	766	GTCCTGTAGGGCCTTT	ekkddddddddddkke	6	5968	5983	681
612055	754	769	CTGGTCCTGTAGGGCC	ekkddddddddddkke	19	5971	5986	682
612056	758	773	CCAGCTGGTCCTGTAG	ekkddddddddddkke	34	5975	5990	683
612057	759	774	ACCAGCTGGTCCTGTA	ekkddddddddddkke	31	5976	5991	684
612058	762	777	AGCACCAGCTGGTCCT	ekkddddddddddkke	56	5979	5994	53
612059	763	778	TAGCACCAGCTGGTCC	ekkddddddddddkke	35	5980	5995	685
612060	764	779	CTAGCACCAGCTGGTC	ekkddddddddddkke	18	5981	5996	686
612061	765	780	ACTAGCACCAGCTGGT	ekkddddddddddkke	10	5982	5997	687
612062	766	781	GACTAGCACCAGCTGG	ekkddddddddddkke	32	5983	5998	688
612063	767	782	CGACTAGCACCAGCTG	ekkddddddddddkke	49	5984	5999	689
612064	768	783	GCGACTAGCACCAGCT	ekkddddddddddkke	39	5985	6000	690
612065	769	784	AGCGACTAGCACCAGC	ekkddddddddddkke	29	5986	6001	691
612066	770	785	CAGCGACTAGCACCAG	ekkddddddddddkke	38	5987	6002	692
612067	771	786	GCAGCGACTAGCACCA	ekkddddddddddkke	39	5988	6003	693
612068	772	787	TGCAGCGACTAGCACC	ekkddddddddddkke	31	5989	6004	54

TABLE 4-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612069	773	788	TTGCAGCGACTAGCAC	ekkddddddddddkke	28	5990	6005	694
612070	774	789	TTTGCAGCGACTAGCA	ekkddddddddddkke	31	5991	6006	695
612071	775	790	TTTTGCAGCGACTAGC	ekkddddddddddkke	28	5992	6007	696
612072	776	791	GTTTTGCAGCGACTAG	ekkddddddddddkke	11	5993	6008	697
612073	777	792	AGTTTTGCAGCGACTA	ekkddddddddddkke	7	5994	6009	698
612074	778	793	AAGTTTTGCAGCGACT	ekkddddddddddkke	10	5995	6010	699
612075	781	796	GTCAAGTTTTGCAGCG	ekkddddddddddkke	49	5998	6013	700
612076	784	799	GGTGTCAAGTTTTGCA	ekkddddddddddkke	39	6001	6016	701
612077	787	802	TTCGGTGTCAAGTTTT	ekkddddddddddkke	53	6004	6019	702
612078	790	805	GTCTTCGGTGTCAAGT	ekkddddddddddkke	39	6007	6022	703
612079	793	808	CTTGTCTTCGGTGTCA	ekkddddddddddkke	35	6010	6025	704
612080	796	811	CAACTTGTCTTCGGTG	ekkddddddddddkke	42	6013	6028	705
612081	799	814	CCTCAACTTGTCTTCG	ekkddddddddddkke	1	6016	6031	706
612082	802	817	GGCCCTCAACTTGTCT	ekkddddddddddkke	0	6019	6034	707
612083	805	820	TGCGGCCCTCAACTTG	ekkddddddddddkke	13	6022	6037	708
612084	808	823	CATTGCGGCCCTCAAC	ekkddddddddddkke	0	6025	6040	709
612085	811	826	GACCATTGCGGCCCTC	ekkddddddddddkke	30	6028	6043	710
612086	814	829	CCCGACCATTGCGGCC	ekkddddddddddkke	32	6031	6046	711
612087	817	832	CATCCCGACCATTGCG	ekkddddddddddkke	49	6034	6049	712
612088	820	835	CAGCATCCCGACCATT	ekkddddddddddkke	17	6037	6052	713
612089	823	838	GGCCAGCATCCCGACC	ekkddddddddddkke	46	6040	6055	714
612090	826	841	GTTGGCCAGCATCCCG	ekkddddddddddkke	10	6043	6058	715
612091	829	844	GAAGTTGGCCAGCATC	ekkddddddddddkke	0	6046	6061	716
612092	832	847	CAAGAAGTTGGCCAGC	ekkddddddddddkke	0	6049	6064	717
612093	835	850	GCCCAAGAAGTTGGCC	ekkddddddddddkke	0	6052	6067	59
612094	838	853	GAAGCCCAAGAAGTTG	ekkddddddddddkke	28	6055	6070	718
612095	841	856	ACGGAAGCCCAAGAAG	ekkddddddddddkke	13	6058	6073	719
612096	844	859	TATACGGAAGCCCAAG	ekkddddddddddkke	18	6061	6076	720
612097	847	862	ATATATACGGAAGCCC	ekkddddddddddkke	0	6064	6079	721
612098	850	865	GCCATATATACGGAAG	ekkddddddddddkke	42	6067	6082	722
612099	853	868	CATGCCATATATACGG	ekkddddddddddkke	20	6070	6085	723
612100	856	871	GTGCATGCCATATATA	ekkddddddddddkke	47	6073	6088	724
612101	859	874	ACTGTGCATGCCATAT	ekkddddddddddkke	52	6076	6091	725
612102	862	877	CTCACTGTGCATGCCA	ekkddddddddddkke	62	6079	6094	726

TABLE 4-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612103	865	880	TAGCTCACTGTGCATG	ekkddddddddddkke	45	6082	6097	727
612104	868	883	CCATAGCTCACTGTGC	ekkddddddddddkke	66	6085	6100	728
612105	871	886	GCCCCATAGCTCACTG	ekkddddddddddkke	16	6088	6103	62
612107	877	892	GACCACGCCCCATAGC	ekkddddddddddkke	0	6094	6109	729
612108	880	895	ATGGACCACGCCCCAT	ekkddddddddddkke	0	6097	6112	730
612109	884	899	CCCCATGGACCACGCC	ekkddddddddddkke	0	6101	6116	731
612110	887	902	TGGCCCCATGGACCAC	ekkddddddddddkke	24	6104	6119	732
612111	890	905	CGGTGGCCCCATGGAC	ekkddddddddddkke	1	6107	6122	733
612112	893	908	GGACGGTGGCCCCATG	ekkddddddddddkke	4	6110	6125	734
612113	896	911	AGAGGACGGTGGCCCC	ekkddddddddddkke	7	6113	6128	735
612114	899	914	GGGAGAGGACGGTGGC	ekkddddddddddkke	28	6116	6131	736
612115	913	928	AAAGACAGCCGTTGGG	ekkddddddddddkke	30	6130	6145	64
612116	916	931	GCCAAAGACAGCCGTT	ekkddddddddddkke	45	6133	6148	737
612117	919	934	GGTGCCAAAGACAGCC	ekkddddddddddkke	52	6136	6151	738
612118	922	937	CAGGGTGCCAAAGACA	ekkddddddddddkke	20	6139	6154	739
612119	926	941	AGGCCAGGGTGCCAAA	ekkddddddddddkke	20	6143	6158	740
612120	937	952	CAGATAGAGAGAGGCC	ekkddddddddddkke	0	6154	6169	66
612121	940	955	TCCCAGATAGAGAGAG	ekkddddddddddkke	0	6157	6172	741
612122	943	958	GGCTCCCAGATAGAGA	ekkddddddddddkke	11	6160	6175	742
612123	946	961	CAAGGCTCCCAGATAG	ekkddddddddddkke	5	6163	6178	743
612124	949	964	GTCCAAGGCTCCCAGA	ekkddddddddddkke	14	6166	6181	744
612125	952	967	GTGGTCCAAGGCTCCC	ekkddddddddddkke	19	6169	6184	745
612126	955	970	TGTGTGGTCCAAGGCT	ekkddddddddddkke	25	6172	6187	746
612127	958	973	AGCTGTGTGGTCCAAG	ekkddddddddddkke	40	6175	6190	747
612128	961	976	GTCAGCTGTGTGGTCC	ekkddddddddddkke	22	6178	6193	748
612281	1547	1562	CAGAGGCATAGTGAGG	ekkddddddddddkke	25	10558	10573	749
612282	1550	1565	GGTCAGAGGCATAGTG	ekkddddddddddkke	20	10561	10576	103
612283	1553	1568	CCAGGTCAGAGGCATA	ekkddddddddddkke	36	10564	10579	750
612284	1557	1572	TTGTCCAGGTCAGAGG	ekkddddddddddkke	24	10568	10583	751
612285	1560	1575	ACCTTGTCCAGGTCAG	ekkddddddddddkke	37	10571	10586	752
612286	1566	1581	CCCTCCACCTTGTCCA	ekkddddddddddkke	9	10577	10592	753
612287	1570	1585	GAGACCCTCCACCTTG	ekkddddddddddkke	31	10581	10596	754
612288	1574	1589	AAGTGAGACCCTCCAC	ekkddddddddddkke	5	10585	10600	755
612289	1578	1593	TGGAAAGTGAGACCCT	ekkddddddddddkke	13	10589	10604	104
612290	1581	1596	TGCTGGAAAGTGAGAC	ekkddddddddddkke	27	10592	10607	756

TABLE 4-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612291	1584	1599	TTTTGCTGGAAAGTGA	ekkddddddddddkke	0	10595	10610	757
612292	1587	1602	GAGTTTTGCTGGAAAG	ekkddddddddddkke	15	10598	10613	758
612293	1590	1605	AGGGAGTTTTGCTGGA	ekkddddddddddkke	27	10601	10616	759
612294	1594	1609	GTTGAGGGAGTTTTGC	ekkddddddddddkke	0	10605	10620	760
612295	1597	1612	CCAGTTGAGGGAGTTT	ekkddddddddddkke	6	10608	10623	761
612296	1600	1615	CATCCAGTTGAGGGAG	ekkddddddddddkke	8	10611	10626	762
612297	1603	1618	CTTCATCCAGTTGAGG	ekkddddddddddkke	11	10614	10629	763
612298	1612	1627	AGATAGTTTCTTCATC	ekkddddddddddkke	0	10623	10638	764
612299	1629	1644	AGGTGGATGGTCCGGG	ekkddddddddddkke	36	N/A	N/A	765
612300	1632	1647	GTCAGGTGGATGGTCC	ekkddddddddddkke	25	12238	12253	766
612301	1636	1651	CATGGTCAGGTGGATG	ekkddddddddddkke	26	12242	12257	767
612302	1639	1654	GGGCATGGTCAGGTGG	ekkddddddddddkke	40	12245	12260	768
612303	1653	1668	TGCAGCACCAGTTGGG	ekkddddddddddkke	33	12259	12274	109
612304	1656	1671	CCTTGCAGCACCAGTT	ekkddddddddddkke	3	12262	12277	769
612305	1659	1674	GATCCTTGCAGCACCA	ekkddddddddddkke	12	12265	12280	770
612306	1662	1677	TAAGATCCTTGCAGCA	ekkddddddddddkke	8	12268	12283	771
612307	1665	1680	TCATAAGATCCTTGCA	ekkddddddddddkke	8	12271	12286	772
612308	1669	1684	CAGGTCATAAGATCCT	ekkddddddddddkke	8	12275	12290	773
612309	1672	1687	CTGCAGGTCATAAGAT	ekkddddddddddkke	0	12278	12293	774
612310	1675	1690	GTCCTGCAGGTCATAA	ekkddddddddddkke	10	12281	12296	775
612311	1682	1697	CGAGCAGGTCCTGCAG	ekkddddddddddkke	32	12288	12303	776
612312	1685	1700	GGGCGAGCAGGTCCTG	ekkddddddddddkke	11	12291	12306	777
612313	1688	1703	CCTGGGCGAGCAGGTC	ekkddddddddddkke	22	12294	12309	778
612314	1700	1715	CGGGCAGCTCAGCCTG	ekkddddddddddkke	0	12306	12321	112
612315	1703	1718	TGGCGGGCAGCTCAGC	ekkddddddddddkke	55	12309	12324	779
612316	1706	1721	GAATGGCGGGCAGCTC	ekkddddddddddkke	16	12312	12327	780
612317	1709	1724	GCAGAATGGCGGGCAG	ekkddddddddddkke	16	12315	12330	781
612318	1712	1727	TGTGCAGAATGGCGGG	ekkddddddddddkke	24	12318	12333	782
612319	1715	1730	CGGTGTGCAGAATGGC	ekkddddddddddkke	35	12321	12336	783
612320	1718	1733	GCTCGGTGTGCAGAAT	ekkddddddddddkke	13	12324	12339	784
612321	1721	1736	TCAGCTCGGTGTGCAG	ekkddddddddddkke	28	12327	12342	785
612322	1724	1739	GGTTCAGCTCGGTGTG	ekkddddddddddkke	49	12330	12345	786
612323	1727	1742	GCAGGTTCAGCTCGGT	ekkddddddddddkke	53	12333	12348	787
612324	1732	1747	TTTTTGCAGGTTCAGC	ekkddddddddddkke	8	12338	12353	788

TABLE 4-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612325	1735	1750	CAATTTTTGCAGGTTC	ekkddddddddddkke	14	12341	12356	789
612326	1738	1753	GCTCAATTTTTGCAGG	ekkddddddddddkke	38	12344	12359	790
612327	1741	1756	ATTGCTCAATTTTTGC	ekkddddddddddkke	2	12347	12362	791
612328	1744	1759	GTCATTGCTCAATTTT	ekkddddddddddkke	38	12350	12365	792
612329	1747	1762	GCGGTCATTGCTCAAT	ekkddddddddddkke	32	12353	12368	793
612330	1750	1765	GATGCGGTCATTGCTC	ekkddddddddddkke	27	12356	12371	794
612331	1753	1768	CCTGATGCGGTCATTG	ekkddddddddddkke	15	12359	12374	795
612332	1756	1771	CACCCTGATGCGGTCA	ekkddddddddddkke	1	12362	12377	796
612333	1759	1774	CCCCACCCTGATGCGG	ekkddddddddddkke	11	12365	12380	797
612334	1762	1777	CTCCCCCACCCTGATG	ekkddddddddddkke	0	12368	12383	798
612335	1771	1786	GTTCAGCACCTCCCCC	ekkddddddddddkke	12	N/A	N/A	799
612336	1774	1789	GCTGTTCAGCACCTCC	ekkddddddddddkke	57	N/A	N/A	800
612337	1777	1792	AATGCTGTTCAGCACC	ekkddddddddddkke	29	13246	13261	801
612338	1780	1795	AAAAATGCTGTTCAGC	ekkddddddddddkke	38	13249	13264	802
612339	1793	1808	CTTCAAGCTCAAAAAA	ekkddddddddddkke	0	13262	13277	803
612340	1796	1811	CCGCTTCAAGCTCAAA	ekkddddddddddkke	41	13265	13280	804
612341	1799	1814	CATCCGCTTCAAGCTC	ekkddddddddddkke	27	13268	13283	805
612342	1802	1817	TCTCATCCGCTTCAAG	ekkddddddddddkke	32	13271	13286	806
612343	1805	1820	CTCTCTCATCCGCTTC	ekkddddddddddkke	26	13274	13289	807
612344	1808	1823	GCTCTCTCTCATCCGC	ekkddddddddddkke	44	13277	13292	808
612345	1812	1827	GTGGGCTCTCTCTCAT	ekkddddddddddkke	15	13281	13296	809
612346	1817	1832	ACTCTGTGGGCTCTCT	ekkddddddddddkke	42	13286	13301	810
612347	1820	1835	TAGACTCTGTGGGCTC	ekkddddddddddkke	55	13289	13304	811
612348	1824	1839	TGGGTAGACTCTGTGG	ekkddddddddddkke	23	13293	13308	812
612349	1827	1842	TGTTGGGTAGACTCTG	ekkddddddddddkke	30	13296	13311	119
612350	1830	1845	AGCTGTTGGGTAGACT	ekkddddddddddkke	34	13299	13314	813
612351	1833	1848	TTAAGCTGTTGGGTAG	ekkddddddddddkke	13	13302	13317	814
612352	1836	1851	TTGTTAAGCTGTTGGG	ekkddddddddddkke	33	13305	13320	815
612353	1839	1854	GGCTTGTTAAGCTGTT	ekkddddddddddkke	30	13308	13323	816
612354	1842	1857	TCAGGCTTGTTAAGCT	ekkddddddddddkke	10	13311	13326	817
612355	1845	1860	ACCTCAGGCTTGTTAA	ekkddddddddddkke	17	13314	13329	818
612356	1848	1863	AAGACCTCAGGCTTGT	ekkddddddddddkke	33	13317	13332	819
612609	2409	2424	ACACGGAGGTCATGTT	ekkddddddddddkke	20	13878	13893	820
612610	2410	2425	TACACGGAGGTCATGT	ekkddddddddddkke	25	13879	13894	821
612611	2411	2426	CTACACGGAGGTCATG	ekkddddddddddkke	24	13880	13895	822

TABLE 4-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612612	2412	2427	ACTACACGGAGGTCAT	ekkddddddddddkke	26	13881	13896	145
612613	2413	2428	CACTACACGGAGGTCA	ekkddddddddddkke	30	13882	13897	823
612614	2414	2429	ACACTACACGGAGGTC	ekkddddddddddkke	49	13883	13898	824
612615	2415	2430	GACACTACACGGAGGT	ekkddddddddddkke	56	13884	13899	825
612616	2416	2431	AGACACTACACGGAGG	ekkddddddddddkke	40	13885	13900	826
612617	2417	2432	CAGACACTACACGGAG	ekkddddddddddkke	48	13886	13901	827
612618	2418	2433	ACAGACACTACACGGA	ekkddddddddddkke	44	13887	13902	828
612619	2419	2434	TACAGACACTACACGG	ekkddddddddddkke	39	13888	13903	829
612620	2420	2435	TTACAGACACTACACG	ekkddddddddddkke	28	13889	13904	830
612621	2421	2436	ATTACAGACACTACAC	ekkddddddddddkke	21	13890	13905	831
612622	2422	2437	TATTACAGACACTACA	ekkddddddddddkke	0	13891	13906	146
612623	2423	2438	GTATTACAGACACTAC	ekkddddddddddkke	35	13892	13907	832
612624	2428	2443	CTAAGGTATTACAGAC	ekkddddddddddkke	8	13897	13912	833
612625	2429	2444	ACTAAGGTATTACAGA	ekkddddddddddkke	14	13898	13913	834
612626	2430	2445	AACTAAGGTATTACAG	ekkddddddddddkke	14	13899	13914	835
612627	2431	2446	AAACTAAGGTATTACA	ekkddddddddddkke	12	13900	13915	836
612628	2432	2447	AAAACTAAGGTATTAC	ekkddddddddddkke	3	13901	13916	837
612629	2438	2453	GTGGAAAAAACTAAGG	ekkddddddddddkke	0	13907	13922	838
612630	2447	2462	CAAGCATCTGTGGAAA	ekkddddddddddkke	0	13916	13931	839
612631	2449	2464	CACAAGCATCTGTGGA	ekkddddddddddkke	20	13918	13933	840
612632	2450	2465	TCACAAGCATCTGTGG	ekkddddddddddkke	1	13919	13934	841
612633	2451	2466	ATCACAAGCATCTGTG	ekkddddddddddkke	20	13920	13935	842
612634	2452	2467	AATCACAAGCATCTGT	ekkddddddddddkke	2	13921	13936	843
612635	2464	2479	GTATTGTTCAAAAATC	ekkddddddddddkke	16	13933	13948	844
612636	2465	2480	CGTATTGTTCAAAAAT	ekkddddddddddkke	0	13934	13949	845
612637	2482	2497	GGTGCTTGCATCTTTC	ekkddddddddddkke	21	13951	13966	147
612638	2483	2498	AGGTGCTTGCATCTTT	ekkddddddddddkke	13	13952	13967	846
612639	2484	2499	CAGGTGCTTGCATCTT	ekkddddddddddkke	19	13953	13968	847
612640	2485	2500	TCAGGTGCTTGCATCT	ekkddddddddddkke	38	13954	13969	848
612641	2486	2501	TTCAGGTGCTTGCATC	ekkddddddddddkke	29	13955	13970	849
612642	2487	2502	ATTCAGGTGCTTGCAT	ekkddddddddddkke	19	13956	13971	850
612643	2488	2503	AATTCAGGTGCTTGCA	ekkddddddddddkke	34	13957	13972	851
612644	2489	2504	AAATTCAGGTGCTTGC	ekkddddddddddkke	24	13958	13973	852
612645	2490	2505	GAAATTCAGGTGCTTG	ekkddddddddddkke	2	13959	13974	853

TABLE 4-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612646	2491	2506	AGAAATTCAGGTGCTT	ekkddddddddddkke	5	13960	13975	854
612647	2493	2508	ACAGAAATTCAGGTGC	ekkddddddddddkke	0	13962	13977	855
612648	2502	2517	CGCATTCAAACAGAAA	ekkddddddddddkke	22	13971	13986	856
612649	2503	2518	CCGCATTCAAACAGAA	ekkddddddddddkke	50	13972	13987	149
612650	2504	2519	TCCGCATTCAAACAGA	ekkddddddddddkke	35	13973	13988	857
612651	2505	2520	TTCCGCATTCAAACAG	ekkddddddddddkke	29	13974	13989	858
612652	2506	2521	GTTCCGCATTCAAACA	ekkddddddddddkke	25	13975	13990	859
612653	2507	2522	GGTTCCGCATTCAAAC	ekkddddddddddkke	28	13976	13991	860
612654	2508	2523	TGGTTCCGCATTCAAA	ekkddddddddddkke	38	13977	13992	861
612655	2509	2524	ATGGTTCCGCATTCAA	ekkddddddddddkke	45	13978	13993	862
612656	2510	2525	TATGGTTCCGCATTCA	ekkddddddddddkke	42	13979	13994	863
612657	2511	2526	CTATGGTTCCGCATTC	ekkddddddddddkke	41	13980	13995	864
612658	2512	2527	GCTATGGTTCCGCATT	ekkddddddddddkke	58	13981	13996	865
612659	2513	2528	AGCTATGGTTCCGCAT	ekkddddddddddkke	32	13982	13997	150
612660	2514	2529	CAGCTATGGTTCCGCA	ekkddddddddddkke	46	13983	13998	866
612661	2515	2530	CCAGCTATGGTTCCGC	ekkddddddddddkke	47	13984	13999	867
612662	2516	2531	ACCAGCTATGGTTCCG	ekkddddddddddkke	60	13985	14000	868
612663	2517	2532	AACCAGCTATGGTTCC	ekkddddddddddkke	36	13986	14001	869
612664	2518	2533	TAACCAGCTATGGTTC	ekkddddddddddkke	0	13987	14002	870
612665	2519	2534	ATAACCAGCTATGGTT	ekkddddddddddkke	17	13988	14003	871
612666	2521	2536	AAATAACCAGCTATGG	ekkddddddddddkke	3	13990	14005	872
612667	2522	2537	GAAATAACCAGCTATG	ekkddddddddddkke	2	13991	14006	873
612668	2523	2538	AGAAATAACCAGCTAT	ekkddddddddddkke	4	13992	14007	874
612669	2535	2550	CTAACACAAGGGAGAA	ekkddddddddddkke	23	14004	14019	875
612670	2536	2551	ACTAACACAAGGGAGA	ekkddddddddddkke	13	14005	14020	876
612671	2537	2552	TACTAACACAAGGGAG	ekkddddddddddkke	9	14006	14021	151
612672	2538	2553	TTACTAACACAAGGGA	ekkddddddddddkke	51	14007	14022	877
612673	2539	2554	ATTACTAACACAAGGG	ekkddddddddddkke	47	14008	14023	878
612674	2540	2555	TATTACTAACACAAGG	ekkddddddddddkke	16	14009	14024	879
612675	2541	2556	TTATTACTAACACAAG	ekkddddddddddkke	0	14010	14025	880
612676	2543	2558	GTTTATTACTAACACA	ekkddddddddddkke	0	14012	14027	881
612677	2544	2559	CGTTTATTACTAACAC	ekkddddddddddkke	35	14013	14028	882
612678	2558	2573	TTATTGTGGCAAGACG	ekkddddddddddkke	28	14027	14042	152
612679	2559	2574	CTTATTGTGGCAAGAC	ekkddddddddddkke	21	14028	14043	883
612680	2560	2575	GCTTATTGTGGCAAGA	ekkddddddddddkke	16	14029	14044	884

TABLE 4-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612681	2561	2576	GGCTTATTGTGGCAAG	ekkddddddddddkke	35	14030	14045	885
612682	2562	2577	AGGCTTATTGTGGCAA	ekkddddddddddkke	34	14031	14046	886
612683	2563	2578	GAGGCTTATTGTGGCA	ekkddddddddddkke	23	14032	14047	887
612684	2564	2579	GGAGGCTTATTGTGGC	ekkddddddddddkke	0	14033	14048	888

[0515] Table 5 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 1000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 5

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	87	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	90	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	95	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	94	13515	13530	129
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	6	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	83	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	86	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	85	13496	13511	163
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	0	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	64	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	74	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	70	13516	13531	165
612129	965	980	GCCTGTCAGCTGTGTG	ekkddddddddddkke	29	6182	6197	889
612130	968	983	GTAGCCTGTCAGCTGT	ekkddddddddddkke	44	6185	6200	890
612131	971	986	CCTGTAGCCTGTCAGC	ekkddddddddddkke	21	6188	6203	891
612132	974	989	TTGCCTGTAGCCTGTC	ekkddddddddddkke	38	6191	6206	892
612133	977	992	GGATTGCCTGTAGCCT	ekkddddddddddkke	14	6194	6209	893
612134	980	995	CCAGGATTGCCTGTAG	ekkddddddddddkke	46	6197	6212	894
612135	983	998	CACCCAGGATTGCCTG	ekkddddddddddkke	23	6200	6215	68
612136	986	1001	GAACACCCAGGATTGC	ekkddddddddddkke	16	6203	6218	895

TABLE 5-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612137	993	1008	TTCCAAGGAACACCCA	ekkddddddddddkke	26	6210	6225	69
612138	997	1012	GTCCTTCCAAGGAACA	ekkddddddddddkke	27	6214	6229	896
612139	1000	1015	CTTGTCCTTCCAAGGA	ekkddddddddddkke	57	6217	6232	897
612140	1003	1018	GTTCTTGTCCTTCCAA	ekkddddddddddkke	22	6220	6235	898
612141	1006	1021	GCAGTTCTTGTCCTTC	ekkddddddddddkke	42	6223	6238	899
612142	1009	1024	GGTGCAGTTCTTGTCC	ekkddddddddddkke	0	6226	6241	900
612143	1012	1027	GGAGGTGCAGTTCTTG	ekkddddddddddkke	0	6229	6244	901
612144	1015	1030	CCGGGAGGTGCAGTTC	ekkddddddddddkke	34	6232	6247	902
612145	1018	1033	CAGCCGGGAGGTGCAG	ekkddddddddddkke	30	6235	6250	903
612146	1021	1036	ATCCAGCCGGGAGGTG	ekkddddddddddkke	43	6238	6253	904
612147	1024	1039	CGCATCCAGCCGGGAG	ekkddddddddddkke	63	6241	6256	905
612148	1027	1042	GTGCGCATCCAGCCGG	ekkddddddddddkke	64	6244	6259	906
612149	1030	1045	CTTGTGCGCATCCAGC	ekkddddddddddkke	4	6247	6262	907
612150	1033	1048	GACCTTGTGCGCATCC	ekkddddddddddkke	0	6250	6265	908
612151	1036	1051	CAGGACCTTGTGCGCA	ekkddddddddddkke	46	6253	6268	909
612152	1039	1054	AGACAGGACCTTGTGC	ekkddddddddddkke	12	6256	6271	910
612153	1042	1057	GGCAGACAGGACCTTG	ekkddddddddddkke	24	6259	6274	911
612154	1060	1075	GCCCTGTACAGCCTGC	ekkddddddddddkke	36	6277	6292	912
612155	1064	1079	GCAGGCCCTGTACAGC	ekkddddddddddkke	19	6281	6296	913
612156	1067	1082	CTAGCAGGCCCTGTAC	ekkddddddddddkke	1	6284	6299	914
612157	1071	1086	GCCACTAGCAGGCCCT	ekkddddddddddkke	0	6288	6303	915
612158	1074	1089	TGGGCCACTAGCAGGC	ekkddddddddddkke	0	6291	6306	916
612159	1077	1092	CCCTGGGCCACTAGCA	ekkddddddddddkke	27	6294	6309	917
612160	1080	1095	CTGCCCTGGGCCACTA	ekkddddddddddkke	42	6297	6312	918
612161	1088	1103	TATCAGCCCTGCCCTG	ekkddddddddddkke	28	6305	6320	74
612162	1091	1106	GGCTATCAGCCCTGCC	ekkddddddddddkke	38	6308	6323	919
612163	1094	1109	CCTGGCTATCAGCCCT	ekkddddddddddkke	38	6311	6326	920
612164	1097	1112	GGGCCTGGCTATCAGC	ekkddddddddddkke	24	6314	6329	921
612165	1100	1115	GCTGGGCCTGGCTATC	ekkddddddddddkke	0	6317	6332	922
612166	1115	1130	CCGTGGACAGCAGCAG	ekkddddddddddkke	0	6332	6347	923
612167	1118	1133	CCACCGTGGACAGCAG	ekkddddddddddkke	28	6335	6350	924
612168	1121	1136	CCACCACCGTGGACAG	ekkddddddddddkke	27	6338	6353	925
612169	1124	1139	CGCCCACCACCGTGGA	ekkddddddddddkke	11	6341	6356	926
612170	1127	1142	ACACGCCCACCACCGT	ekkddddddddddkke	18	6344	6359	927

TABLE 5-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612171	1130	1145	TGAACACGCCCACCAC	ekkddddddddddkke	34	6347	6362	928
612172	1133	1148	CTGTGAACACGCCCAC	ekkddddddddddkke	37	6350	6365	929
612173	1136	1151	GGGCTGTGAACACGCC	ekkddddddddddkke	0	6353	6368	930
612174	1151	1166	TCAGGTGCAGGCCTGG	ekkddddddddddkke	5	6368	6383	78
612175	1154	1169	GCTTCAGGTGCAGGCC	ekkddddddddddkke	45	6371	6386	931
612176	1157	1172	GCTGCTTCAGGTGCAG	ekkddddddddddkke	30	6374	6389	932
612177	1160	1175	ACGGCTGCTTCAGGTG	ekkddddddddddkke	45	6377	6392	933
612178	1163	1178	CAAACGGCTGCTTCAG	ekkddddddddddkke	17	6380	6395	934
612179	1166	1181	GCACAAACGGCTGCTT	ekkddddddddddkke	34	6383	6398	935
612180	1169	1184	CCTGCACAAACGGCTG	ekkddddddddddkke	0	6386	6401	936
612181	1172	1187	GGCCCTGCACAAACGG	ekkddddddddddkke	0	6389	6404	937
612182	1182	1197	TAGAGAGCCAGGCCCT	ekkddddddddddkke	38	6399	6414	80
612183	1185	1200	GTATAGAGAGCCAGGC	ekkddddddddddkke	19	6402	6417	938
612184	1203	1218	CGTGGGAGGACCACAG	ekkddddddddddkke	26	6420	6435	81
612185	1217	1232	TGAAGTCCAGAGAGCG	ekkddddddddddkke	5	6434	6449	82
612186	1220	1235	CTGTGAAGTCCAGAGA	ekkddddddddddkke	45	6437	6452	939
612187	1223	1238	GTTCTGTGAAGTCCAG	ekkddddddddddkke	49	6440	6455	940
612188	1226	1241	CCAGTTCTGTGAAGTC	ekkddddddddddkke	23	6443	6458	941
612189	1229	1244	CATCCAGTTCTGTGAA	ekkddddddddddkke	31	6446	6461	942
612190	1232	1247	CAACATCCAGTTCTGT	ekkddddddddddkke	30	6449	6464	943
612191	1235	1250	CAGCAACATCCAGTTC	ekkddddddddddkke	35	6452	6467	944
612192	1244	1259	TCTTCTCAGCAGCAAC	ekkddddddddddkke	61	6461	6476	84
612193	1247	1262	CAATCTTCTCAGCAGC	ekkddddddddddkke	34	6464	6479	945
612194	1250	1265	TGTCAATCTTCTCAGC	ekkddddddddddkke	44	6467	6482	946
612195	1253	1268	ACCTGTCAATCTTCTC	ekkddddddddddkke	47	6470	6485	947
612196	1256	1271	TGAACCTGTCAATCTT	ekkddddddddddkke	18	6473	6488	948
612197	1259	1274	GCATGAACCTGTCAAT	ekkddddddddddkke	39	6476	6491	949
612198	1262	1277	CCTGCATGAACCTGTC	ekkddddddddddkke	35	6479	6494	950
612199	1265	1280	CAGCCTGCATGAACCT	ekkddddddddddkke	47	6482	6497	951
612200	1267	1282	CACAGCCTGCATGAAC	ekkddddddddddkke	26	6484	6499	952
612201	1268	1283	TCACAGCCTGCATGAA	ekkddddddddddkke	36	6485	6500	953
612202	1274	1289	ATCCTGTCACAGCCTG	ekkddddddddddkke	68	6491	6506	954
612203	1276	1291	CCATCCTGTCACAGCC	ekkddddddddddkke	50	6493	6508	955
612204	1277	1292	TCCATCCTGTCACAGC	ekkddddddddddkke	7	6494	6509	956
612205	1279	1294	CTTCCATCCTGTCACA	ekkddddddddddkke	33	6496	6511	957

TABLE 5-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612206	1282	1297	AGTCTTCCATCCTGTC	ekkddddddddddkke	54	6499	6514	958
612207	1286	1301	AGCCAGTCTTCCATCC	ekkddddddddddkke	58	6503	6518	959
612233	1399	1414	CACCCAGAACTCCTGG	ekkddddddddddkke	7	10410	10425	960
612234	1402	1417	GTCCACCCAGAACTCC	ekkddddddddddkke	66	10413	10428	961
612235	1405	1420	GTTGTCCACCCAGAAC	ekkddddddddddkke	73	10416	10431	962
612236	1408	1423	GCTGTTGTCCACCCAG	ekkddddddddddkke	76	10419	10434	963
612237	1411	1426	GGTGCTGTTGTCCACC	ekkddddddddddkke	25	10422	10437	964
612238	1414	1429	TGAGGTGCTGTTGTCC	ekkddddddddddkke	77	10425	10440	965
612239	1417	1432	CACTGAGGTGCTGTTG	ekkddddddddddkke	92	10428	10443	966
612240	1421	1436	CAGACACTGAGGTGCT	ekkddddddddddkke	50	10432	10447	93
612241	1429	1444	CATGGGAACAGACACT	ekkddddddddddkke	0	10440	10455	967
612242	1432	1447	GAGCATGGGAACAGAC	ekkddddddddddkke	0	10443	10458	968
612243	1435	1450	AGAGAGCATGGGAACA	ekkddddddddddkke	6	10446	10461	969
612244	1438	1453	GCCAGAGAGCATGGGA	ekkddddddddddkke	52	10449	10464	970
612245	1441	1456	CATGCCAGAGAGCATG	ekkddddddddddkke	63	10452	10467	971
612246	1444	1459	GCCCATGCCAGAGAGC	ekkddddddddddkke	59	10455	10470	972
612247	1447	1462	GGTGCCCATGCCAGAG	ekkddddddddddkke	76	10458	10473	973
612248	1450	1465	GAAGGTGCCCATGCCA	ekkddddddddddkke	0	10461	10476	974
612249	1453	1468	CTGGAAGGTGCCCATG	ekkddddddddddkke	47	10464	10479	975
612250	1457	1472	AGTGCTGGAAGGTGCC	ekkddddddddddkke	0	10468	10483	976
612251	1460	1475	TCCAGTGCTGGAAGGT	ekkddddddddddkke	11	10471	10486	977
612252	1462	1477	ACTCCAGTGCTGGAAG	ekkddddddddddkke	85	10473	10488	96
612253	1463	1478	CACTCCAGTGCTGGAA	ekkddddddddddkke	31	10474	10489	978
612254	1465	1480	GTCACTCCAGTGCTGG	ekkddddddddddkke	77	10476	10491	97
612255	1466	1481	TGTCACTCCAGTGCTG	ekkddddddddddkke	58	10477	10492	979
612256	1467	1482	ATGTCACTCCAGTGCT	ekkddddddddddkke	8	10478	10493	980
612257	1468	1483	GATGTCACTCCAGTGC	ekkddddddddddkke	35	10479	10494	981
612258	1469	1484	GGATGTCACTCCAGTG	ekkddddddddddkke	2	10480	10495	982
612259	1470	1485	TGGATGTCACTCCAGT	ekkddddddddddkke	15	10481	10496	983
612260	1472	1487	CCTGGATGTCACTCCA	ekkddddddddddkke	40	10483	10498	984
612261	1475	1490	TGTCCTGGATGTCACT	ekkddddddddddkke	46	10486	10501	985
612262	1478	1493	AGTTGTCCTGGATGTC	ekkddddddddddkke	63	10489	10504	986
612263	1481	1496	AGAAGTTGTCCTGGAT	ekkddddddddddkke	65	10492	10507	987
612264	1484	1499	CCGAGAAGTTGTCCTG	ekkddddddddddkke	59	10495	10510	99

TABLE 5-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612265	1487	1502	TCACCGAGAAGTTGTC	ekkddddddddddkke	0	10498	10513	988
612266	1490	1505	GAGTCACCGAGAAGTT	ekkddddddddddkke	68	10501	10516	989
612267	1493	1508	CTTGAGTCACCGAGAA	ekkddddddddddkke	76	10504	10519	990
612268	1496	1511	GCACTTGAGTCACCGA	ekkddddddddddkke	77	10507	10522	991
612269	1499	1514	AGGGCACTTGAGTCAC	ekkddddddddddkke	43	10510	10525	992
612270	1502	1517	TGAAGGGCACTTGAGT	ekkddddddddddkke	42	10513	10528	993
612271	1505	1520	CAGTGAAGGGCACTTG	ekkddddddddddkke	65	10516	10531	994
612272	1508	1523	TCTCAGTGAAGGGCAC	ekkddddddddddkke	0	10519	10534	995
612273	1511	1526	CGCTCTCAGTGAAGGG	ekkddddddddddkke	35	10522	10537	996
612274	1524	1539	AGCAGCAGGCAGGCGC	ekkddddddddddkke	77	10535	10550	997
612275	1528	1543	GATCAGCAGCAGGCAG	ekkddddddddddkke	64	10539	10554	998
612276	1532	1547	GCTGGATCAGCAGCAG	ekkddddddddddkke	33	10543	10558	999
612277	1535	1550	GAGGCTGGATCAGCAG	ekkddddddddddkke	81	10546	10561	1000
612278	1538	1553	AGTGAGGCTGGATCAG	ekkddddddddddkke	79	10549	10564	1001
612279	1541	1556	CATAGTGAGGCTGGAT	ekkddddddddddkke	58	10552	10567	1002
612280	1544	1559	AGGCATAGTGAGGCTG	ekkddddddddddkke	20	10555	10570	1003
612688	N/A	N/A	CGGCTTACCTTCTGCT	ekkddddddddddkke	0	2483	2498	565
612799	N/A	N/A	AGACACACAGGCCGCC	ekkddddddddddkke	0	10783	10798	1004
612800	N/A	N/A	ACACTAACTGGAGAGC	ekkddddddddddkke	29	10830	10845	1005
612801	N/A	N/A	AGAGGGCGGATTGCAA	ekkddddddddddkke	39	10939	10954	1006
612802	N/A	N/A	CAGAGGGCGGATTGCA	ekkddddddddddkke	37	10940	10955	1007
612803	N/A	N/A	TCTCAGAGGGCGGATT	ekkddddddddddkke	36	10943	10958	1008
612804	N/A	N/A	CTCTCAGAGGGCGGAT	ekkddddddddddkke	55	10944	10959	1009
612805	N/A	N/A	TCTCTCAGAGGGCGGA	ekkddddddddddkke	34	10945	10960	1010
612806	N/A	N/A	GCTGTGTGTCAGGTGT	ekkddddddddddkke	71	10977	10992	1011
612807	N/A	N/A	AAGAAGCTCTTGGATG	ekkddddddddddkke	0	11003	11018	1012
612808	N/A	N/A	TCCAAGAAGCTCTTGG	ekkddddddddddkke	52	11006	11021	1013
612809	N/A	N/A	CCAGCCGCCAGCCGCC	ekkddddddddddkke	28	11109	11124	1014
612810	N/A	N/A	TTAGTGTTTCAGCAGG	ekkddddddddddkke	69	11451	11466	1015
612811	N/A	N/A	AGTTAGTGTTTCAGCA	ekkddddddddddkke	35	11453	11468	1016
612812	N/A	N/A	AACCTCGAGGACATCG	ekkddddddddddkke	37	11506	11521	1017
612813	N/A	N/A	ACTTATAAGAGCTGAC	ekkddddddddddkke	7	11696	11711	1018
612814	N/A	N/A	AGCACTTATAAGAGCT	ekkddddddddddkke	21	11699	11714	1019
612815	N/A	N/A	GCAGTGTTCTTGATGA	ekkddddddddddkke	27	11866	11881	1020
612816	N/A	N/A	ACAGCAGTGTTCTTGA	ekkddddddddddkke	67	11869	11884	1021

TABLE 5-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612817	N/A	N/A	ATAATGCACTGTGTCT	ekkddddddddddkke	57	11895	11910	1022
612818	N/A	N/A	GATGAGGACCTAGGAA	ekkddddddddddkke	48	11996	12011	1023
612819	N/A	N/A	CCGATGAGGACCTAGG	ekkddddddddddkke	67	11998	12013	1024
612820	N/A	N/A	ACGACAGGGATGTTTG	ekkddddddddddkke	21	12128	12143	1025
612821	N/A	N/A	GGTCAGGCACAGACAC	ekkddddddddddkke	0	12398	12413	1026
612822	N/A	N/A	ATCCCGGTTTCAACTC	ekkddddddddddkke	45	12671	12686	1027
612823	N/A	N/A	TCCCGCTGGCCCCCGT	ekkddddddddddkke	21	12866	12881	1028
612824	N/A	N/A	CTAACTTAGCACAGAG	ekkddddddddddkke	13	12888	12903	1029
612825	N/A	N/A	CCATGGCCCACCAGTG	ekkddddddddddkke	44	12915	12930	1030
612826	N/A	N/A	TTGGCCATGGCCCACC	ekkddddddddddkke	30	12919	12934	1031
612827	N/A	N/A	GGCAGAATTCCTGGCT	ekkddddddddddkke	0	12938	12953	1032
612828	N/A	N/A	GCAAGGGTGTGTCTGT	ekkddddddddddkke	13	13059	13074	1033
612829	N/A	N/A	GGCAAGGGTGTGTCTG	ekkddddddddddkke	23	13060	13075	1034
612830	N/A	N/A	CTCAGTGTAGGCAAGG	ekkddddddddddkke	60	13069	13084	1035
612831	N/A	N/A	GAGGATGCACAGTGTA	ekkddddddddddkke	12	13094	13109	1036
612832	N/A	N/A	GCTCAGGACCTCTGTG	ekkddddddddddkke	22	13151	13166	1037
612833	N/A	N/A	GGCTCAGGACCTCTGT	ekkddddddddddkke	34	13152	13167	1038
612834	N/A	N/A	GGCGCACTGGGTGACC	ekkddddddddddkke	38	13198	13213	1039
612835	N/A	N/A	TCTGAGGGCGCACTGG	ekkddddddddddkke	9	13204	13219	1040
612836	N/A	N/A	TCATTCTGAGGGCGCA	ekkddddddddddkke	1	13208	13223	1041
612838	N/A	N/A	GCTCCTACCGGGGAGA	ekkddddddddddkke	33	10636	10651	1042
612839	N/A	N/A	ACACATACCTCCCCCA	ekkddddddddddkke	0	12376	12391	1043
612840	N/A	N/A	CGCATACCCTGAAATA	ekkddddddddddkke	0	5715	5730	1044
612842	N/A	N/A	GGATGGTCCTGGGGAG	ekkddddddddddkke	13	12231	12246	1045
612843	N/A	N/A	TTCAGCACCTGCAAAG	ekkddddddddddkke	0	13239	13254	1046
612844	N/A	N/A	CCGGCTTACCTTCTGC	ekkddddddddddkke	9	2484	2499	1047
612845	N/A	N/A	CCCCCGGCTTACCTTC	ekkddddddddddkke	0	2487	2502	1048
612846	N/A	N/A	GGGCCCCCGGCTTACC	ekkddddddddddkke	0	2490	2505	1049
612847	N/A	N/A	GTGAATGTGAGCCCCG	ekkddddddddddkke	14	3361	3376	1050
612848	N/A	N/A	TCCCTCCTTATAACCC	ekkddddddddddkke	0	3435	3450	1051
612849	N/A	N/A	CCGGGCACTCTCAACT	ekkddddddddddkke	4	3471	3486	1052
612850	N/A	N/A	AGTAATGGTGCTCTGG	ekkddddddddddkke	4	3752	3767	1053
612851	N/A	N/A	TCCTGGGAGTAATGGT	ekkddddddddddkke	30	3759	3774	1054
612852	N/A	N/A	TCTCAGTTGTGATCTG	ekkddddddddddkke	31	3817	3832	1055

TABLE 5-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612853	N/A	N/A	TCCAGAGACGCAATTC	ekkddddddddddkke	0	3868	3883	1056
612854	N/A	N/A	TCTCCAGAGACGCAAT	ekkddddddddddkke	11	3870	3885	1057
612855	N/A	N/A	ACCTGTGGGAACCGAC	ekkddddddddddkke	4	3983	3998	1058
612856	N/A	N/A	AAACCTGTGGGAACCG	ekkddddddddddkke	0	3985	4000	1059
612857	N/A	N/A	CCTAGATTTTTCTGCT	ekkddddddddddkke	27	4340	4355	1060
612858	N/A	N/A	GCCTTTTCTGTCCCCC	ekkddddddddddkke	57	4420	4435	1061
612859	N/A	N/A	CATTTCTTGTGGAGGG	ekkddddddddddkke	12	4464	4479	1062
612860	N/A	N/A	TGGGCTGGCCCTGCTA	ekkddddddddddkke	2	4569	4584	1063
612861	N/A	N/A	GAGCCCCAAAGGCATG	ekkddddddddddkke	33	4822	4837	1064
612862	N/A	N/A	TCTAATATGACCTGTG	ekkddddddddddkke	43	5357	5372	1065
612863	N/A	N/A	TGATCTAATATGACCT	ekkddddddddddkke	13	5360	5375	1066
612864	N/A	N/A	GTCCTCAACCCCAGGA	ekkddddddddddkke	0	5455	5470	1067
612865	N/A	N/A	GCTCCATGGAAAATAT	ekkddddddddddkke	4	5553	5568	1068
612866	N/A	N/A	TCCATTCATGTCTACA	ekkddddddddddkke	19	5593	5608	1069
612867	N/A	N/A	TTAAGTGCCATCTAAC	ekkddddddddddkke	17	5660	5675	1070
612868	N/A	N/A	GCATACCCTGAAATAT	ekkddddddddddkke	0	5714	5729	1071
612893	N/A	N/A	TGTCTACTCCCCACCC	ekkddddddddddkke	42	10707	10722	1072
612894	N/A	N/A	ACAGACACTAACTGGA	ekkddddddddddkke	28	10834	10849	1073
612895	N/A	N/A	GTGTGTCAGGTGTGGG	ekkddddddddddkke	37	10974	10989	1074
612896	N/A	N/A	GCAAGTCAGTTCCAAG	ekkddddddddddkke	35	11016	11031	1075
612897	N/A	N/A	CTCGAAAATGGTTACG	ekkddddddddddkke	55	11336	11351	1076
612898	N/A	N/A	GGTGGTAACCACATGC	ekkddddddddddkke	53	11583	11598	1077
612899	N/A	N/A	ATGCACTGTGTCTTAC	ekkddddddddddkke	31	11892	11907	1078
612900	N/A	N/A	AATAATGCACTGTGTC	ekkddddddddddkke	39	11896	11911	1079
612901	N/A	N/A	GTTACTTGGGTAATTT	ekkddddddddddkke	68	11930	11945	1080
612902	N/A	N/A	TCCTTTGGTGCATTCT	ekkddddddddddkke	19	11974	11989	1081
612903	N/A	N/A	CTAGGAATGGTTGTCC	ekkddddddddddkke	0	11987	12002	1082
612904	N/A	N/A	GACGACAGGGATGTTT	ekkddddddddddkke	20	12129	12144	1083
612905	N/A	N/A	CTGACGACAGGGATGT	ekkddddddddddkke	25	12131	12146	1084
612906	N/A	N/A	GCACAGTTAGGAAGGC	ekkddddddddddkke	60	12210	12225	1085
612907	N/A	N/A	TTAGCTAACTTAGCAC	ekkddddddddddkke	8	12892	12907	1086
612908	N/A	N/A	CATGGCCCACCAGTGC	ekkddddddddddkke	41	12914	12929	1087
612909	N/A	N/A	CACAGTGTATGCCTGC	ekkddddddddddkke	52	13087	13102	1088

TABLE 5-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612910	N/A	N/A	GCACTGGGTGACCCAG	ekkddddddddddkke	0	13195	13210	1089
612911	N/A	N/A	TCAGCACCTGCAAAGC	ekkddddddddddkke	0	13238	13253	1090

[0516] Table 6 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 1000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS4039 (forward sequence GGACAAGGTGGAGGGTCTCA, designated herein as SEQ ID NO: 11; reverse sequence AGATCCTTGCAGCACCAGTTG, designated herein as SEQ ID NO: 12; and probe sequence ATGAAGAAACTATCTCCCCGGACCATCCAX, where X is a fluorescent label, designated herein as SEQ ID NO: 13) was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 6

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	74	13515	13530	129
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	64	13496	13511	163
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	57	13516	13531	165
612205	1279	1294	CTTCCATCCTGTCACA	ekkddddddddddkke	0	6496	6511	957
612206	1282	1297	AGTCTTCCATCCTGTC	ekkddddddddddkke	9	6499	6514	958
612207	1286	1301	AGCCAGTCTTCCATCC	ekkddddddddddkke	0	6503	6518	959
612208	1290	1305	GAGCAGCCAGTCTTCC	ekkddddddddddkke	0	6507	6522	1091
612209	1293	1308	AGGGAGCAGCCAGTCT	ekkddddddddddkke	0	6510	6525	1092
612210	1296	1311	ATCAGGGAGCAGCCAG	ekkddddddddddkke	0	6513	6528	1093
612211	1300	1315	TCCCATCAGGGAGCAG	ekkddddddddddkke	0	6517	6532	1094
612212	1303	1318	GGCTCCCATCAGGGAG	ekkddddddddddkke	0	6520	6535	1095
612213	1306	1321	ACTGGCTCCCATCAGG	ekkddddddddddkke	16	6523	6538	1096
612214	1310	1325	CCACACTGGCTCCCAT	ekkddddddddddkke	0	6527	6542	1097
612215	1315	1330	GCTGTCCACACTGGCT	ekkddddddddddkke	13	6532	6547	1098
612216	1318	1333	GGTGCTGTCCACACTG	ekkddddddddddkke	20	6535	6550	1099
612217	1321	1336	CAGGGTGCTGTCCACA	ekkddddddddddkke	0	6538	6553	1100
612218	1324	1339	AGCCAGGGTGCTGTCC	ekkddddddddddkke	14	6541	6556	1101
612219	1327	1342	GAAAGCCAGGGTGCTG	ekkddddddddddkke	0	6544	6559	1102
612220	1330	1345	GTTGAAAGCCAGGGTG	ekkddddddddddkke	6	6547	6562	1103
612221	1333	1348	GGTGTTGAAAGCCAGG	ekkddddddddddkke	34	6550	6565	1104
612222	1336	1351	GTAGGTGTTGAAAGCC	ekkddddddddddkke	9	6553	6568	1105

TABLE 6-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612223	1351	1366	CCCTTGGAAGTGGACG	ekkddddddddddkke	0	N/A	N/A	1106
612224	1354	1369	CTTCCCTTGGAAGTGG	ekkddddddddddkke	17	N/A	N/A	1107
612225	1357	1372	CATCTTCCCTTGGAAG	ekkddddddddddkke	11	N/A	N/A	1108
612226	1360	1375	CTTCATCTTCCCTTGG	ekkddddddddddkke	0	N/A	N/A	1109
612227	1364	1379	AGCCCTTCATCTTCCC	ekkddddddddddkke	5	10375	10390	1110
612228	1367	1382	AGAAGCCCTTCATCTT	ekkddddddddddkke	0	10378	10393	1111
612229	1370	1385	GGGAGAAGCCCTTCAT	ekkddddddddddkke	0	10381	10396	1112
612230	1373	1388	GCAGGGAGAAGCCCTT	ekkddddddddddkke	25	10384	10399	1113
612231	1380	1395	TCGGCCAGCAGGGAGA	ekkddddddddddkke	32	10391	10406	1114
612232	1383	1398	GGCTCGGCCAGCAGGG	ekkddddddddddkke	24	10394	10409	1115
612233	1399	1414	CACCCAGAACTCCTGG	ekkddddddddddkke	5	10410	10425	960
612234	1402	1417	GTCCACCCAGAACTCC	ekkddddddddddkke	0	10413	10428	961
612235	1405	1420	GTTGTCCACCCAGAAC	ekkddddddddddkke	0	10416	10431	962
612236	1408	1423	GCTGTTGTCCACCCAG	ekkddddddddddkke	14	10419	10434	963
612237	1411	1426	GGTGCTGTTGTCCACC	ekkddddddddddkke	20	10422	10437	964
612238	1414	1429	TGAGGTGCTGTTGTCC	ekkddddddddddkke	32	10425	10440	965
612239	1417	1432	CACTGAGGTGCTGTTG	ekkddddddddddkke	36	10428	10443	966
612240	1421	1436	CAGACACTGAGGTGCT	ekkddddddddddkke	1	10432	10447	93
612241	1429	1444	CATGGGAACAGACACT	ekkddddddddddkke	9	10440	10455	967
612242	1432	1447	GAGCATGGGAACAGAC	ekkddddddddddkke	0	10443	10458	968
612243	1435	1450	AGAGAGCATGGGAACA	ekkddddddddddkke	0	10446	10461	969
612244	1438	1453	GCCAGAGAGCATGGGA	ekkddddddddddkke	5	10449	10464	970
612245	1441	1456	CATGCCAGAGAGCATG	ekkddddddddddkke	27	10452	10467	971
612246	1444	1459	GCCCATGCCAGAGAGC	ekkddddddddddkke	0	10455	10470	972
612247	1447	1462	GGTGCCCATGCCAGAG	ekkddddddddddkke	36	10458	10473	973
612248	1450	1465	GAAGGTGCCCATGCCA	ekkddddddddddkke	0	10461	10476	974
612249	1453	1468	CTGGAAGGTGCCCATG	ekkddddddddddkke	24	10464	10479	975
612250	1457	1472	AGTGCTGGAAGGTGCC	ekkddddddddddkke	0	10468	10483	976
612251	1460	1475	TCCAGTGCTGGAAGGT	ekkddddddddddkke	3	10471	10486	977
612252	1462	1477	ACTCCAGTGCTGGAAG	ekkddddddddddkke	72	10473	10488	96

TABLE 6-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612253	1463	1478	CACTCCAGTGCTGGAA	ekkddddddddddkke	19	10474	10489	978
612254	1465	1480	GTCACTCCAGTGCTGG	ekkddddddddddkke	45	10476	10491	97
612255	1466	1481	TGTCACTCCAGTGCTG	ekkddddddddddkke	15	10477	10492	979
612256	1467	1482	ATGTCACTCCAGTGCT	ekkddddddddddkke	0	10478	10493	980
612257	1468	1483	GATGTCACTCCAGTGC	ekkddddddddddkke	16	10479	10494	981
612258	1469	1484	GGATGTCACTCCAGTG	ekkddddddddddkke	0	10480	10495	982
612259	1470	1485	TGGATGTCACTCCAGT	ekkddddddddddkke	3	10481	10496	983
612260	1472	1487	CCTGGATGTCACTCCA	ekkddddddddddkke	10	10483	10498	984
612261	1475	1490	TGTCCTGGATGTCACT	ekkddddddddddkke	8	10486	10501	985
612262	1478	1493	AGTTGTCCTGGATGTC	ekkddddddddddkke	0	10489	10504	986
612263	1481	1496	AGAAGTTGTCCTGGAT	ekkddddddddddkke	14	10492	10507	987
612264	1484	1499	CCGAGAAGTTGTCCTG	ekkddddddddddkke	10	10495	10510	99
612265	1487	1502	TCACCGAGAAGTTGTC	ekkddddddddddkke	0	10498	10513	988
612266	1490	1505	GAGTCACCGAGAAGTT	ekkddddddddddkke	33	10501	10516	989
612267	1493	1508	CTTGAGTCACCGAGAA	ekkddddddddddkke	35	10504	10519	990
612268	1496	1511	GCACTTGAGTCACCGA	ekkddddddddddkke	37	10507	10522	991
612269	1499	1514	AGGGCACTTGAGTCAC	ekkddddddddddkke	0	10510	10525	992
612270	1502	1517	TGAAGGGCACTTGAGT	ekkddddddddddkke	8	10513	10528	993
612271	1505	1520	CAGTGAAGGGCACTTG	ekkddddddddddkke	8	10516	10531	994
612272	1508	1523	TCTCAGTGAAGGGCAC	ekkddddddddddkke	0	10519	10534	995
612273	1511	1526	CGCTCTCAGTGAAGGG	ekkddddddddddkke	18	10522	10537	996
612274	1524	1539	AGCAGCAGGCAGGCGC	ekkddddddddddkke	27	10535	10550	997
612275	1528	1543	GATCAGCAGCAGGCAG	ekkddddddddddkke	39	10539	10554	998
612276	1532	1547	GCTGGATCAGCAGCAG	ekkddddddddddkke	21	10543	10558	999
612277	1535	1550	GAGGCTGGATCAGCAG	ekkddddddddddkke	34	10546	10561	1000
612278	1538	1553	AGTGAGGCTGGATCAG	ekkddddddddddkke	28	10549	10564	1001
612279	1541	1556	CATAGTGAGGCTGGAT	ekkddddddddddkke	13	10552	10567	1002
612280	1544	1559	AGGCATAGTGAGGCTG	ekkddddddddddkke	0	10555	10570	1003

[0517] Table 7 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 1000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS4039 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 7

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	93	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	90	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	84	13515	13530	129
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	86	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	85	13496	13511	163
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	5	13496	13511	163
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	75	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	70	13516	13531	165
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	0	13516	13531	165
612129	965	980	GCCTGTCAGCTGTGTG	ekkddddddddddkke	35	6182	6197	889
612130	968	983	GTAGCCTGTCAGCTGT	ekkddddddddddkke	35	6185	6200	890
612131	971	986	CCTGTAGCCTGTCAGC	ekkddddddddddkke	30	6188	6203	891
612132	974	989	TTGCCTGTAGCCTGTC	ekkddddddddddkke	37	6191	6206	892
612133	977	992	GGATTGCCTGTAGCCT	ekkddddddddddkke	30	6194	6209	893
612134	980	995	CCAGGATTGCCTGTAG	ekkddddddddddkke	56	6197	6212	894
612135	983	998	CACCCAGGATTGCCTG	ekkddddddddddkke	0	6200	6215	68
612136	986	1001	GAACACCCAGGATTGC	ekkddddddddddkke	8	6203	6218	895
612137	993	1008	TTCCAAGGAACACCCA	ekkddddddddddkke	27	6210	6225	69
612138	997	1012	GTCCTTCCAAGGAACA	ekkddddddddddkke	26	6214	6229	896
612139	1000	1015	CTTGTCCTTCCAAGGA	ekkddddddddddkke	47	6217	6232	897
612140	1003	1018	GTTCTTGTCCTTCCAA	ekkddddddddddkke	36	6220	6235	898
612141	1006	1021	GCAGTTCTTGTCCTTC	ekkddddddddddkke	28	6223	6238	899
612142	1009	1024	GGTGCAGTTCTTGTCC	ekkddddddddddkke	13	6226	6241	900
612143	1012	1027	GGAGGTGCAGTTCTTG	ekkddddddddddkke	0	6229	6244	901
612144	1015	1030	CCGGGAGGTGCAGTTC	ekkddddddddddkke	27	6232	6247	902
612145	1018	1033	CAGCCGGGAGGTGCAG	ekkddddddddddkke	39	6235	6250	903
612146	1021	1036	ATCCAGCCGGGAGGTG	ekkddddddddddkke	24	6238	6253	904
612147	1024	1039	CGCATCCAGCCGGGAG	ekkddddddddddkke	55	6241	6256	905
612148	1027	1042	GTGCGCATCCAGCCGG	ekkddddddddddkke	37	6244	6259	906
612149	1030	1045	CTTGTGCGCATCCAGC	ekkddddddddddkke	11	6247	6262	907
612150	1033	1048	GACCTTGTGCGCATCC	ekkddddddddddkke	12	6250	6265	908
612151	1036	1051	CAGGACCTTGTGCGCA	ekkddddddddddkke	41	6253	6268	909
612152	1039	1054	AGACAGGACCTTGTGC	ekkddddddddddkke	9	6256	6271	910
612153	1042	1057	GGCAGACAGGACCTTG	ekkddddddddddkke	30	6259	6274	911
612154	1060	1075	GCCCTGTACAGCCTGC	ekkddddddddddkke	19	6277	6292	912

TABLE 7-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612155	1064	1079	GCAGGCCCTGTACAGC	ekkddddddddddkke	0	6281	6296	913
612156	1067	1082	CTAGCAGGCCCTGTAC	ekkddddddddddkke	21	6284	6299	914
612157	1071	1086	GCCACTAGCAGGCCCT	ekkddddddddddkke	0	6288	6303	915
612158	1074	1089	TGGGCCACTAGCAGGC	ekkddddddddddkke	13	6291	6306	916
612159	1077	1092	CCCTGGGCCACTAGCA	ekkddddddddddkke	23	6294	6309	917
612160	1080	1095	CTGCCCTGGGCCACTA	ekkddddddddddkke	28	6297	6312	918
612161	1088	1103	TATCAGCCCTGCCCTG	ekkddddddddddkke	0	6305	6320	74
612162	1091	1106	GGCTATCAGCCCTGCC	ekkddddddddddkke	27	6308	6323	919
612163	1094	1109	CCTGGCTATCAGCCCT	ekkddddddddddkke	13	6311	6326	920
612164	1097	1112	GGGCCTGGCTATCAGC	ekkddddddddddkke	3	6314	6329	921
612165	1100	1115	GCTGGGCCTGGCTATC	ekkddddddddddkke	10	6317	6332	922
612166	1115	1130	CCGTGGACAGCAGCAG	ekkddddddddddkke	12	6332	6347	923
612167	1118	1133	CCACCGTGGACAGCAG	ekkddddddddddkke	42	6335	6350	924
612168	1121	1136	CCACCACCGTGGACAG	ekkddddddddddkke	27	6338	6353	925
612169	1124	1139	CGCCCACCACCGTGGA	ekkddddddddddkke	29	6341	6356	926
612170	1127	1142	ACACGCCCACCACCGT	ekkddddddddddkke	9	6344	6359	927
612171	1130	1145	TGAACACGCCCACCAC	ekkddddddddddkke	25	6347	6362	928
612172	1133	1148	CTGTGAACACGCCCAC	ekkddddddddddkke	32	6350	6365	929
612173	1136	1151	GGGCTGTGAACACGCC	ekkddddddddddkke	0	6353	6368	930
612174	1151	1166	TCAGGTGCAGGCCTGG	ekkddddddddddkke	8	6368	6383	78
612175	1154	1169	GCTTCAGGTGCAGGCC	ekkddddddddddkke	30	6371	6386	931
612176	1157	1172	GCTGCTTCAGGTGCAG	ekkddddddddddkke	21	6374	6389	932
612177	1160	1175	ACGGCTGCTTCAGGTG	ekkddddddddddkke	46	6377	6392	933
612178	1163	1178	CAAACGGCTGCTTCAG	ekkddddddddddkke	7	6380	6395	934
612179	1166	1181	GCACAAACGGCTGCTT	ekkddddddddddkke	31	6383	6398	935
612180	1169	1184	CCTGCACAAACGGCTG	ekkddddddddddkke	10	6386	6401	936
612181	1172	1187	GGCCCTGCACAAACGG	ekkddddddddddkke	5	6389	6404	937
612182	1182	1197	TAGAGAGCCAGGCCCT	ekkddddddddddkke	29	6399	6414	80
612183	1185	1200	GTATAGAGAGCCAGGC	ekkddddddddddkke	0	6402	6417	938
612184	1203	1218	CGTGGGAGGACCACAG	ekkddddddddddkke	16	6420	6435	81
612185	1217	1232	TGAAGTCCAGAGAGCG	ekkddddddddddkke	27	6434	6449	82
612186	1220	1235	CTGTGAAGTCCAGAGA	ekkddddddddddkke	26	6437	6452	939
612187	1223	1238	GTTCTGTGAAGTCCAG	ekkddddddddddkke	44	6440	6455	940
612188	1226	1241	CCAGTTCTGTGAAGTC	ekkddddddddddkke	29	6443	6458	941

TABLE 7-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612189	1229	1244	CATCCAGTTCTGTGAA	ekkddddddddddkke	14	6446	6461	942
612190	1232	1247	CAACATCCAGTTCTGT	ekkddddddddddkke	0	6449	6464	943
612191	1235	1250	CAGCAACATCCAGTTC	ekkddddddddddkke	24	6452	6467	944
612192	1244	1259	TCTTCTCAGCAGCAAC	ekkddddddddddkke	62	6461	6476	84
612193	1247	1262	CAATCTTCTCAGCAGC	ekkddddddddddkke	27	6464	6479	945
612194	1250	1265	TGTCAATCTTCTCAGC	ekkddddddddddkke	18	6467	6482	946
612195	1253	1268	ACCTGTCAATCTTCTC	ekkddddddddddkke	33	6470	6485	947
612196	1256	1271	TGAACCTGTCAATCTT	ekkddddddddddkke	25	6473	6488	948
612197	1259	1274	GCATGAACCTGTCAAT	ekkddddddddddkke	27	6476	6491	949
612198	1262	1277	CCTGCATGAACCTGTC	ekkddddddddddkke	15	6479	6494	950
612199	1265	1280	CAGCCTGCATGAACCT	ekkddddddddddkke	42	6482	6497	951
612200	1267	1282	CACAGCCTGCATGAAC	ekkddddddddddkke	39	6484	6499	952
612201	1268	1283	TCACAGCCTGCATGAA	ekkddddddddddkke	27	6485	6500	953
612202	1274	1289	ATCCTGTCACAGCCTG	ekkddddddddddkke	44	6491	6506	954
612203	1276	1291	CCATCCTGTCACAGCC	ekkddddddddddkke	39	6493	6508	955
612204	1277	1292	TCCATCCTGTCACAGC	ekkddddddddddkke	27	6494	6509	956
612688	N/A	N/A	CGGCTTACCTTCTGCT	ekkddddddddddkke	7	2483	2498	565
612761	N/A	N/A	CGAAGGGAGACCCATT	ekkddddddddddkke	24	8270	8285	1116
612762	N/A	N/A	TTCGAAGGGAGACCCA	ekkddddddddddkke	9	8272	8287	1117
612763	N/A	N/A	CTTTCGAAGGGAGACC	ekkddddddddddkke	12	8274	8289	1118
612764	N/A	N/A	CCGATCTCCTCACTGG	ekkddddddddddkke	9	8497	8512	1119
612765	N/A	N/A	CCCCGATCTCCTCACT	ekkddddddddddkke	6	8499	8514	1120
612766	N/A	N/A	ACAGCCCCCGATCTCC	ekkddddddddddkke	35	8504	8519	1121
612767	N/A	N/A	GAGACAGCCCCCGATC	ekkddddddddddkke	3	8507	8522	1122
612768	N/A	N/A	CCGAGACAGCCCCCGA	ekkddddddddddkke	7	8509	8524	1123
612769	N/A	N/A	CTAGCTGCCTGCTGAG	ekkddddddddddkke	27	8569	8584	1124
612770	N/A	N/A	TCTAGCTGCCTGCTGA	ekkddddddddddkke	22	8570	8585	1125
612771	N/A	N/A	GTGGGACACATCTAGC	ekkddddddddddkke	16	8580	8595	1126
612772	N/A	N/A	TCTAGTGGGACACATC	ekkddddddddddkke	27	8584	8599	1127
612773	N/A	N/A	TCTCTAGTGGGACACA	ekkddddddddddkke	17	8586	8601	1128
612774	N/A	N/A	CATGAGAGTGGCTGCC	ekkddddddddddkke	29	8789	8804	1129
612775	N/A	N/A	CTTTTAGTTTAGAGGG	ekkddddddddddkke	25	8883	8898	1130
612776	N/A	N/A	ATGTGAGCGGGAAACT	ekkddddddddddkke	16	8961	8976	1131
612777	N/A	N/A	CATGTGAGCGGGAAAC	ekkddddddddddkke	38	8962	8977	1132
612778	N/A	N/A	CGGAGCACTCAGTCTC	ekkddddddddddkke	38	8985	9000	1133

TABLE 7-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612779	N/A	N/A	GTCCTCAGTCCTCGGA	ekkddddddddddkke	8	8997	9012	1134
612780	N/A	N/A	CGTCCTCAGTCCTCGG	ekkddddddddddkke	53	8998	9013	1135
612781	N/A	N/A	GCAGTGGCAGACCTGG	ekkddddddddddkke	23	9023	9038	1136
612782	N/A	N/A	TAGAGATGGTTCAGAA	ekkddddddddddkke	13	9166	9181	1137
612783	N/A	N/A	TGAGTAGAGATGGTTC	ekkddddddddddkke	25	9170	9185	1138
612784	N/A	N/A	GGAGTCTGAGTAGAGA	ekkddddddddddkke	21	9176	9191	1139
612785	N/A	N/A	GCCCTCGGCTGTCCTC	ekkddddddddddkke	24	9294	9309	1140
612786	N/A	N/A	CTCGACCTTACACTAG	ekkddddddddddkke	29	9319	9334	1141
612787	N/A	N/A	CCTCTGCCTCGACCTT	ekkddddddddddkke	49	9326	9341	1142
612788	N/A	N/A	AACTCGGGAGAGCCCG	ekkddddddddddkke	41	9410	9425	1143
612789	N/A	N/A	AACGAGGGCTCCATTC	ekkddddddddddkke	22	9557	9572	1144
612790	N/A	N/A	GACACACTCACTTTTT	ekkddddddddddkke	25	9999	10014	1145
612791	N/A	N/A	CTGCCAGGTCAACTCA	ekkddddddddddkke	39	10050	10065	1146
612792	N/A	N/A	GTACCTGCCAGGTCAA	ekkddddddddddkke	25	10054	10069	1147
612793	N/A	N/A	CTGGTACCTGCCAGGT	ekkddddddddddkke	32	10057	10072	1148
612794	N/A	N/A	AGTTCACTGAGGCAGC	ekkddddddddddkke	37	10156	10171	1149
612795	N/A	N/A	CCATTTGAGTTCACTG	ekkddddddddddkke	61	10163	10178	1150
612796	N/A	N/A	GCAGCCATTTGAGTTC	ekkddddddddddkke	42	10167	10182	1151
612797	N/A	N/A	AAGGCCCAGATCCTGC	ekkddddddddddkke	0	10286	10301	1152
612798	N/A	N/A	GAAATCCAGACAGGAG	ekkddddddddddkke	11	10358	10373	1153
612821	N/A	N/A	GGTCAGGCACAGACAC	ekkddddddddddkke	0	12398	12413	1026
612822	N/A	N/A	ATCCCGGTTTCAACTC	ekkddddddddddkke	14	12671	12686	1027
612823	N/A	N/A	TCCCGCTGGCCCCCGT	ekkddddddddddkke	36	12866	12881	1028
612824	N/A	N/A	CTAACTTAGCACAGAG	ekkddddddddddkke	22	12888	12903	1029
612825	N/A	N/A	CCATGGCCCACCAGTG	ekkddddddddddkke	35	12915	12930	1030
612826	N/A	N/A	TTGGCCATGGCCCACC	ekkddddddddddkke	23	12919	12934	1031
612827	N/A	N/A	GGCAGAATTCCTGGCT	ekkddddddddddkke	0	12938	12953	1032
612828	N/A	N/A	GCAAGGGTGTGTCTGT	ekkddddddddddkke	23	13059	13074	1033
612829	N/A	N/A	GGCAAGGGTGTGTCTG	ekkddddddddddkke	29	13060	13075	1034
612830	N/A	N/A	CTCAGTGTAGGCAAGG	ekkddddddddddkke	37	13069	13084	1035
612831	N/A	N/A	GAGGATGCACAGTGTA	ekkddddddddddkke	15	13094	13109	1036
612832	N/A	N/A	GCTCAGGACCTCTGTG	ekkddddddddddkke	20	13151	13166	1037
612833	N/A	N/A	GGCTCAGGACCTCTGT	ekkddddddddddkke	48	13152	13167	1038
612834	N/A	N/A	GGCGCACTGGGTGACC	ekkddddddddddkke	32	13198	13213	1039

TABLE 7-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612835	N/A	N/A	TCTGAGGGCGCACTGG	ekkddddddddddkke	24	13204	13219	1040
612836	N/A	N/A	TCATTCTGAGGGCGCA	ekkddddddddddkke	18	13208	13223	1041
612837	N/A	N/A	TGCCTTACCTTGGAAG	ekkddddddddddkke	1	6574	6589	1154
612839	N/A	N/A	ACACATACCTCCCCCA	ekkddddddddddkke	4	12376	12391	1043
612840	N/A	N/A	CGCATACCCTGAAATA	ekkddddddddddkke	1	5715	5730	1044
612841	N/A	N/A	CATCTTCCCTGAAATC	ekkddddddddddkke	0	10368	10383	1155
612843	N/A	N/A	TTCAGCACCTGCAAAG	ekkddddddddddkke	0	13239	13254	1046
612844	N/A	N/A	CCGGCTTACCTTCTGC	ekkddddddddddkke	21	2484	2499	1047
612845	N/A	N/A	CCCCCGGCTTACCTTC	ekkddddddddddkke	0	2487	2502	1048
612846	N/A	N/A	GGGCCCCCGGCTTACC	ekkddddddddddkke	9	2490	2505	1049
612847	N/A	N/A	GTGAATGTGAGCCCCG	ekkddddddddddkke	9	3361	3376	1050
612848	N/A	N/A	TCCCTCCTTATAACCC	ekkddddddddddkke	5	3435	3450	1051
612849	N/A	N/A	CCGGGCACTCTCAACT	ekkddddddddddkke	6	3471	3486	1052
612850	N/A	N/A	AGTAATGGTGCTCTGG	ekkddddddddddkke	13	3752	3767	1053
612851	N/A	N/A	TCCTGGGAGTAATGGT	ekkddddddddddkke	16	3759	3774	1054
612852	N/A	N/A	TCTCAGTTGTGATCTG	ekkddddddddddkke	19	3817	3832	1055
612853	N/A	N/A	TCCAGAGACGCAATTC	ekkddddddddddkke	0	3868	3883	1056
612854	N/A	N/A	TCTCCAGAGACGCAAT	ekkddddddddddkke	15	3870	3885	1057
612855	N/A	N/A	ACCTGTGGGAACCGAC	ekkddddddddddkke	17	3983	3998	1058
612856	N/A	N/A	AAACCTGTGGGAACCG	ekkddddddddddkke	7	3985	4000	1059
612857	N/A	N/A	CCTAGATTTTTCTGCT	ekkddddddddddkke	15	4340	4355	1060
612858	N/A	N/A	GCCTTTTCTGTCCCCC	ekkddddddddddkke	24	4420	4435	1061
612859	N/A	N/A	CATTTCTTGTGGAGGG	ekkddddddddddkke	3	4464	4479	1062
612860	N/A	N/A	TGGGCTGGCCCTGCTA	ekkddddddddddkke	0	4569	4584	1063
612861	N/A	N/A	GAGCCCCAAAGGCATG	ekkddddddddddkke	0	4822	4837	1064
612862	N/A	N/A	TCTAATATGACCTGTG	ekkddddddddddkke	25	5357	5372	1065
612863	N/A	N/A	TGATCTAATATGACCT	ekkddddddddddkke	6	5360	5375	1066
612864	N/A	N/A	GTCCTCAACCCCAGGA	ekkddddddddddkke	9	5455	5470	1067
612865	N/A	N/A	GCTCCATGGAAAATAT	ekkddddddddddkke	0	5553	5568	1068
612866	N/A	N/A	TCCATTCATGTCTACA	ekkddddddddddkke	11	5593	5608	1069
612867	N/A	N/A	TTAAGTGCCATCTAAC	ekkddddddddddkke	23	5660	5675	1070
612868	N/A	N/A	GCATACCCTGAAATAT	ekkddddddddddkke	0	5714	5729	1071
612869	N/A	N/A	AGGTATGTCCGCAGGG	ekkddddddddddkke	35	6679	6694	1156

TABLE 7-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
612870	N/A	N/A	TAGTAGGGCAGCAGGT	ekkddddddddddkke	7	6765	6780	1157
612871	N/A	N/A	TTGTTTCTCCGAGTCT	ekkddddddddddkke	42	6879	6894	1158
612872	N/A	N/A	AGGCACTTTGTTTCTC	ekkddddddddddkke	5	6886	6901	1159
612873	N/A	N/A	CAAGGCACTTTGTTTC	ekkddddddddddkke	0	6888	6903	1160
612874	N/A	N/A	TAGAACTGGGCTGTGG	ekkddddddddddkke	0	6962	6977	1161
612875	N/A	N/A	CCCTCCTAACATGAAA	ekkddddddddddkke	0	7071	7086	1162
612876	N/A	N/A	CTTACAAGTAGCAAAT	ekkddddddddddkke	11	7332	7347	1163
612877	N/A	N/A	GCCAGGCTTAAAGTCT	ekkddddddddddkke	10	7346	7361	1164
612878	N/A	N/A	ATTGACCTTTAAAAGC	ekkddddddddddkke	5	7407	7422	1165
612879	N/A	N/A	TCTGGTTCAACACTCA	ekkddddddddddkke	39	7640	7655	1166
612880	N/A	N/A	TTCCCGTGACTGTGTG	ekkddddddddddkke	25	7813	7828	1167
612881	N/A	N/A	CGAGCTGCTCCCTGAG	ekkddddddddddkke	15	7835	7850	1168
612882	N/A	N/A	CACCCCACCCATGGAT	ekkddddddddddkke	0	7855	7870	1169
612883	N/A	N/A	TCTCTGTCCCTCACGA	ekkddddddddddkke	20	7925	7940	1170
612884	N/A	N/A	TTTCGAAGGGAGACCC	ekkddddddddddkke	9	8273	8288	1171
612885	N/A	N/A	CATCTAGCTGCCTGCT	ekkddddddddddkke	0	8572	8587	1172
612886	N/A	N/A	TGGGACACATCTAGCT	ekkddddddddddkke	9	8579	8594	1173
612887	N/A	N/A	ATCCTCAGGTCCTCTC	ekkddddddddddkke	14	8598	8613	1174
612888	N/A	N/A	ATGGTTCAGAAACAGT	ekkddddddddddkke	28	9161	9176	1175
612889	N/A	N/A	GATTTGCACACTGGGC	ekkddddddddddkke	0	9489	9504	1176
612890	N/A	N/A	CCCCGTGATCAACATC	ekkddddddddddkke	0	9874	9889	1177
612891	N/A	N/A	ATCGAGCAGAAAGTAC	ekkddddddddddkke	24	9932	9947	1178
612892	N/A	N/A	ACTGGTACCTGCCAGG	ekkddddddddddkke	0	10058	10073	1179
612907	N/A	N/A	TTAGCTAACTTAGCAC	ekkddddddddddkke	10	12892	12907	1086
612908	N/A	N/A	CATGGCCCACCAGTGC	ekkddddddddddkke	13	12914	12929	1087
612909	N/A	N/A	CACAGTGTATGCCTGC	ekkddddddddddkke	15	13087	13102	1088
612910	N/A	N/A	GCACTGGGTGACCCAG	ekkddddddddddkke	0	13195	13210	1089
612911	N/A	N/A	TCAGCACCTGCAAAGC	ekkddddddddddkke	0	13238	13253	1090

[0518] Table 8 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 4000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 8

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	91	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	87	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	81	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	43	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	91	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	88	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	91	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	87	13518	13537	239
619461	1	20	CTGCTGCCCGCTCATGGGAT	eeeeeddddddddddeeeee	5	1986	2005	1180
619462	7	26	CTGACCCTGCTGCCCGCTCA	eeeeeddddddddddeeeee	30	1992	2011	1181
619463	13	32	CCACTTCTGACCCTGCTGCC	eeeeeddddddddddeeeee	31	1998	2017	1182
619464	35	54	TCTTGCTTAGGCAACACGGG	eeeeeddddddddddeeeee	31	2020	2039	1183
619465	41	60	GGAGAGTCTTGCTTAGGCAA	eeeeeddddddddddeeeee	16	2026	2045	1184
619466	66	85	GGAGGTGCAGAGGGCAGAGG	eeeeeddddddddddeeeee	5	2051	2070	1185
619467	72	91	CAGGCCGGAGGTGCAGAGGG	eeeeeddddddddddeeeee	11	2057	2076	1186
619468	78	97	GACATGCAGGCCGGAGGTGC	eeeeeddddddddddeeeee	15	2063	2082	1187
619469	84	103	CACAGGGACATGCAGGCCGG	eeeeeddddddddddeeeee	19	2069	2088	1188
619470	90	109	AGAGGCCACAGGGACATGCA	eeeeeddddddddddeeeee	26	2075	2094	1189
619471	96	115	CCCCCAAGAGGCCACAGGGA	eeeeeddddddddddeeeee	10	2081	2100	1190
619472	102	121	GATGTACCCCCAAGAGGCCA	eeeeeddddddddddeeeee	31	2087	2106	1191
619473	108	127	CCGGGAGATGTACCCCCAAG	eeeeeddddddddddeeeee	34	2093	2112	1192
619474	114	133	CCAGCCCCGGGAGATGTACC	eeeeeddddddddddeeeee	11	2099	2118	1193
619475	120	139	TCTGACCCAGCCCCGGGAGA	eeeeeddddddddddeeeee	35	2105	2124	1194
619476	126	145	AGGCCTTCTGACCCAGCCCC	eeeeeddddddddddeeeee	21	2111	2130	1195
619477	132	151	CCACCCAGGCCTTCTGACCC	eeeeeddddddddddeeeee	0	2117	2136	1196
619478	138	157	GGCCAACCACCCAGGCCTTC	eeeeeddddddddddeeeee	31	2123	2142	1197
619479	144	163	GCCTGAGGCCAACCACCCAG	eeeeeddddddddddeeeee	36	2129	2148	1198
619480	150	169	GTGACAGCCTGAGGCCAACC	eeeeeddddddddddeeeee	8	2135	2154	1199
619481	156	175	AGGTGTGTGACAGCCTGAGG	eeeeeddddddddddeeeee	45	2141	2160	1200
619482	162	181	CTCCCTAGGTGTGTGACAGC	eeeeeddddddddddeeeee	27	2147	2166	1201
619483	168	187	GAGCATCTCCCTAGGTGTGT	eeeeeddddddddddeeeee	21	2153	2172	1202
619484	174	193	AAACGGGAGCATCTCCCTAG	eeeeeddddddddddeeeee	27	2159	2178	1203
619485	180	199	TCCCAGAAACGGGAGCATCT	eeeeeddddddddddeeeee	29	2165	2184	1204
619486	186	205	CAAGGTTCCCAGAAACGGGA	eeeeeddddddddddeeeee	0	2171	2190	1205
619487	208	227	CGAAGTTTGCAGGAGTCGGG	eeeeeddddddddddeeeee	27	2193	2212	1206

TABLE 8-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619488	214	233	ATTTACCGAAGTTTGCAGGA	eeeeeddddddddddeeeee	40	2199	2218	1207
619489	220	239	TTACACATTTACCGAAGTTT	eeeeeddddddddddeeeee	10	2205	2224	1208
619490	226	245	GTCGAGTTACACATTTACCG	eeeeeddddddddddeeeee	29	2211	2230	1209
619491	232	251	TGCAGGGTCGAGTTACACAT	eeeeeddddddddddeeeee	24	2217	2236	1210
619492	238	257	AGCCGGTGCAGGGTCGAGTT	eeeeeddddddddddeeeee	20	2223	2242	1211
619493	244	263	AGAGTGAGCCGGTGCAGGGT	eeeeeddddddddddeeeee	20	2229	2248	1212
619494	250	269	CTGAACAGAGTGAGCCGGTG	eeeeeddddddddddeeeee	25	2235	2254	1213
619495	256	275	TCACTGCTGAACAGAGTGAG	eeeeeddddddddddeeeee	41	2241	2260	1214
619496	262	281	AGAGTTTCACTGCTGAACAG	eeeeeddddddddddeeeee	13	2247	2266	1215
619497	268	287	CGATGCAGAGTTTCACTGCT	eeeeeddddddddddeeeee	29	2253	2272	1216
619498	274	293	AGTGATCGATGCAGAGTTTC	eeeeeddddddddddeeeee	28	2259	2278	1217
619499	280	299	AGTCTTAGTGATCGATGCAG	eeeeeddddddddddeeeee	26	2265	2284	1218
619500	286	305	CCAGGAAGTCTTAGTGATCG	eeeeeddddddddddeeeee	26	2271	2290	1219
619501	292	311	CCTCTTCCAGGAAGTCTTAG	eeeeeddddddddddeeeee	28	2277	2296	1220
619502	298	317	CTGGGACCTCTTCCAGGAAG	eeeeeddddddddddeeeee	20	2283	2302	1221
619503	304	323	CTCACGCTGGGACCTCTTCC	eeeeeddddddddddeeeee	12	2289	2308	1222
619504	310	329	GCGACACTCACGCTGGGACC	eeeeeddddddddddeeeee	25	2295	2314	1223
619505	316	335	CCAGAAGCGACACTCACGCT	eeeeeddddddddddeeeee	13	2301	2320	1224
619506	322	341	CAGATGCCAGAAGCGACACT	eeeeeddddddddddeeeee	24	2307	2326	1225
619507	328	347	GAAGGACAGATGCCAGAAGC	eeeeeddddddddddeeeee	40	2313	2332	1226
619508	334	353	TGGCCAGAAGGACAGATGCC	eeeeeddddddddddeeeee	3	2319	2338	1227
619509	340	359	ACAGGCTGGCCAGAAGGACA	eeeeeddddddddddeeeee	31	2325	2344	1228
619510	346	365	CAGACCACAGGCTGGCCAGA	eeeeeddddddddddeeeee	17	2331	2350	1229
619511	352	371	CTTGGCCAGACCACAGGCTG	eeeeeddddddddddeeeee	20	2337	2356	1230
619512	358	377	ACATCACTTGGCCAGACCAC	eeeeeddddddddddeeeee	7	2343	2362	1231
619513	364	383	AGGGTTACATCACTTGGCCA	eeeeeddddddddddeeeee	19	2349	2368	1232
619514	370	389	GAGAGGAGGGTTACATCACT	eeeeeddddddddddeeeee	28	2355	2374	1233
619515	376	395	AGGCTGGAGAGGAGGGTTAC	eeeeeddddddddddeeeee	31	2361	2380	1234
619516	382	401	GTGCACAGGCTGGAGAGGAG	eeeeeddddddddddeeeee	5	2367	2386	1235
619517	388	407	CTGCCTGTGCACAGGCTGGA	eeeeeddddddddddeeeee	15	2373	2392	1236
619518	394	413	CCCAGGCTGCCTGTGCACAG	eeeeeddddddddddeeeee	23	2379	2398	1237
619519	400	419	GCTGTTCCCAGGCTGCCTGT	eeeeeddddddddddeeeee	40	2385	2404	1238
619520	406	425	GATGGAGCTGTTCCCAGGCT	eeeeeddddddddddeeeee	12	2391	2410	1239
619521	431	450	GCCCTATTTATAGCTGAGGG	eeeeeddddddddddeeeee	23	2416	2435	1240

TABLE 8-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619522	437	456	CACGATGCCCTATTTATAGC	eeeeeddddddddddeeeee	10	2422	2441	1241
619523	443	462	CCGGGTCACGATGCCCTATT	eeeeeddddddddddeeeee	24	2428	2447	1242
619524	449	468	CCCCGGCCGGGTCACGATGC	eeeeeddddddddddeeeee	37	2434	2453	1243
619525	452	471	TTCCCCCGGCCGGGTCACGA	eeeeeddddddddddeeeee	24	2437	2456	1244
619526	455	474	TTCTTCCCCCGGCCGGGTCA	eeeeeddddddddddeeeee	19	2440	2459	1245
619527	458	477	AGCTTCTTCCCCCGGCCGGG	eeeeeddddddddddeeeee	7	2443	2462	1246
619528	461	480	GGCAGCTTCTTCCCCCGGCC	eeeeeddddddddddeeeee	38	2446	2465	1247
619529	464	483	AACGGCAGCTTCTTCCCCCG	eeeeeddddddddddeeeee	31	2449	2468	1248
619530	467	486	AACAACGGCAGCTTCTTCCC	eeeeeddddddddddeeeee	40	2452	2471	1249
619531	470	489	CAGAACAACGGCAGCTTCTT	eeeeeddddddddddeeeee	53	2455	2474	1250
619532	473	492	ACCCAGAACAACGGCAGCTT	eeeeeddddddddddeeeee	56	2458	2477	1251
619533	476	495	AGTACCCAGAACAACGGCAG	eeeeeddddddddddeeeee	50	2461	2480	1252
619534	479	498	TGTAGTACCCAGAACAACGG	eeeeeddddddddddeeeee	31	2464	2483	1253
619535	482	501	TGCTGTAGTACCCAGAACAA	eeeeeddddddddddeeeee	39	2467	2486	1254
619536	485	504	TTCTGCTGTAGTACCCAGAA	eeeeeddddddddddeeeee	52	2470	2489	1255
619537	488	507	CCCTTCTGCTGTAGTACCCA	eeeeeddddddddddeeeee	55	N/A	N/A	1256
619538	491	510	ATACCCTTCTGCTGTAGTAC	eeeeeddddddddddeeeee	39	N/A	N/A	1257
619539	494	513	CGCATACCCTTCTGCTGTAG	eeeeeddddddddddeeeee	69	N/A	N/A	1258
619540	497	516	TTCCGCATACCCTTCTGCTG	eeeeeddddddddddeeeee	65	N/A	N/A	1259
619541	500	519	CGCTTCCGCATACCCTTCTG	eeeeeddddddddddeeeee	60	N/A	N/A	1260
619542	503	522	GCTCGCTTCCGCATACCCTT	eeeeeddddddddddeeeee	78	N/A	N/A	1261
619543	506	525	GGTGCTCGCTTCCGCATACC	eeeeeddddddddddeeeee	69	5723	5742	1262
619544	525	544	AGGAGCCATCTCAGACTGGG	eeeeeddddddddddeeeee	53	5742	5761	1263
619545	528	547	GGCAGGAGCCATCTCAGACT	eeeeeddddddddddeeeee	56	5745	5764	1264
619546	531	550	ACCGGCAGGAGCCATCTCAG	eeeeeddddddddddeeeee	47	5748	5767	1265
619547	534	553	CACACCGGCAGGAGCCATCT	eeeeeddddddddddeeeee	39	5751	5770	1266
619548	537	556	GCTCACACCGGCAGGAGCCA	eeeeeddddddddddeeeee	47	5754	5773	1267
619549	540	559	CAGGCTCACACCGGCAGGAG	eeeeeddddddddddeeeee	42	5757	5776	1268
619550	543	562	CCTCAGGCTCACACCGGCAG	eeeeeddddddddddeeeee	58	5760	5779	1269
619551	546	565	GGCCCTCAGGCTCACACCGG	eeeeeddddddddddeeeee	53	5763	5782	1270
619552	549	568	GGTGGCCCTCAGGCTCACAC	eeeeeddddddddddeeeee	31	5766	5785	1271
619553	552	571	GATGGTGGCCCTCAGGCTCA	eeeeeddddddddddeeeee	8	5769	5788	1272
619554	555	574	GAGGATGGTGGCCCTCAGGC	eeeeeddddddddddeeeee	35	5772	5791	1273
619555	558	577	GCAGAGGATGGTGGCCCTCA	eeeeeddddddddddeeeee	54	5775	5794	1274
619556	561	580	GAGGCAGAGGATGGTGGCCC	eeeeeddddddddddeeeee	37	5778	5797	1275

TABLE 8-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619557	564	583	CAGGAGGCAGAGGATGGTGG	eeeeeddddddddddeeeee	13	5781	5800	1276
619558	572	591	GCCCAGGCCAGGAGGCAGAG	eeeeeddddddddddeeeee	43	5789	5808	1277
619559	575	594	CCAGCCCAGGCCAGGAGGCA	eeeeeddddddddddeeeee	44	5792	5811	1278
619560	578	597	AGGCCAGCCCAGGCCAGGAG	eeeeeddddddddddeeeee	50	5795	5814	1279
619561	581	600	GCCAGGCCAGCCCAGGCCAG	eeeeeddddddddddeeeee	55	5798	5817	1280
619562	584	603	GCAGCCAGGCCAGCCCAGGC	eeeeeddddddddddeeeee	43	5801	5820	1281
619563	587	606	CCTGCAGCCAGGCCAGCCCA	eeeeeddddddddddeeeee	38	5804	5823	1282
619564	590	609	TCACCTGCAGCCAGGCCAGC	eeeeeddddddddddeeeee	33	5807	5826	1283
619565	593	612	CGGTCACCTGCAGCCAGGCC	eeeeeddddddddddeeeee	45	5810	5829	1284
619566	596	615	ACCCGGTCACCTGCAGCCAG	eeeeeddddddddddeeeee	42	5813	5832	1285
619567	599	618	TACACCCGGTCACCTGCAGC	eeeeeddddddddddeeeee	22	5816	5835	1286
619568	602	621	ATGTACACCCGGTCACCTGC	eeeeeddddddddddeeeee	37	5819	5838	1287
619569	605	624	TGTATGTACACCCGGTCACC	eeeeeddddddddddeeeee	18	5822	5841	1288
619570	608	627	GGGTGTATGTACACCCGGTC	eeeeeddddddddddeeeee	26	5825	5844	1289
619571	626	645	TGGATGACGAGGTGGAAGGG	eeeeeddddddddddeeeee	44	5843	5862	1290
619572	629	648	TTGTGGATGACGAGGTGGAA	eeeeeddddddddddeeeee	35	5846	5865	1291
619573	632	651	TCATTGTGGATGACGAGGTG	eeeeeddddddddddeeeee	39	5849	5868	1292
619574	635	654	CTCTCATTGTGGATGACGAG	eeeeeddddddddddeeeee	68	5852	5871	1293
619575	638	657	GTACTCTCATTGTGGATGAC	eeeeeddddddddddeeeee	65	5855	5874	1294
619576	641	660	CAGGTACTCTCATTGTGGAT	eeeeeddddddddddeeeee	54	5858	5877	1295
619577	644	663	TCACAGGTACTCTCATTGTG	eeeeeddddddddddeeeee	42	5861	5880	1296
619578	647	666	TGCTCACAGGTACTCTCATT	eeeeeddddddddddeeeee	59	5864	5883	1297
619579	650	669	AGCTGCTCACAGGTACTCTC	eeeeeddddddddddeeeee	57	5867	5886	1298
619580	653	672	GCCAGCTGCTCACAGGTACT	eeeeeddddddddddeeeee	70	5870	5889	1299
619581	656	675	TTTGCCAGCTGCTCACAGGT	eeeeeddddddddddeeeee	47	5873	5892	1300
619582	659	678	GCCTTTGCCAGCTGCTCACA	eeeeeddddddddddeeeee	49	5876	5895	1301
619583	662	681	TTGGCCTTTGCCAGCTGCTC	eeeeeddddddddddeeeee	58	5879	5898	1302
619584	665	684	GCATTGGCCTTTGCCAGCTG	eeeeeddddddddddeeeee	56	5882	5901	1303
619585	668	687	CCGGCATTGGCCTTTGCCAG	eeeeeddddddddddeeeee	45	5885	5904	1304
619586	671	690	TTCCCGGCATTGGCCTTTGC	eeeeeddddddddddeeeee	46	5888	5907	1305
619587	674	693	GGCTTCCCGGCATTGGCCTT	eeeeeddddddddddeeeee	39	5891	5910	1306
619588	677	696	TTGGGCTTCCCGGCATTGGC	eeeeeddddddddddeeeee	41	5894	5913	1307
619589	680	699	TCTTTGGGCTTCCCGGCATT	eeeeeddddddddddeeeee	28	5897	5916	1308
619590	701	720	GGAGCAGGTATGAAGGTGGG	eeeeeddddddddddeeeee	35	5918	5937	1309

TABLE 8-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619591	704	723	ATTGGAGCAGGTATGAAGGT	eeeeeddddddddddeeeee	49	5921	5940	1310
619592	707	726	TGAATTGGAGCAGGTATGAA	eeeeeddddddddddeeeee	32	5924	5943	1311
619593	710	729	GCCTGAATTGGAGCAGGTAT	eeeeeddddddddddeeeee	57	5927	5946	1312
619594	713	732	TTGGCCTGAATTGGAGCAGG	eeeeeddddddddddeeeee	51	5930	5949	1313
619595	716	735	GTCTTGGCCTGAATTGGAGC	eeeeeddddddddddeeeee	42	5933	5952	1314
619596	719	738	GATGTCTTGGCCTGAATTGG	eeeeeddddddddddeeeee	24	5936	5955	1315
619597	740	759	AGGGCCTTTTCATCCACAGG	eeeeeddddddddddeeeee	17	5957	5976	1316
619598	743	762	TGTAGGGCCTTTTCATCCAC	eeeeeddddddddddeeeee	33	5960	5979	1317
619599	746	765	TCCTGTAGGGCCTTTTCATC	eeeeeddddddddddeeeee	6	5963	5982	1318
619600	749	768	TGGTCCTGTAGGGCCTTTTC	eeeeeddddddddddeeeee	42	5966	5985	1319
619601	752	771	AGCTGGTCCTGTAGGGCCTT	eeeeeddddddddddeeeee	51	5969	5988	1320
619602	755	774	ACCAGCTGGTCCTGTAGGGC	eeeeeddddddddddeeeee	37	5972	5991	1321
619603	758	777	AGCACCAGCTGGTCCTGTAG	eeeeeddddddddddeeeee	44	5975	5994	1322
619604	761	780	ACTAGCACCAGCTGGTCCTG	eeeeeddddddddddeeeee	37	5978	5997	1323
619605	764	783	GCGACTAGCACCAGCTGGTC	eeeeeddddddddddeeeee	52	5981	6000	1324
619606	767	786	GCAGCGACTAGCACCAGCTG	eeeeeddddddddddeeeee	67	5984	6003	1325
619607	770	789	TTTGCAGCGACTAGCACCAG	eeeeeddddddddddeeeee	60	5987	6006	1326
619608	773	792	AGTTTTGCAGCGACTAGCAC	eeeeeddddddddddeeeee	43	5990	6009	1327
619609	776	795	TCAAGTTTTGCAGCGACTAG	eeeeeddddddddddeeeee	38	5993	6012	1328
619610	779	798	GTGTCAAGTTTTGCAGCGAC	eeeeeddddddddddeeeee	57	5996	6015	1329
619611	782	801	TCGGTGTCAAGTTTTGCAGC	eeeeeddddddddddeeeee	55	5999	6018	1330
619612	785	804	TCTTCGGTGTCAAGTTTTGC	eeeeeddddddddddeeeee	45	6002	6021	1331
619613	788	807	TTGTCTTCGGTGTCAAGTTT	eeeeeddddddddddeeeee	50	6005	6024	1332
619614	791	810	AACTTGTCTTCGGTGTCAAG	eeeeeddddddddddeeeee	48	6008	6027	1333
619615	794	813	CTCAACTTGTCTTCGGTGTC	eeeeeddddddddddeeeee	59	6011	6030	1334
619616	797	816	GCCCTCAACTTGTCTTCGGT	eeeeeddddddddddeeeee	41	6014	6033	1335
619617	800	819	GCGGCCCTCAACTTGTCTTC	eeeeeddddddddddeeeee	42	6017	6036	1336
619618	803	822	ATTGCGGCCCTCAACTTGTC	eeeeeddddddddddeeeee	32	6020	6039	1337
619619	806	825	ACCATTGCGGCCCTCAACTT	eeeeeddddddddddeeeee	34	6023	6042	1338
619620	809	828	CCGACCATTGCGGCCCTCAA	eeeeeddddddddddeeeee	55	6026	6045	1339
619621	812	831	ATCCCGACCATTGCGGCCCT	eeeeeddddddddddeeeee	37	6029	6048	1340
619622	815	834	AGCATCCCGACCATTGCGGC	eeeeeddddddddddeeeee	50	6032	6051	1341
619623	818	837	GCCAGCATCCCGACCATTGC	eeeeeddddddddddeeeee	58	6035	6054	1342
619624	821	840	TTGGCCAGCATCCCGACCAT	eeeeeddddddddddeeeee	38	6038	6057	1343
619625	824	843	AAGTTGGCCAGCATCCCGAC	eeeeeddddddddddeeeee	46	6041	6060	1344

TABLE 8-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619626	827	846	AAGAAGTTGGCCAGCATCCC	eeeeeddddddddddeeeee	24	6044	6063	1345
619627	830	849	CCCAAGAAGTTGGCCAGCAT	eeeeeddddddddddeeeee	55	6047	6066	1346
619628	833	852	AAGCCCAAGAAGTTGGCCAG	eeeeeddddddddddeeeee	48	6050	6069	1347
619629	836	855	CGGAAGCCCAAGAAGTTGGC	eeeeeddddddddddeeeee	36	6053	6072	1348
619630	839	858	ATACGGAAGCCCAAGAAGTT	eeeeeddddddddddeeeee	40	6056	6075	1349
619631	842	861	TATATACGGAAGCCCAAGAA	eeeeeddddddddddeeeee	29	6059	6078	1350
619632	845	864	CCATATATACGGAAGCCCAA	eeeeeddddddddddeeeee	48	6062	6081	1351
619633	848	867	ATGCCATATATACGGAAGCC	eeeeeddddddddddeeeee	58	6065	6084	1352
619634	851	870	TGCATGCCATATATACGGAA	eeeeeddddddddddeeeee	59	6068	6087	1353
619635	854	873	CTGTGCATGCCATATATACG	eeeeeddddddddddeeeee	66	6071	6090	1354
619636	857	876	TCACTGTGCATGCCATATAT	eeeeeddddddddddeeeee	72	6074	6093	1355
619637	860	879	AGCTCACTGTGCATGCCATA	eeeeeddddddddddeeeee	74	6077	6096	1356
619638	863	882	CATAGCTCACTGTGCATGCC	eeeeeddddddddddeeeee	69	6080	6099	1357
619639	866	885	CCCCATAGCTCACTGTGCAT	eeeeeddddddddddeeeee	43	6083	6102	1358
619640	869	888	ACGCCCCATAGCTCACTGTG	eeeeeddddddddddeeeee	48	6086	6105	1359
619641	872	891	ACCACGCCCCATAGCTCACT	eeeeeddddddddddeeeee	56	6089	6108	1360
619642	875	894	TGGACCACGCCCCATAGCTC	eeeeeddddddddddeeeee	40	6092	6111	1361
619643	878	897	CCATGGACCACGCCCCATAG	eeeeeddddddddddeeeee	24	6095	6114	1362
619644	881	900	GCCCCATGGACCACGCCCCA	eeeeeddddddddddeeeee	40	6098	6117	1363
619645	884	903	GTGGCCCCATGGACCACGCC	eeeeeddddddddddeeeee	26	6101	6120	1364
619646	887	906	ACGGTGGCCCCATGGACCAC	eeeeeddddddddddeeeee	35	6104	6123	1365
619647	890	909	AGGACGGTGGCCCCATGGAC	eeeeeddddddddddeeeee	35	6107	6126	1366
619648	893	912	GAGAGGACGGTGGCCCCATG	eeeeeddddddddddeeeee	44	6110	6129	1367
619649	913	932	TGCCAAAGACAGCCGTTGGG	eeeeeddddddddddeeeee	53	6130	6149	1368
619650	916	935	GGGTGCCAAAGACAGCCGTT	eeeeeddddddddddeeeee	40	6133	6152	1369
619651	919	938	CCAGGGTGCCAAAGACAGCC	eeeeeddddddddddeeeee	62	6136	6155	1370
619652	922	941	AGGCCAGGGTGCCAAAGACA	eeeeeddddddddddeeeee	44	6139	6158	1371
619653	925	944	GAGAGGCCAGGGTGCCAAAG	eeeeeddddddddddeeeee	58	6142	6161	1372
619654	928	947	AGAGAGAGGCCAGGGTGCCA	eeeeeddddddddddeeeee	34	6145	6164	1373
619655	931	950	GATAGAGAGAGGCCAGGGTG	eeeeeddddddddddeeeee	16	6148	6167	1374
619656	934	953	CCAGATAGAGAGAGGCCAGG	eeeeeddddddddddeeeee	41	6151	6170	1375
619657	937	956	CTCCCAGATAGAGAGAGGCC	eeeeeddddddddddeeeee	58	6154	6173	1376
619658	940	959	AGGCTCCCAGATAGAGAGAG	eeeeeddddddddddeeeee	21	6157	6176	1377
619659	943	962	CCAAGGCTCCCAGATAGAGA	eeeeeddddddddddeeeee	21	6160	6179	1378

TABLE 8-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619660	946	965	GGTCCAAGGCTCCCAGATAG	eeeeeddddddddddeeeee	43	6163	6182	1379
619661	949	968	TGTGGTCCAAGGCTCCCAGA	eeeeeddddddddddeeeee	45	6166	6185	1380
619662	952	971	CTGTGTGGTCCAAGGCTCCC	eeeeeddddddddddeeeee	33	6169	6188	1381
619663	955	974	CAGCTGTGTGGTCCAAGGCT	eeeeeddddddddddeeeee	52	6172	6191	1382
619664	958	977	TGTCAGCTGTGTGGTCCAAG	eeeeeddddddddddeeeee	44	6175	6194	1383
619665	961	980	GCCTGTCAGCTGTGTGGTCC	eeeeeddddddddddeeeee	66	6178	6197	1384
619666	964	983	GTAGCCTGTCAGCTGTGTGG	eeeeeddddddddddeeeee	47	6181	6200	1385
619667	967	986	CCTGTAGCCTGTCAGCTGTG	eeeeeddddddddddeeeee	59	6184	6203	1386
619668	970	989	TTGCCTGTAGCCTGTCAGCT	eeeeeddddddddddeeeee	57	6187	6206	1387
619669	973	992	GGATTGCCTGTAGCCTGTCA	eeeeeddddddddddeeeee	53	6190	6209	1388
619670	976	995	CCAGGATTGCCTGTAGCCTG	eeeeeddddddddddeeeee	57	6193	6212	1389
619671	979	998	CACCCAGGATTGCCTGTAGC	eeeeeddddddddddeeeee	52	6196	6215	1390
619672	982	1001	GAACACCCAGGATTGCCTGT	eeeeeddddddddddeeeee	63	6199	6218	1391
619673	985	1004	AAGGAACACCCAGGATTGCC	eeeeeddddddddddeeeee	47	6202	6221	1392
619674	988	1007	TCCAAGGAACACCCAGGATT	eeeeeddddddddddeeeee	63	6205	6224	1393
619675	991	1010	CCTTCCAAGGAACACCCAGG	eeeeeddddddddddeeeee	60	6208	6227	1394
619676	994	1013	TGTCCTTCCAAGGAACACCC	eeeeeddddddddddeeeee	62	6211	6230	1395
619677	997	1016	TCTTGTCCTTCCAAGGAACA	eeeeeddddddddddeeeee	48	6214	6233	1396
619678	1000	1019	AGTTCTTGTCCTTCCAAGGA	eeeeeddddddddddeeeee	35	6217	6236	1397
619679	1003	1022	TGCAGTTCTTGTCCTTCCAA	eeeeeddddddddddeeeee	56	6220	6239	1398
619680	1006	1025	AGGTGCAGTTCTTGTCCTTC	eeeeeddddddddddeeeee	41	6223	6242	1399
619681	1009	1028	GGGAGGTGCAGTTCTTGTCC	eeeeeddddddddddeeeee	26	6226	6245	1400
619682	1012	1031	GCCGGGAGGTGCAGTTCTTG	eeeeeddddddddddeeeee	44	6229	6248	1401
619683	1015	1034	CCAGCCGGGAGGTGCAGTTC	eeeeeddddddddddeeeee	36	6232	6251	1402
619684	1018	1037	CATCCAGCCGGGAGGTGCAG	eeeeeddddddddddeeeee	32	6235	6254	1403
619685	1021	1040	GCGCATCCAGCCGGGAGGTG	eeeeeddddddddddeeeee	21	6238	6257	1404
619686	1024	1043	TGTGCGCATCCAGCCGGGAG	eeeeeddddddddddeeeee	44	6241	6260	1405
619687	1027	1046	CCTTGTGCGCATCCAGCCGG	eeeeeddddddddddeeeee	60	6244	6263	1406
619688	1030	1049	GGACCTTGTGCGCATCCAGC	eeeeeddddddddddeeeee	61	6247	6266	1407
619689	1033	1052	ACAGGACCTTGTGCGCATCC	eeeeeddddddddddeeeee	65	6250	6269	1408
619690	1036	1055	CAGACAGGACCTTGTGCGCA	eeeeeddddddddddeeeee	59	6253	6272	1409
619691	1039	1058	GGGCAGACAGGACCTTGTGC	eeeeeddddddddddeeeee	45	6256	6275	1410
619692	1042	1061	GCAGGGCAGACAGGACCTTG	eeeeeddddddddddeeeee	46	6259	6278	1411
619693	1045	1064	CCTGCAGGGCAGACAGGACC	eeeeeddddddddddeeeee	38	6262	6281	1412
619694	1048	1067	CAGCCTGCAGGGCAGACAGG	eeeeeddddddddddeeeee	41	6265	6284	1413

TABLE 8-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619695	1051	1070	GTACAGCCTGCAGGGCAGAC	eeeeeddddddddddeeeee	43	6268	6287	1414
619696	1054	1073	CCTGTACAGCCTGCAGGGCA	eeeeeddddddddddeeeee	48	6271	6290	1415
619697	1057	1076	GGCCCTGTACAGCCTGCAGG	eeeeeddddddddddeeeee	35	6274	6293	1416
619698	1060	1079	GCAGGCCCTGTACAGCCTGC	eeeeeddddddddddeeeee	22	6277	6296	1417
619699	1063	1082	CTAGCAGGCCCTGTACAGCC	eeeeeddddddddddeeeee	1	6280	6299	1418
619700	1066	1085	CCACTAGCAGGCCCTGTACA	eeeeeddddddddddeeeee	29	6283	6302	1419
619701	1069	1088	GGGCCACTAGCAGGCCCTGT	eeeeeddddddddddeeeee	2	6286	6305	1420
619702	1072	1091	CCTGGGCCACTAGCAGGCCC	eeeeeddddddddddeeeee	25	6289	6308	1421
619703	1075	1094	TGCCCTGGGCCACTAGCAGG	eeeeeddddddddddeeeee	23	6292	6311	1422
619704	1078	1097	CCCTGCCCTGGGCCACTAGC	eeeeeddddddddddeeeee	46	6295	6314	1423
619705	1081	1100	CAGCCCTGCCCTGGGCCACT	eeeeeddddddddddeeeee	59	6298	6317	1424
619706	1084	1103	TATCAGCCCTGCCCTGGGCC	eeeeeddddddddddeeeee	36	6301	6320	1425
619707	1087	1106	GGCTATCAGCCCTGCCCTGG	eeeeeddddddddddeeeee	51	6304	6323	1426
619708	1090	1109	CCTGGCTATCAGCCCTGCCC	eeeeeddddddddddeeeee	34	6307	6326	1427
619709	1093	1112	GGGCCTGGCTATCAGCCCTG	eeeeeddddddddddeeeee	17	6310	6329	1428
619710	1096	1115	GCTGGGCCTGGCTATCAGCC	eeeeeddddddddddeeeee	31	6313	6332	1429
619711	1099	1118	GCAGCTGGGCCTGGCTATCA	eeeeeddddddddddeeeee	44	6316	6335	1430
619712	1102	1121	GCAGCAGCTGGGCCTGGCTA	eeeeeddddddddddeeeee	38	6319	6338	1431
619713	1105	1124	ACAGCAGCAGCTGGGCCTGG	eeeeeddddddddddeeeee	29	6322	6341	1432
619714	1108	1127	TGGACAGCAGCAGCTGGGCC	eeeeeddddddddddeeeee	50	6325	6344	1433
619715	1111	1130	CCGTGGACAGCAGCAGCTGG	eeeeeddddddddddeeeee	53	6328	6347	1434
619716	1114	1133	CCACCGTGGACAGCAGCAGC	eeeeeddddddddddeeeee	24	6331	6350	1435
619717	1117	1136	CCACCACCGTGGACAGCAGC	eeeeeddddddddddeeeee	34	6334	6353	1436
619718	1120	1139	CGCCCACCACCGTGGACAGC	eeeeeddddddddddeeeee	56	6337	6356	1437
619719	1123	1142	ACACGCCCACCACCGTGGAC	eeeeeddddddddddeeeee	27	6340	6359	1438
619720	1126	1145	TGAACACGCCCACCACCGTG	eeeeeddddddddddeeeee	16	6343	6362	1439
619721	1129	1148	CTGTGAACACGCCCACCACC	eeeeeddddddddddeeeee	40	6346	6365	1440
619722	1132	1151	GGGCTGTGAACACGCCCACC	eeeeeddddddddddeeeee	25	6349	6368	1441
619723	1150	1169	GCTTCAGGTGCAGGCCTGGG	eeeeeddddddddddeeeee	36	6367	6386	1442
619724	1153	1172	GCTGCTTCAGGTGCAGGCCT	eeeeeddddddddddeeeee	47	6370	6389	1443
619725	1156	1175	ACGGCTGCTTCAGGTGCAGG	eeeeeddddddddddeeeee	14	6373	6392	1444
619726	1159	1178	CAAACGGCTGCTTCAGGTGC	eeeeeddddddddddeeeee	37	6376	6395	1445
619727	1162	1181	GCACAAACGGCTGCTTCAGG	eeeeeddddddddddeeeee	19	6379	6398	1446

TABLE 8-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619728	1165	1184	CCTGCACAAACGGCTGCTTC	eeeeeddddddddddeeeee	33	6382	6401	1447
619729	1168	1187	GGCCCTGCACAAACGGCTGC	eeeeeddddddddddeeeee	48	6385	6404	1448
619730	1171	1190	CCAGGCCCTGCACAAACGGC	eeeeeddddddddddeeeee	27	6388	6407	1449
619731	1174	1193	GAGCCAGGCCCTGCACAAAC	eeeeeddddddddddeeeee	35	6391	6410	1450
619732	1177	1196	AGAGAGCCAGGCCCTGCACA	eeeeeddddddddddeeeee	51	6394	6413	1451
619733	1180	1199	TATAGAGAGCCAGGCCCTGC	eeeeeddddddddddeeeee	27	6397	6416	1452
619734	1183	1202	GGGTATAGAGAGCCAGGCCC	eeeeeddddddddddeeeee	41	6400	6419	1453
619735	1217	1236	TCTGTGAAGTCCAGAGAGCG	eeeeeddddddddddeeeee	22	6434	6453	1454
619736	1220	1239	AGTTCTGTGAAGTCCAGAGA	eeeeeddddddddddeeeee	48	6437	6456	1455
619737	1223	1242	TCCAGTTCTGTGAAGTCCAG	eeeeeddddddddddeeeee	26	6440	6459	1456
619738	1226	1245	ACATCCAGTTCTGTGAAGTC	eeeeeddddddddddeeeee	35	6443	6462	1457
619739	1229	1248	GCAACATCCAGTTCTGTGAA	eeeeeddddddddddeeeee	28	6446	6465	1458
619740	1232	1251	GCAGCAACATCCAGTTCTGT	eeeeeddddddddddeeeee	40	6449	6468	1459
619741	1235	1254	TCAGCAGCAACATCCAGTTC	eeeeeddddddddddeeeee	41	6452	6471	1460
619742	1238	1257	TTCTCAGCAGCAACATCCAG	eeeeeddddddddddeeeee	29	6455	6474	1461
619743	1241	1260	ATCTTCTCAGCAGCAACATC	eeeeeddddddddddeeeee	32	6458	6477	1462
619744	1244	1263	TCAATCTTCTCAGCAGCAAC	eeeeeddddddddddeeeee	38	6461	6480	1463
619745	1247	1266	CTGTCAATCTTCTCAGCAGC	eeeeeddddddddddeeeee	39	6464	6483	1464
619746	1250	1269	AACCTGTCAATCTTCTCAGC	eeeeeddddddddddeeeee	20	6467	6486	1465
619747	1253	1272	ATGAACCTGTCAATCTTCTC	eeeeeddddddddddeeeee	50	6470	6489	1466
619748	1256	1275	TGCATGAACCTGTCAATCTT	eeeeeddddddddddeeeee	61	6473	6492	1467
619749	1259	1278	GCCTGCATGAACCTGTCAAT	eeeeeddddddddddeeeee	62	6476	6495	1468
619750	1262	1281	ACAGCCTGCATGAACCTGTC	eeeeeddddddddddeeeee	56	6479	6498	1469
619751	1265	1284	GTCACAGCCTGCATGAACCT	eeeeeddddddddddeeeee	75	6482	6501	1470
619752	1268	1287	CCTGTCACAGCCTGCATGAA	eeeeeddddddddddeeeee	46	6485	6504	1471
619753	1271	1290	CATCCTGTCACAGCCTGCAT	eeeeeddddddddddeeeee	74	6488	6507	1472
619754	1274	1293	TTCCATCCTGTCACAGCCTG	eeeeeddddddddddeeeee	71	6491	6510	1473
619755	1277	1296	GTCTTCCATCCTGTCACAGC	eeeeeddddddddddeeeee	65	6494	6513	1474
619756	1280	1299	CCAGTCTTCCATCCTGTCAC	eeeeeddddddddddeeeee	56	6497	6516	1475
619757	1283	1302	CAGCCAGTCTTCCATCCTGT	eeeeeddddddddddeeeee	63	6500	6519	1476

[0519] Table 9 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 4000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS4039 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 9

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	91	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	86	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	88	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	85	13518	13537	239
619692	1042	1061	GCAGGGCAGACAGGACCTTG	eeeeeddddddddddeeeee	17	6259	6278	1411
619693	1045	1064	CCTGCAGGGCAGACAGGACC	eeeeeddddddddddeeeee	25	6262	6281	1412
619694	1048	1067	CAGCCTGCAGGGCAGACAGG	eeeeeddddddddddeeeee	32	6265	6284	1413
619695	1051	1070	GTACAGCCTGCAGGGCAGAC	eeeeeddddddddddeeeee	25	6268	6287	1414
619696	1054	1073	CCTGTACAGCCTGCAGGGCA	eeeeeddddddddddeeeee	48	6271	6290	1415
619697	1057	1076	GGCCCTGTACAGCCTGCAGG	eeeeeddddddddddeeeee	32	6274	6293	1416
619698	1060	1079	GCAGGCCCTGTACAGCCTGC	eeeeeddddddddddeeeee	17	6277	6296	1417
619699	1063	1082	CTAGCAGGCCCTGTACAGCC	eeeeeddddddddddeeeee	13	6280	6299	1418
619700	1066	1085	CCACTAGCAGGCCCTGTACA	eeeeeddddddddddeeeee	36	6283	6302	1419
619701	1069	1088	GGGCCACTAGCAGGCCCTGT	eeeeeddddddddddeeeee	6	6286	6305	1420
619702	1072	1091	CCTGGGCCACTAGCAGGCCC	eeeeeddddddddddeeeee	16	6289	6308	1421
619703	1075	1094	TGCCCTGGGCCACTAGCAGG	eeeeeddddddddddeeeee	26	6292	6311	1422
619704	1078	1097	CCCTGCCCTGGGCCACTAGC	eeeeeddddddddddeeeee	41	6295	6314	1423
619705	1081	1100	CAGCCCTGCCCTGGGCCACT	eeeeeddddddddddeeeee	36	6298	6317	1424
619706	1084	1103	TATCAGCCCTGCCCTGGGCC	eeeeeddddddddddeeeee	21	6301	6320	1425
619707	1087	1106	GGCTATCAGCCCTGCCCTGG	eeeeeddddddddddeeeee	27	6304	6323	1426
619708	1090	1109	CCTGGCTATCAGCCCTGCCC	eeeeeddddddddddeeeee	30	6307	6326	1427
619709	1093	1112	GGGCCTGGCTATCAGCCCTG	eeeeeddddddddddeeeee	9	6310	6329	1428
619710	1096	1115	GCTGGGCCTGGCTATCAGCC	eeeeeddddddddddeeeee	15	6313	6332	1429
619711	1099	1118	GCAGCTGGGCCTGGCTATCA	eeeeeddddddddddeeeee	26	6316	6335	1430
619712	1102	1121	GCAGCAGCTGGGCCTGGCTA	eeeeeddddddddddeeeee	61	6319	6338	1431
619713	1105	1124	ACAGCAGCAGCTGGGCCTGG	eeeeeddddddddddeeeee	44	6322	6341	1432
619714	1108	1127	TGGACAGCAGCAGCTGGGCC	eeeeeddddddddddeeeee	47	6325	6344	1433
619715	1111	1130	CCGTGGACAGCAGCAGCTGG	eeeeeddddddddddeeeee	41	6328	6347	1434
619716	1114	1133	CCACCGTGGACAGCAGCAGC	eeeeeddddddddddeeeee	35	6331	6350	1435
619717	1117	1136	CCACCACCGTGGACAGCAGC	eeeeeddddddddddeeeee	34	6334	6353	1436
619718	1120	1139	CGCCCACCACCGTGGACAGC	eeeeeddddddddddeeeee	37	6337	6356	1437
619719	1123	1142	ACACGCCCACCACCGTGGAC	eeeeeddddddddddeeeee	17	6340	6359	1438
619720	1126	1145	TGAACACGCCCACCACCGTG	eeeeeddddddddddeeeee	20	6343	6362	1439
619721	1129	1148	CTGTGAACACGCCCACCACC	eeeeeddddddddddeeeee	36	6346	6365	1440
619722	1132	1151	GGGCTGTGAACACGCCCACC	eeeeeddddddddddeeeee	14	6349	6368	1441

TABLE 9-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619723	1150	1169	GCTTCAGGTGCAGGCCTGGG	eeeeeddddddddddeeeee	32	6367	6386	1442
619724	1153	1172	GCTGCTTCAGGTGCAGGCCT	eeeeeddddddddddeeeee	47	6370	6389	1443
619725	1156	1175	ACGGCTGCTTCAGGTGCAGG	eeeeeddddddddddeeeee	27	6373	6392	1444
619726	1159	1178	CAAACGGCTGCTTCAGGTGC	eeeeeddddddddddeeeee	20	6376	6395	1445
619727	1162	1181	GCACAAACGGCTGCTTCAGG	eeeeeddddddddddeeeee	13	6379	6398	1446
619728	1165	1184	CCTGCACAAACGGCTGCTTC	eeeeeddddddddddeeeee	25	6382	6401	1447
619729	1168	1187	GGCCCTGCACAAACGGCTGC	eeeeeddddddddddeeeee	29	6385	6404	1448
619730	1171	1190	CCAGGCCCTGCACAAACGGC	eeeeeddddddddddeeeee	27	6388	6407	1449
619731	1174	1193	GAGCCAGGCCCTGCACAAAC	eeeeeddddddddddeeeee	18	6391	6410	1450
619732	1177	1196	AGAGAGCCAGGCCCTGCACA	eeeeeddddddddddeeeee	33	6394	6413	1451
619733	1180	1199	TATAGAGAGCCAGGCCCTGC	eeeeeddddddddddeeeee	0	6397	6416	1452
619734	1183	1202	GGGTATAGAGAGCCAGGCCC	eeeeeddddddddddeeeee	14	6400	6419	1453
619735	1217	1236	TCTGTGAAGTCCAGAGAGCG	eeeeeddddddddddeeeee	17	6434	6453	1454
619736	1220	1239	AGTTCTGTGAAGTCCAGAGA	eeeeeddddddddddeeeee	41	6437	6456	1455
619737	1223	1242	TCCAGTTCTGTGAAGTCCAG	eeeeeddddddddddeeeee	31	6440	6459	1456
619738	1226	1245	ACATCCAGTTCTGTGAAGTC	eeeeeddddddddddeeeee	35	6443	6462	1457
619739	1229	1248	GCAACATCCAGTTCTGTGAA	eeeeeddddddddddeeeee	29	6446	6465	1458
619740	1232	1251	GCAGCAACATCCAGTTCTGT	eeeeeddddddddddeeeee	35	6449	6468	1459
619741	1235	1254	TCAGCAGCAACATCCAGTTC	eeeeeddddddddddeeeee	35	6452	6471	1460
619742	1238	1257	TTCTCAGCAGCAACATCCAG	eeeeeddddddddddeeeee	5	6455	6474	1461
619743	1241	1260	ATCTTCTCAGCAGCAACATC	eeeeeddddddddddeeeee	22	6458	6477	1462
619744	1244	1263	TCAATCTTCTCAGCAGCAAC	eeeeeddddddddddeeeee	45	6461	6480	1463
619745	1247	1266	CTGTCAATCTTCTCAGCAGC	eeeeeddddddddddeeeee	21	6464	6483	1464
619746	1250	1269	AACCTGTCAATCTTCTCAGC	eeeeeddddddddddeeeee	8	6467	6486	1465
619747	1253	1272	ATGAACCTGTCAATCTTCTC	eeeeeddddddddddeeeee	43	6470	6489	1466
619748	1256	1275	TGCATGAACCTGTCAATCTT	eeeeeddddddddddeeeee	31	6473	6492	1467
619749	1259	1278	GCCTGCATGAACCTGTCAAT	eeeeeddddddddddeeeee	44	6476	6495	1468
619750	1262	1281	ACAGCCTGCATGAACCTGTC	eeeeeddddddddddeeeee	41	6479	6498	1469
619751	1265	1284	GTCACAGCCTGCATGAACCT	eeeeeddddddddddeeeee	69	6482	6501	1470
619752	1268	1287	CCTGTCACAGCCTGCATGAA	eeeeeddddddddddeeeee	43	6485	6504	1471
619753	1271	1290	CATCCTGTCACAGCCTGCAT	eeeeeddddddddddeeeee	59	6488	6507	1472
619754	1274	1293	TTCCATCCTGTCACAGCCTG	eeeeeddddddddddeeeee	49	6491	6510	1473
619755	1277	1296	GTCTTCCATCCTGTCACAGC	eeeeeddddddddddeeeee	42	6494	6513	1474
619756	1280	1299	CCAGTCTTCCATCCTGTCAC	eeeeeddddddddddeeeee	20	6497	6516	1475

TABLE 9-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619757	1283	1302	CAGCCAGTCTTCCATCCTGT	eeeeeddddddddddeeeee	41	6500	6519	1476
619758	1286	1305	GAGCAGCCAGTCTTCCATCC	eeeeeddddddddddeeeee	41	6503	6522	1477
619759	1289	1308	AGGGAGCAGCCAGTCTTCCA	eeeeeddddddddddeeeee	29	6506	6525	1478
619760	1292	1311	ATCAGGGAGCAGCCAGTCTT	eeeeeddddddddddeeeee	29	6509	6528	1479
619761	1295	1314	CCCATCAGGGAGCAGCCAGT	eeeeeddddddddddeeeee	7	6512	6531	1480
619762	1298	1317	GCTCCCATCAGGGAGCAGCC	eeeeeddddddddddeeeee	4	6515	6534	1481
619763	1301	1320	CTGGCTCCCATCAGGGAGCA	eeeeeddddddddddeeeee	8	6518	6537	1482
619764	1304	1323	ACACTGGCTCCCATCAGGGA	eeeeeddddddddddeeeee	0	6521	6540	1483
619765	1307	1326	TCCACACTGGCTCCCATCAG	eeeeeddddddddddeeeee	27	6524	6543	1484
619766	1310	1329	CTGTCCACACTGGCTCCCAT	eeeeeddddddddddeeeee	27	6527	6546	1485
619767	1313	1332	GTGCTGTCCACACTGGCTCC	eeeeeddddddddddeeeee	42	6530	6549	1486
619768	1316	1335	AGGGTGCTGTCCACACTGGC	eeeeeddddddddddeeeee	39	6533	6552	1487
619769	1319	1338	GCCAGGGTGCTGTCCACACT	eeeeeddddddddddeeeee	65	6536	6555	1488
619770	1322	1341	AAAGCCAGGGTGCTGTCCAC	eeeeeddddddddddeeeee	65	6539	6558	1489
619771	1325	1344	TTGAAAGCCAGGGTGCTGTC	eeeeeddddddddddeeeee	48	6542	6561	1490
619772	1328	1347	GTGTTGAAAGCCAGGGTGCT	eeeeeddddddddddeeeee	44	6545	6564	1491
619773	1331	1350	TAGGTGTTGAAAGCCAGGGT	eeeeeddddddddddeeeee	16	6548	6567	1492
619774	1351	1370	TCTTCCCTTGGAAGTGGACG	eeeeeddddddddddeeeee	40	N/A	N/A	1493
619775	1354	1373	TCATCTTCCCTTGGAAGTGG	eeeeeddddddddddeeeee	41	N/A	N/A	1494
619776	1357	1376	CCTTCATCTTCCCTTGGAAG	eeeeeddddddddddeeeee	30	N/A	N/A	1495
619777	1360	1379	AGCCCTTCATCTTCCCTTGG	eeeeeddddddddddeeeee	53	N/A	N/A	1496
619778	1363	1382	AGAAGCCCTTCATCTTCCCT	eeeeeddddddddddeeeee	33	10374	10393	1497
619779	1366	1385	GGGAGAAGCCCTTCATCTTC	eeeeeddddddddddeeeee	56	10377	10396	1498
619780	1369	1388	GCAGGGAGAAGCCCTTCATC	eeeeeddddddddddeeeee	42	10380	10399	1499
619781	1372	1391	CCAGCAGGGAGAAGCCCTTC	eeeeeddddddddddeeeee	63	10383	10402	1500
619782	1375	1394	CGGCCAGCAGGGAGAAGCCC	eeeeeddddddddddeeeee	52	10386	10405	1501
619783	1378	1397	GCTCGGCCAGCAGGGAGAAG	eeeeeddddddddddeeeee	37	10389	10408	1502
619784	1398	1417	GTCCACCCAGAACTCCTGGG	eeeeeddddddddddeeeee	67	10409	10428	1503
619785	1401	1420	GTTGTCCACCCAGAACTCCT	eeeeeddddddddddeeeee	65	10412	10431	1504
619786	1404	1423	GCTGTTGTCCACCCAGAACT	eeeeeddddddddddeeeee	43	10415	10434	1505
619787	1407	1426	GGTGCTGTTGTCCACCCAGA	eeeeeddddddddddeeeee	49	10418	10437	1506
619788	1410	1429	TGAGGTGCTGTTGTCCACCC	eeeeeddddddddddeeeee	50	10421	10440	1507
619789	1413	1432	CACTGAGGTGCTGTTGTCCA	eeeeeddddddddddeeeee	47	10424	10443	1508
619790	1416	1435	AGACACTGAGGTGCTGTTGT	eeeeeddddddddddeeeee	50	10427	10446	1509
619791	1419	1438	AACAGACACTGAGGTGCTGT	eeeeeddddddddddeeeee	58	10430	10449	1510

TABLE 9-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619792	1422	1441	GGGAACAGACACTGAGGTGC	eeeeeddddddddddeeeee	56	10433	10452	1511
619793	1425	1444	CATGGGAACAGACACTGAGG	eeeeeddddddddddeeeee	45	10436	10455	1512
619794	1428	1447	GAGCATGGGAACAGACACTG	eeeeeddddddddddeeeee	49	10439	10458	1513
619795	1431	1450	AGAGAGCATGGGAACAGACA	eeeeeddddddddddeeeee	32	10442	10461	1514
619796	1434	1453	GCCAGAGAGCATGGGAACAG	eeeeeddddddddddeeeee	32	10445	10464	1515
619797	1437	1456	CATGCCAGAGAGCATGGGAA	eeeeeddddddddddeeeee	35	10448	10467	1516
619798	1440	1459	GCCCATGCCAGAGAGCATGG	eeeeeddddddddddeeeee	23	10451	10470	1517
619799	1443	1462	GGTGCCCATGCCAGAGAGCA	eeeeeddddddddddeeeee	48	10454	10473	1518
619800	1446	1465	GAAGGTGCCCATGCCAGAGA	eeeeeddddddddddeeeee	46	10457	10476	1519
619801	1449	1468	CTGGAAGGTGCCCATGCCAG	eeeeeddddddddddeeeee	55	10460	10479	1520
619802	1452	1471	GTGCTGGAAGGTGCCCATGC	eeeeeddddddddddeeeee	43	10463	10482	1521
619803	1455	1474	CCAGTGCTGGAAGGTGCCCA	eeeeeddddddddddeeeee	58	10466	10485	1522
619804	1458	1477	ACTCCAGTGCTGGAAGGTGC	eeeeeddddddddddeeeee	50	10469	10488	1523
619805	1461	1480	GTCACTCCAGTGCTGGAAGG	eeeeeddddddddddeeeee	53	10472	10491	1524
619806	1464	1483	GATGTCACTCCAGTGCTGGA	eeeeeddddddddddeeeee	46	10475	10494	1525
619807	1467	1486	CTGGATGTCACTCCAGTGCT	eeeeeddddddddddeeeee	70	10478	10497	1526
619808	1470	1489	GTCCTGGATGTCACTCCAGT	eeeeeddddddddddeeeee	49	10481	10500	1527
619809	1473	1492	GTTGTCCTGGATGTCACTCC	eeeeeddddddddddeeeee	51	10484	10503	1528
619810	1476	1495	GAAGTTGTCCTGGATGTCAC	eeeeeddddddddddeeeee	51	10487	10506	1529
619811	1479	1498	CGAGAAGTTGTCCTGGATGT	eeeeeddddddddddeeeee	33	10490	10509	1530
619812	1482	1501	CACCGAGAAGTTGTCCTGGA	eeeeeddddddddddeeeee	49	10493	10512	1531
619813	1485	1504	AGTCACCGAGAAGTTGTCCT	eeeeeddddddddddeeeee	53	10496	10515	1532
619814	1488	1507	TTGAGTCACCGAGAAGTTGT	eeeeeddddddddddeeeee	41	10499	10518	1533
619815	1491	1510	CACTTGAGTCACCGAGAAGT	eeeeeddddddddddeeeee	32	10502	10521	1534
619816	1494	1513	GGGCACTTGAGTCACCGAGA	eeeeeddddddddddeeeee	69	10505	10524	1535
619817	1497	1516	GAAGGGCACTTGAGTCACCG	eeeeeddddddddddeeeee	63	10508	10527	1536
619818	1500	1519	AGTGAAGGGCACTTGAGTCA	eeeeeddddddddddeeeee	37	10511	10530	1537
619819	1503	1522	CTCAGTGAAGGGCACTTGAG	eeeeeddddddddddeeeee	35	10514	10533	1538
619820	1506	1525	GCTCTCAGTGAAGGGCACTT	eeeeeddddddddddeeeee	65	10517	10536	1539
619821	1524	1543	GATCAGCAGCAGGCAGGCGC	eeeeeddddddddddeeeee	58	10535	10554	1540
619822	1527	1546	CTGGATCAGCAGCAGGCAGG	eeeeeddddddddddeeeee	55	10538	10557	1541
619823	1530	1549	AGGCTGGATCAGCAGCAGGC	eeeeeddddddddddeeeee	72	10541	10560	1542
619824	1533	1552	GTGAGGCTGGATCAGCAGCA	eeeeeddddddddddeeeee	70	10544	10563	1543
619825	1536	1555	ATAGTGAGGCTGGATCAGCA	eeeeeddddddddddeeeee	17	10547	10566	1544

TABLE 9-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619826	1539	1558	GGCATAGTGAGGCTGGATCA	eeeeeddddddddddeeeee	67	10550	10569	1545
619827	1542	1561	AGAGGCATAGTGAGGCTGGA	eeeeeddddddddddeeeee	51	10553	10572	1546
619828	1545	1564	GTCAGAGGCATAGTGAGGCT	eeeeeddddddddddeeeee	46	10556	10575	1547

[0520] Table 10 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 4000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 10

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	91	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	84	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	91	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	78	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	89	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	81	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	92	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	90	13518	13537	239
619784	1398	1417	GTCCACCCAGAACTCCTGGG	eeeeeddddddddddeeeee	73	10409	10428	1503
619785	1401	1420	GTTGTCCACCCAGAACTCCT	eeeeeddddddddddeeeee	76	10412	10431	1504
619786	1404	1423	GCTGTTGTCCACCCAGAACT	eeeeeddddddddddeeeee	54	10415	10434	1505
619787	1407	1426	GGTGCTGTTGTCCACCCAGA	eeeeeddddddddddeeeee	55	10418	10437	1506
619788	1410	1429	TGAGGTGCTGTTGTCCACCC	eeeeeddddddddddeeeee	51	10421	10440	1507
619789	1413	1432	CACTGAGGTGCTGTTGTCCA	eeeeeddddddddddeeeee	46	10424	10443	1508
619790	1416	1435	AGACACTGAGGTGCTGTTGT	eeeeeddddddddddeeeee	51	10427	10446	1509
619791	1419	1438	AACAGACACTGAGGTGCTGT	eeeeeddddddddddeeeee	36	10430	10449	1510
619792	1422	1441	GGGAACAGACACTGAGGTGC	eeeeeddddddddddeeeee	57	10433	10452	1511
619793	1425	1444	CATGGGAACAGACACTGAGG	eeeeeddddddddddeeeee	42	10436	10455	1512
619794	1428	1447	GAGCATGGGAACAGACACTG	eeeeeddddddddddeeeee	45	10439	10458	1513
619795	1431	1450	AGAGAGCATGGGAACAGACA	eeeeeddddddddddeeeee	25	10442	10461	1514
619796	1434	1453	GCCAGAGAGCATGGGAACAG	eeeeeddddddddddeeeee	45	10445	10464	1515

TABLE 10-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619797	1437	1456	CATGCCAGAGAGCATGGGAA	eeeeeddddddddddeeeee	38	10448	10467	1516
619798	1440	1459	GCCCATGCCAGAGAGCATGG	eeeeeddddddddddeeeee	27	10451	10470	1517
619799	1443	1462	GGTGCCCATGCCAGAGAGCA	eeeeeddddddddddeeeee	50	10454	10473	1518
619800	1446	1465	GAAGGTGCCCATGCCAGAGA	eeeeeddddddddddeeeee	39	10457	10476	1519
619801	1449	1468	CTGGAAGGTGCCCATGCCAG	eeeeeddddddddddeeeee	54	10460	10479	1520
619802	1452	1471	GTGCTGGAAGGTGCCCATGC	eeeeeddddddddddeeeee	42	10463	10482	1521
619803	1455	1474	CCAGTGCTGGAAGGTGCCCA	eeeeeddddddddddeeeee	83	10466	10485	1522
619804	1458	1477	ACTCCAGTGCTGGAAGGTGC	eeeeeddddddddddeeeee	42	10469	10488	1523
619805	1461	1480	GTCACTCCAGTGCTGGAAGG	eeeeeddddddddddeeeee	66	10472	10491	1524
619806	1464	1483	GATGTCACTCCAGTGCTGGA	eeeeeddddddddddeeeee	55	10475	10494	1525
619807	1467	1486	CTGGATGTCACTCCAGTGCT	eeeeeddddddddddeeeee	68	10478	10497	1526
619808	1470	1489	GTCCTGGATGTCACTCCAGT	eeeeeddddddddddeeeee	49	10481	10500	1527
619809	1473	1492	GTTGTCCTGGATGTCACTCC	eeeeeddddddddddeeeee	61	10484	10503	1528
619810	1476	1495	GAAGTTGTCCTGGATGTCAC	eeeeeddddddddddeeeee	47	10487	10506	1529
619811	1479	1498	CGAGAAGTTGTCCTGGATGT	eeeeeddddddddddeeeee	44	10490	10509	1530
619812	1482	1501	CACCGAGAAGTTGTCCTGGA	eeeeeddddddddddeeeee	56	10493	10512	1531
619813	1485	1504	AGTCACCGAGAAGTTGTCCT	eeeeeddddddddddeeeee	48	10496	10515	1532
619814	1488	1507	TTGAGTCACCGAGAAGTTGT	eeeeeddddddddddeeeee	45	10499	10518	1533
619815	1491	1510	CACTTGAGTCACCGAGAAGT	eeeeeddddddddddeeeee	33	10502	10521	1534
619816	1494	1513	GGGCACTTGAGTCACCGAGA	eeeeeddddddddddeeeee	70	10505	10524	1535
619817	1497	1516	GAAGGGCACTTGAGTCACCG	eeeeeddddddddddeeeee	72	10508	10527	1536
619818	1500	1519	AGTGAAGGGCACTTGAGTCA	eeeeeddddddddddeeeee	41	10511	10530	1537
619819	1503	1522	CTCAGTGAAGGGCACTTGAG	eeeeeddddddddddeeeee	39	10514	10533	1538
619820	1506	1525	GCTCTCAGTGAAGGGCACTT	eeeeeddddddddddeeeee	57	10517	10536	1539
619821	1524	1543	GATCAGCAGCAGGCAGGCGC	eeeeeddddddddddeeeee	58	10535	10554	1540
619822	1527	1546	CTGGATCAGCAGCAGGCAGG	eeeeeddddddddddeeeee	59	10538	10557	1541
619823	1530	1549	AGGCTGGATCAGCAGCAGGC	eeeeeddddddddddeeeee	82	10541	10560	1542
619824	1533	1552	GTGAGGCTGGATCAGCAGCA	eeeeeddddddddddeeeee	65	10544	10563	1543
619825	1536	1555	ATAGTGAGGCTGGATCAGCA	eeeeeddddddddddeeeee	7	10547	10566	1544
619826	1539	1558	GGCATAGTGAGGCTGGATCA	eeeeeddddddddddeeeee	71	10550	10569	1545
619827	1542	1561	AGAGGCATAGTGAGGCTGGA	eeeeeddddddddddeeeee	58	10553	10572	1546
619828	1545	1564	GTCAGAGGCATAGTGAGGCT	eeeeeddddddddddeeeee	57	10556	10575	1547
619829	1548	1567	CAGGTCAGAGGCATAGTGAG	eeeeeddddddddddeeeee	46	10559	10578	1548
619830	1551	1570	GTCCAGGTCAGAGGCATAGT	eeeeeddddddddddeeeee	6	10562	10581	1549
619831	1554	1573	CTTGTCCAGGTCAGAGGCAT	eeeeeddddddddddeeeee	54	10565	10584	1550

TABLE 10-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619832	1557	1576	CACCTTGTCCAGGTCAGAGG	eeeeeddddddddddeeeee	47	10568	10587	1551
619833	1560	1579	CTCCACCTTGTCCAGGTCAG	eeeeeddddddddddeeeee	33	10571	10590	1552
619834	1563	1582	ACCCTCCACCTTGTCCAGGT	eeeeeddddddddddeeeee	59	10574	10593	1553
619835	1566	1585	GAGACCCTCCACCTTGTCCA	eeeeeddddddddddeeeee	47	10577	10596	1554
619836	1569	1588	AGTGAGACCCTCCACCTTGT	eeeeeddddddddddeeeee	52	10580	10599	1555
619837	1572	1591	GAAAGTGAGACCCTCCACCT	eeeeeddddddddddeeeee	40	10583	10602	1556
619838	1575	1594	CTGGAAAGTGAGACCCTCCA	eeeeeddddddddddeeeee	55	10586	10605	1557
619839	1578	1597	TTGCTGGAAAGTGAGACCCT	eeeeeddddddddddeeeee	44	10589	10608	1558
619840	1581	1600	GTTTTGCTGGAAAGTGAGAC	eeeeeddddddddddeeeee	50	10592	10611	1559
619841	1584	1603	GGAGTTTTGCTGGAAAGTGA	eeeeeddddddddddeeeee	54	10595	10614	1560
619842	1587	1606	GAGGGAGTTTTGCTGGAAAG	eeeeeddddddddddeeeee	35	10598	10617	1561
619843	1590	1609	GTTGAGGGAGTTTTGCTGGA	eeeeeddddddddddeeeee	40	10601	10620	1562
619844	1593	1612	CCAGTTGAGGGAGTTTTGCT	eeeeeddddddddddeeeee	32	10604	10623	1563
619845	1596	1615	CATCCAGTTGAGGGAGTTTT	eeeeeddddddddddeeeee	52	10607	10626	1564
619846	1599	1618	CTTCATCCAGTTGAGGGAGT	eeeeeddddddddddeeeee	56	10610	10629	1565
619847	1602	1621	TTTCTTCATCCAGTTGAGGG	eeeeeddddddddddeeeee	38	10613	10632	1566
619848	1605	1624	TAGTTTCTTCATCCAGTTGA	eeeeeddddddddddeeeee	29	10616	10635	1567
619849	1608	1627	AGATAGTTTCTTCATCCAGT	eeeeeddddddddddeeeee	12	10619	10638	1568
619850	1611	1630	GGGAGATAGTTTCTTCATCC	eeeeeddddddddddeeeee	32	10622	10641	1569
619851	1629	1648	GGTCAGGTGGATGGTCCGGG	eeeeeddddddddddeeeee	43	N/A	N/A	1570
619852	1632	1651	CATGGTCAGGTGGATGGTCC	eeeeeddddddddddeeeee	41	12238	12257	1571
619853	1635	1654	GGGCATGGTCAGGTGGATGG	eeeeeddddddddddeeeee	57	12241	12260	1572
619854	1653	1672	TCCTTGCAGCACCAGTTGGG	eeeeeddddddddddeeeee	46	12259	12278	1573
619855	1656	1675	AGATCCTTGCAGCACCAGTT	eeeeeddddddddddeeeee	36	12262	12281	1574
619856	1659	1678	ATAAGATCCTTGCAGCACCA	eeeeeddddddddddeeeee	37	12265	12284	1575
619857	1662	1681	GTCATAAGATCCTTGCAGCA	eeeeeddddddddddeeeee	35	12268	12287	1576
619858	1665	1684	CAGGTCATAAGATCCTTGCA	eeeeeddddddddddeeeee	41	12271	12290	1577
619859	1668	1687	CTGCAGGTCATAAGATCCTT	eeeeeddddddddddeeeee	32	12274	12293	1578
619860	1671	1690	GTCCTGCAGGTCATAAGATC	eeeeeddddddddddeeeee	47	12277	12296	1579
619861	1674	1693	CAGGTCCTGCAGGTCATAAG	eeeeeddddddddddeeeee	33	12280	12299	1580
619862	1677	1696	GAGCAGGTCCTGCAGGTCAT	eeeeeddddddddddeeeee	53	12283	12302	1581
619863	1680	1699	GGCGAGCAGGTCCTGCAGGT	eeeeeddddddddddeeeee	51	12286	12305	1582
619864	1683	1702	CTGGGCGAGCAGGTCCTGCA	eeeeeddddddddddeeeee	50	12289	12308	1583
619865	1686	1705	AGCCTGGGCGAGCAGGTCCT	eeeeeddddddddddeeeee	49	12292	12311	1584

TABLE 10-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619866	1689	1708	CTCAGCCTGGGCGAGCAGGT	eeeeeddddddddddeeeee	63	12295	12314	1585
619867	1692	1711	CAGCTCAGCCTGGGCGAGCA	eeeeeddddddddddeeeee	45	12298	12317	1586
619868	1699	1718	TGGCGGGCAGCTCAGCCTGG	eeeeeddddddddddeeeee	46	12305	12324	1587
619869	1702	1721	GAATGGCGGGCAGCTCAGCC	eeeeeddddddddddeeeee	46	12308	12327	1588
619870	1705	1724	GCAGAATGGCGGGCAGCTCA	eeeeeddddddddddeeeee	41	12311	12330	1589
619871	1708	1727	TGTGCAGAATGGCGGGCAGC	eeeeeddddddddddeeeee	44	12314	12333	1590
619872	1711	1730	CGGTGTGCAGAATGGCGGGC	eeeeeddddddddddeeeee	36	12317	12336	1591
619873	1714	1733	GCTCGGTGTGCAGAATGGCG	eeeeeddddddddddeeeee	63	12320	12339	1592
619874	1717	1736	TCAGCTCGGTGTGCAGAATG	eeeeeddddddddddeeeee	42	12323	12342	1593
619875	1720	1739	GGTTCAGCTCGGTGTGCAGA	eeeeeddddddddddeeeee	62	12326	12345	1594
619876	1723	1742	GCAGGTTCAGCTCGGTGTGC	eeeeeddddddddddeeeee	73	12329	12348	1595
619877	1726	1745	TTTGCAGGTTCAGCTCGGTG	eeeeeddddddddddeeeee	52	12332	12351	1596
619878	1729	1748	ATTTTTGCAGGTTCAGCTCG	eeeeeddddddddddeeeee	43	12335	12354	1597
619879	1732	1751	TCAATTTTTGCAGGTTCAGC	eeeeeddddddddddeeeee	29	12338	12357	1598
619880	1735	1754	TGCTCAATTTTTGCAGGTTC	eeeeeddddddddddeeeee	72	12341	12360	1599
619881	1738	1757	CATTGCTCAATTTTTGCAGG	eeeeeddddddddddeeeee	36	12344	12363	1600
619882	1741	1760	GGTCATTGCTCAATTTTTGC	eeeeeddddddddddeeeee	56	12347	12366	1601
619883	1744	1763	TGCGGTCATTGCTCAATTTT	eeeeeddddddddddeeeee	45	12350	12369	1602
619884	1747	1766	TGATGCGGTCATTGCTCAAT	eeeeeddddddddddeeeee	51	12353	12372	1603
619885	1750	1769	CCCTGATGCGGTCATTGCTC	eeeeeddddddddddeeeee	77	12356	12375	1604
619886	1753	1772	CCACCCTGATGCGGTCATTG	eeeeeddddddddddeeeee	56	12359	12378	1605
619887	1756	1775	CCCCCACCCTGATGCGGTCA	eeeeeddddddddddeeeee	52	12362	12381	1606
619888	1759	1778	CCTCCCCCACCCTGATGCGG	eeeeeddddddddddeeeee	36	12365	12384	1607
619889	1762	1781	GCACCTCCCCCACCCTGATG	eeeeeddddddddddeeeee	36	N/A	N/A	1608
619890	1765	1784	TCAGCACCTCCCCCACCCTG	eeeeeddddddddddeeeee	57	N/A	N/A	1609
619891	1768	1787	TGTTCAGCACCTCCCCCACC	eeeeeddddddddddeeeee	60	N/A	N/A	1610
619892	1771	1790	TGCTGTTCAGCACCTCCCCC	eeeeeddddddddddeeeee	65	N/A	N/A	1611
619893	1774	1793	AAATGCTGTTCAGCACCTCC	eeeeeddddddddddeeeee	68	N/A	N/A	1612
619894	1777	1796	AAAAAATGCTGTTCAGCACC	eeeeeddddddddddeeeee	41	13246	13265	1613
619895	1780	1799	CAAAAAAAATGCTGTTCAGC	eeeeeddddddddddeeeee	40	13249	13268	1614
619896	1783	1802	GCTCAAAAAAAATGCTGTTC	eeeeeddddddddddeeeee	64	13252	13271	1615
619897	1786	1805	CAAGCTCAAAAAAAATGCTG	eeeeeddddddddddeeeee	44	13255	13274	1616
619898	1789	1808	CTTCAAGCTCAAAAAAAATG	eeeeeddddddddddeeeee	15	13258	13277	1617
619899	1792	1811	CCGCTTCAAGCTCAAAAAAA	eeeeeddddddddddeeeee	62	13261	13280	1618
619900	1795	1814	CATCCGCTTCAAGCTCAAAA	eeeeeddddddddddeeeee	62	13264	13283	1619

TABLE 10-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619901	1798	1817	TCTCATCCGCTTCAAGCTCA	eeeeeddddddddddeeeee	72	13267	13286	1620
619902	1801	1820	CTCTCTCATCCGCTTCAAGC	eeeeeddddddddddeeeee	66	13270	13289	1621
619903	1804	1823	GCTCTCTCTCATCCGCTTCA	eeeeeddddddddddeeeee	68	13273	13292	1622
619904	1807	1826	TGGGCTCTCTCTCATCCGCT	eeeeeddddddddddeeeee	83	13276	13295	1623
619905	1810	1829	CTGTGGGCTCTCTCTCATCC	eeeeeddddddddddeeeee	80	13279	13298	1624
619906	1813	1832	ACTCTGTGGGCTCTCTCTCA	eeeeeddddddddddeeeee	54	13282	13301	1625
619907	1816	1835	TAGACTCTGTGGGCTCTCTC	eeeeeddddddddddeeeee	75	13285	13304	1626
619908	1824	1843	CTGTTGGGTAGACTCTGTGG	eeeeeddddddddddeeeee	46	13293	13312	1627
619909	1827	1846	AAGCTGTTGGGTAGACTCTG	eeeeeddddddddddeeeee	63	13296	13315	1628
619910	1830	1849	GTTAAGCTGTTGGGTAGACT	eeeeeddddddddddeeeee	61	13299	13318	1629
619911	1833	1852	CTTGTTAAGCTGTTGGGTAG	eeeeeddddddddddeeeee	47	13302	13321	1630
619912	1836	1855	AGGCTTGTTAAGCTGTTGGG	eeeeeddddddddddeeeee	69	13305	13324	1631
619913	1839	1858	CTCAGGCTTGTTAAGCTGTT	eeeeeddddddddddeeeee	62	13308	13327	1632
619914	1842	1861	GACCTCAGGCTTGTTAAGCT	eeeeeddddddddddeeeee	55	13311	13330	1633
619915	1845	1864	CAAGACCTCAGGCTTGTTAA	eeeeeddddddddddeeeee	50	13314	13333	1634
619916	1848	1867	CTCCAAGACCTCAGGCTTGT	eeeeeddddddddddeeeee	60	13317	13336	1635
619917	1851	1870	CACCTCCAAGACCTCAGGCT	eeeeeddddddddddeeeee	61	13320	13339	1636
619918	1854	1873	GGTCACCTCCAAGACCTCAG	eeeeeddddddddddeeeee	67	13323	13342	1637
619919	1857	1876	CAGGGTCACCTCCAAGACCT	eeeeeddddddddddeeeee	54	13326	13345	1638
619920	1860	1879	GTTCAGGGTCACCTCCAAGA	eeeeeddddddddddeeeee	54	13329	13348	1639
619921	1863	1882	GCGGTTCAGGGTCACCTCCA	eeeeeddddddddddeeeee	70	13332	13351	1640
619922	1873	1892	ACAGGAATGGGCGGTTCAGG	eeeeeddddddddddeeeee	34	13342	13361	1641
619926	1876	1895	CAAACAGGAATGGGCGGTTC	eeeeeddddddddddeeeee	40	13345	13364	1642
619927	1879	1898	CAGCAAACAGGAATGGGCGG	eeeeeddddddddddeeeee	49	13348	13367	1643
619928	1882	1901	ACACAGCAAACAGGAATGGG	eeeeeddddddddddeeeee	28	13351	13370	1644
619929	1885	1904	CATACACAGCAAACAGGAAT	eeeeeddddddddddeeeee	29	13354	13373	1645
619930	1888	1907	GATCATACACAGCAAACAGG	eeeeeddddddddddeeeee	49	13357	13376	1646
619931	1891	1910	TTTGATCATACACAGCAAAC	eeeeeddddddddddeeeee	22	13360	13379	1647
619932	1894	1913	CGCTTTGATCATACACAGCA	eeeeeddddddddddeeeee	56	13363	13382	1648
619933	1911	1930	GAAGTGCAGGGCAGTGGCGC	eeeeeddddddddddeeeee	44	13380	13399	1649
619934	1914	1933	CAGGAAGTGCAGGGCAGTGG	eeeeeddddddddddeeeee	39	13383	13402	1650
619935	1917	1936	GCCCAGGAAGTGCAGGGCAG	eeeeeddddddddddeeeee	20	13386	13405	1651
619936	1920	1939	GCGGCCCAGGAAGTGCAGGG	eeeeeddddddddddeeeee	19	13389	13408	1652
619937	1923	1942	CACGCGGCCCAGGAAGTGCA	eeeeeddddddddddeeeee	34	13392	13411	1653

TABLE 10-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619938	1926	1945	GGCCACGCGGCCCAGGAAGT	eeeeeddddddddddeeeee	21	13395	13414	1654
619939	1929	1948	GTTGGCCACGCGGCCCAGGA	eeeeeddddddddddeeeee	34	13398	13417	1655
619940	1932	1951	CGGGTTGGCCACGCGGCCCA	eeeeeddddddddddeeeee	38	13401	13420	1656
619941	1935	1954	CAGCGGGTTGGCCACGCGGC	eeeeeddddddddddeeeee	42	13404	13423	1657
619942	1938	1957	GCTCAGCGGGTTGGCCACGC	eeeeeddddddddddeeeee	64	13407	13426	1658
619943	1941	1960	TGTGCTCAGCGGGTTGGCCA	eeeeeddddddddddeeeee	43	13410	13429	1659
619944	1944	1963	TGCTGTGCTCAGCGGGTTGG	eeeeeddddddddddeeeee	29	13413	13432	1660
619945	1947	1966	TCATGCTGTGCTCAGCGGGT	eeeeeddddddddddeeeee	49	13416	13435	1661
619946	1950	1969	GCCTCATGCTGTGCTCAGCG	eeeeeddddddddddeeeee	74	13419	13438	1662
619947	1953	1972	CTGGCCTCATGCTGTGCTCA	eeeeeddddddddddeeeee	56	13422	13441	1663
619948	1956	1975	GCCCTGGCCTCATGCTGTGC	eeeeeddddddddddeeeee	44	13425	13444	1664
619949	1976	1995	GCCAGGCACTGTGTTCTGGG	eeeeeddddddddddeeeee	65	13445	13464	1665
619950	1979	1998	CTTGCCAGGCACTGTGTTCT	eeeeeddddddddddeeeee	71	13448	13467	1666
619951	1982	2001	GGCCTTGCCAGGCACTGTGT	eeeeeddddddddddeeeee	80	13451	13470	1667
619952	2114	2133	AGGAGAAACGGCTGCTTTCC	eeeeeddddddddddeeeee	61	13583	13602	1668
619953	2117	2136	CCAAGGAGAAACGGCTGCTT	eeeeeddddddddddeeeee	75	13586	13605	1669
619954	2120	2139	AGACCAAGGAGAAACGGCTG	eeeeeddddddddddeeeee	76	13589	13608	1670
619955	2123	2142	CTTAGACCAAGGAGAAACGG	eeeeeddddddddddeeeee	67	13592	13611	1671
619956	2126	2145	ACACTTAGACCAAGGAGAAA	eeeeeddddddddddeeeee	45	13595	13614	1672
619957	2129	2148	AGCACACTTAGACCAAGGAG	eeeeeddddddddddeeeee	74	13598	13617	1673
619958	2132	2151	TGCAGCACACTTAGACCAAG	eeeeeddddddddddeeeee	55	13601	13620	1674
619959	2135	2154	CCATGCAGCACACTTAGACC	eeeeeddddddddddeeeee	56	13604	13623	1675
619960	2138	2157	ACTCCATGCAGCACACTTAG	eeeeeddddddddddeeeee	66	13607	13626	1676
619961	2141	2160	CTCACTCCATGCAGCACACT	eeeeeddddddddddeeeee	63	13610	13629	1677
619962	2159	2178	CGCTGCAGGCTTCTACTGCT	eeeeeddddddddddeeeee	64	13628	13647	1678
619963	2162	2181	TGCCGCTGCAGGCTTCTACT	eeeeeddddddddddeeeee	60	13631	13650	1679
619964	2165	2184	TTGTGCCGCTGCAGGCTTCT	eeeeeddddddddddeeeee	45	13634	13653	1680
619965	2168	2187	CATTTGTGCCGCTGCAGGCT	eeeeeddddddddddeeeee	62	13637	13656	1681
619966	2171	2190	GTGCATTTGTGCCGCTGCAG	eeeeeddddddddddeeeee	85	13640	13659	1682
619967	2174	2193	GAGGTGCATTTGTGCCGCTG	eeeeeddddddddddeeeee	80	13643	13662	1683
619968	2177	2196	TGGGAGGTGCATTTGTGCCG	eeeeeddddddddddeeeee	53	13646	13665	1684
619969	2180	2199	AACTGGGAGGTGCATTTGTG	eeeeeddddddddddeeeee	34	13649	13668	1685
619970	2183	2202	GCAAACTGGGAGGTGCATTT	eeeeeddddddddddeeeee	62	13652	13671	1686
619971	2186	2205	CCAGCAAACTGGGAGGTGCA	eeeeeddddddddddeeeee	76	13655	13674	1687
619972	2189	2208	AACCCAGCAAACTGGGAGGT	eeeeeddddddddddeeeee	56	13658	13677	1688

TABLE 10-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
619973	2192	2211	ATAAACCCAGCAAACTGGGA	eeeeeddddddddddeeeee	56	13661	13680	1689
619974	2195	2214	AAAATAAACCCAGCAAACTG	eeeeeddddddddddeeeee	33	13664	13683	1690
619975	2198	2217	TCTAAAATAAACCCAGCAAA	eeeeeddddddddddeeeee	29	13667	13686	1691
619976	2201	2220	TTCTCTAAAATAAACCCAGC	eeeeeddddddddddeeeee	58	13670	13689	1692
619977	2204	2223	CCATTCTCTAAAATAAACCC	eeeeeddddddddddeeeee	55	13673	13692	1693
619978	2207	2226	CCCCCATTCTCTAAAATAAA	eeeeeddddddddddeeeee	49	13676	13695	1694
619979	2210	2229	CCACCCCCATTCTCTAAAAT	eeeeeddddddddddeeeee	19	13679	13698	1695
619980	2213	2232	TCCCCACCCCCATTCTCTAA	eeeeeddddddddddeeeee	41	13682	13701	1696
619981	2216	2235	GCCTCCCCACCCCCATTCTC	eeeeeddddddddddeeeee	53	13685	13704	1697
619982	2219	2238	CTTGCCTCCCCACCCCCATT	eeeeeddddddddddeeeee	56	13688	13707	1698
619983	2222	2241	GTTCTTGCCTCCCCACCCCC	eeeeeddddddddddeeeee	72	13691	13710	1699
619984	2225	2244	CTGGTTCTTGCCTCCCCACC	eeeeeddddddddddeeeee	82	13694	13713	1700
619985	2228	2247	ACACTGGTTCTTGCCTCCCC	eeeeeddddddddddeeeee	74	13697	13716	1701
619986	2231	2250	TAAACACTGGTTCTTGCCTC	eeeeeddddddddddeeeee	72	13700	13719	1702
619987	2234	2253	CGCTAAACACTGGTTCTTGC	eeeeeddddddddddeeeee	93	13703	13722	1703
619988	2237	2256	CCGCGCTAAACACTGGTTCT	eeeeeddddddddddeeeee	82	13706	13725	1704
619989	2240	2259	GTCCCGCGCTAAACACTGGT	eeeeeddddddddddeeeee	75	13709	13728	1705
619990	2243	2262	GTAGTCCCGCGCTAAACACT	eeeeeddddddddddeeeee	73	13712	13731	1706
619991	2246	2265	ACAGTAGTCCCGCGCTAAAC	eeeeeddddddddddeeeee	64	13715	13734	1707
619992	2249	2268	GGAACAGTAGTCCCGCGCTA	eeeeeddddddddddeeeee	85	13718	13737	1708
619993	2252	2271	TTTGGAACAGTAGTCCCGCG	eeeeeddddddddddeeeee	65	13721	13740	1709
619994	2255	2274	CTTTTTGGAACAGTAGTCCC	eeeeeddddddddddeeeee	69	13724	13743	1710
619995	2258	2277	ATTCTTTTTGGAACAGTAGT	eeeeeddddddddddeeeee	53	13727	13746	1711
619996	2261	2280	GGAATTCTTTTTGGAACAGT	eeeeeddddddddddeeeee	70	13730	13749	1712
619997	2264	2283	GTTGGAATTCTTTTTGGAAC	eeeeeddddddddddeeeee	57	13733	13752	1713
619998	2267	2286	TCGGTTGGAATTCTTTTTGG	eeeeeddddddddddeeeee	83	13736	13755	1714
619999	2270	2289	TGGTCGGTTGGAATTCTTTT	eeeeeddddddddddeeeee	74	13739	13758	1715
620000	2273	2292	AGCTGGTCGGTTGGAATTCT	eeeeeddddddddddeeeee	78	13742	13761	1716
620001	2276	2295	ACAAGCTGGTCGGTTGGAAT	eeeeeddddddddddeeeee	61	13745	13764	1717
620002	2279	2298	CAAACAAGCTGGTCGGTTGG	eeeeeddddddddddeeeee	61	13748	13767	1718
620003	2282	2301	TCACAAACAAGCTGGTCGGT	eeeeeddddddddddeeeee	88	13751	13770	1719
620004	2285	2304	GTTTCACAAACAAGCTGGTC	eeeeeddddddddddeeeee	91	13754	13773	1720
620005	2288	2307	TTTGTTTCACAAACAAGCTG	eeeeeddddddddddeeeee	73	13757	13776	1721
620006	2304	2323	GAAAAGGGAACACTTTTTTG	eeeeeddddddddddeeeee	59	13773	13792	1722

TABLE 10-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
620007	2307	2326	CTTGAAAAGGGAACACTTTT	eeeeeddddddddddeeeee	57	13776	13795	1723
620008	2310	2329	CAACTTGAAAAGGGAACACT	eeeeeddddddddddeeeee	88	13779	13798	1724
620009	2313	2332	TCTCAACTTGAAAAGGGAAC	eeeeeddddddddddeeeee	88	13782	13801	1725
620010	2316	2335	TGTTCTCAACTTGAAAAGGG	eeeeeddddddddddeeeee	93	13785	13804	1726
620011	2319	2338	TTTTGTTCTCAACTTGAAAA	eeeeeddddddddddeeeee	49	13788	13807	1727
620012	2322	2341	AATTTTTGTTCTCAACTTGA	eeeeeddddddddddeeeee	63	13791	13810	1728
620013	2325	2344	CCCAATTTTTGTTCTCAACT	eeeeeddddddddddeeeee	89	13794	13813	1729
620014	2328	2347	AAACCCAATTTTTGTTCTCA	eeeeeddddddddddeeeee	78	13797	13816	1730
620015	2331	2350	TTAAAACCCAATTTTTGTTC	eeeeeddddddddddeeeee	68	13800	13819	1731
620016	2334	2353	ATTTTAAAACCCAATTTTTG	eeeeeddddddddddeeeee	15	13803	13822	1732
620017	2337	2356	TTAATTTTAAAACCCAATTT	eeeeeddddddddddeeeee	15	13806	13825	1733
620018	2353	2372	TGCAAAAATGTATACTTTAA	eeeeeddddddddddeeeee	32	13822	13841	1734
620019	2356	2375	CAATGCAAAAATGTATACTT	eeeeeddddddddddeeeee	64	13825	13844	1735
620020	2359	2378	AGGCAATGCAAAAATGTATA	eeeeeddddddddddeeeee	76	13828	13847	1736
620021	2362	2381	CGAAGGCAATGCAAAAATGT	eeeeeddddddddddeeeee	50	13831	13850	1737
620022	2365	2384	AACCGAAGGCAATGCAAAAA	eeeeeddddddddddeeeee	55	13834	13853	1738
620023	2368	2387	ACAAACCGAAGGCAATGCAA	eeeeeddddddddddeeeee	68	13837	13856	1739
620024	2371	2390	AATACAAACCGAAGGCAATG	eeeeeddddddddddeeeee	68	13840	13859	1740
620025	2374	2393	CTAAATACAAACCGAAGGCA	eeeeeddddddddddeeeee	64	13843	13862	1741
620026	2377	2396	ACACTAAATACAAACCGAAG	eeeeeddddddddddeeeee	49	13846	13865	1742
620027	2380	2399	AAGACACTAAATACAAACCG	eeeeeddddddddddeeeee	53	13849	13868	1743
620028	2383	2402	TTCAAGACACTAAATACAAA	eeeeeddddddddddeeeee	31	13852	13871	1744
620029	2386	2405	ACATTCAAGACACTAAATAC	eeeeeddddddddddeeeee	35	13855	13874	1745
620030	2389	2408	CTTACATTCAAGACACTAAA	eeeeeddddddddddeeeee	57	13858	13877	1746
620031	2392	2411	GTTCTTACATTCAAGACACT	eeeeeddddddddddeeeee	54	13861	13880	1747
620032	2395	2414	CATGTTCTTACATTCAAGAC	eeeeeddddddddddeeeee	39	13864	13883	1748
620033	2398	2417	GGTCATGTTCTTACATTCAA	eeeeeddddddddddeeeee	58	13867	13886	1749
620034	2401	2420	GGAGGTCATGTTCTTACATT	eeeeeddddddddddeeeee	51	13870	13889	1750
620035	2404	2423	CACGGAGGTCATGTTCTTAC	eeeeeddddddddddeeeee	61	13873	13892	1751
620036	2407	2426	CTACACGGAGGTCATGTTCT	eeeeeddddddddddeeeee	53	13876	13895	1752
620037	2410	2429	ACACTACACGGAGGTCATGT	eeeeeddddddddddeeeee	44	13879	13898	1753
620038	2413	2432	CAGACACTACACGGAGGTCA	eeeeeddddddddddeeeee	50	13882	13901	1754
620039	2416	2435	TTACAGACACTACACGGAGG	eeeeeddddddddddeeeee	66	13885	13904	1755
620040	2419	2438	GTATTACAGACACTACACGG	eeeeeddddddddddeeeee	53	13888	13907	1756
620041	2422	2441	AAGGTATTACAGACACTACA	eeeeeddddddddddeeeee	57	13891	13910	1757

TABLE 10-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
620042	2425	2444	ACTAAGGTATTACAGACACT	eeeeeddddddddddeeeee	50	13894	13913	1758
620043	2428	2447	AAAACTAAGGTATTACAGAC	eeeeeddddddddddeeeee	28	13897	13916	1759
620044	2431	2450	GAAAAAACTAAGGTATTACA	eeeeeddddddddddeeeee	19	13900	13919	1760
620045	2434	2453	GTGGAAAAAACTAAGGTATT	eeeeeddddddddddeeeee	36	13903	13922	1761
620046	2437	2456	TCTGTGGAAAAAACTAAGGT	eeeeeddddddddddeeeee	38	13906	13925	1762
620047	2440	2459	GCATCTGTGGAAAAAACTAA	eeeeeddddddddddeeeee	29	13909	13928	1763
620048	2443	2462	CAAGCATCTGTGGAAAAAAC	eeeeeddddddddddeeeee	21	13912	13931	1764
620049	2446	2465	TCACAAGCATCTGTGGAAAA	eeeeeddddddddddeeeee	30	13915	13934	1765
620050	2449	2468	AAATCACAAGCATCTGTGGA	eeeeeddddddddddeeeee	36	13918	13937	1766
620051	2452	2471	CAAAAATCACAAGCATCTGT	eeeeeddddddddddeeeee	19	13921	13940	1767
620052	2455	2474	GTTCAAAAATCACAAGCATC	eeeeeddddddddddeeeee	32	13924	13943	1768
620053	2458	2477	ATTGTTCAAAAATCACAAGC	eeeeeddddddddddeeeee	16	13927	13946	1769
620054	2461	2480	CGTATTGTTCAAAAATCACA	eeeeeddddddddddeeeee	30	13930	13949	1770
620055	2479	2498	AGGTGCTTGCATCTTTCACG	eeeeeddddddddddeeeee	61	13948	13967	1771
620056	2482	2501	TTCAGGTGCTTGCATCTTTC	eeeeeddddddddddeeeee	58	13951	13970	1772
620057	2485	2504	AAATTCAGGTGCTTGCATCT	eeeeeddddddddddeeeee	35	13954	13973	1773
620058	2488	2507	CAGAAATTCAGGTGCTTGCA	eeeeeddddddddddeeeee	58	13957	13976	1774
620059	2491	2510	AAACAGAAATTCAGGTGCTT	eeeeeddddddddddeeeee	51	13960	13979	1775
620060	2494	2513	TTCAAACAGAAATTCAGGTG	eeeeeddddddddddeeeee	46	13963	13982	1776
620061	2497	2516	GCATTCAAACAGAAATTCAG	eeeeeddddddddddeeeee	40	13966	13985	1777
620062	2500	2519	TCCGCATTCAAACAGAAATT	eeeeeddddddddddeeeee	73	13969	13988	1778
620063	2503	2522	GGTTCCGCATTCAAACAGAA	eeeeeddddddddddeeeee	54	13972	13991	1779
620064	2506	2525	TATGGTTCCGCATTCAAACA	eeeeeddddddddddeeeee	44	13975	13994	1780
620065	2509	2528	AGCTATGGTTCCGCATTCAA	eeeeeddddddddddeeeee	67	13978	13997	1781
620066	2512	2531	ACCAGCTATGGTTCCGCATT	eeeeeddddddddddeeeee	60	13981	14000	1782
620067	2515	2534	ATAACCAGCTATGGTTCCGC	eeeeeddddddddddeeeee	70	13984	14003	1783
620068	2518	2537	GAAATAACCAGCTATGGTTC	eeeeeddddddddddeeeee	50	13987	14006	1784
620069	2521	2540	GGAGAAATAACCAGCTATGG	eeeeeddddddddddeeeee	50	13990	14009	1785
620070	2524	2543	AAGGGAGAAATAACCAGCTA	eeeeeddddddddddeeeee	56	13993	14012	1786
620071	2527	2546	CACAAGGGAGAAATAACCAG	eeeeeddddddddddeeeee	53	13996	14015	1787
620072	2530	2549	TAACACAAGGGAGAAATAAC	eeeeeddddddddddeeeee	27	13999	14018	1788
620073	2533	2552	TACTAACACAAGGGAGAAAT	eeeeeddddddddddeeeee	39	14002	14021	1789
620074	2536	2555	TATTACTAACACAAGGGAGA	eeeeeddddddddddeeeee	52	14005	14024	1790
620075	2539	2558	GTTTATTACTAACACAAGGG	eeeeeddddddddddeeeee	56	14008	14027	1791

TABLE 10-continued

Inhibition of AGT mRNA by MOE containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
620076	2558	2577	AGGCTTATTGTGGCAAGACG	eeeeeddddddddddeeeee	50	14027	14046	1792
620077	2561	2580	TGGAGGCTTATTGTGGCAAG	eeeeeddddddddddeeeee	38	14030	14049	1793
620078	2564	2583	TTTTGGAGGCTTATTGTGGC	eeeeeddddddddddeeeee	22	14033	14052	1794
620079	2567	2586	TTTTTTTGGAGGCTTATTGT	eeeeeddddddddddeeeee	48	N/A	N/A	1795

[0521] Table 11 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 500 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 11

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	62	13515	13530	129
594621	2022	2037	CTGCTGCTGGCCTTTG	kkkddddddddddkkk	16	13491	13506	162
594622	2027	2042	GTTATCTGCTGCTGGC	kkkddddddddddkkk	44	13496	13511	163
594623	2032	2047	GGGTTGTTATCTGCTG	kkkddddddddddkkk	32	13501	13516	164
594624	2046	2061	CGCTGATTTGTCCGGG	kkkddddddddddkkk	62	13515	13530	129
594625	2047	2062	TCGCTGATTTGTCCGG	kkkddddddddddkkk	49	13516	13531	165
594626	2049	2064	CATCGCTGATTTGTCC	kkkddddddddddkkk	36	13518	13533	166
594627	2053	2068	GACACATCGCTGATTT	kkkddddddddddkkk	0	13522	13537	167
594628	2073	2088	AAAGGTGGGAGACTGG	kkkddddddddddkkk	51	13542	13557	168
609078	2020	2035	GCTGCTGGCCTTTGCC	kkkddddddddddkkk	26	13489	13504	173
609079	2021	2036	TGCTGCTGGCCTTTGC	kkkddddddddddkkk	31	13490	13505	174
609080	2023	2038	TCTGCTGCTGGCCTTT	kkkddddddddddkkk	41	13492	13507	175
609081	2024	2039	ATCTGCTGCTGGCCTT	kkkddddddddddkkk	29	13493	13508	176
609082	2025	2040	TATCTGCTGCTGGCCT	kkkddddddddddkkk	43	13494	13509	177
609083	2026	2041	TTATCTGCTGCTGGCC	kkkddddddddddkkk	19	13495	13510	178
609084	2028	2043	TGTTATCTGCTGCTGG	kkkddddddddddkkk	0	13497	13512	179
609085	2029	2044	TTGTTATCTGCTGCTG	kkkddddddddddkkk	40	13498	13513	180
609086	2030	2045	GTTGTTATCTGCTGCT	kkkddddddddddkkk	67	13499	13514	181
609087	2031	2046	GGTTGTTATCTGCTGC	kkkddddddddddkkk	73	13500	13515	182
609088	2048	2063	ATCGCTGATTTGTCCG	kkkddddddddddkkk	59	13517	13532	183

TABLE 11-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
609089	2050	2065	ACATCGCTGATTTGTC	kkkddddddddddkkk	47	13519	13534	184
609090	2051	2066	CACATCGCTGATTTGT	kkkddddddddddkkk	34	13520	13535	185
609091	2052	2067	ACACATCGCTGATTTG	kkkddddddddddkkk	59	13521	13536	186
609092	2054	2069	TGACACATCGCTGATT	kkkddddddddddkkk	27	13523	13538	187
609093	2055	2070	GTGACACATCGCTGAT	kkkddddddddddkkk	38	13524	13539	188
609094	2056	2071	GGTGACACATCGCTGA	kkkddddddddddkkk	51	13525	13540	130
609095	2057	2072	GGGTGACACATCGCTG	kkkddddddddddkkk	59	13526	13541	189
609096	2074	2089	AAAAGGTGGGAGACTG	kkkddddddddddkkk	20	13543	13558	190
609097	2075	2090	GAAAAGGTGGGAGACT	kkkddddddddddkkk	19	13544	13559	131
609098	2076	2091	AGAAAAGGTGGGAGAC	kkkddddddddddkkk	12	13545	13560	191
622201	2020	2035	GCTGCTGGCCTTTGCC	ekkddddddddddkke	29	13489	13504	173
622202	2021	2036	TGCTGCTGGCCTTTGC	ekkddddddddddkke	17	13490	13505	174
622203	2022	2037	CTGCTGCTGGCCTTTG	ekkddddddddddkke	28	13491	13506	162
622204	2023	2038	TCTGCTGCTGGCCTTT	ekkddddddddddkke	23	13492	13507	175
622205	2024	2039	ATCTGCTGCTGGCCTT	ekkddddddddddkke	0	13493	13508	176
622206	2025	2040	TATCTGCTGCTGGCCT	ekkddddddddddkke	22	13494	13509	177
622207	2026	2041	TTATCTGCTGCTGGCC	ekkddddddddddkke	16	13495	13510	178
622208	2027	2042	GTTATCTGCTGCTGGC	ekkddddddddddkke	29	13496	13511	163
622209	2028	2043	TGTTATCTGCTGCTGG	ekkddddddddddkke	37	13497	13512	179
622210	2029	2044	TTGTTATCTGCTGCTG	ekkddddddddddkke	44	13498	13513	180
622211	2030	2045	GTTGTTATCTGCTGCT	ckkddddddddddkke	61	13499	13514	181
622212	2031	2046	GGTTGTTATCTGCTGC	ekkddddddddddkke	51	13500	13515	182
622213	2032	2047	GGGTTGTTATCTGCTG	ekkddddddddddkke	44	13501	13516	164
622214	2046	2061	CGCTGATTTGTCCGGG	ekkddddddddddkke	62	13515	13530	129
622215	2047	2062	TCGCTGATTTGTCCGG	ekkddddddddddkke	47	13516	13531	165
622216	2048	2063	ATCGCTGATTTGTCCG	ekkddddddddddkke	55	13517	13532	183
622217	2049	2064	CATCGCTGATTTGTCC	ekkddddddddddkke	11	13518	13533	166
622218	2050	2065	ACATCGCTGATTTGTC	ekkddddddddddkke	33	13519	13534	184
622219	2051	2066	CACATCGCTGATTTGT	ekkddddddddddkke	41	13520	13535	185
622220	2052	2067	ACACATCGCTGATTTG	ekkddddddddddkke	49	13521	13536	186

TABLE 11-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
622221	2053	2068	GACACATCGCTGATTT	ekkddddddddddkke	52	13522	13537	167
622222	2054	2069	TGACACATCGCTGATT	ekkddddddddddkke	34	13523	13538	187
622223	2055	2070	GTGACACATCGCTGAT	ekkddddddddddkke	32	13524	13539	188
622224	2056	2071	GGTGACACATCGCTGA	ekkddddddddddkke	45	13525	13540	130
622225	2057	2072	GGGTGACACATCGCTG	ekkddddddddddkke	58	13526	13541	189
622226	2073	2088	AAAGGTGGGAGACTGG	ekkddddddddddkke	18	13542	13557	168
622227	2074	2089	AAAAGGTGGGAGACTG	ekkddddddddddkke	0	13543	13558	190
622228	2075	2090	GAAAAGGTGGGAGACT	ekkddddddddddkke	0	13544	13559	131
622229	2076	2091	AGAAAAGGTGGGAGAC	ekkddddddddddkke	0	13545	13560	191
622230	2080	2095	TAGAAGAAAAGGTGGG	ekkddddddddddkke	12	13549	13564	192
622231	2081	2096	TTAGAAGAAAAGGTGG	ekkddddddddddkke	22	13550	13565	193
622232	2082	2097	ATTAGAAGAAAAGGTG	ekkddddddddddkke	7	13551	13566	169
622233	2083	2098	CATTAGAAGAAAAGGT	ekkddddddddddkke	0	13552	13567	194
622234	2084	2099	TCATTAGAAGAAAAGG	ekkddddddddddkke	20	13553	13568	195
622235	2085	2100	CTCATTAGAAGAAAAG	ekkddddddddddkke	4	13554	13569	196
622236	2086	2101	ACTCATTAGAAGAAAA	ekkddddddddddkke	0	13555	13570	197
622237	2087	2102	GACTCATTAGAAGAAA	ekkddddddddddkke	22	13556	13571	198
622238	2088	2103	CGACTCATTAGAAGAA	ekkddddddddddkke	46	13557	13572	132
622239	2089	2104	TCGACTCATTAGAAGA	ekkddddddddddkke	33	13558	13573	199
622240	2090	2105	GTCGACTCATTAGAAG	ekkddddddddddkke	6	13559	13574	170
622241	2091	2106	AGTCGACTCATTAGAA	ekkddddddddddkke	33	13560	13575	200
622242	2092	2107	AAGTCGACTCATTAGA	ekkddddddddddkke	31	13561	13576	201
622243	2093	2108	AAAGTCGACTCATTAG	ekkddddddddddkke	16	13562	13577	202
622244	2094	2109	CAAAGTCGACTCATTA	ekkddddddddddkke	28	13563	13578	203
622245	2095	2110	TCAAAGTCGACTCATT	ekkddddddddddkke	16	13564	13579	171
622246	2096	2111	CTCAAAGTCGACTCAT	ekkddddddddddkke	25	13565	13580	204
622247	2097	2112	GCTCAAAGTCGACTCA	ekkddddddddddkke	43	13566	13581	205
622248	2098	2113	AGCTCAAAGTCGACTC	ekkddddddddddkke	39	13567	13582	206
622249	2099	2114	CAGCTCAAAGTCGACT	ekkddddddddddkke	17	13568	13583	172

[0522] Table 12 shows the percent inhibition of AGT mRNA by antisense oligonucleotides. Cultured HepG2 cells at a density of about 20,000 cells per well were transfected using electroporation with 3,000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 12

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	88	13515	13530	129
610006	2023	2042	GTTATCTGCTGCTGGCCTTT	eeeeeddddddddddeeeee	60	13492	13511	230
610009	2026	2045	GTTGTTATCTGCTGCTGGCC	eeeeeddddddddddeeeee	38	13495	13514	233
610010	2027	2046	GGTTGTTATCTGCTGCTGGC	eeeeeddddddddddeeeee	66	13496	13515	234
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	72	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	25	13518	13537	239
654354	636	655	ACTCTCATTGTGGATGACGA	eeeeeddddddddddeeeee	41	5853	5872	1796
654355	640	659	AGGTACTCTCATTGTGGATG	eeeeeddddddddddeeeee	26	5857	5876	1797
654356	642	661	ACAGGTACTCTCATTGTGGA	eeeeeddddddddddeeeee	16	5859	5878	1798
654357	646	665	GCTCACAGGTACTCTCATTG	eeeeeddddddddddeeeee	18	5863	5882	1799
654358	757	776	GCACCAGCTGGTCCTGTAGG	eeeeeddddddddddeeeee	10	5974	5993	1800
654359	759	778	TAGCACCAGCTGGTCCTGTA	eeeeeddddddddddeeeee	24	5976	5995	1801
654360	760	779	CTAGCACCAGCTGGTCCTGT	eeeeeddddddddddeeeee	13	5977	5996	1802
654361	763	782	CGACTAGCACCAGCTGGTCC	eeeeeddddddddddeeeee	1	5980	5999	1803
654362	765	784	AGCGACTAGCACCAGCTGGT	eeeeeddddddddddeeeee	28	5982	6001	1804
654363	769	788	TTGCAGCGACTAGCACCAGC	eeeeeddddddddddeeeee	18	5986	6005	1805
654364	771	790	TTTTGCAGCGACTAGCACCA	eeeeeddddddddddeeeee	9	5988	6007	1806
654365	775	794	CAAGTTTTGCAGCGACTAGC	eeeeeddddddddddeeeee	0	5992	6011	1807
654366	1267	1286	CTGTCACAGCCTGCATGAAC	eeeeeddddddddddeeeee	15	6484	6503	1808
654367	1269	1288	TCCTGTCACAGCCTGCATGA	eeeeeddddddddddeeeee	34	6486	6505	1809
654368	1270	1289	ATCCTGTCACAGCCTGCATG	eeeeeddddddddddeeeee	34	6487	6506	1810
654369	1273	1292	TCCATCCTGTCACAGCCTGC	eeeeeddddddddddeeeee	32	6490	6509	1811
654370	1275	1294	CTTCCATCCTGTCACAGCCT	eeeeeddddddddddeeeee	50	6492	6511	1812
654371	1460	1479	TCACTCCAGTGCTGGAAGGT	eeeeeddddddddddeeeee	0	10471	10490	1813
654372	1462	1481	TGTCACTCCAGTGCTGGAAG	eeeeeddddddddddeeeee	18	10473	10492	1814
654373	1463	1482	ATGTCACTCCAGTGCTGGAA	eeeeeddddddddddeeeee	6	10474	10493	1815
654374	1466	1485	TGGATGTCACTCCAGTGCTG	eeeeeddddddddddeeeee	26	10477	10496	1816
654375	1468	1487	CCTGGATGTCACTCCAGTGC	eeeeeddddddddddeeeee	20	10479	10498	1817
654376	2115	2134	AAGGAGAAACGGCTGCTTTC	eeeeeddddddddddeeeee	19	13584	13603	1818
654377	2116	2135	CAAGGAGAAACGGCTGCTTT	eeeeeddddddddddeeeee	40	13585	13604	1819
654378	2118	2137	ACCAAGGAGAAACGGCTGCT	eeeeeddddddddddeeeee	48	13587	13606	1820
654379	2119	2138	GACCAAGGAGAAACGGCTGC	eeeeeddddddddddeeeee	57	13588	13607	1821
654380	2121	2140	TAGACCAAGGAGAAACGGCT	eeeeeddddddddddeeeee	46	13590	13609	1822
654381	2122	2141	TTAGACCAAGGAGAAACGGC	eeeeeddddddddddeeeee	32	13591	13610	1823
654382	2124	2143	ACTTAGACCAAGGAGAAACG	eeeeeddddddddddeeeee	42	13593	13612	1824

TABLE 12-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654383	2125	2144	CACTTAGACCAAGGAGAAAC	eeeeeddddddddddeeeee	29	13594	13613	1825
654384	2127	2146	CACACTTAGACCAAGGAGAA	eeeeeddddddddddeeeee	21	13596	13615	1826
654385	2128	2147	GCACACTTAGACCAAGGAGA	eeeeeddddddddddeeeee	65	13597	13616	1827
654386	2130	2149	CAGCACACTTAGACCAAGGA	eeeeeddddddddddeeeee	39	13599	13618	1828
654387	2131	2150	GCAGCACACTTAGACCAAGG	eeeeeddddddddddeeeee	39	13600	13619	1829
654388	2133	2152	ATGCAGCACACTTAGACCAA	eeeeeddddddddddeeeee	27	13602	13621	1830
654389	2134	2153	CATGCAGCACACTTAGACCA	eeeeeddddddddddeeeee	26	13603	13622	1831
654390	2136	2155	TCCATGCAGCACACTTAGAC	eeeeeddddddddddeeeee	2	13605	13624	1832
654391	2137	2156	CTCCATGCAGCACACTTAGA	eeeeeddddddddddeeeee	48	13606	13625	1833
654392	2139	2158	CACTCCATGCAGCACACTTA	eeeeeddddddddddeeeee	60	13608	13627	1834
654393	2140	2159	TCACTCCATGCAGCACACTT	eeeeeddddddddddeeeee	45	13609	13628	1835
654394	2142	2161	GCTCACTCCATGCAGCACAC	eeeeeddddddddddeeeee	72	13611	13630	1836
654395	2160	2179	CCGCTGCAGGCTTCTACTGC	eeeeeddddddddddeeeee	34	13629	13648	1837
654396	2161	2180	GCCGCTGCAGGCTTCTACTG	eeeeeddddddddddeeeee	32	13630	13649	1838
654397	2163	2182	GTGCCGCTGCAGGCTTCTAC	eeeeeddddddddddeeeee	38	13632	13651	1839
654398	2164	2183	TGTGCCGCTGCAGGCTTCTA	eeeeeddddddddddeeeee	17	13633	13652	1840
654399	2166	2185	TTTGTGCCGCTGCAGGCTTC	eeeeeddddddddddeeeee	16	13635	13654	1841
654400	2167	2186	ATTTGTGCCGCTGCAGGCTT	eeeeeddddddddddeeeee	27	13636	13655	1842
654401	2169	2188	GCATTTGTGCCGCTGCAGGC	eeeeeddddddddddeeeee	75	13638	13657	1843
654402	2170	2189	TGCATTTGTGCCGCTGCAGG	eeeeeddddddddddeeeee	64	13639	13658	1844
654403	2172	2191	GGTGCATTTGTGCCGCTGCA	eeeeeddddddddddeeeee	64	13641	13660	1845
654404	2173	2192	AGGTGCATTTGTGCCGCTGC	eeeeeddddddddddeeeee	68	13642	13661	1846
654405	2175	2194	GGAGGTGCATTTGTGCCGCT	eeeeeddddddddddeeeee	42	13644	13663	1847
654406	2176	2195	GGGAGGTGCATTTGTGCCGC	eeeeeddddddddddeeeee	36	13645	13664	1848
654407	2178	2197	CTGGGAGGTGCATTTGTGCC	eeeeeddddddddddeeeee	26	13647	13666	1849
654408	2179	2198	ACTGGGAGGTGCATTTGTGC	eeeeeddddddddddeeeee	10	13648	13667	1850
654409	2181	2200	AAACTGGGAGGTGCATTTGT	eeeeeddddddddddeeeee	15	13650	13669	1851
654410	2182	2201	CAAACTGGGAGGTGCATTTG	eeeeeddddddddddeeeee	7	13651	13670	1852
654411	2184	2203	AGCAAACTGGGAGGTGCATT	eeeeeddddddddddeeeee	34	13653	13672	1853
654412	2185	2204	CAGCAAACTGGGAGGTGCAT	eeeeeddddddddddeeeee	33	13654	13673	1854
654413	2187	2206	CCCAGCAAACTGGGAGGTGC	eeeeeddddddddddeeeee	57	13656	13675	1855
654414	2188	2207	ACCCAGCAAACTGGGAGGTG	eeeeeddddddddddeeeee	53	13657	13676	1856
654415	2193	2212	AATAAACCCAGCAAACTGGG	eeeeeddddddddddeeeee	17	13662	13681	1857
654416	2194	2213	AAATAAACCCAGCAAACTGG	eeeeeddddddddddeeeee	20	13663	13682	1858
654417	2196	2215	TAAAATAAACCCAGCAAACT	eeeeeddddddddddeeeee	13	13665	13684	1859

TABLE 12-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654418	2197	2216	CTAAAATAAACCCAGCAAAC	eeeeeddddddddddeeeee	2	13666	13685	1860
654419	2199	2218	CTCTAAAATAAACCCAGCAA	eeeeeddddddddddeeeee	12	13668	13687	1861
654420	2200	2219	TCTCTAAAATAAACCCAGCA	eeeeeddddddddddeeeee	47	13669	13688	1862
654421	2202	2221	ATTCTCTAAAATAAACCCAG	eeeeeddddddddddeeeee	23	13671	13690	1863
654422	2203	2222	CATTCTCTAAAATAAACCCA	eeeeeddddddddddeeeee	22	13672	13691	1864
654423	2205	2224	CCCATTCTCTAAAATAAACC	eeeeeddddddddddeeeee	12	13674	13693	1865
654424	2206	2225	CCCCATTCTCTAAAATAAAC	eeeeeddddddddddeeeee	20	13675	13694	1866
654425	2208	2227	ACCCCCATTCTCTAAAATAA	eeeeeddddddddddeeeee	21	13677	13696	1867
654426	2209	2228	CACCCCCATTCTCTAAAATA	eeeeeddddddddddeeeee	32	13678	13697	1868
654427	2211	2230	CCCACCCCCATTCTCTAAAA	eeeeeddddddddddeeeee	18	13680	13699	1869

[0523] Table 13 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 4000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 13

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	95	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	98	13515	13530	129
610012	2046	2065	ACATCGCTGATTTGTCCGGG	eeeeeddddddddddeeeee	74	13515	13534	236
610013	2047	2066	CACATCGCTGATTTGTCCGG	eeeeeddddddddddeeeee	76	13516	13535	237
610014	2048	2067	ACACATCGCTGATTTGTCCG	eeeeeddddddddddeeeee	85	13517	13536	238
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	85	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	3	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	17	13518	13537	239
610043	2096	2115	CCAGCTCAAAGTCGACTCAT	eeeeeddddddddddeeeee	87	13565	13584	267
619998	2267	2286	TCGGTTGGAATTCTTTTTGG	eeeeeddddddddddeeeee	80	13736	13755	1714
620000	2273	2292	AGCTGGTCGGTTGGAATTCT	eeeeeddddddddddeeeee	69	13742	13761	1716
654428	2212	2231	CCCCACCCCCATTCTCTAAA	eeeeeddddddddddeeeee	49	13681	13700	1870
654429	2214	2233	CTCCCCACCCCCATTCTCTA	eeeeeddddddddddeeeee	58	13683	13702	1871
654430	2217	2236	TGCCTCCCCACCCCCATTCT	eeeeeddddddddddeeeee	61	13686	13705	1872
654431	2218	2237	TTGCCTCCCCACCCCCATTC	eeeeeddddddddddeeeee	51	13687	13706	1873

TABLE 13-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654432	2220	2239	TCTTGCCTCCCCACCCCCAT	eeeeeddddddddddeeeee	67	13689	13708	1874
654433	2223	2242	GGTTCTTGCCTCCCCACCCC	eeeeeddddddddddeeeee	84	13692	13711	1875
654434	2224	2243	TGGTTCTTGCCTCCCCACCC	eeeeeddddddddddeeeee	83	13693	13712	1876
654435	2226	2245	ACTGGTTCTTGCCTCCCCAC	eeeeeddddddddddeeeee	75	13695	13714	1877
654436	2227	2246	CACTGGTTCTTGCCTCCCCA	eeeeeddddddddddeeeee	84	13696	13715	1878
654437	2229	2248	AACACTGGTTCTTGCCTCCC	eeeeeddddddddddeeeee	76	13698	13717	1879
654438	2230	2249	AAACACTGGTTCTTGCCTCC	eeeeeddddddddddeeeee	75	13699	13718	1880
654439	2232	2251	CTAAACACTGGTTCTTGCCT	eeeeeddddddddddeeeee	70	13701	13720	1881
654440	2233	2252	GCTAAACACTGGTTCTTGCC	eeeeeddddddddddeeeee	79	13702	13721	1882
654441	2235	2254	GCGCTAAACACTGGTTCTTG	eeeeeddddddddddeeeee	79	13704	13723	1883
654442	2236	2255	CGCGCTAAACACTGGTTCTT	eeeeeddddddddddeeeee	81	13705	13724	1884
654443	2238	2257	CCCGCGCTAAACACTGGTTC	eeeeeddddddddddeeeee	80	13707	13726	1885
654444	2239	2258	TCCCGCGCTAAACACTGGTT	eeeeeddddddddddeeeee	89	13708	13727	1886
654445	2241	2260	AGTCCCGCGCTAAACACTGG	eeeeeddddddddddeeeee	75	13710	13729	1887
654446	2242	2261	TAGTCCCGCGCTAAACACTG	eeeeeddddddddddeeeee	73	13711	13730	1888
654447	2244	2263	AGTAGTCCCGCGCTAAACAC	eeeeeddddddddddeeeee	59	13713	13732	1889
654448	2245	2264	CAGTAGTCCCGCGCTAAACA	eeeeeddddddddddeeeee	67	13714	13733	1890
654449	2247	2266	AACAGTAGTCCCGCGCTAAA	eeeeeddddddddddeeeee	60	13716	13735	1891
654450	2248	2267	GAACAGTAGTCCCGCGCTAA	eeeeeddddddddddeeeee	69	13717	13736	1892
654451	2250	2269	TGGAACAGTAGTCCCGCGCT	eeeeeddddddddddeeeee	87	13719	13738	1893
654452	2251	2270	TTGGAACAGTAGTCCCGCGC	eeeeeddddddddddeeeee	87	13720	13739	1894
654453	2253	2272	TTTTGGAACAGTAGTCCCGC	eeeeeddddddddddeeeee	73	13722	13741	1895
654454	2254	2273	TTTTTGGAACAGTAGTCCCG	eeeeeddddddddddeeeee	51	13723	13742	1896
654455	2256	2275	TCTTTTTGGAACAGTAGTCC	eeeeeddddddddddeeeee	74	13725	13744	1897
654456	2257	2276	TTCTTTTTGGAACAGTAGTC	eeeeeddddddddddeeeee	66	13726	13745	1898
654457	2259	2278	AATTCTTTTTGGAACAGTAG	eeeeeddddddddddeeeee	46	13728	13747	1899
654458	2260	2279	GAATTCTTTTTGGAACAGTA	eeeeeddddddddddeeeee	74	13729	13748	1900
654459	2262	2281	TGGAATTCTTTTTGGAACAG	eeeeeddddddddddeeeee	41	13731	13750	1901
654460	2263	2282	TTGGAATTCTTTTTGGAACA	eeeeeddddddddddeeeee	34	13732	13751	1902
654461	2265	2284	GGTTGGAATTCTTTTTGGAA	eeeeeddddddddddeeeee	58	13734	13753	1903
654462	2266	2285	CGGTTGGAATTCTTTTTGGA	eeeeeddddddddddeeeee	77	13735	13754	1904
654463	2268	2287	GTCGGTTGGAATTCTTTTTG	eeeeeddddddddddeeeee	74	13737	13756	1905
654464	2269	2288	GGTCGGTTGGAATTCTTTTT	eeeeeddddddddddeeeee	81	13738	13757	1906
654465	2271	2290	CTGGTCGGTTGGAATTCTTT	eeeeeddddddddddeeeee	78	13740	13759	1907
654466	2272	2291	GCTGGTCGGTTGGAATTCTT	eeeeeddddddddddeeeee	81	13741	13760	1908

TABLE 13-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654467	2274	2293	AAGCTGGTCGGTTGGAATTC	eeeeeddddddddddeeeee	61	13743	13762	1909
654468	2275	2294	CAAGCTGGTCGGTTGGAATT	eeeeeddddddddddeeeee	62	13744	13763	1910
654469	2277	2296	AACAAGCTGGTCGGTTGGAA	eeeeeddddddddddeeeee	70	13746	13765	1911
654470	2278	2297	AAACAAGCTGGTCGGTTGGA	eeeeeddddddddddeeeee	62	13747	13766	1912
654471	2280	2299	ACAAACAAGCTGGTCGGTTG	eeeeeddddddddddeeeee	62	13749	13768	1913
654472	2281	2300	CACAAACAAGCTGGTCGGTT	eeeeeddddddddddeeeee	88	13750	13769	1914
654473	2283	2302	TTCACAAACAAGCTGGTCGG	eeeeeddddddddddeeeee	76	13752	13771	1915
654474	2284	2303	TTTCACAAACAAGCTGGTCG	eeeeeddddddddddeeeee	77	13753	13772	1916
654475	2286	2305	TGTTTCACAAACAAGCTGGT	eeeeeddddddddddeeeee	80	13755	13774	1917
654476	2287	2306	TTGTTTCACAAACAAGCTGG	eeeeeddddddddddeeeee	83	13756	13775	1918
654477	2289	2308	TTTTGTTTCACAAACAAGCT	eeeeeddddddddddeeeee	66	13758	13777	1919
654478	2290	2309	TTTTTGTTTCACAAACAAGC	eeeeeddddddddddeeeee	70	13759	13778	1920
654479	2309	2328	AACTTGAAAAGGGAACACTT	eeeeeddddddddddeeeee	69	13778	13797	1921
654480	2311	2330	TCAACTTGAAAAGGGAACAC	eeeeeddddddddddeeeee	84	13780	13799	1922
654481	2312	2331	CTCAACTTGAAAAGGGAACA	eeeeeddddddddddeeeee	90	13781	13800	1923
654482	2314	2333	TTCTCAACTTGAAAAGGGAA	eeeeeddddddddddeeeee	67	13783	13802	1924
654483	2315	2334	GTTCTCAACTTGAAAAGGGA	eeeeeddddddddddeeeee	92	13784	13803	1925
654484	2317	2336	TTGTTCTCAACTTGAAAAGG	eeeeeddddddddddeeeee	82	13786	13805	1926
654485	2318	2337	TTTGTTCTCAACTTGAAAAG	eeeeeddddddddddeeeee	61	13787	13806	1927
654486	2320	2339	TTTTTGTTCTCAACTTGAAA	eeeeeddddddddddeeeee	35	13789	13808	1928
654487	2321	2340	ATTTTTGTTCTCAACTTGAA	eeeeeddddddddddeeeee	44	13790	13809	1929
654488	2323	2342	CAATTTTTGTTCTCAACTTG	eeeeeddddddddddeeeee	54	13792	13811	1930
654489	2324	2343	CCAATTTTTGTTCTCAACTT	eeeeeddddddddddeeeee	79	13793	13812	1931
654490	2326	2345	ACCCAATTTTTGTTCTCAAC	eeeeeddddddddddeeeee	85	13795	13814	1932
654491	2327	2346	AACCCAATTTTTGTTCTCAA	eeeeeddddddddddeeeee	82	13796	13815	1933
654492	2330	2349	TAAAACCCAATTTTTGTTCT	eeeeeddddddddddeeeee	52	13799	13818	1934
654493	2332	2351	TTTAAAACCCAATTTTTGTT	eeeeeddddddddddeeeee	13	13801	13820	1935
654494	2355	2374	AATGCAAAAATGTATACTTT	eeeeeddddddddddeeeee	53	13824	13843	1936
654495	2357	2376	GCAATGCAAAAATGTATACT	eeeeeddddddddddeeeee	73	13826	13845	1937
654496	2360	2379	AAGGCAATGCAAAAATGTAT	eeeeeddddddddddeeeee	56	13829	13848	1938
654497	2361	2380	GAAGGCAATGCAAAAATGTA	eeeeeddddddddddeeeee	70	13830	13849	1939
654498	2363	2382	CCGAAGGCAATGCAAAAATG	eeeeeddddddddddeeeee	60	13832	13851	1940
654521	495	511	CATACCCTTCTGCTGTA	eeeddddddddddeeee	46	N/A	N/A	1941
654522	498	514	CCGCATACCCTTCTGCT	eeeddddddddddeeee	44	N/A	N/A	1942
654523	504	520	TCGCTTCCGCATACCCT	eeeddddddddddeeee	69	5721	5737	1943

TABLE 13-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654524	507	523	TGCTCGCTTCCGCATAC	eeeddddddddddeeee	58	5724	5740	1944
654525	636	652	CTCATTGTGGATGACGA	eeeddddddddddeeee	53	5853	5869	1945
654526	639	655	ACTCTCATTGTGGATGA	eeeddddddddddeeee	48	5856	5872	1946
654527	654	670	CAGCTGCTCACAGGTAC	eeeddddddddddeeee	47	5871	5887	1947
654528	768	784	AGCGACTAGCACCAGCT	eeeddddddddddeeee	56	5985	6001	1948
654529	1266	1282	CACAGCCTGCATGAACC	eeeddddddddddeeee	48	6483	6499	1949
654530	1272	1288	TCCTGTCACAGCCTGCA	eeeddddddddddeeee	68	6489	6505	1950
654531	1275	1291	CCATCCTGTCACAGCCT	eeeddddddddddeeee	65	6492	6508	1951
654532	1456	1472	AGTGCTGGAAGGTGCCC	eeeddddddddddeeee	41	10467	10483	1952
654533	1531	1547	GCTGGATCAGCAGCAGG	eeeddddddddddeeee	61	10542	10558	1953
654534	1751	1767	CTGATGCGGTCATTGCT	eeeddddddddddeeee	52	12357	12373	1954
654535	1808	1824	GGCTCTCTCTCATCCGC	eeeddddddddddeeee	74	13277	13293	1955
654536	1811	1827	GTGGGCTCTCTCTCATC	eeeddddddddddeeee	60	13280	13296	1956
654537	1983	1999	CCTTGCCAGGCACTGTG	eeeddddddddddeeee	69	13452	13468	1957
654538	1984	2000	GCCTTGCCAGGCACTGT	eeeddddddddddeeee	77	13453	13469	1958
654539	1986	2002	AGGCCTTGCCAGGCACT	eeeddddddddddeeee	78	13455	13471	1959
654540	1987	2003	GAGGCCTTGCCAGGCAC	eeeddddddddddeeee	44	13456	13472	1960
654541	2019	2035	GCTGCTGGCCTTTGCCT	eeeddddddddddeeee	54	13488	13504	1961
654542	2024	2040	TATCTGCTGCTGGCCTT	eeeddddddddddeeee	59	13493	13509	1962
654543	2025	2041	TTATCTGCTGCTGGCCT	eeeddddddddddeeee	4	13494	13510	1963
654544	2027	2043	TGTTATCTGCTGCTGGC	eeeddddddddddeeee	67	13496	13512	1964
654545	2028	2044	TTGTTATCTGCTGCTGG	eeeddddddddddeeee	56	13497	13513	1965
654546	2029	2045	GTTGTTATCTGCTGCTG	eeeddddddddddeeee	77	13498	13514	1966
654547	2047	2063	ATCGCTGATTTGTCCGG	eeeddddddddddeeee	80	13516	13532	1967
654548	2048	2064	CATCGCTGATTTGTCCG	eeeddddddddddeeee	59	13517	13533	1968
654549	2049	2065	ACATCGCTGATTTGTCC	eeeddddddddddeeee	65	13518	13534	1969
654550	2050	2066	CACATCGCTGATTTGTC	eeeddddddddddeeee	81	13519	13535	1970
654551	2051	2067	ACACATCGCTGATTTGT	eeeddddddddddeeee	74	13520	13536	1971
654552	2053	2069	TGACACATCGCTGATTT	eeeddddddddddeeee	53	13522	13538	1972
654553	2054	2070	GTGACACATCGCTGATT	eeeddddddddddeeee	74	13523	13539	1973
654554	2082	2098	CATTAGAAGAAAAGGTG	eeeddddddddddeeee	18	13551	13567	1974
654555	2083	2099	TCATTAGAAGAAAAGGT	eeeddddddddddeeee	23	13552	13568	1975
654556	2087	2103	CGACTCATTAGAAGAAA	eeeddddddddddeeee	51	13556	13572	1976
654557	2096	2112	GCTCAAAGTCGACTCAT	eeeddddddddddeeee	70	13565	13581	1977
654558	2097	2113	AGCTCAAAGTCGACTCA	eeeddddddddddeeee	82	13566	13582	1978

TABLE 13-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654559	2098	2114	CAGCTCAAAGTCGACTC	eeeddddddddddeeee	88	13567	13583	1979
654560	2099	2115	CCAGCTCAAAGTCGACT	eeeddddddddddeeee	84	13568	13584	1980
654561	2100	2116	TCCAGCTCAAAGTCGAC	eeeddddddddddeeee	81	13569	13585	1981
654562	2103	2119	CTTTCCAGCTCAAAGTC	eeeddddddddddeeee	53	13572	13588	1982
654563	2114	2130	AGAAACGGCTGCTTTCC	eeeddddddddddeeee	54	13583	13599	1983
654564	2121	2137	ACCAAGGAGAAACGGCT	eeeddddddddddeeee	66	13590	13606	1984
654565	2172	2188	GCATTTGTGCCGCTGCA	eeeddddddddddeeee	82	13641	13657	1985
654566	2175	2191	GGTGCATTTGTGCCGCT	eeeddddddddddeeee	85	13644	13660	1986
654567	2187	2203	AGCAAACTGGGAGGTGC	eeeddddddddddeeee	70	13656	13672	1987
654568	2226	2242	GGTTCTTGCCTCCCCAC	eeeddddddddddeeee	88	13695	13711	1988
654569	2235	2251	CTAAACACTGGTTCTTG	eeeddddddddddeeee	64	13704	13720	1989
654570	2238	2254	GCGCTAAACACTGGTTC	eeeddddddddddeeee	85	13707	13723	1990
654571	2250	2266	AACAGTAGTCCCGCGCT	eeeddddddddddeeee	83	13719	13735	1991
654572	2268	2284	GGTTGGAATTCTTTTTG	eeeddddddddddeeee	38	13737	13753	1992
654573	2274	2290	CTGGTCGGTTGGAATTC	eeeddddddddddeeee	67	13743	13759	1993
654574	2283	2299	ACAAACAAGCTGGTCGG	eeeddddddddddeeee	70	13752	13768	1994
654575	2286	2302	TTCACAAACAAGCTGGT	eeeddddddddddeeee	67	13755	13771	1995
654576	2311	2327	ACTTGAAAAGGGAACAC	eeeddddddddddeeee	72	13780	13796	1996
654577	2314	2330	TCAACTTGAAAAGGGAA	eeeddddddddddeeee	29	13783	13799	1997
654578	2317	2333	TTCTCAACTTGAAAAGG	eeeddddddddddeeee	46	13786	13802	1998
654579	2326	2342	CAATTTTTGTTCTCAAC	eeeddddddddddeeee	11	13795	13811	1999
654580	2329	2345	ACCCAATTTTTGTTCTC	eeeddddddddddeeee	70	13798	13814	2000
654582	2024	2040	TATCTGCTGCTGGCCTT	eeeddddddddeeeeee	58	13493	13509	1962
654585	2027	2043	TGTTATCTGCTGCTGGC	eeeddddddddeeeeee	66	13496	13512	1964
654586	2028	2044	TTGTTATCTGCTGCTGG	eeeddddddddeeeeee	66	13497	13513	1965
654609	2024	2040	TATCTGCTGCTGGCCTT	eeeeddddddddeeeee	62	13493	13509	1962
654612	2027	2043	TGTTATCTGCTGCTGGC	eeeeddddddddeeeee	61	13496	13512	1964
654636	2024	2040	TATCTGCTGCTGGCCTT	eeeeeddddddddeeee	70	13493	13509	1962
654639	2027	2043	TGTTATCTGCTGCTGGC	eeeeeddddddddeeee	69	13496	13512	1964
654689	2023	2039	ATCTGCTGCTGGCCTTT	eeeddddddddddeeee	60	13492	13508	2001
654690	2026	2042	GTTATCTGCTGCTGGCC	eeeddddddddddeeee	77	13495	13511	2002
654691	2046	2062	TCGCTGATTTGTCCGGG	eeeddddddddddeeee	90	13515	13531	2003
654692	2249	2265	ACAGTAGTCCCGCGCTA	eeeddddddddddeeee	76	13718	13734	2004
654693	2251	2267	GAACAGTAGTCCCGCGC	eeeddddddddddeeee	74	13720	13736	2005
654694	2267	2283	GTTGGAATTCTTTTTGG	eeeddddddddddeeee	41	13736	13752	2006

TABLE 13-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654695	2269	2285	CGGTTGGAATTCTTTTT	eeeddddddddddeeee	65	13738	13754	2007
654696	2273	2289	TGGTCGGTTGGAATTCT	eeeddddddddddeeee	61	13742	13758	2008
654697	2275	2291	GCTGGTCGGTTGGAATT	eeeddddddddddeeee	61	13744	13760	2009
654698	2282	2298	CAAACAAGCTGGTCGGT	eeeddddddddddeeee	84	13751	13767	2010
654699	2284	2300	CACAAACAAGCTGGTCG	eeeddddddddddeeee	78	13753	13769	2011

[0524] Table 14 shows inhibition of AGT mRNA in HepG2 cells cultured at a density of 20,000 cells per well which were transfected using electroporation with 4000 nM antisense oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 14

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	96	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	98	13515	13530	129
568637	2046	2061	CGCTGATTTGTCCGGG	eekddddddddddkke	97	13515	13530	129
610006	2023	2042	GTTATCTGCTGCTGGCCTTT	eeeeeddddddddddeeeee	77	13492	13511	230
610009	2026	2045	GTTGTTATCTGCTGCTGGCC	eeeeeddddddddddeeeee	73	13495	13514	233
610010	2027	2046	GGTTGTTATCTGCTGCTGGC	eeeeeddddddddddeeeee	82	13496	13515	234
610012	2046	2065	ACATCGCTGATTTGTCCGGG	eeeeeddddddddddeeeee	84	13515	13534	236
610013	2047	2066	CACATCGCTGATTTGTCCGG	eeeeeddddddddddeeeee	76	13516	13535	237
610014	2048	2067	ACACATCGCTGATTTGTCCG	eeeeeddddddddddeeeee	89	13517	13536	238
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	35	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	28	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	89	13518	13537	239
610015	2049	2068	GACACATCGCTGATTTGTCC	eeeeeddddddddddeeeee	22	13518	13537	239
610043	2096	2115	CCAGCTCAAAGTCGACTCAT	eeeeeddddddddddeeeee	82	13565	13584	267
619992	2249	2268	GGAACAGTAGTCCCGCGCTA	eeeeeddddddddddeeeee	88	13718	13737	1708
619998	2267	2286	TCGGTTGGAATTCTTTTTGG	eeeeeddddddddddeeeee	69	13736	13755	1714
620000	2273	2292	AGCTGGTCGGTTGGAATTCT	eeeeeddddddddddeeeee	76	13742	13761	1716
620003	2282	2301	TCACAAACAAGCTGGTCGGT	eeeeeddddddddddeeeee	85	13751	13770	1719
654701	2024	2041	TTATCTGCTGCTGGCCTT	eeeeddddddddddeeee	50	13493	13510	2012
654704	2027	2044	TTGTTATCTGCTGCTGGC	eeeeddddddddddeeee	46	13496	13513	2013

TABLE 14-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654705	2028	2045	GTTGTTATCTGCTGCTGG	eeeeddddddddddeeee	60	13497	13514	2014
654707	2046	2063	ATCGCTGATTTGTCCGGG	eeeeddddddddddeeee	91	13515	13532	2015
654708	2047	2064	CATCGCTGATTTGTCCGG	eeeeddddddddddeeee	78	13516	13533	2016
654709	2048	2065	ACATCGCTGATTTGTCCG	eeeeddddddddddeeee	66	13517	13534	2017
654710	2049	2066	CACATCGCTGATTTGTCC	eeeeddddddddddeeee	80	13518	13535	2018
654711	2050	2067	ACACATCGCTGATTTGTC	eeeeddddddddddeeee	77	13519	13536	2019
654713	2097	2114	CAGCTCAAAGTCGACTCA	eeeeddddddddddeeee	77	13566	13583	2020
654716	2250	2267	GAACAGTAGTCCCGCGCT	eeeeddddddddddeeee	80	13719	13736	2021
654719	2268	2285	CGGTTGGAATTCTTTTTG	eeeeddddddddddeeee	65	13737	13754	2022
654722	2274	2291	GCTGGTCGGTTGGAATTC	eeeeddddddddddeeee	74	13743	13760	2023
654724	2282	2299	ACAAACAAGCTGGTCGGT	eeeeddddddddddeeee	81	13751	13768	2024
654725	2283	2300	CACAAACAAGCTGGTCGG	eeeeddddddddddeeee	80	13752	13769	2025
654728	2024	2041	TTATCTGCTGCTGGCCTT	eeeeddddddddeeeeee	53	13493	13510	2012
654731	2027	2044	TTGTTATCTGCTGCTGGC	eeeeddddddddeeeeee	56	13496	13513	2013
654732	2028	2045	GTTGTTATCTGCTGCTGG	eeeeddddddddeeeeee	71	13497	13514	2014
654735	2047	2064	CATCGCTGATTTGTCCGG	eeeeddddddddeeeeee	71	13516	13533	2016
654736	2048	2065	ACATCGCTGATTTGTCCG	eeeeddddddddeeeeee	72	13517	13534	2017
654737	2049	2066	CACATCGCTGATTTGTCC	eeeeddddddddeeeeee	82	13518	13535	2018
654740	2097	2114	CAGCTCAAAGTCGACTCA	eeeeddddddddeeeeee	88	13566	13583	2020
654743	2250	2267	GAACAGTAGTCCCGCGCT	eeeeddddddddeeeeee	75	13719	13736	2021
654745	2267	2284	GGTTGGAATTCTTTTTGG	eeeeddddddddeeeeee	49	13736	13753	2026
654746	2268	2285	CGGTTGGAATTCTTTTTG	eeeeddddddddeeeeee	62	13737	13754	2022
654749	2274	2291	GCTGGTCGGTTGGAATTC	eeeeddddddddeeeeee	55	13743	13760	2023
654752	2283	2300	CACAAACAAGCTGGTCGG	eeeeddddddddeeeeee	74	13752	13769	2025
654755	2024	2041	TTATCTGCTGCTGGCCTT	eeeeeddddddddeeeee	47	13493	13510	2012
654758	2027	2044	TTGTTATCTGCTGCTGGC	eeeeeddddddddeeeee	51	13496	13513	2013
654759	2028	2045	GTTGTTATCTGCTGCTGG	eeeeeddddddddeeeee	56	13497	13514	2014
654761	2046	2063	ATCGCTGATTTGTCCGGG	eeeeeddddddddeeeee	74	13515	13532	2015
654762	2047	2064	CATCGCTGATTTGTCCGG	eeeeeddddddddeeeee	62	13516	13533	2016
654763	2048	2065	ACATCGCTGATTTGTCCG	eeeeeddddddddeeeee	61	13517	13534	2017
654764	2049	2066	CACATCGCTGATTTGTCC	eeeeeddddddddeeeee	68	13518	13535	2018
654765	2050	2067	ACACATCGCTGATTTGTC	eeeeeddddddddeeeee	72	13519	13536	2019
654767	2097	2114	CAGCTCAAAGTCGACTCA	eeeeeddddddddeeeee	63	13566	13583	2020
654768	2098	2115	CCAGCTCAAAGTCGACTC	eeeeeddddddddeeeee	86	13567	13584	2027
654770	2250	2267	GAACAGTAGTCCCGCGCT	eeeeeddddddddeeeee	55	13719	13736	2021

TABLE 14-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654771	2251	2268	GGAACAGTAGTCCCGCGC	eeeeeddddddddeeeee	82	13720	13737	2028
654773	2268	2285	CGGTTGGAATTCTTTTTG	eeeeeddddddddeeeee	58	13737	13754	2022
654776	2274	2291	GCTGGTCGGTTGGAATTC	eeeeeddddddddeeeee	37	13743	13760	2023
654778	2282	2299	ACAAACAAGCTGGTCGGT	eeeeeddddddddeeeee	71	13751	13768	2024
654779	2283	2300	CACAAACAAGCTGGTCGG	eeeeeddddddddeeeee	63	13752	13769	2025
654781	2023	2040	TATCTGCTGCTGGCCTTT	eeeeeeddddddddeeee	56	13492	13509	2029
654782	2024	2041	TTATCTGCTGCTGGCCTT	eeeeeeddddddddeeee	63	13493	13510	2012
654784	2026	2043	TGTTATCTGCTGCTGGCC	eeeeeeddddddddeeee	65	13495	13512	2030
654785	2027	2044	TTGTTATCTGCTGCTGGC	eeeeeeddddddddeeee	55	13496	13513	2013
654786	2028	2045	GTTGTTATCTGCTGCTGG	eeeeeeddddddddeeee	48	13497	13514	2014
654789	2047	2064	CATCGCTGATTTGTCCGG	eeeeeeddddddddeeee	73	13516	13533	2016
654790	2048	2065	ACATCGCTGATTTGTCCG	eeeeeeddddddddeeee	69	13517	13534	2017
654791	2049	2066	CACATCGCTGATTTGTCC	eeeeeeddddddddeeee	61	13518	13535	2018
654794	2097	2114	CAGCTCAAAGTCGACTCA	eeeeeeddddddddeeee	79	13566	13583	2020
654797	2250	2267	GAACAGTAGTCCCGCGCT	eeeeeeddddddddeeee	37	13719	13736	2021
654800	2268	2285	CGGTTGGAATTCTTTTTG	eeeeeeddddddddeeee	63	13737	13754	2022
654801	2269	2286	TCGGTTGGAATTCTTTTT	eeeeeeddddddddeeee	59	13738	13755	2031
654803	2274	2291	GCTGGTCGGTTGGAATTC	eeeeeeddddddddeeee	61	13743	13760	2023
654806	2283	2300	CACAAACAAGCTGGTCGG	eeeeeeddddddddeeee	54	13752	13769	2025
654809	2023	2041	TTATCTGCTGCTGGCCTTT	eeeeddddddddddeeeee	45	13492	13510	2032
654812	2026	2044	TTGTTATCTGCTGCTGGCC	eeeeddddddddddeeeee	57	13495	13513	2033
654813	2027	2045	GTTGTTATCTGCTGCTGGC	eeeeddddddddddeeeee	64	13496	13514	2034
654815	2046	2064	CATCGCTGATTTGTCCGGG	eeeeddddddddddeeeee	83	13515	13533	2035
654816	2047	2065	ACATCGCTGATTTGTCCGG	eeeeddddddddddeeeee	68	13516	13534	2036
654817	2048	2066	CACATCGCTGATTTGTCCG	eeeeddddddddddeeeee	82	13517	13535	2037
654818	2049	2067	ACACATCGCTGATTTGTCC	eeeeddddddddddeeeee	44	13518	13536	2038
654820	2096	2114	CAGCTCAAAGTCGACTCAT	eeeeddddddddddeeeee	80	13565	13583	2039
654822	2248	2266	AACAGTAGTCCCGCGCTAA	eeeeddddddddddeeeee	63	13717	13735	2040
654823	2249	2267	GAACAGTAGTCCCGCGCTA	eeeeddddddddddeeeee	77	13718	13736	2041
654826	2267	2285	CGGTTGGAATTCTTTTTGG	eeeeddddddddddeeeee	76	13736	13754	2042
654829	2273	2291	GCTGGTCGGTTGGAATTCT	eeeeddddddddddeeeee	78	13742	13760	2043
654832	2282	2300	CACAAACAAGCTGGTCGGT	eeeeddddddddddeeeee	82	13751	13769	2044
654833	2283	2301	TCACAAACAAGCTGGTCGG	eeeeddddddddddeeeee	28	13752	13770	2045
654834	2022	2040	TATCTGCTGCTGGCCTTTG	eeeeddddddddeeeeeee	3	13491	13509	2046
654835	2023	2041	TTATCTGCTGCTGGCCTTT	eeeeddddddddeeeeeee	48	13492	13510	2032

TABLE 14-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654837	2025	2043	TGTTATCTGCTGCTGGCCT	eeeeddddddddeeeeeee	64	13494	13512	2047
654838	2026	2044	TTGTTATCTGCTGCTGGCC	eeeeddddddddeeeeeee	38	13495	13513	2033
654839	2027	2045	GTTGTTATCTGCTGCTGGC	eeeeddddddddeeeeeee	60	13496	13514	2034
654841	2046	2064	CATCGCTGATTTGTCCGGG	eeeeddddddddeeeeeee	72	13515	13533	2035
654842	2047	2065	ACATCGCTGATTTGTCCGG	eeeeddddddddeeeeeee	70	13516	13534	2036
654843	2048	2066	CACATCGCTGATTTGTCCG	eeeeddddddddeeeeeee	85	13517	13535	2037
654845	2095	2113	AGCTCAAAGTCGACTCATT	eeeeddddddddeeeeeee	44	13564	13582	2048
654846	2096	2114	CAGCTCAAAGTCGACTCAT	eeeeddddddddeeeeeee	84	13565	13583	2039
654849	2249	2267	GAACAGTAGTCCCGCGCTA	eeeeddddddddeeeeeee	43	13718	13736	2041
654852	2267	2285	CGGTTGGAATTCTTTTTGG	eeeeddddddddeeeeeee	73	13736	13754	2042
654855	2273	2291	GCTGGTCGGTTGGAATTCT	eeeeddddddddeeeeeee	59	13742	13760	2043
654858	2282	2300	CACAAACAAGCTGGTCGGT	eeeeddddddddeeeeeee	72	13751	13769	2044
654861	2023	2041	TTATCTGCTGCTGGCCTTT	eeeeeddddddddeeeeee	40	13492	13510	2032
654864	2026	2044	TTGTTATCTGCTGCTGGCC	eeeeeddddddddeeeeee	57	13495	13513	2033
654865	2027	2045	GTTGTTATCTGCTGCTGGC	eeeeeddddddddeeeeee	52	13496	13514	2034
654867	2046	2064	CATCGCTGATTTGTCCGGG	eeeeeddddddddeeeeee	71	13515	13533	2035
654868	2047	2065	ACATCGCTGATTTGTCCGG	eeeeeddddddddeeeeee	69	13516	13534	2036
654869	2048	2066	CACATCGCTGATTTGTCCG	eeeeeddddddddeeeeee	69	13517	13535	2037
654872	2096	2114	CAGCTCAAAGTCGACTCAT	eeeeeddddddddeeeeee	63	13565	13583	2039
654875	2249	2267	GAACAGTAGTCCCGCGCTA	eeeeeddddddddeeeeee	55	13718	13736	2041
654877	2266	2284	GGTTGGAATTCTTTTTGGA	eeeeeddddddddeeeeee	43	13735	13753	2049
654878	2267	2285	CGGTTGGAATTCTTTTTGG	eeeeeddddddddeeeeee	61	13736	13754	2042
654881	2273	2291	GCTGGTCGGTTGGAATTCT	eeeeeddddddddeeeeee	49	13742	13760	2043
654883	2281	2299	ACAAACAAGCTGGTCGGTT	eeeeeddddddddeeeeee	40	13750	13768	2050
654884	2282	2300	CACAAACAAGCTGGTCGGT	eeeeeddddddddeeeeee	73	13751	13769	2044
654887	2023	2041	TTATCTGCTGCTGGCCTTT	eeeeeeddddddddeeeee	60	13492	13510	2032
654890	2026	2044	TTGTTATCTGCTGCTGGCC	eeeeeeddddddddeeeee	44	13495	13513	2033
654891	2027	2045	GTTGTTATCTGCTGCTGGC	eeeeeeddddddddeeeee	60	13496	13514	2034
654893	2046	2064	CATCGCTGATTTGTCCGGG	eeeeeeddddddddeeeee	74	13515	13533	2035
654894	2047	2065	ACATCGCTGATTTGTCCGG	eeeeeeddddddddeeeee	64	13516	13534	2036
654895	2048	2066	CACATCGCTGATTTGTCCG	eeeeeeddddddddeeeee	62	13517	13535	2037
654898	2096	2114	CAGCTCAAAGTCGACTCAT	eeeeeeddddddddeeeee	67	13565	13583	2039
654899	2097	2115	CCAGCTCAAAGTCGACTCA	eeeeeeddddddddeeeee	63	13566	13584	2051
654901	2249	2267	GAACAGTAGTCCCGCGCTA	eeeeeeddddddddeeeee	55	13718	13736	2041
654904	2267	2285	CGGTTGGAATTCTTTTTGG	eeeeeeddddddddeeeee	45	13736	13754	2042

TABLE 14-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654907	2273	2291	GCTGGTCGGTTGGAATTCT	eeeeeeddddddddeeeee	51	13742	13760	2043
654910	2282	2300	CACAAACAAGCTGGTCGGT	eeeeeeddddddddeeeee	47	13751	13769	2044
654911	2283	2301	TCACAAACAAGCTGGTCGG	eeeeeeddddddddeeeee	72	13752	13770	2045
654917	2027	2045	GTTGTTATCTGCTGCTGGC	eeeeeeeddddddddeeee	45	13496	13514	2034
654920	2047	2065	ACATCGCTGATTTGTCCGG	eeeeeeeddddddddeeee	77	13516	13534	2036
654939	2023	2042	GTTATCTGCTGCTGGCCTTT	eeeeeeeddddddddeeeee	65	13492	13511	230
654941	2025	2044	TTGTTATCTGCTGCTGGCCT	eeeeeeeddddddddeeeee	55	13494	13513	232
654942	2026	2045	GTTGTTATCTGCTGCTGGCC	eeeeeeeddddddddeeeee	48	13495	13514	233
654943	2027	2046	GGTTGTTATCTGCTGCTGGC	eeeeeeeddddddddeeeee	67	13496	13515	234
654944	2028	2047	GGGTTGTTATCTGCTGCTGG	eeeeeeeddddddddeeeee	56	13497	13516	235
654945	2046	2065	ACATCGCTGATTTGTCCGGG	eeeeeeeddddddddeeeee	77	13515	13534	236
654946	2047	2066	CACATCGCTGATTTGTCCGG	eeeeeeeddddddddeeeee	67	13516	13535	237
654947	2048	2067	ACACATCGCTGATTTGTCCG	eeeeeeeddddddddeeeee	56	13517	13536	238
654950	2096	2115	CCAGCTCAAAGTCGACTCAT	eeeeeeeddddddddeeeee	75	13565	13584	267
654951	2097	2116	TCCAGCTCAAAGTCGACTCA	eeeeeeeddddddddeeeee	50	13566	13585	268
654952	2248	2267	GAACAGTAGTCCCGCGCTAA	eeeeeeeddddddddeeeee	53	13717	13736	1892
654953	2249	2268	GGAACAGTAGTCCCGCGCTA	eeeeeeeddddddddeeeee	44	13718	13737	1708
654955	2266	2285	CGGTTGGAATTCTTTTTGGA	eeeeeeeddddddddeeeee	58	13735	13754	1904
654956	2267	2286	TCGGTTGGAATTCTTTTTGG	eeeeeeeddddddddeeeee	66	13736	13755	1714
654959	2273	2292	AGCTGGTCGGTTGGAATTCT	eeeeeeeddddddddeeeee	56	13742	13761	1716
654962	2282	2301	TCACAAACAAGCTGGTCGGT	eeeeeeeddddddddeeeee	55	13751	13770	1719
654963	2283	2302	TTCACAAACAAGCTGGTCGG	eeeeeeeddddddddeeeee	63	13752	13771	1915
654964	2022	2041	TTATCTGCTGCTGGCCTTTG	eeeeeeddddddddeeeeee	43	13491	13510	229
654965	2023	2042	GTTATCTGCTGCTGGCCTTT	eeeeeeddddddddeeeeee	65	13492	13511	230
654968	2026	2045	GTTGTTATCTGCTGCTGGCC	eeeeeeddddddddeeeeee	44	13495	13514	233
654969	2027	2046	GGTTGTTATCTGCTGCTGGC	eeeeeeddddddddeeeeee	64	13496	13515	234
654970	2028	2047	GGGTTGTTATCTGCTGCTGG	eeeeeeddddddddeeeeee	76	13497	13516	235
654971	2046	2065	ACATCGCTGATTTGTCCGGG	eeeeeeddddddddeeeeee	60	13515	13534	236
654972	2047	2066	CACATCGCTGATTTGTCCGG	eeeeeeddddddddeeeeee	74	13516	13535	237
654973	2048	2067	ACACATCGCTGATTTGTCCG	eeeeeeddddddddeeeeee	54	13517	13536	238
654974	2049	2068	GACACATCGCTGATTTGTCC	eeeeeeddddddddeeeeee	78	13518	13537	239
654976	2096	2115	CCAGCTCAAAGTCGACTCAT	eeeeeeddddddddeeeeee	62	13565	13584	267
654979	2249	2268	GGAACAGTAGTCCCGCGCTA	eeeeeeddddddddeeeeee	59	13718	13737	1708
654982	2267	2286	TCGGTTGGAATTCTTTTTGG	eeeeeeddddddddeeeeee	63	13736	13755	1714
654985	2273	2292	AGCTGGTCGGTTGGAATTCT	eeeeeeddddddddeeeeee	57	13742	13761	1716

TABLE 14-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
654988	2282	2301	TCACAAACAAGCTGGTCGGT	eeeeeeddddddddeeeeee	70	13751	13770	1719
654989	2283	2302	TTCACAAACAAGCTGGTCGG	eeeeeeddddddddeeeeee	77	13752	13771	1915
654990	2022	2041	TTATCTGCTGCTGGCCTTTG	eeeeeddddddddeeeeeee	41	13491	13510	229
654991	2023	2042	GTTATCTGCTGCTGGCCTTT	eeeeeddddddddeeeeeee	70	13492	13511	230
654994	2026	2045	GTTGTTATCTGCTGCTGGCC	eeeeeddddddddeeeeeee	33	13495	13514	233
654995	2027	2046	GGTTGTTATCTGCTGCTGGC	eeeeeddddddddeeeeeee	79	13496	13515	234
654997	2046	2065	ACATCGCTGATTTGTCCGGG	eeeeeddddddddeeeeeee	64	13515	13534	236
654998	2047	2066	CACATCGCTGATTTGTCCGG	eeeeeddddddddeeeeeee	70	13516	13535	237
654999	2048	2067	ACACATCGCTGATTTGTCCG	eeeeeddddddddeeeeeee	85	13517	13536	238
655002	2096	2115	CCAGCTCAAAGTCGACTCAT	eeeeeddddddddeeeeeee	85	13565	13584	267
655005	2249	2268	GGAACAGTAGTCCCGCGCTA	eeeeeddddddddeeeeeee	73	13718	13737	1708
655008	2267	2286	TCGGTTGGAATTCTTTTTGG	eeeeeddddddddeeeeeee	67	13736	13755	1714
655011	2273	2292	AGCTGGTCGGTTGGAATTCT	eeeeeddddddddeeeeeee	31	13742	13761	1716
655014	2282	2301	TCACAAACAAGCTGGTCGGT	eeeeeddddddddeeeeeee	76	13751	13770	1719
655044	2024	2041	TTATCTGCTGCTGGCCTT	eeeeddddddddeeeee	55	13493	13510	2012
655045	2027	2044	TTGTTATCTGCTGCTGGC	eeeeddddddddeeeee	46	13496	13513	2013
655046	2028	2045	GTTGTTATCTGCTGCTGG	eeeeddddddddeeeee	54	13497	13514	2014
655047	2047	2064	CATCGCTGATTTGTCCGG	eeeeddddddddeeeee	61	13516	13533	2016
655048	2048	2065	ACATCGCTGATTTGTCCG	eeeeddddddddeeeee	59	13517	13534	2017
655049	2049	2066	CACATCGCTGATTTGTCC	eeeeddddddddeeeee	84	13518	13535	2018
655050	2097	2114	CAGCTCAAAGTCGACTCA	eeeeddddddddeeeee	75	13566	13583	2020
655051	2250	2267	GAACAGTAGTCCCGCGCT	eeeeddddddddeeeee	74	13719	13736	2021
655052	2268	2285	CGGTTGGAATTCTTTTTG	eeeeddddddddeeeee	58	13737	13754	2022
655053	2274	2291	GCTGGTCGGTTGGAATTC	eeeeddddddddeeeee	58	13743	13760	2023
655054	2283	2300	CACAAACAAGCTGGTCGG	eeeeddddddddeeeee	76	13752	13769	2025
655055	2024	2041	TTATCTGCTGCTGGCCTT	eeeeeddddddddeeee	57	13493	13510	2012
655056	2027	2044	TTGTTATCTGCTGCTGGC	eeeeeddddddddeeee	50	13496	13513	2013
655057	2028	2045	GTTGTTATCTGCTGCTGG	eeeeeddddddddeeee	63	13497	13514	2014
655058	2047	2064	CATCGCTGATTTGTCCGG	eeeeeddddddddeeee	80	13516	13533	2016
655059	2048	2065	ACATCGCTGATTTGTCCG	eeeeeddddddddeeee	60	13517	13534	2017
655060	2049	2066	CACATCGCTGATTTGTCC	eeeeeddddddddeeee	68	13518	13535	2018
655061	2097	2114	CAGCTCAAAGTCGACTCA	eeeeeddddddddeeee	79	13566	13583	2020
655062	2250	2267	GAACAGTAGTCCCGCGCT	eeeeeddddddddeeee	51	13719	13736	2021
655063	2268	2285	CGGTTGGAATTCTTTTTG	eeeeeddddddddeeee	74	13737	13754	2022
655064	2274	2291	GCTGGTCGGTTGGAATTC	eeeeeddddddddeeee	65	13743	13760	2023

TABLE 14-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
655065	2283	2300	CACAAACAAGCTGGTCGG	eeeeeddddddddeeee	69	13752	13769	2025
655066	2023	2041	TTATCTGCTGCTGGCCTTT	eeeeddddddddeeeeee	50	13492	13510	2032
655067	2026	2044	TTGTTATCTGCTGCTGGCC	eeeeddddddddeeeeee	60	13495	13513	2033
655068	2027	2045	GTTGTTATCTGCTGCTGGC	eeeeddddddddeeeeee	65	13496	13514	2034
655069	2046	2064	CATCGCTGATTTGTCCGGG	eeeeddddddddeeeeee	71	13515	13533	2035
655070	2047	2065	ACATCGCTGATTTGTCCGG	eeeeddddddddeeeeee	65	13516	13534	2036
655071	2048	2066	CACATCGCTGATTTGTCCG	eeeeddddddddeeeeee	87	13517	13535	2037
655072	2096	2114	CAGCTCAAAGTCGACTCAT	eeeeddddddddeeeeee	75	13565	13583	2039
655073	2249	2267	GAACAGTAGTCCCGCGCTA	eeeeddddddddeeeeee	73	13718	13736	2041
655074	2267	2285	CGGTTGGAATTCTTTTTGG	eeeeddddddddeeeeee	70	13736	13754	2042
655075	2273	2291	GCTGGTCGGTTGGAATTCT	eeeeddddddddeeeeee	65	13742	13760	2043
655076	2282	2300	CACAAACAAGCTGGTCGGT	eeeeddddddddeeeeee	65	13751	13769	2044
655077	2023	2041	TTATCTGCTGCTGGCCTTT	eeeeeddddddddeeeee	40	13492	13510	2032
655078	2026	2044	TTGTTATCTGCTGCTGGCC	eeeeeddddddddeeeee	57	13495	13513	2033
655079	2027	2045	GTTGTTATCTGCTGCTGGC	eeeeeddddddddeeeee	66	13496	13514	2034
655080	2046	2064	CATCGCTGATTTGTCCGGG	eeeeeddddddddeeeee	70	13515	13533	2035
655081	2047	2065	ACATCGCTGATTTGTCCGG	eeeeeddddddddeeeee	66	13516	13534	2036
655082	2048	2066	CACATCGCTGATTTGTCCG	eeeeeddddddddeeeee	73	13517	13535	2037
655083	2096	2114	CAGCTCAAAGTCGACTCAT	eeeeeddddddddeeeee	81	13565	13583	2039
655084	2249	2267	GAACAGTAGTCCCGCGCTA	eeeeeddddddddeeeee	65	13718	13736	2041
655085	2267	2285	CGGTTGGAATTCTTTTTGG	eeeeeddddddddeeeee	70	13736	13754	2042
655086	2273	2291	GCTGGTCGGTTGGAATTCT	eeeeeddddddddeeeee	69	13742	13760	2043
655087	2282	2300	CACAAACAAGCTGGTCGGT	eeeeeddddddddeeeee	79	13751	13769	2044
655088	2023	2042	GTTATCTGCTGCTGGCCTTT	eeeeeddddddddeeeeee	70	13492	13511	230
655089	2026	2045	GTTGTTATCTGCTGCTGGCC	eeeeeddddddddeeeeee	42	13495	13514	233
655090	2027	2046	GGTTGTTATCTGCTGCTGGC	eeeeeddddddddeeeeee	82	13496	13515	234
655091	2046	2065	ACATCGCTGATTTGTCCGGG	eeeeeddddddddeeeeee	66	13515	13534	236
655092	2047	2066	CACATCGCTGATTTGTCCGG	eeeeeddddddddeeeeee	78	13516	13535	237
655093	2048	2067	ACACATCGCTGATTTGTCCG	eeeeeddddddddeeeeee	90	13517	13536	238
655094	2096	2115	CCAGCTCAAAGTCGACTCAT	eeeeeddddddddeeeeee	80	13565	13584	267
655095	2249	2268	GGAACAGTAGTCCCGCGCTA	eeeeeddddddddeeeeee	84	13718	13737	1708
655096	2267	2286	TCGGTTGGAATTCTTTTTGG	eeeeeddddddddeeeeee	76	13736	13755	1714
655097	2273	2292	AGCTGGTCGGTTGGAATTCT	eeeeeddddddddeeeeee	63	13742	13761	1716
655098	2282	2301	TCACAAACAAGCTGGTCGGT	eeeeeddddddddeeeeee	79	13751	13770	1719
655099	2023	2042	GTTATCTGCTGCTGGCCTTT	eeeeeddddddddeeeee	75	13492	13511	230

TABLE 14-continued

Inhibition of AGT mRNA by MOE and/or cEt containing gapmers targeting SEQ ID NO: 1 and/or 2

ISIS NO	SEQ ID: 1 Start Site	SEQ ID: 1 Stop Site	Sequence	Chemistry	% Inhibition	SEQ ID: 2 Start Site	SEQ ID 2: Stop Site	SEQ ID NO
655100	2026	2045	GTTGTTATCTGCTGCTGGCC	eeeeeeddddddddeeeee	67	13495	13514	233
655101	2027	2046	GGTTGTTATCTGCTGCTGGC	eeeeeeddddddddeeeee	78	13496	13515	234
655102	2046	2065	ACATCGCTGATTTGTCCGGG	eeeeeeddddddddeeeee	82	13515	13534	236
655103	2047	2066	CACATCGCTGATTTGTCCGG	eeeeeeddddddddeeeee	74	13516	13535	237
655104	2048	2067	ACACATCGCTGATTTGTCCG	eeeeeeddddddddeeeee	71	13517	13536	238
655105	2096	2115	CCAGCTCAAAGTCGACTCAT	eeeeeeddddddddeeeee	82	13565	13584	267
655106	2249	2268	GGAACAGTAGTCCCGCGCTA	eeeeeeddddddddeeeee	68	13718	13737	1708
655107	2267	2286	TCGGTTGGAATTCTTTTTGG	eeeeeeddddddddeeeee	79	13736	13755	1714
655108	2273	2292	AGCTGGTCGGTTGGAATTCT	eeeeeeddddddddeeeee	65	13742	13761	1716
655109	2282	2301	TCACAAACAAGCTGGTCGGT	eeeeeeddddddddeeeee	82	13751	13770	1719

Example 2: Dose-Dependent Antisense Inhibition of Human Angiotensinogen (AGT) in HepG2 Cells

[0525] Of over 2000 antisense oligonucleotides designed and tested in single dose in vitro assays described in Example 1, several of those exhibiting significant inhibition of AGT mRNA were selected and further tested at various doses in HepG2 cells. The results for exemplary antisense oligonucleotides tested in several series of experiment are presented in tables shown below.

[0526] Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.406 μM, 0.813 μM, 1.63 μM, 3.25 μM, 6.5 μM and 13.0 μM concentrations of antisense oligonucleotide, as specified in Table 15 below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells. The half maximal inhibitory concentration (IC_{50}) of each oligonucleotide is also presented. AGT mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE 15

ISIS NO	0.406 μM	0.813 μM	1.63 μM	3.25 μM	6.5 μM	13.0 μM	IC_{50} (μM)	SEQ ID NO
568550	34	36	55	68	78	83	1.3	46
568557	32	42	61	71	69	72	1.2	53
568558	30	31	54	67	72	80	1.6	54
568565	19	32	45	60	72	75	2.2	61
568572	29	17	56	53	65	63	2.9	68
568580	13	12	51	56	67	69	3	76
568589	32	46	61	69	78	88	1.1	85
568601	23	16	40	56	71	73	2.8	93
568605	37	45	61	68	76	77	1	97
568617	12	28	52	57	76	76	2.3	109
568635	21	27	40	61	82	90	2	127

TABLE 15-continued

ISIS NO	0.406 μM	0.813 μM	1.63 μM	3.25 μM	6.5 μM	13.0 μM	IC_{50} (μM)	SEQ ID NO
568637	69	82	95	94	98	97	<0.4	129
568637	15	9	35	43	59	67	4.6	129
568638	31	60	74	86	93	90	0.6	130
568640	41	47	61	84	90	97	0.8	132
568642	30	41	71	83	94	97	0.9	134
568643	33	51	74	83	92	93	0.7	135
568645	26	38	55	74	88	92	1.3	137
568646	15	37	57	72	88	94	1.4	138
568647	32	50	71	85	94	96	0.8	139
568650	44	51	70	79	87	90	0.6	142

[0527] Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 39.1 nM, 156.3 nM, 625.0 nM, 2500 nM and 10,000 nM concentrations of antisense oligonucleotide, as specified in Table 16 below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells, and are an average of two trials. The half maximal inhibitory concentration (IC_{50}) of each oligonucleotide is also presented. AGT mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE 16

ISIS NO	39.1 nM	156.3 nM	625.0 nM	2500 nM	10,000 nM	IC_{50} (μM)	SEQ ID NO
568637	-2	33	77	92	98	0.4	129
594622	15	52	84	96	97	0.3	163
594623	16	30	65	87	96	0.4	164
594624	13	37	74	92	96	0.4	129

TABLE 16-continued

ISIS NO	39.1 nM	156.3 nM	625.0 nM	2500 nM	10,000 nM	IC_{50} (μM)	SEQ ID NO
594625	14	31	74	90	95	0.4	165
594626	11	20	58	84	94	0.6	166
594627	11	36	72	93	95	0.3	167
594628	−30	4	51	78	87	1.1	168
594629	−20	−1	39	67	94	1.4	169
594630	−10	13	35	52	78	2.4	170
594631	13	13	49	81	94	0.6	171
594632	2	27	60	85	97	0.6	172

[0528] Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 312.5 nM, 625 nM, 1250 nM, 2500 nM and 5000 nM concentrations of antisense oligonucleotide, as specified in Tables 17 and 18 below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells, and are an average of two trials. The half maximal inhibitory concentration (IC_{50}) of each oligonucleotide is also presented. AGT mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE 17

ISIS NO	312.5 nM	625 nM	1250 nM	2500 nM	5000 nM	IC_{50} (μM)	SEQ ID NO
568637	51	71	84	89	93	0.2	129
594625	52	73	84	92	95	0.2	165
611933	−7	7	5	1	−2	>5	313
612024	22	39	48	73	80	1.1	46
612025	21	15	36	59	64	2.2	654
612058	49	52	53	72	74	0.5	53
612063	22	38	56	63	65	1.3	689
612077	35	37	45	67	74	1.1	702
612101	32	59	68	83	93	0.6	725
612102	53	67	80	85	91	0.2	726
612104	41	51	50	72	83	0.7	728
612117	25	47	56	68	73	1.0	738
612134	40	43	49	67	79	0.9	894
612147	30	48	74	76	82	0.7	905
612151	33	38	51	71	81	1.0	909
612202	33	49	62	83	87	0.7	954
612315	7	33	55	72	76	1.3	779
612322	29	48	61	78	87	0.8	786
612323	42	60	82	87	91	0.4	787
612336	31	59	72	83	89	0.5	800
612344	31	39	69	76	85	0.8	808
612346	13	42	55	74	86	1.1	810
612347	29	46	71	83	90	0.7	811
612448	15	26	59	76	86	1.1	411
612491	16	14	33	29	49	8.0	452
612502	28	37	58	75	89	0.9	463
612503	44	55	75	83	91	0.4	464
612504	17	44	63	68	88	1.0	465
612505	43	50	66	76	90	0.5	466
612506	32	44	70	81	91	0.7	467
612507	24	45	49	70	81	1.0	468
612509	25	43	60	77	88	0.9	470
612514	44	41	59	79	92	0.6	475
612515	21	38	48	61	78	1.3	476
612516	38	47	74	79	93	0.6	477
612517	33	37	60	75	86	0.8	478
612519	14	16	38	54	64	2.4	480

TABLE 17-continued

ISIS NO	312.5 nM	625 nM	1250 nM	2500 nM	5000 nM	IC_{50} (μM)	SEQ ID NO
612540	38	53	76	80	91	0.5	500
612541	38	51	58	83	90	0.6	501
612542	43	61	73	83	94	0.4	502
612543	34	53	64	81	91	0.6	503
612553	44	64	78	87	91	0.3	512
612559	36	59	74	89	95	0.5	517
612560	49	57	68	80	95	0.4	518
612567	38	50	57	83	85	0.6	524
612568	32	67	73	86	92	0.5	525
612569	27	54	71	78	93	0.7	526
612615	44	64	65	70	75	0.3	825
612658	19	23	43	57	58	2.3	865
612662	39	47	62	77	75	0.6	868

TABLE 18

ISIS NO	312.5 nM	625 nM	1250 nM	2500 nM	5000 nM	IC_{50} (μM)	SEQ ID NO
568637	57	79	89	95	97	<0.3	129
594625	72	80	91	97	97	<0.3	165
610015	41	70	72	84	92	0.3	239
612129	28	40	67	71	84	0.9	889
612135	41	40	47	62	73	1.0	68
612145	22	48	54	61	65	1.3	903
612185	16	29	36	45	62	2.7	83
612239	42	57	65	66	72	0.5	966
612252	23	22	30	61	60	2.4	96
612806	52	73	67	76	73	<0.3	1011
612810	24	36	57	73	79	1.1	1015
612816	14	30	24	51	61	2.9	1021
612819	31	40	53	64	67	1.2	1024
612901	40	44	54	72	80	0.8	1080
612906	4	9	21	37	39	8.8	1085

[0529] Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 37 nM, 111 nM, 333 nM, 1,000 nM and 3,000 nM concentrations of antisense oligonucleotide, as specified in Table 19 below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells, and are an average of two trials. The half maximal inhibitory concentration (IC_{50}) of each oligonucleotide is also presented. AGT mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE 19

ISIS NO	37 nM	111 nM	333 nM	1000 nM	3000 nM	IC_{50} (μM)	SEQ ID NO
568637	10	59	74	88	98	0.1	129
594622	46	58	65	89	96	0.1	163
594625	24	46	68	85	94	0.1	165
594628	13	48	53	74	91	0.2	168
609089	44	27	61	72	92	0.2	184
609094	−3	41	67	87	96	0.2	130
622210	18	36	51	74	95	0.3	180

TABLE 19-continued

ISIS NO	37 nM	111 nM	333 nM	1000 nM	3000 nM	IC_{50} (μM)	SEQ ID NO
622212	38	51	85	88	97	0.1	182
622213	41	51	69	89	97	0.1	164
622215	36	40	61	84	89	0.1	165
622216	18	51	60	85	96	0.2	183
622220	48	51	63	81	90	0.1	186
622221	28	46	62	76	88	0.2	167
622224	8	32	55	77	91	0.3	130
622238	45	33	60	67	91	0.2	132

[0530] Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 12.3 nM, 37 nM, 111 nM, 333 nM, 1,000 nM and 3,000 nM concentrations of antisense oligonucleotide, as specified in Table 20 below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells, and are an average of two trials. The half maximal inhibitory concentration (IC_{50}) of each oligonucleotide is also presented. AGT mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE 20

ISIS NO	12.3 nM	37 nM	111 nM	333 nM	1000 nM	3000 nM	IC_{50} (μM)	SEQ ID NO
568637	−5	6	24	69	86	95	0.2	129
594622	1	−1	32	63	88	97	0.2	163
594624	9	0	54	57	87	92	0.2	129
594625	14	11	6	47	81	93	0.3	165
609086	26	3	35	72	92	97	0.1	181
609087	−9	16	38	63	81	90	0.2	182
609088	11	9	44	61	86	97	0.2	183
609091	3	7	27	58	75	92	0.3	186
609095	−4	−15	20	67	88	98	0.3	189
622211	21	7	3	50	85	94	0.3	181
622214	8	19	39	69	89	96	0.1	129
622225	5	19	30	59	82	97	0.2	189

[0531] Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.33 μM, 1.0 μM, 3.0 μM and 9.0 μM concentrations of antisense oligonucleotide, as specified in Table 21 below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells. The half maximal inhibitory concentration (IC_{50}) of each oligonucleotide is also presented. AGT mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE 21

ISIS NO	0.33 μM	1.0 μM	3.0 μM	9.0 μM	IC_{50} (μM)	SEQ ID NO
568637	74	74	95	97	0.03	129
568637	52	80	89	95	0.2	129
610015	47	64	90	92	0.4	239
610015	25	51	79	92	1	239
654385	2	38	71	82	1.9	1827
654394	24	47	80	90	1.1	1836
654401	27	57	85	89	0.8	1843
654402	11	38	72	90	1.5	1844
654404	16	47	79	82	1.3	1846
654444	18	48	78	91	1.2	1886
654451	34	59	83	93	0.7	1893
654452	35	50	82	92	0.8	1894
654472	23	49	79	93	1	1914
654481	22	53	79	93	1	1923
654483	28	63	80	95	0.8	1925
654490	31	55	68	95	0.9	1932
654559	16	44	75	92	1.3	1979
654566	20	40	78	84	1.3	1986
654568	37	58	81	92	0.6	1988
654570	19	39	71	89	1.4	1990
654691	31	57	86	92	0.7	2003
654707	32	72	90	95	0.5	2015
654737	31	69	83	96	0.6	2018
654740	36	67	82	94	0.5	2020
654768	29	64	82	95	0.7	2027
654771	43	72	84	89	0.3	2028
654815	25	51	78	91	1	2035
654817	23	55	89	95	0.9	2037
654832	12	46	75	94	1.3	2044
654843	20	57	85	87	1	2037
654846	26	57	84	92	0.8	2039
654999	48	63	82	93	0.4	238
655002	29	64	86	94	0.7	267
655049	38	67	88	95	0.5	2018
655071	47	64	84	96	0.4	2037
655093	35	71	86	93	0.5	238
655095	28	54	80	86	0.9	1708
655102	42	54	77	90	0.6	236

[0532] Cells were plated at a density of 20,000 cells per well and transfected using electroporation with 0.44 μM, 1.33 μM, 4.0 μM and 12.0 μM concentrations of antisense oligonucleotide, as specified in Table 22 below. After a treatment period of approximately 16 hours, RNA was isolated from the cells and AGT mRNA levels were measured by quantitative real-time PCR. Human primer probe set RTS3721 was used to measure mRNA levels. AGT mRNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells. The half maximal inhibitory concentration (IC_{50}) of each oligonucleotide is also presented. AGT mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

TABLE 22

ISIS NO	0.44 μM	1.33 μM	4.0 μM	12.0 μM	IC_{50} (μM)	SEQ ID NO
610010	15	67	84	96	1.2	234
610010	20	64	85	97	1.1	234
610015	39	76	90	94	0.5	239
610015	43	73	91	98	0.5	239
619539	21	33	45	74	3.5	1258
619540	7	22	40	70	5.3	1259
619542	22	34	69	84	2.1	1261

TABLE 22-continued

ISIS NO	0.44 μM	1.33 μM	4.0 μM	12.0 μM	IC_{50} (μM)	SEQ ID NO
619543	29	33	59	70	2.8	1262
619574	34	43	56	80	1.8	1293
619575	20	35	59	74	2.8	1294
619580	19	37	53	79	2.8	1299
619606	24	39	48	57	5.1	1325
619751	2	20	49	77	4.0	1470
619753	6	23	57	83	3.2	1472
619754	7	22	52	72	4.1	1473
619803	74	82	87	92	<0.4	1522
619823	47	64	72	86	0.5	1542
619885	20	34	61	80	2.4	1604
619904	30	45	70	87	1.5	1623
619905	11	34	65	78	2.7	1624
619951	49	68	94	99	0.4	1667
619954	7	68	82	95	1.4	1670
619966	33	73	90	96	0.7	1682
619967	42	67	89	92	0.6	1683
619971	1	44	76	90	2.1	1687
619984	35	63	91	95	0.8	1700
619987	73	84	96	98	<0.4	1703
619988	40	71	92	95	0.6	1704
619992	42	71	90	97	0.5	1708
619998	31	64	90	98	0.8	1714
620000	29	61	82	94	1.0	1716
620003	45	77	93	98	0.4	1719
620004	52	78	93	98	0.3	1720
620008	46	72	88	96	0.4	1724
620009	61	82	96	98	<0.4	1725
620010	58	83	97	96	<0.4	1726
620013	46	77	90	98	0.4	1729
620014	26	31	76	92	1.7	1730

Example 3: Tolerability and Efficacy of Single Dose Treatment of Antisense Oligonucleotides Targeting Human AGT in Transgenic Mouse Model

[0533] A transgenic (Tg) mouse model "huAGT" was generated and the efficacy of antisense oligonucleotides was evaluated in this huAGT Tg model. Selected AGT antisense oligonucleotides from the in vitro studies were assessed in huAGT mice.

[0534] The huAGT transgenic mice were maintained on a 12-hour light/dark cycle and were fed ad libitum normal mouse chow. Animals were acclimated for at least 7 days in the research facility before initiation of the experiment. Antisense oligonucleotides (ASOs) were prepared in buffered saline (PBS) and sterilized by filtering through a 0.2 micron filter. Oligonucleotides were dissolved in 0.9% PBS for injection.

Treatment #1

[0535] Transgenic huAGT female mice, 10 weeks old, were divided into groups of 4 mice each. Eight groups received subcutaneous injections of antisense oligonucleotide at a dose of 20 mg/kg once per week over a course of 2.5 weeks (for three treatments). One group of mice received subcutaneous injections of PBS once per week for 2.5 weeks. The saline-injected group served as the control group to which oligonucleotide-treated groups were compared.

RNA Analysis, Treatment #1

[0536] On day 17, total RNA was extracted from liver and kidney of the transgenic mice for real-time PCR analysis and measurement of human AGT mRNA expression. Results are presented as percent inhibition, relative to PBS control, normalized with RIBOGREEN®. As shown in Table 23, treatment with most antisense oligonucleotides resulted in significant reduction of human AGT mRNA in comparison to the PBS control.

TABLE 23

Percent inhibition of huAGT mRNA in transgenic mouse liver and kidney relative to PBS control

ISIS NO	liver	kidney	SEQ ID NO
568605	42	20	97
568637	77	39	129
568638	56	11	130
568640	38	49	132
568642	0	7	134
568643	41	8	135
568647	49	32	139
568650	34	13	142

Plasma Chemistry Markers, Treatment #1

[0537] To evaluate the effect of antisense oligonucleotides on liver and kidney function, plasma levels of transaminases, total bilirubin and blood urea nitrogen (BUN) were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.). The results are presented in Table 24. Antisense oligonucleotides causing changes in the levels of any of the liver or kidney function markers outside the expected range for antisense oligonucleotides were excluded from further studies.

TABLE 24

Plasma chemistry markers in female transgenic huAGT mice

ISIS NO	ALT (U/L)	AST (U/L)	T. Bilirubin (mg/dL)	BUN (mg/dL)
PBS	22	54	0.18	28
568605	40	82	0.19	28
568637	30	57	0.19	30
568638	39	67	0.21	27
568640	78	141	0.28	31
568642	127	227	0.39	25
568643	35	66	0.16	31
568647	26	46	0.18	27
568650	71	105	0.18	27

Body and Organ Weights, Treatment #1

[0538] Body weights of transgenic mice were measured at day 15 and the average body weight for each group is presented in the table below. Liver, spleen and kidney weights were measured at the end of the study, and are presented in Table 25. Antisense oligonucleotides that caused any changes in organ weights outside the expected range for antisense oligonucleotides were excluded from further studies.

TABLE 25

Body and organ weights (in grams)

ISIS NO	body (g)	kidney (g)	liver (g)	spleen (g)
PBS	18.8	0.3	0.9	0.08
568605	19.0	0.2	1.0	0.09

TABLE 25-continued

Body and organ weights (in grams)

ISIS NO	body (g)	kidney (g)	liver (g)	spleen (g)
568637	19.3	0.3	1.0	0.08
568638	20.5	0.3	0.9	0.11
568640	19.7	0.3	1.0	0.09
568642	19.3	0.3	1.0	0.08
568643	19.9	0.3	1.0	0.09
568647	20.6	0.3	1.0	0.09
568650	20.0	0.3	1.0	0.09

Treatment #2

[0539] Groups of two huAGT mice each received subcutaneous injections of antisense oligonucleotide at doses of 25 mg/kg/wk over the course of two weeks. One group of huAGT mice received subcutaneous injections of PBS as the control group to which oligonucleotide-treated groups were compared.

RNA Analysis, Treatment #2

[0540] On day 10, total RNA was extracted from livers of the transgenic mice for real-time PCR analysis and measurement of human AGT mRNA expression. The results were averaged for each group of two mice, and are presented as percent inhibition, relative to PBS control, normalized with RIBOGREEN®. As shown in Table 26, treatment with most antisense antisense oligonucleotides resulted in significant reduction of human AGT mRNA in comparison to the PBS control.

TABLE 26

Percent inhibition of human AGT mRNA in the transgenic mouse liver relative to the PBS control

ISIS NO	% inhibit	SEQ ID NO
568637	96	129
610010	66	234
610015	29	239
619967	59	1683
619984	56	1700
619987	25	1703
619988	38	1704
619992	70	1708
619998	75	1714
620000	75	1716
620003	56	1719
620004	27	1720
620008	4	1724
620009	41	1725
620010	72	1726
620013	65	1729

Plasma Chemistry Markers, Treatment #2

[0541] To evaluate the effect of antisense oligonucleotides on liver function, plasma levels of transaminases were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.). The results were averaged for each group of two mice, and are presented in Table 27. Antisense oligonucleotides causing changes in the levels of any of the liver function markers outside the expected range for antisense oligonucleotides were excluded from further studies.

TABLE 27

Plasma chemistry markers in female transgenic huAGT mice

ISIS NO	ALT (U/L)	AST (U/L)
PBS	29	58
568637	29	82
610010	28	72
610015	71	103
619967	58	179
619984	23	41
619987	24	39
619988	29	107
619992	26	43
619998	25	71
620000	31	106
620003	24	46
620004	24	105
620008	24	51
620009	28	53
620010	24	38
620013	41	130

Body and Organ Weights, Treatment #2

[0542] Body weights of all treatment groups of huAGT mice were measured at day 1 and day 8, and animals were sacrificed and their livers harvested and weighed at day 10. The results were averaged for each group of two mice, and are presented in Table 28. Antisense oligonucleotides that caused any changes in organ weights outside the expected range for antisense oligonucleotides were excluded from further studies.

TABLE 28

Body and liver weights (in grams)

ISIS NO	Day 1 body (g)	Day 8 body (g)	liver (g)
PBS	18.3	18.8	1.0
568637	20.0	20.5	1.1
610010	19.0	19.4	1.1
610015	19.9	20.7	1.2
619967	19.8	19.9	1.0
619984	18.9	19.3	1.0
619987	20.2	20.5	1.2
619988	17.3	18.2	0.9
619992	18.3	19.4	1.0
619998	18.8	19.0	1.0
620000	19.7	20.4	1.1
620003	19.8	20.2	1.0
620004	21.0	21.6	1.1
620008	20.0	19.8	1.0
620009	18.9	19.0	1.0
620010	18.9	19.6	1.0
620013	19.7	20.3	1.1

Treatment #3

[0543] Groups of two huAGT mice each received subcutaneous injections of antisense oligonucleotide at doses of 25 mg/kg/wk over the course of two weeks. One group of four huAGT mice received subcutaneous injections of PBS as the control group to which oligonucleotide-treated groups were compared.

RNA Analysis, Treatment #3

[0544] On day 10, total RNA was extracted from livers of the transgenic mice for real-time PCR analysis and mea-

surement of human AGT mRNA expression. The results were averaged for each group of two mice, and are presented as percent inhibition, relative to PBS control, normalized with RIBOGREEN®. As shown in Table 29, treatment with most antisense antisense oligonucleotides resulted in significant reduction of human AGT mRNA in comparison to the PBS control.

TABLE 29

Percent inhibition of human AGT mRNA in the transgenic mouse liver relative to the PBS control

ISIS NO	% inhibit	SEQ ID NO
568637	93	129
654401	63	1843
654451	43	1893
654452	48	1894
654472	69	1914
654481	0	1923
654483	58	1925
654490	80	1932
654568	70	1988
654691	81	2003
654707	32	2015
654740	0	2020
654771	0	2028
654999	76	238
655049	75	2018
655071	81	2037
655093	59	238

Plasma Chemistry Markers, Treatment #3

[0545] To evaluate the effect of antisense oligonucleotides on liver function, plasma levels of transaminases were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.). The results were averaged for each group of two mice, and are presented in Table 30. Antisense oligonucleotides causing changes in the levels of any of the liver function markers outside the expected range for antisense oligonucleotides were excluded from further studies.

TABLE 30

Plasma chemistry markers in female transgenic huAGT mice

ISIS NO	ALT (U/L)	AST (U/L)
PBS	32	44
568637	36	41
654401	34	44
654451	52	82
654452	29	54
654472	50	78
654481	35	43
654483	28	62
654490	28	75
654568	35	60
654691	32	54
654707	48	65
654740	43	55
654771	59	166
654999	31	60
655049	27	61
655071	42	67
655093	26	50

Body and Organ Weights, Treatment #3

[0546] Body weights of all treatment groups of huAGT mice were measured at day 1 and day 8, and animals were sacrificed and their livers harvested and weighed at day 10. The results were averaged for each group of two mice, and are presented in Table 31. Antisense oligonucleotides that caused any changes in weights outside the expected range for antisense oligonucleotides were excluded from further studies.

TABLE 31

Body and liver weights

ISIS NO	Day 1 body (g)	Day 8 body (g)	liver (g)
PBS	26.7	27.4	1.5
568637	28.6	29.9	1.7
654401	29.1	30.9	1.9
654451	27.0	27.4	1.4
654452	26.6	27.2	1.4
654472	29.7	30.8	1.8
654481	28.3	29.4	1.6
654483	25.8	26.4	1.3
654490	28.6	28.7	1.5
654568	28.6	29.6	1.7
654691	29.6	31.1	1.7
654707	29.3	30.4	1.9
654740	29.1	29.8	1.7
654771	29.1	30.3	1.7
654999	28.2	29.0	1.6
655049	29.8	32.2	1.8
655071	28.5	30.4	1.8
655093	28.0	29.7	1.6

Treatment #4

[0547] Transgenic huAGT male mice, six weeks old, were divided into groups of 3-4 mice each. Eight groups received subcutaneous injections of antisense oligonucleotide at a dose of 5 mg/kg once per week over a course of 2 weeks. One group of mice received subcutaneous injections of PBS once per week for 2 weeks. The saline-injected group served as the control group to which oligonucleotide-treated groups were compared.

RNA Analysis, Treatment #4

[0548] On day 17, total RNA was extracted from liver and kidney of the transgenic mice for real-time PCR analysis and measurement of human AGT mRNA expression. Results are presented as percent inhibition, relative to PBS control, normalized with RIBOGREEN®. As shown in Table 32, treatment with most antisense oligonucleotides resulted in significant reduction of human AGT mRNA in comparison to the PBS control.

TABLE 32

Percent inhibition of huAGT mRNA in transgenic mouse liver and kidney relative to PBS control

ISIS NO	liver	kidney	SEQ ID NO
594622	81	90	163
594623	32	55	164
594624	79	67	129
594625	91	70	165
594626	76	81	166
594627	82	88	167
594628	28	22	168
594629	17	20	169

TABLE 32-continued

Percent inhibition of huAGT mRNA in transgenic mouse liver and kidney relative to PBS control

ISIS NO	liver	kidney	SEQ ID NO
594630	37	35	170
594631	45	75	171
594632	50	51	172
568637	67	54	129

Plasma Chemistry Markers, Treatment #4

[0549] On day 15, to evaluate the effect of antisense oligonucleotides on liver and kidney function, plasma levels of transaminases, total bilirubin and blood urea nitrogen (BUN) were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.). The results are presented in Table 33. Antisense oligonucleotides causing changes in the levels of any of the liver function markers outside the expected range for antisense oligonucleotides were excluded from further studies.

TABLE 33

Plasma chemistry markers in female transgenic huAGT mice

ISIS NO	ALT (U/L)	AST (U/L)	T. Bilirubin (mg/dL)	BUN (mg/dL)
PBS	77	118	0.18	40
594622	71	152	0.24	34
594623	57	92	0.18	36
594624	53	72	0.14	40
594625	92	116	0.17	36
594626	43	68	0.15	37
594627	50	67	0.17	35
594628	86	210	0.24	34
594629	55	68	0.16	31
594630	55	59	0.16	32
594631	32	44	0.15	36
594632	58	59	0.15	35
568637	110	371	0.22	31

Body and Organ Weights, Treatment #4

[0550] Body weights of transgenic mice were measured at days 1, 8 and 13 and the averages for each group are presented in the table below. On day 15, liver, spleen and kidney weights were also measured, and are presented in Table 34. Antisense oligonucleotides that caused any changes in weights outside the expected range for antisense oligonucleotides were excluded from further studies.

TABLE 34

Body and organ weights (in grams)

ISIS	body (g)			kidney	liver	spleen
NO	Day 1	Day 8	Day 13	(g)	(g)	(g)
PBS	20.4	21.6	21.5	0.3	1.2	0.08
594622	18.5	21.6	21.6	0.3	1.5	0.11
594623	18.1	20.4	20.3	0.3	1.2	0.06
594624	19.8	22.8	22.6	0.3	1.3	0.08
594625	20.3	22.2	22.1	0.3	1.3	0.06
594626	21.6	22.9	22.7	0.3	1.2	0.07

TABLE 34-continued

Body and organ weights (in grams)

ISIS	body (g)			kidney	liver	spleen
NO	Day 1	Day 8	Day 13	(g)	(g)	(g)
594627	21.9	22.8	22.7	0.3	1.2	0.07
594628	20.6	22.2	21.9	0.3	1.2	0.07
594629	20.8	22.1	22.0	0.3	1.2	0.07
594630	22.2	24.0	23.7	0.3	1.2	0.08
594631	20.2	21.9	21.6	0.3	1.1	0.07
594632	21.3	22.5	22.4	0.3	1.3	0.07
568637	20.1	21.4	21.5	0.3	1.2	0.05

Example 4: Tolerability and Efficacy of Multiple Dose Treatment of Antisense Oligonucleotides Targeting Human AGT in Transgenic Mouse Model

[0551] Selected AGT antisense oligonucleotides from the single dose studies in huAGT transgenic mice were further assessed in dose response studies in huAGT transgenic mice.

[0552] The huAGT transgenic mice were maintained on a 12-hour light/dark cycle and were fed ad libitum normal mouse chow. Animals were acclimated for at least 7 days in the research facility before initiation of the experiment. Antisense oligonucleotides (ASOs) were prepared in buffered saline (PBS) and sterilized by filtering through a 0.2 micron filter. Oligonucleotides were dissolved in 0.9% PBS for injection.

Treatment #1

[0553] For a four point dose-response study, male huAGT mice were divided into 37 groups of four mice each. 36 groups received subcutaneous injections of antisense oligonucleotide at doses of 5, 10, 25 and 50 mg/kg/week for 2.5 weeks (three doses in total). One group of huAGT mice received subcutaneous injections of saline as a control group, to which oligonucleotide-treated groups were compared.

RNA Analysis, Treatment #1

[0554] On day 17, the huAGT mice were sacrificed, and total RNA was extracted from liver and kidney for real-time PCR analysis and measurement of human AGT mRNA expression. RT-PCR results are presented as average percent inhibition relative to the saline-treated control group, and normalized with RIBOGREEN®. As shown in Table 35, treatment with the selected antisense oligonucleotides resulted in significant reduction of human AGT mRNA in comparison to the saline control.

TABLE 35

Percent inhibition of human AGT mRNA in organs of huAGT mice treated with nine lead ASOs

ISIS NO	mg/kg/wk	ED50	Liver	Kidney	SEQ ID NO
619998	50	7	96	71	1714
	25		89	59	
	10		80	69	
	5		45	47	

TABLE 35-continued

Percent inhibition of human AGT mRNA in organs of huAGT mice treated with nine lead ASOs

ISIS NO	mg/kg/wk	ED50	Liver	Kidney	SEQ ID NO
620003	50	10	80	69	1719
	25		91	68	
	10		51	52	
	5		35	58	
654451	50	9	94	56	1893
	25		81	48	
	10		36	43	
	5		11	48	
654452	50	8	82	53	1894
	25		77	59	
	10		69	62	
	5		0	54	
654472	50	5	81	41	1914
	25		82	62	
	10		51	50	
	5		46	51	
654481	50	~47	84	70	1923
	25		31	54	
	10		47	59	
	5		52	67	
654483	50	18	78	33	1925
	25		77	45	
	10		84	73	
	5		11	41	
654691	50	6	93	70	2003
	25		87	78	
	10		43	70	
	5		54	70	
654999	50	1	99	87	238
	25		95	76	
	10		74	78	
	5		69	81	

Body and Organ Weights, Treatment #1

[0555] Body weights of all treatment groups of huAGT mice were measured at days 1, 8 and 15 of the experiment. The results were averaged for each group of mice, and are presented in Table 36.

TABLE 36

Body Weight (BW) of huAGT mice treated with nine lead ASOs

		BW (grams)		
ISIS NO	mg/kg/wk	day 1	day 8	day 15
saline	n/a	28	29	29
619998	50	30	30	31
	25	29	29	30
	10	32	32	32
	5	32	32	31
620003	50	31	32	32
	25	32	32	33
	10	30	30	30
	5	32	32	32
654451	50	27	28	28
	25	28	28	28
	10	26	27	27
	5	27	28	29
654452	50	28	29	28
	25	27	28	28
	10	27	28	27
	5	28	28	29
654472	50	26	27	28
	25	27	27	28

TABLE 36-continued

Body Weight (BW) of huAGT mice treated with nine lead ASOs

		BW (grams)		
ISIS NO	mg/kg/wk	day 1	day 8	day 15
	10	25	26	27
	5	27	27	28
654481	50	28	29	29
	25	33	34	34
	10	32	33	33
	5	30	32	32
654483	50	34	36	36
	25	31	31	32
	10	29	30	30
	5	31	32	32
654691	50	29	30	30
	25	30	31	31
	10	30	31	32
	5	30	30	30
654999	50	33	33	34
	25	33	32	32
	10	31	31	31
	5	31	31	31

Treatment #2

[0556] Five potent antisense oligonucleotides targeting human AGT from previous studies (ISIS NOs. 620003, 654451, 654472, 654691 and 654999) were selected for another four point dose-response study and compared to ISIS 568637 which had been potent in vitro and potent and tolerable in single dose huAGT transgenic mice studies. In this study, huAGT mice were divided into 25 groups of three mice each. Groups received subcutaneous injections of antisense oligonucleotide at doses of 1, 4, 10 and 40 mg/kg for two injections over ten days. One group of three huAGT mice received subcutaneous injections of saline as a control group, to which oligonucleotide-treated groups were compared.

RNA Analysis, Treatment #2

[0557] On day 10, the antisense oligonucleotide treated huAGT mice were sacrificed, and total RNA was extracted from liver and kidney for real-time PCR analysis and measurement of human AGT mRNA expression. Results are presented as average percent inhibition of mRNA, relative to the PBS control group, and normalized with RIBOGREEN®. As shown in Table 37, treatment with the antisense oligonucleotides resulted in significant reduction of human AGT mRNA in comparison to the saline control.

TABLE 37

Percent inhibition of human AGT mRNA in organs of huAGT mice treated with five lead ASOs

ISIS NO	mg/kg	ED50 AGT mRNA	Liver	Kidney	SEQ ID NO
568637	1	4.1	16	54	129
	4		42	69	
	10		82	82	
	40		96	90	
620003	1	9.5	22	25	1719
	4		29	32	
	10		54	50	
	40		81	49	

TABLE 37-continued

Percent inhibition of human AGT mRNA in organs of huAGT mice treated with five lead ASOs

ISIS NO	mg/kg	ED50 AGT mRNA	Liver	Kidney	SEQ ID NO
654451	1	8.0	18	31	1893
	4		24	32	
	10		59	49	
	40		87	58	
654472	1	5.6	15	13	1914
	4		26	38	
	10		64	59	
	40		82	66	
654691	1	7.2	10	18	2003
	4		28	53	
	10		63	61	
	40		95	65	
654999	1	3.4	0	13	238
	4		37	62	
	10		70	65	
	40		93	67	

Plasma Chemistry Markers, Treatment #2

[0558] On day 10, plasma levels of transaminases, bilirubin and BUN were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.) to evaluate the effect of antisense oligonucleotides on liver and kidney function. The results are presented in Table 38.

TABLE 38

Plasma chemistry markers in transgenic huAGT mice

ISIS NO	mg/kg	ALT (U/L)	AST (U/L)	BUN (mg/dL)	T. Bil (mg/dL)
PBS	n/a	34	62	29	0.12
568637	1	29	50	25	0.08
	4	37	54	31	0.11
	10	44	54	29	0.12
	40	39	52	27	0.12
620003	1	32	59	27	0.16
	4	44	53	34	0.10
	10	40	60	29	0.14
	40	33	34	26	0.11
654451	1	38	49	30	0.12
	4	33	49	30	0.13
	10	35	45	29	0.11
	40	33	38	29	0.12
654472	1	39	69	28	0.17
	4	31	54	30	0.11
	10	30	70	30	0.15
	40	33	41	30	0.10
654691	1	39	79	32	0.11
	4	35	54	29	0.12
	10	34	44	32	0.12
	40	37	43	30	0.14
654999	1	34	56	31	0.11
	4	38	51	32	0.13
	10	29	53	33	0.09
	40	30	42	28	0.09

Treatment #3

[0559] Five potent antisense oligonucleotides targeting human AGT from a previous dose response study (ISIS NOs. 568637, 594622, 594624, 594625 and 594627) were selected for a three-point dose-response study. In this study, huAGT mice were divided into 16 groups of three mice each. Groups received subcutaneous injections of antisense oligonucleotide at doses of 1, 5 and 15 mg/kg for two injections over the course of a week. One group of three huAGT mice received subcutaneous injections of saline as a control group, to which oligonucleotide-treated groups were compared.

RNA Analysis, Treatment #3

[0560] On day 8, total RNA was extracted from liver and kidneys of the transgenic mice for real-time PCR analysis and measurement of human AGT mRNA expression. Results are presented as percent inhibition, relative to PBS control, normalized with RIBOGREEN®. As shown in Table 39, treatment with most antisense oligonucleotides resulted in significant reduction of human AGT mRNA in comparison to the PBS control.

TABLE 39

Percent inhibition of huAGT mRNA in transgenic mouse liver and kidney relative to PBS control

	ISIS NO	Liver ED50 AGT mRNA	mg/kg	liver	kidney	SEQ ID NO
males	594622	2.4	1	35	76	163
			5	84	89	
			15	98	92	
	594624	3.9	1	23	10	129
			5	84	70	
			15	96	83	
	594625	1.8	1	34	15	165
			5	82	59	
			15	96	76	
	594627	1.4	1	17	71	167
			5	78	87	
			15	91	91	
	568637	3.8	1	21	10	129
			5	75	49	
			15	91	74	
females	594625	1.7	1	45	77	165
			5	86	88	
			15	98	96	

Plasma Chemistry Markers, Treatment #3

[0561] On day 8, to evaluate the effect of antisense oligonucleotides on liver and kidney function, plasma levels of transaminases, total bilirubin and blood urea nitrogen (BUN) were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.). The results are presented in Table 40.

TABLE 40

Plasma chemistry markers in male and female transgenic huAGT mice

	ISIS NO	mg/kg	ALT (U/L)	AST (U/L)	BUN (mg/dL)	T. Bilirubin (mg/dL)
males		PBS	67	101	34	0.26
	594622	1	34	43	36	0.14
		5	92	151	32	0.29
		15	132	167	30	0.30

TABLE 40-continued

Plasma chemistry markers in male and female transgenic huAGT mice

	ISIS NO	mg/kg	ALT (U/L)	AST (U/L)	BUN (mg/dL)	T. Bilirubin (mg/dL)
	594624	1	40	57	31	0.15
		5	46	83	35	0.12
		15	37	74	32	0.15
	594625	1	74	166	33	0.23
		5	55	67	34	0.18
		15	63	89	34	0.15
	594627	1	36	96	34	0.12
		5	40	67	33	0.13
		15	57	62	30	0.13
	568637	1	38	69	33	0.14
		5	33	48	32	0.15
		15	74	81	28	0.14
females		PBS	39	61	33	0.17
	594625	1	53	91	29	0.22
		5	276	304	28	0.25
		15	60	77	29	0.21

Body and Organ Weights, Treatment #3

[0562] Body weights of transgenic mice were measured at days 1, 8 and 13 and the averages for each group are presented in the table below. On day 15, liver, spleen and kidney weights were also measured, and are presented in Table 41.

TABLE 41

Body and organ weights (in grams)

	ISIS		body (g)		kidney	liver	spleen
	NO	mg/kg	Day 1	Day 6	(g)	(g)	(g)
males		PBS	23.6	23.9	0.33	1.3	0.08
	594622	1	23.3	23.7	0.30	1.3	0.07
		5	22.7	23.5	0.34	1.4	0.09
		15	23.9	24.4	0.33	1.8	0.08
	594624	1	24.9	26.0	0.35	1.4	0.08
		5	23.8	24.6	0.33	1.4	0.09
		15	23.7	24.2	0.33	1.4	0.07
	594625	1	23.3	23.7	0.31	1.3	0.07
		5	22.1	23.0	0.30	1.4	0.08
		15	23.8	24.6	0.32	1.6	0.09
	594627	1	22.8	23.8	0.31	1.3	0.07
		5	23.8	23.9	0.32	1.4	0.08
		15	21.2	21.7	0.29	1.4	0.07
	568637	1	22.6	23.3	0.30	1.3	0.08
		5	22.7	22.9	0.31	1.2	0.07
		15	23.0	23.6	0.31	1.4	0.08
females		PBS	17.6	18.0	0.25	1.0	0.07
	594625	1	18.2	18.4	0.24	1.0	0.08
		5	18.0	18.8	0.25	1.1	0.08
		15	19.2	19.7	0.28	1.2	0.09

Example 5: Viscosity Assessment of Nine Lead Antisense Oligonucleotides Targeting AGT

[0563] The viscosity of the 9 antisense oligonucleotides was measured with the aim of screening out antisense oligonucleotides which have a viscosity more than 40 cP. Oligonucleotides having a viscosity greater than 40 cP are considered too viscous to be administered to any subject.

[0564] Antisense oligonucleotides (32-35 mg) were weighed into a glass vial, 120 μL of water was added and the antisense oligonucleotide was dissolved into solution by heating the vial at 50° C. Part of (75 μL) the pre-heated sample was pipetted to a micro-viscometer (Cambridge). The temperature of the micro-viscometer was set to 25° C. and the viscosity of the sample was measured. Another part (20 μL) of the pre-heated sample was pipetted into 10 mL of water for UV reading at 260 nM at 85° C. (Cary UV instrument). The results are presented in Table 42 and indicate that the antisense oligonucleotides tested do not exceed a viscosity of 40 cP.

TABLE 42

Viscosity Data for ASOs targeting AGT

ISIS NO	Chemistry	cP
619998	5-10-5 MOE	29
620003	5-10-5 MOE	12
654451	5-10-5 MOE	25
654452	5-10-5 MOE	13
654472	5-10-5 MOE	11
654481	5-10-5 MOE	12
654483	5-10-5 MOE	28
654691	3-10-4 MOE	23
654999	5-8-7 MOE	34

Example 6: Tolerability of Nine Lead Antisense Oligonucleotides (ASOs) Targeting Human AGT in CD1 Mice

[0565] CD1® mice (Charles River, Mass.) are a multipurpose mice model, frequently utilized for safety and efficacy testing. The mice were treated with antisense oligonucleotides selected from studies described above and evaluated for changes in the levels of various plasma chemistry markers.

[0566] The 9 antisense oligonucleotides identified in the examples, above, were tested in CD1 mice for tolerability. The mice were divided into groups of four mice per group, and were injected subcutaneously twice a week for six weeks with 50 mg/kg of antisense oligonucleotides (100 mg/kg/week dose). One group of male CD1 mice was injected subcutaneously twice a week for six weeks with PBS. Mice were euthanized 48 hours after the last dose, and organs and plasma were harvested for further analysis.

Body and Organ Weights

[0567] Body weights of ASO-treated CD1 mice were measured weekly. On day 43, the mice were sacrificed and organs harvested and weighed. The body and organ weights in grams (g) at the end of the study are shown in Table 43.

TABLE 43

Body and organ weights (grams) of CD1 mice treated with nine lead ASOs

ISIS NO	body day 41	liver	kidney	spleen
PBS	39.1	2.2	0.7	0.2
619998	42.5	2.5	0.6	0.3
620003	38.9	2.5	0.6	0.2
654451	31.6	1.8	0.5	0.1
654452	37.1	2.3	0.6	0.2
654472	37.2	2.3	0.6	0.1
654481	37.7	2.2	0.6	0.2
654483	35.1	2.3	0.6	0.2

TABLE 43-continued

Body and organ weights (grams) of CD1 mice treated with nine lead ASOs

ISIS NO	body day 41	liver	kidney	spleen
654691	37.5	2.3	0.7	0.3
654999	35.9	2.2	0.5	0.5

Plasma Chemistry Markers

[0568] To evaluate the effect of the oligonucleotides on liver and kidney function, plasma levels of ALT (alanine transaminase) and AST (aspartate transaminase), bilirubin, creatinine, and BUN were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.).

[0569] The results were averaged for each group, and a selection of these is presented in Table 44.

TABLE 44

Plasma chemistry markers in CD1 mice

ISIS NO	Compound	ALT (U/L)	AST (U/L)	BUN (mg/dL)	Cre (mg/dL)	T. Bilirubin (mg/dL)
PBS	N/A	25	41	28	0.15	0.14
619998	5-10-5 MOE	79	124	28	0.16	0.12
620003	5-10-5 MOE	30	46	29	0.16	0.14
654451	5-10-5 MOE	46	84	22	0.08	0.16
654452	5-10-5 MOE	122	182	25	0.10	0.11
654472	5-10-5 MOE	50	65	29	0.11	0.11
654481	5-10-5 MOE	35	50	25	0.08	0.14
654483	5-10-5 MOE	107	108	25	0.09	0.17
654691	3-10-4 MOE	95	109	25	0.11	0.13
654999	5-8-7 MOE	71	135	28	0.11	0.10

[0570] In a separate study antisense compounds ISIS 568637, 594622, 594624, 594625 and 594627 were also tested in CD1 mice, but exhibited some tolerability issues and the study was terminated early.

Example 7: Tolerability of Nine Lead Antisense Oligonucleotides (ASOs) Targeting Human AGT in Sprague-Dawley Rats

[0571] Sprague-Dawley (SD) rats are a multipurpose model used for safety and efficacy evaluations. The SD rats were treated with 9 antisense oligonucleotides selected from the studies described in the Examples above and evaluated for changes in the levels of various plasma chemistry markers.

Treatment

[0572] Male SD rats were maintained on a 12-hour light/dark cycle and fed ad libitum with Purina normal rat chow. The rats were divided into groups of four rats per group, and each group was injected subcutaneously with 100 mg/kg/week for six weeks. Forty eight hours after the last dose, rats were euthanized and organs and plasma were harvested for further analysis.

Organ Weights

[0573] Liver, spleen and kidney weights of antisense oligonucleotide treated rats were measured at the end of the study. The body and organ weights are shown in grams in Table 45.

TABLE 45

Body and organ weights (grams) of Sprague-Dawley rats treated with nine lead ASOs

ISIS NO	body	kidney	liver	spleen
619998	333	3.0	12.1	2.6
620003	361	2.9	11.7	1.4
654451	316	2.7	13.4	1.5
654452	320	2.5	11.6	0.9
654472	361	3.0	13.1	1.5
654481	370	3.2	11.4	1.3
654483	366	3.3	13.5	1.2
654691	288	3.1	14.3	2.1
654999	344	2.7	11.5	2.0

Liver and Kidney Function

[0574] To evaluate the effect of the 9 antisense oligonucleotides on liver and kidney function, plasma levels of ALT (alanine transaminase) and AST (aspartate transaminase), albumin, BUN, creatinine and bilirubin were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.), and total urine protein and urine creatinine levels were measured, and the ratio of total urine protein to creatinine (P/C Ratio) was determined.

[0575] Results of each group were averaged, and a selection of these is presented in Table 46.

TABLE 46

Liver and kidney function markers in Sprague-Dawley rats

		plasma						urine		
ISIS NO	Compound	ALT (U/L)	AST (U/L)	Albumin (g/dL)	BUN (mg/dL)	Cre (mg/dL)	T. bil (mg/dL)	Cre (mg/dL)	Total protein (mg/dL)	Urine P/C Ratio
PBS	N/A	28	72	3.2	19	0.28	0.08	86	88	1
619998	5-10-5 MOE	57	125	2.8	28	0.31	0.10	76	251	3
620003	5-10-5 MOE	54	106	3.1	25	0.30	0.09	81	356	4
654401	5-10-5 MOE	69	136	3.3	25	0.36	0.12	64	343	6
654451	5-10-5 MOE	62	149	2.8	28	0.28	0.10	37	209	6
654452	5-10-5 MOE	159	196	3.0	30	0.34	0.11	44	356	8
654472	5-10-5 MOE	44	98	3.1	28	0.36	0.09	69	413	6
654481	5-10-5 MOE	43	101	3.2	26	0.37	0.09	56	323	6
654483	5-10-5 MOE	42	87	3.0	28	0.30	0.08	54	360	6

TABLE 46-continued

Liver and kidney function markers in Sprague-Dawley rats

		plasma						urine		
ISIS NO	Compound	ALT (U/L)	AST (U/L)	Albumin (g/dL)	BUN (mg/dL)	Cre (mg/dL)	T. bil (mg/dL)	Cre (mg/dL)	Total protein (mg/dL)	Urine P/C Ratio
654691	3-10-4 MOE	41	94	2.7	31	0.31	0.08	40	237	6
654999	5-8-7 MOE	40	120	2.8	28	0.30	0.09	47	335	7

Histology

[0576] Liver and kidney from antisense oligonucleotide-treated rats were microscopically examined, and no remarkable treatment-related adverse finding was observed.

[0577] In a separate study, antisense compounds ISIS 568637, 594622, 594624, 594625 and 594627 were also tested in SD rats, but exhibited some tolerability issues and the study was terminated early.

Example 8: Potency in Cynomolgus Monkey Hepatocytes of Nine Lead Antisense Oligonucleotides (ASOs) Targeting Human AGT

[0578] At the time this study was undertaken, the cynomolgus monkey genomic sequence was not available in the National Center for Biotechnology Information (NCBI) database; therefore, cross-reactivity with the cynomolgus monkey gene sequence could not be confirmed. Instead, the sequences of the antisense oligonucleotides used in the cynomolgus monkeys were compared to a rhesus monkey sequence for complementarity. It is expected that antisense oligonucleotides with complementarity to the rhesus monkey sequence are fully cross-reactive with the cynomolgus monkey sequence as well.

[0579] The human antisense oligonucleotides tested had at most 3 mismatches with the rhesus genomic sequence (GENBANK Accession NW_001109259.1 truncated from nucleotide 16090000 to Ser. No. 16/106,000, designated herein as SEQ ID NO: 7). The greater the complementarity between the human oligonucleotide and the rhesus monkey sequence, the more likely the human oligonucleotide can cross-react with the rhesus monkey sequence and the cynomolgus monkey sequence. The start and stop sites of each oligonucleotide to SEQ ID NO: 7 is presented in Table 47. "Start site" indicates the 5'-most nucleotide to which the gapmer is targeted in the rhesus monkey gene sequence.

[0580] Nine antisense oligonucleotides exhibiting significant inhibition of AGT mRNA and tolerability in previous studies were selected and tested at various doses in cryopreserved individual male cynomolgus monkey primary hepatocytes. These 9 lead antisense oligonucleotides are described in the table below.

TABLE 47

ASO complementarity to the rhesus AGT genomic sequence (SEQ ID NO: 7)

ISIS NO	Target Start Site	Target Stop Site	Sequence	Chemistry	# mismatches in Rhesus	SEQ ID NO
619998	13777	13796	TCGGTTGGAATTCTTTTTGG	5-10-5 MOE	0	1714
620003	13792	13811	TCACAAACAAGCTGGTCGGT	5-10-5 MOE	0	1719
654451	N/A	N/A	TGGAACAGTAGTCCCGCGCT	5-10-5 MOE	2	1893
654452	N/A	N/A	TTGGAACAGTAGTCCCGCGC	5-10-5 MOE	2	1894
654472	13791	13810	CACAAACAAGCTGGTCGGTT	5-10-5 MOE	0	1914
654481	13822	13841	CTCAACTTGAAAAGGGAACA	5-10-5 MOE	0	1923
654483	13825	13844	GTTCTCAACTTGAAAAGGGA	5-10-5 MOE	0	1925
654691	N/A	N/A	TCGCTGATTTGTCCGGG	3-10-4 MOE	3	2003
654999	N/A	N/A	ACACATCGCTGATTTGTCCG	5-8-7 MOE	3	238

[0581] Cynomolgus monkey primary hepatocytes were plated at a density of 35,000 cells per well and transfected using electroporation with 0.156 μM, 0.313 μM, 0.625 μM, 1.25 μM, 2.5 μM, 5.0 μM, 10.0 μM and 20.0 μM concentrations of antisense oligonucleotide, as specified in Table 48 below. After a treatment period of approximately 24 hours, the cells were washed and lysed, and RNA was isolated. Monkey AGT mRNA levels were measured by quantitative real-time PCR, using primer probe set RTS4039. AGT mRNA target levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of AGT, relative to untreated control cells.

TABLE 48

Dose response in primary hepatocytes from cynomolgus monkeys

ISIS NO	0.156 μM	0.313 μM	0.625 μM	1.25 μM	2.5 μM	5.0 μM	10.0 μM	20.0 μM	IC50 (μM)	SEQ ID NO
619998	3	1	0	13	20	31	36	64	13.9	1714
620003	9	7	15	27	30	62	76	80	3.0	1719
654451	13	24	20	30	38	42	47	29	10.6	1893
654452	13	13	25	47	44	41	62	35	>20	1894
654472	12	24	22	37	39	55	74	78	3.4	1914
654481	0	14	27	26	43	48	53	45	>20	1923
654483	25	24	39	46	61	50	56	61	3.2	1925
654691	0	12	18	0	23	18	19	24	>20	2003
654999	0	19	0	0	9	17	37	42	>20	238

[0582] Most monkey AGT mRNA levels were significantly reduced in a dose-dependent manner in antisense oligonucleotide treated cells.

Example 9: Effect of Antisense Oligonucleotides Targeting Human AGT in Cynomolgus Monkeys

[0583] In a 12-week dose response study, cynomolgus monkeys were treated with the nine antisense oligonucleotides selected from studies described in the Examples above. Antisense oligonucleotide efficacy and tolerability, as well as their pharmacokinetic profile in the liver and kidney, were evaluated.

Treatment

[0584] Prior to the study, the monkeys were kept in quarantine during which the animals were observed daily for general health. The monkeys were two to four years old and weighed 2-4 kg. Ten groups of five randomly assigned male cynomolgus monkeys each were injected subcutaneously with antisense oligonucleotide or PBS. The monkeys were dosed once a week for 12 weeks with 40 mg/kg/wk of antisense oligonucleotide for a total of 15 doses (monkeys received a loading treatment of two doses of 40 mg/kg in weeks 1 and 2). A control group of cynomolgus monkeys was injected with PBS in a similar manner and served as the control group.

[0585] During the study period, the monkeys were observed twice daily for signs of illness or distress. Any animal experiencing more than momentary or slight pain or distress due to the treatment, injury or illness was treated by the veterinary staff with approved analgesics or agents to relieve the pain after consultation with the Study Director. Any animal in poor health or in a possible moribund condition was identified for further monitoring and possible euthanasia. At the end of the 12-week study, the monkeys were sacrificed and organs removed. The protocols described in the Example were approved by the Institutional Animal Care and Use Committee (IACUC).

Body and Organ Weights

[0586] Body weight was assessed weekly, and no remarkable effects of the antisense oligonucleotides on body weight were observed. Body weight at day 77 and organ weights at day 79 were measured and are presented in Table 49 below

TABLE 49

Body and organ weights (grams) of cynomolgus monkeys treated with nine lead ASOs

	Weight (g)				
ISIS NO	body (day 77)	heart	kidney	liver	spleen
PBS	2524	10.2	12.2	51.8	2.7
619998	2520	9.3	23.4	73.9	4.4
620003	2638	9.5	14.5	67.8	3.3
654451	2488	9.4	15.9	68.5	3.0
654452	2510	9.8	14.2	60.8	3.1
654472	2623	9.8	14.8	62.1	4.0
654481	2549	9.6	14.2	59.8	4.0
654483	2525	10.0	15.8	68.6	4.0
654691	2497	8.8	15.3	67.9	4.3
654999	2590	10.1	16.6	69.1	5.7

Pharmacodynamics

[0587] Plasma, serum and urine were collected for analysis during the study. To evaluate the effect of the nine lead antisense oligonucleotides on liver and kidney function, on day 79, plasma levels of ALT (alanine transaminase) and AST (aspartate transaminase), BUN and bilirubin were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.). As shown in Table 50, no significant effects on ALT, AST, BUN and bilirubin were observed.

TABLE 50

Plasma chemistry markers in monkeys treated with antisense oligonucleotides

ISIS NO	ALT (U/L)	AST (U/L)	BUN (mg/dL)	T. Bil (mg/dL)
PBS	54	87	25.1	0.20
619998	74	98	36.6	0.16
620003	53	87	24.9	0.16
654451	61	74	30.1	0.13
654452	63	100	25.8	0.20
654472	62	77	27.1	0.16
654481	58	63	21.7	0.16
654483	70	78	25.0	0.14
654691	57	97	26.4	0.14
654999	62	111	23.2	0.14

[0588] In addition, no significant changes in ECG, blood pressure, plasma electrolytes, proteinuria, inflammatory response (e.g., CRP levels) or renal accumulation were observed. In general, the antisense oligonucleotides were well tolerated.

RNA Analysis

[0589] At the end of the study, RNA was extracted from monkey livers and kidneys for real-time PCR analysis of measurement of mRNA expression of AGT. Primer probe set RTS4039 was used, and the results for each group were averaged and presented as percent inhibition of mRNA, relative to the PBS control, normalized with RIBOGREEN®. As shown in Table 51, treatment with antisense oligonucleotides resulted in variable effects on AGT mRNA levels.

TABLE 51

Percent inhibition of AGT mRNA in the cynomolgus monkey liver relative to the PBS control

ISIS NO	% inhibit	SEQ ID NO
619998	75	1714
620003	40	1719
654451	33	1893
654452	0	1894
654472	9	1914
654481	1	1923
654483	38	1925
654691	36	2003
654999	3	238

Example 10: Tolerability of GalNAc Conjugated Antisense Oligonucleotides in CD-1 Mice

[0590] A lead candidate (ISIS 654472) 5-10-5 full phosphorothioate MOE gapmer was chosen from studies above and used as the basis for design of six 5'-Trishexylamino-(THA)-C6 $GalNAc_3$ (a.k.a. "GalNAc") -conjugated 5-10-5 MOE gapmers having the same nucleotide sequence but differences in the backbone structure, as described in Table 52 below. "s" is a phosphorothioate internucleoside linkage. "o" is a phosphodiester internucleoside linkage. "A" is an adenine nucleobase. "mC" is a 5'-methylcytosine nucleobase. "G" is a guanine nucleobase. "T" is a thymine nucleobase. "e" indicates a MOE modification. "d" indicates deoxyribose.

TABLE 52

GalNAc-conjugated ASOs and unconjugated parent ASO

ISIS NO	Chemistry notation		SEQ ID NO
654472 (parent)	mCes Aes mCes Aes Aes Ads mCds Ads Ads Gds mCds Tds Gds Gds Tds mCes Ges Ges Tes Te	(PS)	1914
757456	mCes Aes mCes Aes Aes Ads mCds Ads Ads Gds mCds Tds Gds Gds Tds mCes Ges Ges Tes Te	(PS) GalNAc	1914
757457	mCes Aeo mCeo Aeo Aeo Ads mCds Ads Ads Gds mCds Tds Gds Gds Tds mCeo Geo Ges Tes Te	(mixed backbone) GalNAc	1914
775493	mCes Aeo mCeo Aeo Aes Ads mCds Ads Ads Gds mCds Tds Gds Gds Tds mCeo Geo Ges Tes Te	(mixed backbone) GalNAc	1914
775494	mCes Aes mCeo Aeo Aes Ads mCds Ads Ads Gds mCds Tds Gds Gds Tds mCeo Geo Ges Tes Te	(mixed backbone) GalNAc	1914
775495	mCes Aeo mCes Aeo Aes Ads mCds Ads Ads Gds mCds Tds Gds Gds Tds mCeo Ges Geo Tes Te	(mixed backbone) GalNAc	1914
775496	mCes Aes mCeo Aes Aes Ads mCds Ads Ads Gds mCds Tds Gds Gds Tds mCes Geo Ges Tes Te	(mixed backbone) GalNAc	1914

[0591] For a three-point dose response study, sixteen groups of four CD1 mice each were subcutaneously injected with 10 mg/kg/week of GalNAc-conjugated antisense oligonucleotide over the course of four weeks. One group of mice was injected subcutaneously twice a week for six weeks with PBS. Body weights of ASO-treated CD1 mice were measured weekly. Mice were euthanized 48 hours after the last dose, and organs and plasma were harvested for further analysis. Plasma and urine were collected and plasma levels of transaminases, bilirubin and BUN were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.) to evaluate the effect of antisense oligonucleotides on liver and kidney function. At the end of the experiment, the livers, kidneys and spleens were harvested and weighed.

[0592] The results were averaged for each group is presented in Table 53.

TABLE 53

Plasma chemistry markers in CD1 mice treated with GalNAc-conjugated ASOs

		Weight (g)							T.
ISIS NO	mg/kg	body (day 29)	kidney	liver	spleen	ALT (U/L)	AST (U/L)	BUN (mg/dL)	Bil (mg/dL)
PBS	n/a	36.1	0.58	2.1	0.1	31	41	25.8	0.14
757456	20	39.3	0.64	2.5	0.1	28	31	24.2	0.18
	10	38.1	0.56	2.3	0.1	38	74	24.8	0.22
	5	36.1	0.54	2.1	0.1	39	59	27.7	0.21
757457	20	40.6	0.59	2.4	0.2	34	38	26.2	0.23
	10	38.7	0.60	2.2	0.1	24	29	23.1	0.26
	5	39.1	0.58	2.2	0.2	39	46	29.0	0.20
775493	20	36.3	0.59	2.0	0.1	36	51	28.3	0.21
	10	38.6	0.58	2.2	0.1	30	45	25.0	0.34
	5	37.1	0.58	2.3	0.1	23	32	26.5	0.15
775494	20	37.9	0.56	2.0	0.2	47	55	29.1	0.31
	10	36.4	0.59	2.1	0.3	25	34	25.4	0.20
	5	38.4	0.59	2.0	0.1	35	69	24.9	0.21
775495	20	39.3	0.67	2.3	0.2	42	86	23.7	0.19
	10	37.0	0.55	2.1	0.1	34	44	25.1	0.21
	5	38.1	0.62	2.2	0.1	20	28	22.5	0.28
775496	20	37.0	0.58	2.1	0.2	32	38	24.7	0.15
	10	36.4	0.59	1.9	0.2	38	42	25.0	0.24
	5	36.7	0.56	2.1	0.1	23	28	25.6	0.34

Example 11: Tolerability of GalNAc Conjugated ASOs in SD Rats

[0593] Twenty-eight male SD rats were divided into seven groups, four rats per group. Rats were subcutaneously injected with PBS as an untreated control or 10 mg/kg/week of a GalNAc conjugated antisense oligonucleotide over the course of four weeks.

[0594] Plasma and urine were collected and analyzed using an automated clinical chemistry analyzer (Hitachi Olympus AU400e, Melville, N.Y.) to evaluate the effect of antisense oligonucleotides on liver and kidney function. At the end of the experiment, the livers, kidneys and spleens were harvested and weighed.

[0595] Results are presented as average of 4 animals in each group and presented in Table 54.

TABLE 54

Tolerability of GalNAc-conjugated ASOs in SD rats

	plasma				urine			Weight (g) body			
Tx	ALT	AST	BUN	T. Bil	Cre	MTP	MTP/ Cre	(day 30)	kidney	liver	spleen
PBS	29	76	17	0.08	95.8	106.8	0.99	395	3.1	11.2	0.70
757456	42	103	19	0.12	55.5	65.3	1.24	387	3.0	11.1	0.84
757457	36	84	18	0.08	82.8	103.0	1.13	396	3.0	11.2	0.69
775493	42	102	16	0.14	66.0	85.3	1.29	407	3.3	12.3	0.78
775494	43	84	17	0.09	91.3	119.0	1.34	396	2.8	10.7	0.89
775495	37	92	17	0.10	65.0	70.3	1.01	387	3.0	10.4	0.81
775496	36	90	16	0.08	58.3	92.8	1.39	397	3.1	11.7	0.91

Example 12: Dose Response Comparison of Unconjugated and GalNAc Conjugated Antisense Oligonucleotides in Male and Female huAGT Mice

[0596] As described in previous examples, huAGT mice are useful in testing the potency of antisense oligonucleotides. A dose response comparison of the parent 5-10-5 MOE gapmer (ISIS 654472) to a GalNAc conjugated compound with the same sequence (ISIS 757456) was performed. The GalNAc conjugated antisense oligonucleotide is 8-fold more potent than the unconjugated antisense oligonucleotide as shown in Table 55.

TABLE 55

Dose response of conjugated versus unconjugated ASO

	mg/kg	ED50 AGT mRNA	Liver % inhib	Kidney % inhib	% reduction plasma AGT protein
Females	Saline	n/a	0	0	0
654472	2.5	24	15	4	0
(parent)	8		17	26	8
	25		48	38	31
	80		76	51	70

TABLE 55-continued

Dose response of conjugated versus unconjugated ASO

	mg/kg	ED50 AGT mRNA	Liver % inhib	Kidney % inhib	% reduction plasma AGT protein
757456	0.3	3	10	0	0
(GalNAc)	1		19	12	15
	3		47	0	53
	10		79	7	75
Males	Saline	n/a	0	0	2
654472	8	24	35	23	33
(parent)	25		48	36	39
	80		79	49	78
757456	1	3	19	1	17
(GalNAc)	3		50	10	39
	10		78	20	71

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agcagggaga agcccttca 19

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ggccagacca caggct 16

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tacatcactt ggccag 16

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gtgcacaggc tggaga 16

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tatttatagc tgaggg 16

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cacgatgccc tattta 16

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gccatctcag actggg 16

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accggcagga gccatc 16

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ggtactctca ttgtgg 16

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agctgctcac aggtac 16

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<400> SEQUENCE: 48

cttcccggca ttggcc 16

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agcaggtatg aaggtg 16

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cctgaattgg agcagg 16

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gatgtcttgg cctgaa 16

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agcaccagct ggtcct 16

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tgcagcgact agcacc 16

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tgtcaagttt tgcagc 16

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cggccctcaa cttgtc 16

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tcccgaccat tgcggc 16

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ttggccagca tcccga 16

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<400> SEQUENCE: 59

gcccaagaag ttggcc 16

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atatacggaa gcccaa 16

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tgcatgccat atatac 16

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gccccatagc tcactg 16

<210> SEQ ID NO 63
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ggccccatgg accacg 16

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aaagacagcc gttggg 16

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ccagggtgcc aaagac 16

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cagatagaga gaggcc 16

<210> SEQ ID NO 67
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gtgtggtcca aggctc 16

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cacccaggat tgcctg 16

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ttccaaggaa caccca 16

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agccgggagg tgcagt 16

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tccagccggg aggtgc 16

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acagcctgca gggcag 16

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ccactagcag gccctg 16

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tatcagccct gccctg 16

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tgggcctggc tatcag 16

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cgtggacagc agcagc 16

<210> SEQ ID NO 77
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<400> SEQUENCE: 77

gtgaacacgc ccacca 16

<210> SEQ ID NO 78
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 78

tcaggtgcag gcctgg 16

<210> SEQ ID NO 79
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 79

gccctgcaca aacggc 16

<210> SEQ ID NO 80
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 80

tagagagcca ggccct 16

<210> SEQ ID NO 81
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 81

cgtgggagga ccacag 16

<210> SEQ ID NO 82
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 82

tgaagtccag agagcg 16

<210> SEQ ID NO 83
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 83

gcaacatcca gttctg 16

<210> SEQ ID NO 84
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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tcttctcagc agcaac 16

<210> SEQ ID NO 85
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 85

cctgtcacag cctgca 16

<210> SEQ ID NO 86
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 86

ttccatcctg tcacag 16

<210> SEQ ID NO 87
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 87

tgtccaccca gaactc 16

<210> SEQ ID NO 88
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 88

tgttgtccac ccagaa 16

<210> SEQ ID NO 89
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 89

tgctgttgtc caccca 16

<210> SEQ ID NO 90
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 90

aggtgctgtt gtccac 16

<210> SEQ ID NO 91
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 91

ctgaggtgct gttgtc 16

<210> SEQ ID NO 92
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<212> TYPE: DNA
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<400> SEQUENCE: 92

acactgaggt gctgtt 16

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<211> LENGTH: 16
<212> TYPE: DNA
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<400> SEQUENCE: 93

cagacactga ggtgct 16

<210> SEQ ID NO 94
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<212> TYPE: DNA
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<400> SEQUENCE: 94

agcatgggaa cagaca 16

<210> SEQ ID NO 95
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<400> SEQUENCE: 95

cccatgccag agagca 16

<210> SEQ ID NO 96
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 96

actccagtgc tggaag 16

<210> SEQ ID NO 97
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<212> TYPE: DNA
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<400> SEQUENCE: 97

gtcactccag tgctgg 16

<210> SEQ ID NO 98
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<212> TYPE: DNA
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<400> SEQUENCE: 98

gtcctggatg tcactc 16

<210> SEQ ID NO 99
<211> LENGTH: 16

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<212> TYPE: DNA
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<400> SEQUENCE: 99

ccgagaagtt gtcctg 16

<210> SEQ ID NO 100
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 100

acttgagtca ccgaga 16

<210> SEQ ID NO 101
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 101

agtgaagggc acttga 16

<210> SEQ ID NO 102
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 102

ctggatcagc agcagg 16

<210> SEQ ID NO 103
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<212> TYPE: DNA
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<400> SEQUENCE: 103

ggtcagaggc atagtg 16

<210> SEQ ID NO 104
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 104

tggaaagtga gaccct 16

<210> SEQ ID NO 105
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 105

ggagttttgc tggaaa 16

<210> SEQ ID NO 106
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 106

tccagttgag ggagtt 16

<210> SEQ ID NO 107
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 107

ggagatagtt tcttca 16

<210> SEQ ID NO 108
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 108

tcaggtggat ggtccg 16

<210> SEQ ID NO 109
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 109

tgcagcacca gttggg 16

<210> SEQ ID NO 110
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 110

ataagatcct tgcagc 16

<210> SEQ ID NO 111
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 111

agcaggtcct gcaggt 16

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<210> SEQ ID NO 112
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 112

cgggcagctc agcctg 16

<210> SEQ ID NO 113
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 113

tgcagaatgg cgggca 16

<210> SEQ ID NO 114
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 114

cagctcggtg tgcaga 16

<210> SEQ ID NO 115
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 115

tttgcaggtt cagctc 16

<210> SEQ ID NO 116
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 116

ggtcattgct caattt 16

<210> SEQ ID NO 117
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 117

accctgatgc ggtcat 16

<210> SEQ ID NO 118
<211> LENGTH: 16

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 118

gcttcaagct caaaaa 16

<210> SEQ ID NO 119
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 119

tgttgggtag actctg 16

<210> SEQ ID NO 120
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 120

caggcttgtt aagctg 16

<210> SEQ ID NO 121
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 121

tccaagacct caggct 16

<210> SEQ ID NO 122
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 122

aggaatgggc ggttca 16

<210> SEQ ID NO 123
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 123

cggcccagga agtgca 16

<210> SEQ ID NO 124
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 124

gttggccacg cggccc 16

<210> SEQ ID NO 125
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 125

tgctcagcgg gttggc 16

<210> SEQ ID NO 126
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 126

ggccctggcc tcatgc 16

<210> SEQ ID NO 127
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 127

ggccttgcca ggcact 16

<210> SEQ ID NO 128
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 128

gcctcaaagg ccaggg 16

<210> SEQ ID NO 129
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 129

cgctgatttg tccggg 16

<210> SEQ ID NO 130
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 130

ggtgacacat cgctga 16

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<210> SEQ ID NO 131
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 131

gaaaaggtgg gagact 16

<210> SEQ ID NO 132
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 132

cgactcatta gaagaa 16

<210> SEQ ID NO 133
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 133

acggctgctt tccagc 16

<210> SEQ ID NO 134
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 134

ccaaggagaa acggct 16

<210> SEQ ID NO 135
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 135

cacacttaga ccaagg 16

<210> SEQ ID NO 136
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 136

tgccgctgca ggcttc 16

<210> SEQ ID NO 137
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 137

ggtgcatttg tgccgc 16

<210> SEQ ID NO 138
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 138

tggtcggttg gaattc 16

<210> SEQ ID NO 139
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 139

acaaacaagc tggtcg 16

<210> SEQ ID NO 140
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 140

cttgaaaagg gaacac 16

<210> SEQ ID NO 141
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 141

aacccaattt ttgttc 16

<210> SEQ ID NO 142
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 142

ggcaatgcaa aaatgt 16

<210> SEQ ID NO 143
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 143

tacattcaag acacta 16

<210> SEQ ID NO 144
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 144

ggtcatgttc ttacat 16

<210> SEQ ID NO 145
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 145

actacacgga ggtcat 16

<210> SEQ ID NO 146
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 146

tattacagac actaca 16

<210> SEQ ID NO 147
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 147

ggtgcttgca tctttc 16

<210> SEQ ID NO 148
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 148

cagaaattca ggtgct 16

<210> SEQ ID NO 149
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 149

ccgcattcaa acagaa 16

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<210> SEQ ID NO 150
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 150

agctatggtt ccgcat 16

<210> SEQ ID NO 151
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 151

tactaacaca agggag 16

<210> SEQ ID NO 152
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 152

ttattgtggc aagacg 16

<210> SEQ ID NO 153
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 153

ttactaatac agccca 16

<210> SEQ ID NO 154
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 154

ggtttccctg atgcag 16

<210> SEQ ID NO 155
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 155

tgatagttgg attcct 16

<210> SEQ ID NO 156
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 156

tgtggtccca acatgc 16

<210> SEQ ID NO 157
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 157

ttgaagtcct caaccc 16

<210> SEQ ID NO 158
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 158

ctcttggatg tcacag 16

<210> SEQ ID NO 159
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 159

gatggcaaat tttgtt 16

<210> SEQ ID NO 160
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 160

tgtgttactt gggtaa 16

<210> SEQ ID NO 161
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 161

gccacacagt gagggc 16

<210> SEQ ID NO 162
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 162

ctgctgctgg cctttg 16

<210> SEQ ID NO 163
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 163

gttatctgct gctggc 16

<210> SEQ ID NO 164
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 164

gggttgttat ctgctg 16

<210> SEQ ID NO 165
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 165

tcgctgattt gtccgg 16

<210> SEQ ID NO 166
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 166

catcgctgat ttgtcc 16

<210> SEQ ID NO 167
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 167

gacacatcgc tgattt 16

<210> SEQ ID NO 168
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 168

aaaggtggga gactgg 16

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<210> SEQ ID NO 169
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 169

attagaagaa aaggtg 16

<210> SEQ ID NO 170
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 170

gtcgactcat tagaag 16

<210> SEQ ID NO 171
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 171

tcaaagtcga ctcatt 16

<210> SEQ ID NO 172
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 172

cagctcaaag tcgact 16

<210> SEQ ID NO 173
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 173

gctgctggcc tttgcc 16

<210> SEQ ID NO 174
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 174

tgctgctggc ctttgc 16

<210> SEQ ID NO 175
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 175

tctgctgctg gccttt 16

<210> SEQ ID NO 176
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 176

atctgctgct ggcctt 16

<210> SEQ ID NO 177
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 177

tatctgctgc tggcct 16

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<220> FEATURE:
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<400> SEQUENCE: 178

ttatctgctg ctggcc 16

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 179

tgttatctgc tgctgg 16

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ttgttatctg ctgctg 16

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gttgttatct gctgct 16

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ggttgttatc tgctgc 16

<210> SEQ ID NO 183
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atcgctgatt tgtccg 16

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acatcgctga tttgtc 16

<210> SEQ ID NO 185
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cacatcgctg atttgt 16

<210> SEQ ID NO 186
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<400> SEQUENCE: 186

acacatcgct gatttg 16

<210> SEQ ID NO 187
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tgacacatcg ctgatt 16

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gtgacacatc gctgat 16

<210> SEQ ID NO 189
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gggtgacaca tcgctg 16

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aaaaggtggg agactg 16

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agaaaaggtg ggagac 16

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tagaagaaaa ggtggg 16

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ttagaagaaa aggtgg 16

<210> SEQ ID NO 194
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cattagaaga aaaggt 16

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<400> SEQUENCE: 195

tcattagaag aaaagg 16

<210> SEQ ID NO 196
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<400> SEQUENCE: 196

ctcattagaa gaaaag 16

<210> SEQ ID NO 197
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<400> SEQUENCE: 197

actcattaga agaaaa 16

<210> SEQ ID NO 198
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<400> SEQUENCE: 198

gactcattag aagaaa 16

<210> SEQ ID NO 199
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<400> SEQUENCE: 199

tcgactcatt agaaga 16

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<400> SEQUENCE: 200

agtcgactca ttagaa 16

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<400> SEQUENCE: 201

aagtcgactc attaga 16

<210> SEQ ID NO 202
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<212> TYPE: DNA
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<400> SEQUENCE: 202

aaagtcgact cattag 16

<210> SEQ ID NO 203
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<400> SEQUENCE: 203

caaagtcgac tcatta 16

<210> SEQ ID NO 204
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<400> SEQUENCE: 204

ctcaaagtcg actcat 16

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 205

gctcaaagtc gactca 16

<210> SEQ ID NO 206
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<400> SEQUENCE: 206

agctcaaagt cgactc 16

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 207

aggccttgcc aggcactgtg 20

<210> SEQ ID NO 208
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 208

gaggccttgc caggcactgt 20

<210> SEQ ID NO 209
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 209

agaggccttg ccaggcactg 20

<210> SEQ ID NO 210
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 210

cagaggcctt gccaggcact 20

<210> SEQ ID NO 211
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 211

gcagaggcct tgccaggcac 20

<210> SEQ ID NO 212
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 212

ggcagaggcc ttgccaggca 20

<210> SEQ ID NO 213
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 213

gggcagaggc cttgccaggc 20

<210> SEQ ID NO 214
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 214

ctttgcctca aaggccaggg 20

<210> SEQ ID NO 215
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 215

cctttgcctc aaaggccagg 20

<210> SEQ ID NO 216
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 216

gcctttgcct caaaggccag 20

<210> SEQ ID NO 217
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 217

ggcctttgcc tcaaaggcca 20

<210> SEQ ID NO 218
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 218

tggcctttgc ctcaaaggcc 20

<210> SEQ ID NO 219
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 219

ctggcctttg cctcaaaggc 20

<210> SEQ ID NO 220
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 220

gctggccttt gcctcaaagg 20

<210> SEQ ID NO 221
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 221

tgctggcctt tgcctcaaag 20

<210> SEQ ID NO 222
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 222

ctgctggcct ttgcctcaaa 20

<210> SEQ ID NO 223
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 223

gctgctggcc tttgcctcaa 20

<210> SEQ ID NO 224
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 224

tgctgctggc ctttgcctca 20

<210> SEQ ID NO 225
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 225

ctgctgctgg cctttgcctc 20

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<210> SEQ ID NO 226
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 226

tctgctgctg gcctttgcct 20

<210> SEQ ID NO 227
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 227

atctgctgct ggcctttgcc 20

<210> SEQ ID NO 228
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 228

tatctgctgc tggcctttgc 20

<210> SEQ ID NO 229
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 229

ttatctgctg ctggcctttg 20

<210> SEQ ID NO 230
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 230

gttatctgct gctggccttt 20

<210> SEQ ID NO 231
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 231

tgttatctgc tgctggcctt 20

<210> SEQ ID NO 232
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 232

ttgttatctg ctgctggcct 20

<210> SEQ ID NO 233
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 233

gttgttatct gctgctggcc 20

<210> SEQ ID NO 234
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 234

ggttgttatc tgctgctggc 20

<210> SEQ ID NO 235
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 235

gggttgttat ctgctgctgg 20

<210> SEQ ID NO 236
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 236

acatcgctga tttgtccggg 20

<210> SEQ ID NO 237
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 237

cacatcgctg atttgtccgg 20

<210> SEQ ID NO 238
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 238

acacatcgct gatttgtccg 20

<210> SEQ ID NO 239
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 239

gacacatcgc tgatttgtcc 20

<210> SEQ ID NO 240
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 240

tgacacatcg ctgatttgtc 20

<210> SEQ ID NO 241
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 241

gtgacacatc gctgatttgt 20

<210> SEQ ID NO 242
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 242

ggtgacacat cgctgatttg 20

<210> SEQ ID NO 243
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 243

gggtgacaca tcgctgattt 20

<210> SEQ ID NO 244
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 244

aagaaaaggt gggagactgg 20

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<210> SEQ ID NO 245
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 245

gaagaaaagg tgggagactg 20

<210> SEQ ID NO 246
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 246

agaagaaaag gtgggagact 20

<210> SEQ ID NO 247
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 247

tagaagaaaa ggtgggagac 20

<210> SEQ ID NO 248
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 248

ttagaagaaa aggtgggaga 20

<210> SEQ ID NO 249
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 249

attagaagaa aaggtgggag 20

<210> SEQ ID NO 250
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 250

cattagaaga aaaggtggga 20

<210> SEQ ID NO 251
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 251

tcattagaag aaaaggtggg 20

<210> SEQ ID NO 252
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 252

ctcattagaa gaaaaggtgg 20

<210> SEQ ID NO 253
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 253

actcattaga agaaaaggtg 20

<210> SEQ ID NO 254
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 254

gactcattag aagaaaaggt 20

<210> SEQ ID NO 255
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 255

cgactcatta gaagaaaagg 20

<210> SEQ ID NO 256
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 256

tcgactcatt agaagaaaag 20

<210> SEQ ID NO 257
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 257

gtcgactcat tagaagaaaa 20

<210> SEQ ID NO 258
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 258

agtcgactca ttagaagaaa 20

<210> SEQ ID NO 259
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 259

aagtcgactc attagaagaa 20

<210> SEQ ID NO 260
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 260

aaagtcgact cattagaaga 20

<210> SEQ ID NO 261
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 261

caaagtcgac tcattagaag 20

<210> SEQ ID NO 262
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 262

tcaaagtcga ctcattagaa 20

<210> SEQ ID NO 263
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 263

ctcaaagtcg actcattaga 20

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<210> SEQ ID NO 264
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 264

gctcaaagtc gactcattag 20

<210> SEQ ID NO 265
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 265

agctcaaagt cgactcatta 20

<210> SEQ ID NO 266
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 266

cagctcaaag tcgactcatt 20

<210> SEQ ID NO 267
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 267

ccagctcaaa gtcgactcat 20

<210> SEQ ID NO 268
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 268

tccagctcaa agtcgactca 20

<210> SEQ ID NO 269
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 269

ttccagctca aagtcgactc 20

<210> SEQ ID NO 270
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 270

tttccagctc aaagtcgact 20

<210> SEQ ID NO 271
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 271

ctttccagct caaagtcgac 20

<210> SEQ ID NO 272
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 272

gctttccagc tcaaagtcga 20

<210> SEQ ID NO 273
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 273

tgctttccag ctcaaagtcg 20

<210> SEQ ID NO 274
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 274

ctgctttcca gctcaaagtc 20

<210> SEQ ID NO 275
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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gctgctttcc agctcaaagt 20

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ggctgctttc cagctcaaag 20

<210> SEQ ID NO 277
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<220> FEATURE:
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<400> SEQUENCE: 277

cggctgcttt ccagctcaaa 20

<210> SEQ ID NO 278
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<400> SEQUENCE: 278

acggctgctt tccagctcaa 20

<210> SEQ ID NO 279
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<213> ORGANISM: Artificial sequence
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aacggctgct ttccagctca 20

<210> SEQ ID NO 280
<211> LENGTH: 20
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aaacggctgc tttccagctc 20

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gaaacggctg ctttccagct 20

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agaaacggct gctttccagc 20

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gagaaacggc tgctttccag 20

<210> SEQ ID NO 284
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ggagaaacgg ctgctttcca 20

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<400> SEQUENCE: 285

cctgctgccc gctcat 16

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ctgaccctgc tgcccg 16

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cacttctgac cctgct 16

<210> SEQ ID NO 288
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gtcttgctta ggcaac 16

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ggtgcagagg gcagag 16

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ccggaggtgc agaggg 16

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gcaggccgga ggtgca 16

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gacatgcagg ccggag 16

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acagggacat gcaggc 16

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aggccacagg gacatg 16

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<400> SEQUENCE: 295

ccaagaggcc acaggg 16

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<400> SEQUENCE: 296

taccccaag aggcca 16

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<400> SEQUENCE: 297

agatgtaccc ccaaga 16

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<400> SEQUENCE: 298

ccgggagatg tacccc 16

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cagccccggg agatgt 16

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<400> SEQUENCE: 300

ccttctgacc cagccc 16

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<400> SEQUENCE: 301

ccaggccttc tgaccc 16

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<400> SEQUENCE: 302

accacccagg ccttct 16

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<400> SEQUENCE: 303

ggccaaccac ccaggc 16

<210> SEQ ID NO 304
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 304

cctgaggcca accacc 16

<210> SEQ ID NO 305
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<212> TYPE: DNA
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<400> SEQUENCE: 305

gacagcctga ggccaa 16

<210> SEQ ID NO 306
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 306

tgtgtgacag cctgag 16

<210> SEQ ID NO 307
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 307

ctaggtgtgt gacagc 16

<210> SEQ ID NO 308
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 308

tctccctagg tgtgtg 16

<210> SEQ ID NO 309
<211> LENGTH: 16
<212> TYPE: DNA
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<400> SEQUENCE: 309

gagcatctcc ctaggt 16

<210> SEQ ID NO 310
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 310

aacgggagca tctccc 16

<210> SEQ ID NO 311
<211> LENGTH: 16
<212> TYPE: DNA
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<400> SEQUENCE: 311

ccagaaacgg gagcat 16

<210> SEQ ID NO 312
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 312

ggttcccaga aacggg 16

<210> SEQ ID NO 313
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 313

gccaaggttc ccagaa 16

<210> SEQ ID NO 314
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 314

gtttgcagga gtcggg 16

<210> SEQ ID NO 315
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 315

ccgaagtttg caggag 16

<210> SEQ ID NO 316
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 316

atttaccgaa gtttgc 16

<210> SEQ ID NO 317
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 317

tacacattta ccgaag 16

<210> SEQ ID NO 318
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 318

cgagttacac atttac 16

<210> SEQ ID NO 319
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 319

agggtcgagt tacaca 16

<210> SEQ ID NO 320
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 320

ggtgcagggt cgagtt 16

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<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 321

gagccggtgc agggtc 16

<210> SEQ ID NO 322
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 322

tgaacagagt gagccg 16

<210> SEQ ID NO 323
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 323

gtttcactgc tgaaca 16

<210> SEQ ID NO 324
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 324

tcgatgcaga gtttca 16

<210> SEQ ID NO 325
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 325

gtcttagtga tcgatg 16

<210> SEQ ID NO 326
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 326

cttccaggaa gtctta 16

<210> SEQ ID NO 327
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 327

ggacctcttc caggaa 16

<210> SEQ ID NO 328
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 328

cgctgggacc tcttcc 16

<210> SEQ ID NO 329
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 329

actcacgctg ggacct 16

<210> SEQ ID NO 330
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 330

gcgacactca cgctgg 16

<210> SEQ ID NO 331
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 331

cagaagcgac actcac 16

<210> SEQ ID NO 332
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 332

gatgccagaa gcgaca 16

<210> SEQ ID NO 333
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 333

ggacagatgc cagaag 16

<210> SEQ ID NO 334
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 334

cagaaggaca gatgcc 16

<210> SEQ ID NO 335
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 335

ctggccagaa ggacag 16

<210> SEQ ID NO 336
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 336

acaggctggc cagaag 16

<210> SEQ ID NO 337
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 337

agaccacagg ctggcc 16

<210> SEQ ID NO 338
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 338

tggccagacc acaggc 16

<210> SEQ ID NO 339
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 339

tcacttggcc agacca 16

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<210> SEQ ID NO 340
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 340

ttacatcact tggcca 16

<210> SEQ ID NO 341
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 341

gagggttaca tcactt 16

<210> SEQ ID NO 342
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 342

gagaggaggg ttacat 16

<210> SEQ ID NO 343
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 343

tgcctgtgca caggct 16

<210> SEQ ID NO 344
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 344

caggctgcct gtgcac 16

<210> SEQ ID NO 345
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 345

gttcccaggc tgcctg 16

<210> SEQ ID NO 346
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 346

gagctgttcc caggct 16

<210> SEQ ID NO 347
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 347

ggatggagct gttccc 16

<210> SEQ ID NO 348
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 348

tgccctattt atagct 16

<210> SEQ ID NO 349
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 349

ctccaagacc tcaggc 16

<210> SEQ ID NO 350
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 350

cacctccaag acctca 16

<210> SEQ ID NO 351
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 351

ggtcacctcc aagacc 16

<210> SEQ ID NO 352
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 352

cagggtcacc tccaag 16

<210> SEQ ID NO 353
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 353

gttcagggtc acctcc 16

<210> SEQ ID NO 354
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 354

gcggttcagg gtcacc 16

<210> SEQ ID NO 355
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 355

gaatgggcgg ttcagg 16

<210> SEQ ID NO 356
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 356

caggaatggg cggttc 16

<210> SEQ ID NO 357
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 357

aaacaggaat gggcgg 16

<210> SEQ ID NO 358
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 358

cagcaaacag gaatgg 16

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<210> SEQ ID NO 359
<211> LENGTH: 16
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 359

tacacagcaa acagga 16

<210> SEQ ID NO 360
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 360

gatcatacac agcaaa 16

<210> SEQ ID NO 361
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 361

tttgatcata cacagc 16

<210> SEQ ID NO 362
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 362

cgctttgatc atacac 16

<210> SEQ ID NO 363
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 363

ggaagtgcag ggcagt 16

<210> SEQ ID NO 364
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 364

acgcggccca ggaagt 16

<210> SEQ ID NO 365
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<212> TYPE: DNA
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<400> SEQUENCE: 365

gccacgcggc ccagga 16

<210> SEQ ID NO 366
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 366

ttggccacgc ggccca 16

<210> SEQ ID NO 367
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 367

gggttggcca cgcggc 16

<210> SEQ ID NO 368
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 368

agcgggttgg ccacgc 16

<210> SEQ ID NO 369
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 369

ctcagcgggt tggcca 16

<210> SEQ ID NO 370
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 370

gtgctcagcg ggttgg 16

<210> SEQ ID NO 371
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 371

gctgtgctca gcgggt 16

<210> SEQ ID NO 372
<211> LENGTH: 16
<212> TYPE: DNA
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<400> SEQUENCE: 372

atgctgtgct cagcgg 16

<210> SEQ ID NO 373
<211> LENGTH: 16
<212> TYPE: DNA
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<400> SEQUENCE: 373

catgctgtgc tcagcg 16

<210> SEQ ID NO 374
<211> LENGTH: 16
<212> TYPE: DNA
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<400> SEQUENCE: 374

tcatgctgtg ctcagc 16

<210> SEQ ID NO 375
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 375

ctcatgctgt gctcag 16

<210> SEQ ID NO 376
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<212> TYPE: DNA
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<400> SEQUENCE: 376

gcctcatgct gtgctc 16

<210> SEQ ID NO 377
<211> LENGTH: 16
<212> TYPE: DNA
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tggcctcatg ctgtgc 16

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<212> TYPE: DNA
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<220> FEATURE:
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ctggcctcat gctgtg 16

<210> SEQ ID NO 379
<211> LENGTH: 16
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ccctggcctc atgctg 16

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<400> SEQUENCE: 380

gccctggcct catgct 16

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<400> SEQUENCE: 381

ggcactgtgt tctggg 16

<210> SEQ ID NO 382
<211> LENGTH: 16
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<400> SEQUENCE: 382

aggccttgcc aggcac 16

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ggcagaggcc ttgcca 16

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tgcctcaaag gccagg 16

<210> SEQ ID NO 385
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<400> SEQUENCE: 385

ttgcctcaaa ggccag 16

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tttgcctcaa aggcca 16

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<400> SEQUENCE: 387

ctttgcctca aaggcc 16

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cctttgcctc aaaggc 16

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gcctttgcct caaagg 16

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<400> SEQUENCE: 390

ggcctttgcc tcaaag 16

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<400> SEQUENCE: 391

tggcctttgc ctcaaa 16

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<212> TYPE: DNA
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<400> SEQUENCE: 392

ctggcctttg cctcaa 16

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<220> FEATURE:
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<400> SEQUENCE: 393

gctggccttt gcctca 16

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<400> SEQUENCE: 394

ccagctcaaa gtcgac 16

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<400> SEQUENCE: 395

tccagctcaa agtcga 16

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<400> SEQUENCE: 396

ttccagctca aagtcg 16

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<400> SEQUENCE: 397

tttccagctc aaagtc 16

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<400> SEQUENCE: 398

gctttccagc tcaaag 16

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<400> SEQUENCE: 399

cggctgcttt ccagct 16

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<400> SEQUENCE: 400

aacggctgct ttccag 16

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<400> SEQUENCE: 401

aaacggctgc tttcca 16

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<400> SEQUENCE: 402

gaaacggctg ctttcc 16

<210> SEQ ID NO 403
<211> LENGTH: 16

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 403

agaaacggct gctttc 16

<210> SEQ ID NO 404
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 404

gagaaacggc tgcttt 16

<210> SEQ ID NO 405
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 405

ggagaaacgg ctgctt 16

<210> SEQ ID NO 406
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 406

aggagaaacg gctgct 16

<210> SEQ ID NO 407
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 407

aaggagaaac ggctgc 16

<210> SEQ ID NO 408
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 408

caaggagaaa cggctg 16

<210> SEQ ID NO 409
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 409

accaaggaga aacggc 16

<210> SEQ ID NO 410
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 410

gaccaaggag aaacgg 16

<210> SEQ ID NO 411
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 411

agaccaagga gaaacg 16

<210> SEQ ID NO 412
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 412

tagaccaagg agaaac 16

<210> SEQ ID NO 413
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 413

ttagaccaag gagaaa 16

<210> SEQ ID NO 414
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 414

acttagacca aggaga 16

<210> SEQ ID NO 415
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 415

cacttagacc aaggag 16

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<210> SEQ ID NO 416
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 416

acacttagac caagga 16

<210> SEQ ID NO 417
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<212> TYPE: DNA
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<400> SEQUENCE: 417

agcacactta gaccaa 16

<210> SEQ ID NO 418
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 418

cagcacactt agacca 16

<210> SEQ ID NO 419
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 419

gcagcacact tagacc 16

<210> SEQ ID NO 420
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 420

tgcagcacac ttagac 16

<210> SEQ ID NO 421
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 421

atgcagcaca cttaga 16

<210> SEQ ID NO 422
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 422

catgcagcac acttag 16

<210> SEQ ID NO 423
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 423

ccatgcagca cactta 16

<210> SEQ ID NO 424
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 424

tccatgcagc acactt 16

<210> SEQ ID NO 425
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 425

ctccatgcag cacact 16

<210> SEQ ID NO 426
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 426

actccatgca gcacac 16

<210> SEQ ID NO 427
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 427

cactccatgc agcaca 16

<210> SEQ ID NO 428
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 428

tcactccatg cagcac 16

<210> SEQ ID NO 429
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 429

gctgcaggct tctact 16

<210> SEQ ID NO 430
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 430

cgctgcaggc ttctac 16

<210> SEQ ID NO 431
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 431

ccgctgcagg cttcta 16

<210> SEQ ID NO 432
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 432

gccgctgcag gcttct 16

<210> SEQ ID NO 433
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 433

gtgccgctgc aggctt 16

<210> SEQ ID NO 434
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 434

tgtgccgctg caggct 16

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<210> SEQ ID NO 435
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 435

ttgtgccgct gcaggc 16

<210> SEQ ID NO 436
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 436

tttgtgccgc tgcagg 16

<210> SEQ ID NO 437
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 437

atttgtgccg ctgcag 16

<210> SEQ ID NO 438
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 438

catttgtgcc gctgca 16

<210> SEQ ID NO 439
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 439

gcatttgtgc cgctgc 16

<210> SEQ ID NO 440
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 440

tgcatttgtg ccgctg 16

<210> SEQ ID NO 441
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 441

gtgcatttgt gccgct 16

<210> SEQ ID NO 442
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 442

aggtgcattt gtgccg 16

<210> SEQ ID NO 443
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 443

gaggtgcatt tgtgcc 16

<210> SEQ ID NO 444
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 444

ggaggtgcat ttgtgc 16

<210> SEQ ID NO 445
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 445

gggaggtgca tttgtg 16

<210> SEQ ID NO 446
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 446

tgggaggtgc atttgt 16

<210> SEQ ID NO 447
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 447

ctgggaggtg catttg 16

<210> SEQ ID NO 448
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 448

actgggaggt gcattt 16

<210> SEQ ID NO 449
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 449

aactgggagg tgcatt 16

<210> SEQ ID NO 450
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 450

aaactgggag gtgcat 16

<210> SEQ ID NO 451
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 451

caaactggga ggtgca 16

<210> SEQ ID NO 452
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 452

gcaaactggg aggtgc 16

<210> SEQ ID NO 453
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 453

agcaaactgg gaggtg 16

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<210> SEQ ID NO 454
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 454

acccagcaaa ctggga 16

<210> SEQ ID NO 455
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 455

aacccagcaa actggg 16

<210> SEQ ID NO 456
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 456

taaacccagc aaactg 16

<210> SEQ ID NO 457
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 457

ataaacccag caaact 16

<210> SEQ ID NO 458
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 458

ccccattctc taaaat 16

<210> SEQ ID NO 459
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 459

cccccattct ctaaaa 16

<210> SEQ ID NO 460
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 460

acccccattc tctaaa 16

<210> SEQ ID NO 461
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 461

caccccccatt ctctaa 16

<210> SEQ ID NO 462
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 462

ccacccccat tctcta 16

<210> SEQ ID NO 463
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 463

gttcttgcct ccccac 16

<210> SEQ ID NO 464
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 464

ggttcttgcc tcccca 16

<210> SEQ ID NO 465
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 465

tggttcttgc ctcccc 16

<210> SEQ ID NO 466
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 466

ctggttcttg cctccc 16

<210> SEQ ID NO 467
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 467

actggttctt gcctcc 16

<210> SEQ ID NO 468
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 468

cactggttct tgcctc 16

<210> SEQ ID NO 469
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 469

acactggttc ttgcct 16

<210> SEQ ID NO 470
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 470

aacactggtt cttgcc 16

<210> SEQ ID NO 471
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 471

aaacactggt tcttgc 16

<210> SEQ ID NO 472
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 472

taaacactgg ttcttg 16

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<210> SEQ ID NO 473
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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ctaaacactg gttctt 16

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gctaaacact ggttct 16

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cgctaaacac tggttc 16

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gcgctaaaca ctggtt 16

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cgcgctaaac actggt 16

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ccgcgctaaa cactgg 16

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cccgcgctaa acactg 16

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tcccgcgcta aacact 16

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gtcccgcgct aaacac 16

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agtcccgcgc taaaca 16

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tagtcccgcg ctaaac 16

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agtagtcccg cgctaa 16

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cagtagtccc gcgcta 16

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acagtagtcc cgcgct 16

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aacagtagtc ccgcgc 16

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gaacagtagt cccgcg 16

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ggaacagtag tcccgc 16

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tggaacagta gtcccg 16

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ttggaacagt agtccc 16

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tttggaacag tagtcc 16

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ttttggaaca gtagtc 16

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tttttggaac agtagt 16

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ctttttggaa cagtag 16

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gaattctttt tggaac 16

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ggaattcttt ttggaa 16

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tggaattctt tttgga 16

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ttggaattct ttttgg 16

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<400> SEQUENCE: 500

cggttggaat tctttt 16

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<400> SEQUENCE: 501

tcggttggaa ttcttt 16

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<400> SEQUENCE: 502

gtcggttgga attctt 16

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ggtcggttgg aattct 16

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<220> FEATURE:
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ctggtcggtt ggaatt 16

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gctggtcggt tggaat 16

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<400> SEQUENCE: 506

agctggtcgg ttggaa 16

<210> SEQ ID NO 507
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<212> TYPE: DNA
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<400> SEQUENCE: 507

aagctggtcg gttgga 16

<210> SEQ ID NO 508
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<400> SEQUENCE: 508

caagctggtc ggttgg 16

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acaagctggt cggttg 16

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<400> SEQUENCE: 510

aacaagctgg tcggtt 16

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aaacaagctg gtcggt 16

<210> SEQ ID NO 512
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<400> SEQUENCE: 512

caaacaagct ggtcgg 16

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<400> SEQUENCE: 513

cacaaacaag ctggtc 16

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<400> SEQUENCE: 514

tcacaaacaa gctggt 16

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ttcacaaaca agctgg 16

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<400> SEQUENCE: 516

tttcacaaac aagctg 16

<210> SEQ ID NO 517
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gtttcacaaa caagct 16

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<400> SEQUENCE: 518

tgtttcacaa acaagc 16

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ttgtttcaca aacaag 16

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<400> SEQUENCE: 520

agggaacact tttttg 16

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acttgaaaag ggaaca 16

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<400> SEQUENCE: 522

aacttgaaaa gggaac 16

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ctcaacttga aaaggg 16

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ttgttctcaa cttgaa 16

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<400> SEQUENCE: 525

tttgttctca acttga 16

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<400> SEQUENCE: 526

cccaattttt gttctc 16

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<400> SEQUENCE: 527

acccaatttt tgttct 16

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<212> TYPE: DNA
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<400> SEQUENCE: 528

cgaaggcaat gcaaaa 16

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 529

ccgaaggcaa tgcaaa 16

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<400> SEQUENCE: 530

accgaaggca atgcaa 16

<210> SEQ ID NO 531
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 531

aaccgaaggc aatgca 16

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<212> TYPE: DNA
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<400> SEQUENCE: 532

aaaccgaagg caatgc 16

<210> SEQ ID NO 533
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 533

caaaccgaag gcaatg 16

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<212> TYPE: DNA
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<400> SEQUENCE: 534

acaaaccgaa ggcaat 16

<210> SEQ ID NO 535
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 535

tacaaaccga aggcaa 16

<210> SEQ ID NO 536
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 536

atacaaaccg aaggca 16

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aatacaaacc gaaggc 16

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aaatacaaac cgaagg 16

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<400> SEQUENCE: 539

taaatacaaa ccgaag 16

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<400> SEQUENCE: 540

ctaaatacaa accgaa 16

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<400> SEQUENCE: 541

actaaataca aaccga 16

<210> SEQ ID NO 542
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 542

cactaaatac aaaccg 16

<210> SEQ ID NO 543
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<400> SEQUENCE: 543

gacactaaat acaaac 16

<210> SEQ ID NO 544
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 544

caagacacta aataca 16

<210> SEQ ID NO 545
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 545

tcaagacact aaatac 16

<210> SEQ ID NO 546
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 546

ttcaagacac taaata 16

<210> SEQ ID NO 547
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 547

attcaagaca ctaaat 16

<210> SEQ ID NO 548
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 548

cattcaagac actaaa 16

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<210> SEQ ID NO 549
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<212> TYPE: DNA
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<400> SEQUENCE: 549

acattcaaga cactaa 16

<210> SEQ ID NO 550
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 550

ttacattcaa gacact 16

<210> SEQ ID NO 551
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 551

cttacattca agacac 16

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<400> SEQUENCE: 552

tcttacattc aagaca 16

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 553

ttcttacatt caagac 16

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 554

atgttcttac attcaa 16

<210> SEQ ID NO 555
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<212> TYPE: DNA
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<400> SEQUENCE: 555

gtcatgttct tacatt 16

<210> SEQ ID NO 556
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 556

aggtcatgtt cttaca 16

<210> SEQ ID NO 557
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 557

gaggtcatgt tcttac 16

<210> SEQ ID NO 558
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 558

ggaggtcatg ttctta 16

<210> SEQ ID NO 559
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 559

cggaggtcat gttctt 16

<210> SEQ ID NO 560
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 560

acggaggtca tgttct 16

<210> SEQ ID NO 561
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 561

cacggaggtc atgttc 16

<210> SEQ ID NO 562
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 562

tggaggctta ttgtgg 16

<210> SEQ ID NO 563
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 563

ttggaggctt attgtg 16

<210> SEQ ID NO 564
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 564

tttggaggct tattgt 16

<210> SEQ ID NO 565
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 565

cggcttacct tctgct 16

<210> SEQ ID NO 566
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 566

cctcccggcc ttttcc 16

<210> SEQ ID NO 567
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 567

tagggtgacc actctg 16

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<210> SEQ ID NO 568
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 568

agcaaatcga ggttca 16

<210> SEQ ID NO 569
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 569

tattagttct cttcag 16

<210> SEQ ID NO 570
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 570

ccttttagct tatccc 16

<210> SEQ ID NO 571
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 571

aatctgcctt ttagct 16

<210> SEQ ID NO 572
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 572

caatctacgc tgccct 16

<210> SEQ ID NO 573
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agcaccaatc tacgct 16

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catcctggag aagtag 16

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gcatcctgga gaagta 16

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atacagccca cattcc 16

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ctgtaccatg tagtta 16

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ccacaccggg cactct 16

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cccaccacac cgggca 16

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<400> SEQUENCE: 580

ttccccacca caccgg 16

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ttcaccctgc agcttt 16

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catagtcctc accttc 16

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gtgaagatga cggctc 16

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tatgtctccc tacttc 16

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gggagtaatg gtgctc 16

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gtcctgggag taatgg 16

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gggaaccgac tgctgg 16

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cctgtgggaa ccgact 16

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cctaatctag acagtc 16

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catccgctgt tctcag 16

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ctccatccgc tgttct 16

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gactccatcc gctgtt 16

<210> SEQ ID NO 593
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tgactccatc cgctgt 16

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gctgaagtac ctggtg 16

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<400> SEQUENCE: 595

gccctcaaca cggtgc 16

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tgccctcaac acggtg 16

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<400> SEQUENCE: 597

gtcattcttc ttacat 16

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gcttccttgg agctgt 16

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<400> SEQUENCE: 599

gtgtactgca atatcg 16

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<400> SEQUENCE: 600

cactcatttc ttgtgg 16

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<400> SEQUENCE: 601

ttgtaccaca tctcac 16

<210> SEQ ID NO 602
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 602

gttctctcaa aggcct 16

<210> SEQ ID NO 603
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 603

gcagggttta gaaccc 16

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 604

tatgtaagca gggttt 16

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<220> FEATURE:
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<400> SEQUENCE: 605

aaaccagctc tcaacc 16

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 606

taagacatgc tcctgc 16

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<400> SEQUENCE: 607

acttatggca gcccaa 16

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 608

tacttatggc agccca 16

<210> SEQ ID NO 609
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 609

ccattatttg gagaca 16

<210> SEQ ID NO 610
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 610

tgccatctaa ccagat 16

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 611

gttttcagta atgccc 16

<210> SEQ ID NO 612
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 612

cgggtcacga tgccct 16

<210> SEQ ID NO 613
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 613

ccggccgggt cacgat 16

<210> SEQ ID NO 614
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 614

cccccggccg ggtcac 16

<210> SEQ ID NO 615
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 615

cttcccccgg ccgggt 16

<210> SEQ ID NO 616
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 616

cttcttcccc cggccg 16

<210> SEQ ID NO 617
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 617

cagcttcttc ccccgg 16

<210> SEQ ID NO 618
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 618

cggcagcttc ttcccc 16

<210> SEQ ID NO 619
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 619

caacggcagc ttcttc 16

<210> SEQ ID NO 620
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 620

gaacaacggc agcttc 16

<210> SEQ ID NO 621
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 621

ccagaacaac ggcagc 16

<210> SEQ ID NO 622
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 622

tagtacccag aacaac 16

<210> SEQ ID NO 623
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 623

ctgtagtacc cagaac 16

<210> SEQ ID NO 624
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 624

ctgctgtagt acccag 16

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<210> SEQ ID NO 625
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 625

cttctgctgt agtacc 16

<210> SEQ ID NO 626
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 626

acccttctgc tgtagt 16

<210> SEQ ID NO 627
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 627

catacccttc tgctgt 16

<210> SEQ ID NO 628
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 628

ccgcataccc ttctgc 16

<210> SEQ ID NO 629
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 629

cttccgcata cccttc 16

<210> SEQ ID NO 630
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 630

tcgcttccgc ataccc 16

<210> SEQ ID NO 631
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 631

tgctcgcttc cgcata 16

<210> SEQ ID NO 632
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 632

gggtgctcgc ttccgc 16

<210> SEQ ID NO 633
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 633

cggcaggagc catctc 16

<210> SEQ ID NO 634
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 634

caccggcagg agccat 16

<210> SEQ ID NO 635
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 635

tcacaccggc aggagc 16

<210> SEQ ID NO 636
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 636

ggctcacacc ggcagg 16

<210> SEQ ID NO 637
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 637

gccctcaggc tcacac 16

<210> SEQ ID NO 638
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 638

gtggccctca ggctca 16

<210> SEQ ID NO 639
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 639

cagaggatgg tggccc 16

<210> SEQ ID NO 640
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 640

cccggtcacc tgcagc 16

<210> SEQ ID NO 641
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 641

acacccggtc acctgc 16

<210> SEQ ID NO 642
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 642

tgtacacccg gtcacc 16

<210> SEQ ID NO 643
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 643

gtatgtacac ccggtc 16

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<210> SEQ ID NO 644
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 644

ggtgtatgta cacccg 16

<210> SEQ ID NO 645
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 645

tgacgaggtg gaaggg 16

<210> SEQ ID NO 646
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 646

ggatgacgag gtggaa 16

<210> SEQ ID NO 647
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 647

tgtggatgac gaggtg 16

<210> SEQ ID NO 648
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 648

cattgtggat gacgag 16

<210> SEQ ID NO 649
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 649

tctcattgtg gatgac 16

<210> SEQ ID NO 650
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 650

ctctcattgt ggatga 16

<210> SEQ ID NO 651
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 651

actctcattg tggatg 16

<210> SEQ ID NO 652
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 652

tactctcatt gtggat 16

<210> SEQ ID NO 653
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 653

gtactctcat tgtgga 16

<210> SEQ ID NO 654
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 654

caggtactct cattgt 16

<210> SEQ ID NO 655
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 655

acaggtactc tcattg 16

<210> SEQ ID NO 656
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 656

cacaggtact ctcatt 16

<210> SEQ ID NO 657
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 657

tcacaggtac tctcat 16

<210> SEQ ID NO 658
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 658

ctcacaggta ctctca 16

<210> SEQ ID NO 659
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 659

ctgctcacag gtactc 16

<210> SEQ ID NO 660
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 660

ttgccagctg ctcaca 16

<210> SEQ ID NO 661
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 661

cctttgccag ctgctc 16

<210> SEQ ID NO 662
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 662

tggcctttgc cagctg 16

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<210> SEQ ID NO 663
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 663

cattggcctt tgccag 16

<210> SEQ ID NO 664
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 664

cggcattggc ctttgc 16

<210> SEQ ID NO 665
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 665

tcccggcatt ggcctt 16

<210> SEQ ID NO 666
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 666

gcttcccggc attggc 16

<210> SEQ ID NO 667
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 667

tgggcttccc ggcatt 16

<210> SEQ ID NO 668
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 668

ctttgggctt cccggc 16

<210> SEQ ID NO 669
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 669

ggtctttggg cttccc 16

<210> SEQ ID NO 670
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 670

caggtatgaa ggtggg 16

<210> SEQ ID NO 671
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<212> TYPE: DNA
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<400> SEQUENCE: 671

gagcaggtat gaaggt 16

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ttggagcagg tatgaa 16

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<400> SEQUENCE: 673

gaattggagc aggtat 16

<210> SEQ ID NO 674
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 674

tggcctgaat tggagc 16

<210> SEQ ID NO 675
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<400> SEQUENCE: 675

tcttggcctg aattgg 16

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<212> TYPE: DNA
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atgtcttggc ctgaat 16

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gggatgtctt ggcctg 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 678

ggccttttca tccaca 16

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<212> TYPE: DNA
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<400> SEQUENCE: 679

tagggccttt tcatcc 16

<210> SEQ ID NO 680
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<212> TYPE: DNA
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<400> SEQUENCE: 680

ctgtagggcc ttttca 16

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<212> TYPE: DNA
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<400> SEQUENCE: 681

gtcctgtagg gccttt 16

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<400> SEQUENCE: 682

ctggtcctgt agggcc 16

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<400> SEQUENCE: 683

ccagctggtc ctgtag 16

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<220> FEATURE:
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<400> SEQUENCE: 684

accagctggt cctgta 16

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 685

tagcaccagc tggtcc 16

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<212> TYPE: DNA
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<400> SEQUENCE: 686

ctagcaccag ctggtc 16

<210> SEQ ID NO 687
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<212> TYPE: DNA
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<400> SEQUENCE: 687

actagcacca gctggt 16

<210> SEQ ID NO 688
<211> LENGTH: 16

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<212> TYPE: DNA
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<400> SEQUENCE: 688

gactagcacc agctgg 16

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cgactagcac cagctg 16

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gcgactagca ccagct 16

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<400> SEQUENCE: 691

agcgactagc accagc 16

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<400> SEQUENCE: 692

cagcgactag caccag 16

<210> SEQ ID NO 693
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<212> TYPE: DNA
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<400> SEQUENCE: 693

gcagcgacta gcacca 16

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<400> SEQUENCE: 694

ttgcagcgac tagcac 16

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<400> SEQUENCE: 695

tttgcagcga ctagca 16

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<212> TYPE: DNA
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<400> SEQUENCE: 696

ttttgcagcg actagc 16

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<400> SEQUENCE: 697

gttttgcagc gactag 16

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<212> TYPE: DNA
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<400> SEQUENCE: 698

agttttgcag cgacta 16

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 699

aagttttgca gcgact 16

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 700

gtcaagtttt gcagcg 16

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<210> SEQ ID NO 701
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 701

ggtgtcaagt tttgca 16

<210> SEQ ID NO 702
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 702

ttcggtgtca agtttt 16

<210> SEQ ID NO 703
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 703

gtcttcggtg tcaagt 16

<210> SEQ ID NO 704
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 704

cttgtcttcg gtgtca 16

<210> SEQ ID NO 705
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 705

caacttgtct tcggtg 16

<210> SEQ ID NO 706
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 706

cctcaacttg tcttcg 16

<210> SEQ ID NO 707
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 707

ggccctcaac ttgtct 16

<210> SEQ ID NO 708
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 708

tgcggccctc aacttg 16

<210> SEQ ID NO 709
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 709

cattgcggcc ctcaac 16

<210> SEQ ID NO 710
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<212> TYPE: DNA
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<400> SEQUENCE: 710

gaccattgcg gccctc 16

<210> SEQ ID NO 711
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 711

cccgaccatt gcggcc 16

<210> SEQ ID NO 712
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 712

catcccgacc attgcg 16

<210> SEQ ID NO 713
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 713

cagcatcccg accatt 16

<210> SEQ ID NO 714
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 714

ggccagcatc ccgacc 16

<210> SEQ ID NO 715
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 715

gttggccagc atcccg 16

<210> SEQ ID NO 716
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 716

gaagttggcc agcatc 16

<210> SEQ ID NO 717
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 717

caagaagttg gccagc 16

<210> SEQ ID NO 718
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 718

gaagcccaag aagttg 16

<210> SEQ ID NO 719
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 719

acggaagccc aagaag 16

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<210> SEQ ID NO 720
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 720

tatacggaag cccaag 16

<210> SEQ ID NO 721
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 721

atatatacgg aagccc 16

<210> SEQ ID NO 722
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 722

gccatatata cggaag 16

<210> SEQ ID NO 723
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 723

catgccatat atacgg 16

<210> SEQ ID NO 724
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 724

gtgcatgcca tatata 16

<210> SEQ ID NO 725
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 725

actgtgcatg ccatat 16

<210> SEQ ID NO 726
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 726

ctcactgtgc atgcca 16

<210> SEQ ID NO 727
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 727

tagctcactg tgcatg 16

<210> SEQ ID NO 728
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 728

ccatagctca ctgtgc 16

<210> SEQ ID NO 729
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 729

gaccacgccc catagc 16

<210> SEQ ID NO 730
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 730

atggaccacg ccccat 16

<210> SEQ ID NO 731
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 731

ccccatggac cacgcc 16

<210> SEQ ID NO 732
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 732

tggccccatg gaccac 16

<210> SEQ ID NO 733
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 733

cggtggcccc atggac 16

<210> SEQ ID NO 734
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 734

ggacggtggc cccatg 16

<210> SEQ ID NO 735
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 735

agaggacggt ggcccc 16

<210> SEQ ID NO 736
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 736

gggagaggac ggtggc 16

<210> SEQ ID NO 737
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 737

gccaaagaca gccgtt 16

<210> SEQ ID NO 738
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 738

ggtgccaaag acagcc 16

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<210> SEQ ID NO 739
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 739

cagggtgcca aagaca 16

<210> SEQ ID NO 740
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 740

aggccagggt gccaaa 16

<210> SEQ ID NO 741
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 741

tcccagatag agagag 16

<210> SEQ ID NO 742
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 742

ggctcccaga tagaga 16

<210> SEQ ID NO 743
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 743

caaggctccc agatag 16

<210> SEQ ID NO 744
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 744

gtccaaggct cccaga 16

<210> SEQ ID NO 745
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 745

gtggtccaag gctccc 16

<210> SEQ ID NO 746
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 746

tgtgtggtcc aaggct 16

<210> SEQ ID NO 747
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 747

agctgtgtgg tccaag 16

<210> SEQ ID NO 748
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 748

gtcagctgtg tggtcc 16

<210> SEQ ID NO 749
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 749

cagaggcata gtgagg 16

<210> SEQ ID NO 750
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 750

ccaggtcaga ggcata 16

<210> SEQ ID NO 751
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 751

ttgtccaggt cagagg 16

<210> SEQ ID NO 752
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 752

accttgtcca ggtcag 16

<210> SEQ ID NO 753
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 753

ccctccacct tgtcca 16

<210> SEQ ID NO 754
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 754

gagaccctcc accttg 16

<210> SEQ ID NO 755
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 755

aagtgagacc ctccac 16

<210> SEQ ID NO 756
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 756

tgctggaaag tgagac 16

<210> SEQ ID NO 757
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 757

ttttgctgga aagtga 16

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<210> SEQ ID NO 758
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 758

gagttttgct ggaaag 16

<210> SEQ ID NO 759
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 759

agggagtttt gctgga 16

<210> SEQ ID NO 760
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 760

gttgagggag ttttgc 16

<210> SEQ ID NO 761
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 761

ccagttgagg gagttt 16

<210> SEQ ID NO 762
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 762

catccagttg agggag 16

<210> SEQ ID NO 763
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 763

cttcatccag ttgagg 16

<210> SEQ ID NO 764
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 764

agatagtttc ttcatc 16

<210> SEQ ID NO 765
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 765

aggtggatgg tccggg 16

<210> SEQ ID NO 766
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 766

gtcaggtgga tggtcc 16

<210> SEQ ID NO 767
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 767

catggtcagg tggatg 16

<210> SEQ ID NO 768
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 768

gggcatggtc aggtgg 16

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<211> LENGTH: 16
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 769

ccttgcagca ccagtt 16

<210> SEQ ID NO 770
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 770

gatccttgca gcacca 16

<210> SEQ ID NO 771
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 771

taagatcctt gcagca 16

<210> SEQ ID NO 772
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 772

tcataagatc cttgca 16

<210> SEQ ID NO 773
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 773

caggtcataa gatcct 16

<210> SEQ ID NO 774
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 774

ctgcaggtca taagat 16

<210> SEQ ID NO 775
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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 775

gtcctgcagg tcataa 16

<210> SEQ ID NO 776
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<212> TYPE: DNA
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<400> SEQUENCE: 776

cgagcaggtc ctgcag 16

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<210> SEQ ID NO 777
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 777

gggcgagcag gtcctg 16

<210> SEQ ID NO 778
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 778

cctgggcgag caggtc 16

<210> SEQ ID NO 779
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<212> TYPE: DNA
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<400> SEQUENCE: 779

tggcgggcag ctcagc 16

<210> SEQ ID NO 780
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 780

gaatggcggg cagctc 16

<210> SEQ ID NO 781
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 781

gcagaatggc gggcag 16

<210> SEQ ID NO 782
<211> LENGTH: 16
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<220> FEATURE:
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<400> SEQUENCE: 782

tgtgcagaat ggcggg 16

<210> SEQ ID NO 783
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 783

cggtgtgcag aatggc 16

<210> SEQ ID NO 784
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 784

gctcggtgtg cagaat 16

<210> SEQ ID NO 785
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<212> TYPE: DNA
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<400> SEQUENCE: 785

tcagctcggt gtgcag 16

<210> SEQ ID NO 786
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 786

ggttcagctc ggtgtg 16

<210> SEQ ID NO 787
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 787

gcaggttcag ctcggt 16

<210> SEQ ID NO 788
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 788

tttttgcagg ttcagc 16

<210> SEQ ID NO 789
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 789

caatttttgc aggttc 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 790

gctcaatttt tgcagg 16

<210> SEQ ID NO 791
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 791

attgctcaat ttttgc 16

<210> SEQ ID NO 792
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 792

gtcattgctc aatttt 16

<210> SEQ ID NO 793
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<212> TYPE: DNA
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<400> SEQUENCE: 793

gcggtcattg ctcaat 16

<210> SEQ ID NO 794
<211> LENGTH: 16
<212> TYPE: DNA
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<400> SEQUENCE: 794

gatgcggtca ttgctc 16

<210> SEQ ID NO 795
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 795

cctgatgcgg tcattg 16

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<210> SEQ ID NO 796
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<212> TYPE: DNA
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<400> SEQUENCE: 796

caccctgatg cggtca 16

<210> SEQ ID NO 797
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 797

ccccaccctg atgcgg 16

<210> SEQ ID NO 798
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 798

ctcccccacc ctgatg 16

<210> SEQ ID NO 799
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 799

gttcagcacc tccccc 16

<210> SEQ ID NO 800
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 800

gctgttcagc acctcc 16

<210> SEQ ID NO 801
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 801

aatgctgttc agcacc 16

<210> SEQ ID NO 802
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 802

aaaaatgctg ttcagc 16

<210> SEQ ID NO 803
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 803

cttcaagctc aaaaaa 16

<210> SEQ ID NO 804
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 804

ccgcttcaag ctcaaa 16

<210> SEQ ID NO 805
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 805

catccgcttc aagctc 16

<210> SEQ ID NO 806
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 806

tctcatccgc ttcaag 16

<210> SEQ ID NO 807
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 807

ctctctcatc cgcttc 16

<210> SEQ ID NO 808
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 808

gctctctctc atccgc 16

<210> SEQ ID NO 809
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 809

gtgggctctc tctcat 16

<210> SEQ ID NO 810
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 810

actctgtggg ctctct 16

<210> SEQ ID NO 811
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 811

tagactctgt gggctc 16

<210> SEQ ID NO 812
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 812

tgggtagact ctgtgg 16

<210> SEQ ID NO 813
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 813

agctgttggg tagact 16

<210> SEQ ID NO 814
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 814

ttaagctgtt gggtag 16

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<210> SEQ ID NO 815
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 815

ttgttaagct gttggg 16

<210> SEQ ID NO 816
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 816

ggcttgttaa gctgtt 16

<210> SEQ ID NO 817
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 817

tcaggcttgt taagct 16

<210> SEQ ID NO 818
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 818

acctcaggct tgttaa 16

<210> SEQ ID NO 819
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 819

aagacctcag gcttgt 16

<210> SEQ ID NO 820
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 820

acacggaggt catgtt 16

<210> SEQ ID NO 821
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 821

tacacggagg tcatgt 16

<210> SEQ ID NO 822
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 822

ctacacggag gtcatg 16

<210> SEQ ID NO 823
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 823

cactacacgg aggtca 16

<210> SEQ ID NO 824
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 824

acactacacg gaggtc 16

<210> SEQ ID NO 825
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 825

gacactacac ggaggt 16

<210> SEQ ID NO 826
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 826

agacactaca cggagg 16

<210> SEQ ID NO 827
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 827

cagacactac acggag 16

<210> SEQ ID NO 828
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 828

acagacacta cacgga 16

<210> SEQ ID NO 829
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 829

tacagacact acacgg 16

<210> SEQ ID NO 830
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 830

ttacagacac tacacg 16

<210> SEQ ID NO 831
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 831

attacagaca ctacac 16

<210> SEQ ID NO 832
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 832

gtattacaga cactac 16

<210> SEQ ID NO 833
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 833

ctaaggtatt acagac 16

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<210> SEQ ID NO 834
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 834

actaaggtat tacaga 16

<210> SEQ ID NO 835
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 835

aactaaggta ttacag 16

<210> SEQ ID NO 836
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 836

aaactaaggt attaca 16

<210> SEQ ID NO 837
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 837

aaaactaagg tattac 16

<210> SEQ ID NO 838
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 838

gtggaaaaaa ctaagg 16

<210> SEQ ID NO 839
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 839

caagcatctg tggaaa 16

<210> SEQ ID NO 840
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 840

cacaagcatc tgtgga 16

<210> SEQ ID NO 841
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 841

tcacaagcat ctgtgg 16

<210> SEQ ID NO 842
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 842

atcacaagca tctgtg 16

<210> SEQ ID NO 843
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 843

aatcacaagc atctgt 16

<210> SEQ ID NO 844
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 844

gtattgttca aaaatc 16

<210> SEQ ID NO 845
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 845

cgtattgttc aaaaat 16

<210> SEQ ID NO 846
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 846

aggtgcttgc atcttt 16

<210> SEQ ID NO 847
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 847

caggtgcttg catctt 16

<210> SEQ ID NO 848
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 848

tcaggtgctt gcatct 16

<210> SEQ ID NO 849
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 849

ttcaggtgct tgcatc 16

<210> SEQ ID NO 850
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 850

attcaggtgc ttgcat 16

<210> SEQ ID NO 851
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 851

aattcaggtg cttgca 16

<210> SEQ ID NO 852
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 852

aaattcaggt gcttgc 16

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<210> SEQ ID NO 853
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 853

gaaattcagg tgcttg 16

<210> SEQ ID NO 854
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 854

agaaattcag gtgctt 16

<210> SEQ ID NO 855
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 855

acagaaattc aggtgc 16

<210> SEQ ID NO 856
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 856

cgcattcaaa cagaaa 16

<210> SEQ ID NO 857
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 857

tccgcattca aacaga 16

<210> SEQ ID NO 858
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 858

ttccgcattc aaacag 16

<210> SEQ ID NO 859
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 859

gttccgcatt caaaca 16

<210> SEQ ID NO 860
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 860

ggttccgcat tcaaac 16

<210> SEQ ID NO 861
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 861

tggttccgca ttcaaa 16

<210> SEQ ID NO 862
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 862

atggttccgc attcaa 16

<210> SEQ ID NO 863
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 863

tatggttccg cattca 16

<210> SEQ ID NO 864
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 864

ctatggttcc gcattc 16

<210> SEQ ID NO 865
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 865

gctatggttc cgcatt 16

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 866

cagctatggt tccgca 16

<210> SEQ ID NO 867
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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ccagctatgg ttccgc 16

<210> SEQ ID NO 868
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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accagctatg gttccg 16

<210> SEQ ID NO 869
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 869

aaccagctat ggttcc 16

<210> SEQ ID NO 870
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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taaccagcta tggttc 16

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ataaccagct atggtt 16

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aaataaccag ctatgg 16

<210> SEQ ID NO 873
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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 873

gaaataacca gctatg 16

<210> SEQ ID NO 874
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<400> SEQUENCE: 874

agaaataacc agctat 16

<210> SEQ ID NO 875
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<400> SEQUENCE: 875

ctaacacaag ggagaa 16

<210> SEQ ID NO 876
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<400> SEQUENCE: 876

actaacacaa gggaga 16

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ttactaacac aaggga 16

<210> SEQ ID NO 878
<211> LENGTH: 16

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<212> TYPE: DNA
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attactaaca caaggg 16

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tattactaac acaagg 16

<210> SEQ ID NO 880
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ttattactaa cacaag 16

<210> SEQ ID NO 881
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gtttattact aacaca 16

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cgtttattac taacac 16

<210> SEQ ID NO 883
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cttattgtgg caagac 16

<210> SEQ ID NO 884
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<400> SEQUENCE: 884

gcttattgtg gcaaga 16

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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 885

ggcttattgt ggcaag 16

<210> SEQ ID NO 886
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 886

aggcttattg tggcaa 16

<210> SEQ ID NO 887
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 887

gaggcttatt gtggca 16

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<400> SEQUENCE: 888

ggaggcttat tgtggc 16

<210> SEQ ID NO 889
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<212> TYPE: DNA
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<400> SEQUENCE: 889

gcctgtcagc tgtgtg 16

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<212> TYPE: DNA
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gtagcctgtc agctgt 16

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<210> SEQ ID NO 891
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 891

cctgtagcct gtcagc 16

<210> SEQ ID NO 892
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<212> TYPE: DNA
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<400> SEQUENCE: 892

ttgcctgtag cctgtc 16

<210> SEQ ID NO 893
<211> LENGTH: 16
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<400> SEQUENCE: 893

ggattgcctg tagcct 16

<210> SEQ ID NO 894
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<400> SEQUENCE: 894

ccaggattgc ctgtag 16

<210> SEQ ID NO 895
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<400> SEQUENCE: 895

gaacacccag gattgc 16

<210> SEQ ID NO 896
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 896

gtccttccaa ggaaca 16

<210> SEQ ID NO 897
<211> LENGTH: 16

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 897

cttgtccttc caagga 16

<210> SEQ ID NO 898
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 898

gttcttgtcc ttccaa 16

<210> SEQ ID NO 899
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 899

gcagttcttg tccttc 16

<210> SEQ ID NO 900
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 900

ggtgcagttc ttgtcc 16

<210> SEQ ID NO 901
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 901

ggaggtgcag ttcttg 16

<210> SEQ ID NO 902
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 902

ccgggaggtg cagttc 16

<210> SEQ ID NO 903
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 903

cagccgggag gtgcag 16

<210> SEQ ID NO 904
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 904

atccagccgg gaggtg 16

<210> SEQ ID NO 905
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 905

cgcatccagc cgggag 16

<210> SEQ ID NO 906
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 906

gtgcgcatcc agccgg 16

<210> SEQ ID NO 907
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 907

cttgtgcgca tccagc 16

<210> SEQ ID NO 908
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 908

gaccttgtgc gcatcc 16

<210> SEQ ID NO 909
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 909

caggaccttg tgcgca 16

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<210> SEQ ID NO 910
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 910

agacaggacc ttgtgc 16

<210> SEQ ID NO 911
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 911

ggcagacagg accttg 16

<210> SEQ ID NO 912
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 912

gccctgtaca gcctgc 16

<210> SEQ ID NO 913
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 913

gcaggccctg tacagc 16

<210> SEQ ID NO 914
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 914

ctagcaggcc ctgtac 16

<210> SEQ ID NO 915
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 915

gccactagca ggccct 16

<210> SEQ ID NO 916
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 916

tgggccacta gcaggc 16

<210> SEQ ID NO 917
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 917

ccctgggcca ctagca 16

<210> SEQ ID NO 918
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 918

ctgccctggg ccacta 16

<210> SEQ ID NO 919
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 919

ggctatcagc cctgcc 16

<210> SEQ ID NO 920
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 920

cctggctatc agccct 16

<210> SEQ ID NO 921
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 921

gggcctggct atcagc 16

<210> SEQ ID NO 922
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 922

gctgggcctg gctatc 16

<210> SEQ ID NO 923
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 923

ccgtggacag cagcag 16

<210> SEQ ID NO 924
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 924

ccaccgtgga cagcag 16

<210> SEQ ID NO 925
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 925

ccaccaccgt ggacag 16

<210> SEQ ID NO 926
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 926

cgcccaccac cgtgga 16

<210> SEQ ID NO 927
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 927

acacgcccac caccgt 16

<210> SEQ ID NO 928
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 928

tgaacacgcc caccac 16

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<210> SEQ ID NO 929
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 929

ctgtgaacac gcccac 16

<210> SEQ ID NO 930
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 930

gggctgtgaa cacgcc 16

<210> SEQ ID NO 931
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 931

gcttcaggtg caggcc 16

<210> SEQ ID NO 932
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 932

gctgcttcag gtgcag 16

<210> SEQ ID NO 933
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 933

acggctgctt caggtg 16

<210> SEQ ID NO 934
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 934

caaacggctg cttcag 16

<210> SEQ ID NO 935
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 935

gcacaaacgg ctgctt 16

<210> SEQ ID NO 936
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 936

cctgcacaaa cggctg 16

<210> SEQ ID NO 937
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 937

ggccctgcac aaacgg 16

<210> SEQ ID NO 938
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 938

gtatagagag ccaggc 16

<210> SEQ ID NO 939
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 939

ctgtgaagtc cagaga 16

<210> SEQ ID NO 940
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 940

gttctgtgaa gtccag 16

<210> SEQ ID NO 941
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 941

ccagttctgt gaagtc 16

<210> SEQ ID NO 942
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 942

catccagttc tgtgaa 16

<210> SEQ ID NO 943
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 943

caacatccag ttctgt 16

<210> SEQ ID NO 944
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 944

cagcaacatc cagttc 16

<210> SEQ ID NO 945
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 945

caatcttctc agcagc 16

<210> SEQ ID NO 946
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 946

tgtcaatctt ctcagc 16

<210> SEQ ID NO 947
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 947

acctgtcaat cttctc 16

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<210> SEQ ID NO 948
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 948

tgaacctgtc aatctt 16

<210> SEQ ID NO 949
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 949

gcatgaacct gtcaat 16

<210> SEQ ID NO 950
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 950

cctgcatgaa cctgtc 16

<210> SEQ ID NO 951
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 951

cagcctgcat gaacct 16

<210> SEQ ID NO 952
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 952

cacagcctgc atgaac 16

<210> SEQ ID NO 953
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 953

tcacagcctg catgaa 16

<210> SEQ ID NO 954
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 954

atcctgtcac agcctg 16

<210> SEQ ID NO 955
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 955

ccatcctgtc acagcc 16

<210> SEQ ID NO 956
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 956

tccatcctgt cacagc 16

<210> SEQ ID NO 957
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 957

cttccatcct gtcaca 16

<210> SEQ ID NO 958
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 958

agtcttccat cctgtc 16

<210> SEQ ID NO 959
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 959

agccagtctt ccatcc 16

<210> SEQ ID NO 960
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 960

cacccagaac tcctgg 16

<210> SEQ ID NO 961
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 961

gtccacccag aactcc 16

<210> SEQ ID NO 962
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 962

gttgtccacc cagaac 16

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 963

gctgttgtcc acccag 16

<210> SEQ ID NO 964
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 964

ggtgctgttg tccacc 16

<210> SEQ ID NO 965
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 965

tgaggtgctg ttgtcc 16

<210> SEQ ID NO 966
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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cactgaggtg ctgttg 16

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catgggaaca gacact 16

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gagcatggga acagac 16

<210> SEQ ID NO 969
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agagagcatg ggaaca 16

<210> SEQ ID NO 970
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gccagagagc atggga 16

<210> SEQ ID NO 971
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catgccagag agcatg 16

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gcccatgcca gagagc 16

<210> SEQ ID NO 973
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ggtgcccatg ccagag 16

<210> SEQ ID NO 974
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<400> SEQUENCE: 974

gaaggtgccc atgcca 16

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<400> SEQUENCE: 975

ctggaaggtg cccatg 16

<210> SEQ ID NO 976
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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agtgctggaa ggtgcc 16

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 977

tccagtgctg gaaggt 16

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<400> SEQUENCE: 978

cactccagtg ctggaa 16

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tgtcactcca gtgctg 16

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 980

atgtcactcc agtgct 16

<210> SEQ ID NO 981
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<212> TYPE: DNA
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<400> SEQUENCE: 981

gatgtcactc cagtgc 16

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<400> SEQUENCE: 982

ggatgtcact ccagtg 16

<210> SEQ ID NO 983
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<212> TYPE: DNA
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<400> SEQUENCE: 983

tggatgtcac tccagt 16

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<400> SEQUENCE: 984

cctggatgtc actcca 16

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 985

tgtcctggat gtcact 16

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<210> SEQ ID NO 986
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 986

agttgtcctg gatgtc 16

<210> SEQ ID NO 987
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 987

agaagttgtc ctggat 16

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 988

tcaccgagaa gttgtc 16

<210> SEQ ID NO 989
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 989

gagtcaccga gaagtt 16

<210> SEQ ID NO 990
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<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 990

cttgagtcac cgagaa 16

<210> SEQ ID NO 991
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 991

gcacttgagt caccga 16

<210> SEQ ID NO 992
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 992

agggcacttg agtcac 16

<210> SEQ ID NO 993
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 993

tgaagggcac ttgagt 16

<210> SEQ ID NO 994
<211> LENGTH: 16
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 994

cagtgaaggg cacttg 16

<210> SEQ ID NO 995
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 995

tctcagtgaa gggcac 16

<210> SEQ ID NO 996
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 996

cgctctcagt gaaggg 16

<210> SEQ ID NO 997
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 997

agcagcaggc aggcgc 16

<210> SEQ ID NO 998
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 998

gatcagcagc aggcag 16

<210> SEQ ID NO 999
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 999

gctggatcag cagcag 16

<210> SEQ ID NO 1000
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1000

gaggctggat cagcag 16

<210> SEQ ID NO 1001
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1001

agtgaggctg gatcag 16

<210> SEQ ID NO 1002
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1002

catagtgagg ctggat 16

<210> SEQ ID NO 1003
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1003

aggcatagtg aggctg 16

<210> SEQ ID NO 1004
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1004

agacacacag gccgcc 16

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<210> SEQ ID NO 1005
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1005

acactaactg gagagc 16

<210> SEQ ID NO 1006
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1006

agagggcgga ttgcaa 16

<210> SEQ ID NO 1007
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1007

cagagggcgg attgca 16

<210> SEQ ID NO 1008
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1008

tctcagaggg cggatt 16

<210> SEQ ID NO 1009
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1009

ctctcagagg gcggat 16

<210> SEQ ID NO 1010
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1010

tctctcagag ggcgga 16

<210> SEQ ID NO 1011
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1011

gctgtgtgtc aggtgt 16

<210> SEQ ID NO 1012
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1012

aagaagctct tggatg 16

<210> SEQ ID NO 1013
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1013

tccaagaagc tcttgg 16

<210> SEQ ID NO 1014
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1014

ccagccgcca gccgcc 16

<210> SEQ ID NO 1015
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1015

ttagtgtttc agcagg 16

<210> SEQ ID NO 1016
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1016

agttagtgtt tcagca 16

<210> SEQ ID NO 1017
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1017

aacctcgagg acatcg 16

<210> SEQ ID NO 1018
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1018

acttataaga gctgac 16

<210> SEQ ID NO 1019
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1019

agcacttata agagct 16

<210> SEQ ID NO 1020
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1020

gcagtgttct tgatga 16

<210> SEQ ID NO 1021
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1021

acagcagtgt tcttga 16

<210> SEQ ID NO 1022
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1022

ataatgcact gtgtct 16

<210> SEQ ID NO 1023
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1023

gatgaggacc taggaa 16

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<210> SEQ ID NO 1024
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1024

ccgatgagga cctagg 16

<210> SEQ ID NO 1025
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1025

acgacaggga tgtttg 16

<210> SEQ ID NO 1026
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1026

ggtcaggcac agacac 16

<210> SEQ ID NO 1027
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1027

atcccggttt caactc 16

<210> SEQ ID NO 1028
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1028

tcccgctggc ccccgt 16

<210> SEQ ID NO 1029
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1029

ctaacttagc acagag 16

<210> SEQ ID NO 1030
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1030

ccatggccca ccagtg 16

<210> SEQ ID NO 1031
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1031

ttggccatgg cccacc 16

<210> SEQ ID NO 1032
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1032

ggcagaattc ctggct 16

<210> SEQ ID NO 1033
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1033

gcaagggtgt gtctgt 16

<210> SEQ ID NO 1034
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1034

ggcaagggtg tgtctg 16

<210> SEQ ID NO 1035
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1035

ctcagtgtag gcaagg 16

<210> SEQ ID NO 1036
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1036

gaggatgcac agtgta 16

<210> SEQ ID NO 1037
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1037

gctcaggacc tctgtg 16

<210> SEQ ID NO 1038
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1038

ggctcaggac ctctgt 16

<210> SEQ ID NO 1039
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1039

ggcgcactgg gtgacc 16

<210> SEQ ID NO 1040
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1040

tctgagggcg cactgg 16

<210> SEQ ID NO 1041
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1041

tcattctgag ggcgca 16

<210> SEQ ID NO 1042
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1042

gctcctaccg gggaga 16

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<210> SEQ ID NO 1043
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1043

acacatacct ccccca 16

<210> SEQ ID NO 1044
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1044

cgcataccct gaaata 16

<210> SEQ ID NO 1045
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1045

ggatggtcct ggggag 16

<210> SEQ ID NO 1046
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1046

ttcagcacct gcaaag 16

<210> SEQ ID NO 1047
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1047

ccggcttacc ttctgc 16

<210> SEQ ID NO 1048
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1048

cccccggctt accttc 16

<210> SEQ ID NO 1049
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1049

gggcccccgg cttacc 16

<210> SEQ ID NO 1050
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1050

gtgaatgtga gccccg 16

<210> SEQ ID NO 1051
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1051

tccctcctta taaccc 16

<210> SEQ ID NO 1052
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1052

ccgggcactc tcaact 16

<210> SEQ ID NO 1053
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1053

agtaatggtg ctctgg 16

<210> SEQ ID NO 1054
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1054

tcctgggagt aatggt 16

<210> SEQ ID NO 1055
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1055

tctcagttgt gatctg 16

<210> SEQ ID NO 1056
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1056

tccagagacg caattc 16

<210> SEQ ID NO 1057
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1057

tctccagaga cgcaat 16

<210> SEQ ID NO 1058
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1058

acctgtggga accgac 16

<210> SEQ ID NO 1059
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1059

aaacctgtgg gaaccg 16

<210> SEQ ID NO 1060
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1060

cctagatttt tctgct 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1061

gccttttctg tccccc 16

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<210> SEQ ID NO 1062
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1062

catttcttgt ggaggg 16

<210> SEQ ID NO 1063
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1063

tgggctggcc ctgcta 16

<210> SEQ ID NO 1064
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1064

gagccccaaa ggcatg 16

<210> SEQ ID NO 1065
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1065

tctaatatga cctgtg 16

<210> SEQ ID NO 1066
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tgatctaata tgacct 16

<210> SEQ ID NO 1067
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gtcctcaacc ccagga 16

<210> SEQ ID NO 1068
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<212> TYPE: DNA
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gctccatgga aaatat 16

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1069

tccattcatg tctaca 16

<210> SEQ ID NO 1070
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<212> TYPE: DNA
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<400> SEQUENCE: 1070

ttaagtgcca tctaac 16

<210> SEQ ID NO 1071
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<212> TYPE: DNA
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gcataccctg aaatat 16

<210> SEQ ID NO 1072
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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1072

tgtctactcc ccaccc 16

<210> SEQ ID NO 1073
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1073

acagacacta actgga 16

<210> SEQ ID NO 1074
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1074

gtgtgtcagg tgtggg 16

<210> SEQ ID NO 1075
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1075

gcaagtcagt tccaag 16

<210> SEQ ID NO 1076
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ctcgaaaatg gttacg 16

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<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1077

ggtggtaacc acatgc 16

<210> SEQ ID NO 1078
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<212> TYPE: DNA
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<400> SEQUENCE: 1078

atgcactgtg tcttac 16

<210> SEQ ID NO 1079
<211> LENGTH: 16
<212> TYPE: DNA
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<400> SEQUENCE: 1079

aataatgcac tgtgtc 16

<210> SEQ ID NO 1080
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1080

gttacttggg taattt 16

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<210> SEQ ID NO 1081
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1081

tcctttggtg cattct 16

<210> SEQ ID NO 1082
<211> LENGTH: 16
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<400> SEQUENCE: 1082

ctaggaatgg ttgtcc 16

<210> SEQ ID NO 1083
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<220> FEATURE:
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<400> SEQUENCE: 1083

gacgacaggg atgttt 16

<210> SEQ ID NO 1084
<211> LENGTH: 16
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1084

ctgacgacag ggatgt 16

<210> SEQ ID NO 1085
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1085

gcacagttag gaaggc 16

<210> SEQ ID NO 1086
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1086

ttagctaact tagcac 16

<210> SEQ ID NO 1087
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1087

catggcccac cagtgc 16

<210> SEQ ID NO 1088
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1088

cacagtgtat gcctgc 16

<210> SEQ ID NO 1089
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1089

gcactgggtg acccag 16

<210> SEQ ID NO 1090
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1090

tcagcacctg caaagc 16

<210> SEQ ID NO 1091
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1091

gagcagccag tcttcc 16

<210> SEQ ID NO 1092
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1092

agggagcagc cagtct 16

<210> SEQ ID NO 1093
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1093

atcagggagc agccag 16

<210> SEQ ID NO 1094
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1094

tcccatcagg gagcag 16

<210> SEQ ID NO 1095
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1095

ggctcccatc agggag 16

<210> SEQ ID NO 1096
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1096

actggctccc atcagg 16

<210> SEQ ID NO 1097
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1097

ccacactggc tcccat 16

<210> SEQ ID NO 1098
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1098

gctgtccaca ctggct 16

<210> SEQ ID NO 1099
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1099

ggtgctgtcc acactg 16

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<210> SEQ ID NO 1100
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1100

cagggtgctg tccaca 16

<210> SEQ ID NO 1101
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1101

agccagggtg ctgtcc 16

<210> SEQ ID NO 1102
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1102

gaaagccagg gtgctg 16

<210> SEQ ID NO 1103
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1103

gttgaaagcc agggtg 16

<210> SEQ ID NO 1104
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1104

ggtgttgaaa gccagg 16

<210> SEQ ID NO 1105
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1105

gtaggtgttg aaagcc 16

<210> SEQ ID NO 1106
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1106

cccttggaag tggacg 16

<210> SEQ ID NO 1107
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1107

cttcccttgg aagtgg 16

<210> SEQ ID NO 1108
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1108

catcttccct tggaag 16

<210> SEQ ID NO 1109
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1109

cttcatcttc ccttgg 16

<210> SEQ ID NO 1110
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1110

agcccttcat cttccc 16

<210> SEQ ID NO 1111
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1111

agaagccctt catctt 16

<210> SEQ ID NO 1112
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1112

gggagaagcc cttcat 16

<210> SEQ ID NO 1113
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1113

gcagggagaa gccctt 16

<210> SEQ ID NO 1114
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1114

tcggccagca gggaga 16

<210> SEQ ID NO 1115
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1115

ggctcggcca gcaggg 16

<210> SEQ ID NO 1116
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1116

cgaagggaga cccatt 16

<210> SEQ ID NO 1117
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1117

ttcgaaggga gaccca 16

<210> SEQ ID NO 1118
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1118

ctttcgaagg gagacc 16

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<210> SEQ ID NO 1119
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1119

ccgatctcct cactgg 16

<210> SEQ ID NO 1120
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1120

ccccgatctc ctcact 16

<210> SEQ ID NO 1121
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1121

acagcccccg atctcc 16

<210> SEQ ID NO 1122
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1122

gagacagccc ccgatc 16

<210> SEQ ID NO 1123
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1123

ccgagacagc ccccga 16

<210> SEQ ID NO 1124
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1124

ctagctgcct gctgag 16

<210> SEQ ID NO 1125
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1125

tctagctgcc tgctga 16

<210> SEQ ID NO 1126
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1126

gtgggacaca tctagc 16

<210> SEQ ID NO 1127
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1127

tctagtggga cacatc 16

<210> SEQ ID NO 1128
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1128

tctctagtgg gacaca 16

<210> SEQ ID NO 1129
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1129

catgagagtg gctgcc 16

<210> SEQ ID NO 1130
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1130

cttttagttt agaggg 16

<210> SEQ ID NO 1131
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1131

atgtgagcgg gaaact 16

<210> SEQ ID NO 1132
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1132

catgtgagcg ggaaac 16

<210> SEQ ID NO 1133
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1133

cggagcactc agtctc 16

<210> SEQ ID NO 1134
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1134

gtcctcagtc ctcgga 16

<210> SEQ ID NO 1135
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1135

cgtcctcagt cctcgg 16

<210> SEQ ID NO 1136
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1136

gcagtggcag acctgg 16

<210> SEQ ID NO 1137
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1137

tagagatggt tcagaa 16

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<210> SEQ ID NO 1138
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1138

tgagtagaga tggttc 16

<210> SEQ ID NO 1139
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1139

ggagtctgag tagaga 16

<210> SEQ ID NO 1140
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1140

gccctcggct gtcctc 16

<210> SEQ ID NO 1141
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1141

ctcgaccttа cactag 16

<210> SEQ ID NO 1142
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1142

cctctgcctc gacctt 16

<210> SEQ ID NO 1143
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1143

aactcgggag agcccg 16

<210> SEQ ID NO 1144
<211> LENGTH: 16

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1144

aacgagggct ccattc 16

<210> SEQ ID NO 1145
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1145

gacacactca cttttt 16

<210> SEQ ID NO 1146
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1146

ctgccaggtc aactca 16

<210> SEQ ID NO 1147
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1147

gtacctgcca ggtcaa 16

<210> SEQ ID NO 1148
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1148

ctggtacctg ccaggt 16

<210> SEQ ID NO 1149
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1149

agttcactga ggcagc 16

<210> SEQ ID NO 1150
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1150

ccatttgagt tcactg 16

<210> SEQ ID NO 1151
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1151

gcagccattt gagttc 16

<210> SEQ ID NO 1152
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1152

aaggcccaga tcctgc 16

<210> SEQ ID NO 1153
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1153

gaaatccaga caggag 16

<210> SEQ ID NO 1154
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1154

tgccttacct tggaag 16

<210> SEQ ID NO 1155
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1155

catcttccct gaaatc 16

<210> SEQ ID NO 1156
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1156

aggtatgtcc gcaggg 16

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<400> SEQUENCE: 1157

tagtagggca gcaggt 16

<210> SEQ ID NO 1158
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<400> SEQUENCE: 1158

ttgtttctcc gagtct 16

<210> SEQ ID NO 1159
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<400> SEQUENCE: 1159

aggcactttg tttctc 16

<210> SEQ ID NO 1160
<211> LENGTH: 16
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1160

caaggcactt tgtttc 16

<210> SEQ ID NO 1161
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1161

tagaactggg ctgtgg 16

<210> SEQ ID NO 1162
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ccctcctaac atgaaa 16

<210> SEQ ID NO 1163
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<212> TYPE: DNA
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cttacaagta gcaaat 16

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<400> SEQUENCE: 1164

gccaggctta aagtct 16

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1165

attgaccttt aaaagc 16

<210> SEQ ID NO 1166
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<400> SEQUENCE: 1166

tctggttcaa cactca 16

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ttcccgtgac tgtgtg 16

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<400> SEQUENCE: 1168

cgagctgctc cctgag 16

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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1169

caccccaccc atggat 16

<210> SEQ ID NO 1170
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<220> FEATURE:
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tctctgtccc tcacga 16

<210> SEQ ID NO 1171
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tttcgaaggg agaccc 16

<210> SEQ ID NO 1172
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catctagctg cctgct 16

<210> SEQ ID NO 1173
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tgggacacat ctagct 16

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atcctcaggt cctctc 16

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atggttcaga aacagt 16

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gatttgcaca ctgggc 16

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ccccgtgatc aacatc 16

<210> SEQ ID NO 1178
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1178

atcgagcaga aagtac 16

<210> SEQ ID NO 1179
<211> LENGTH: 16
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actggtacct gccagg 16

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<400> SEQUENCE: 1180

ctgctgcccg ctcatgggat 20

<210> SEQ ID NO 1181
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ctgaccctgc tgcccgctca 20

<210> SEQ ID NO 1182
<211> LENGTH: 20

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<212> TYPE: DNA
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ccacttctga ccctgctgcc 20

<210> SEQ ID NO 1183
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tcttgcttag gcaacacggg 20

<210> SEQ ID NO 1184
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ggagagtctt gcttaggcaa 20

<210> SEQ ID NO 1185
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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ggaggtgcag agggcagagg 20

<210> SEQ ID NO 1186
<211> LENGTH: 20
<212> TYPE: DNA
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caggccggag gtgcagaggg 20

<210> SEQ ID NO 1187
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1187

gacatgcagg ccggaggtgc 20

<210> SEQ ID NO 1188
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1188

cacagggaca tgcaggccgg 20

<210> SEQ ID NO 1189
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1189

agaggccaca gggacatgca 20

<210> SEQ ID NO 1190
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1190

cccccaagag gccacaggga 20

<210> SEQ ID NO 1191
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
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gatgtacccc caagaggcca 20

<210> SEQ ID NO 1192
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 1192

ccgggagatg taccccaag 20

<210> SEQ ID NO 1193
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 1193

ccagccccgg gagatgtacc 20

<210> SEQ ID NO 1194
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<212> TYPE: DNA
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<400> SEQUENCE: 1194

tctgacccag ccccgggaga 20

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<210> SEQ ID NO 1195
<211> LENGTH: 20
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<400> SEQUENCE: 1195

aggccttctg acccagcccc 20

<210> SEQ ID NO 1196
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1196

ccacccaggc cttctgaccc 20

<210> SEQ ID NO 1197
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1197

ggccaaccac ccaggccttc 20

<210> SEQ ID NO 1198
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1198

gcctgaggcc aaccacccag 20

<210> SEQ ID NO 1199
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1199

gtgacagcct gaggccaacc 20

<210> SEQ ID NO 1200
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 1200

aggtgtgtga cagcctgagg 20

<210> SEQ ID NO 1201
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1201

ctccctaggt gtgtgacagc 20

<210> SEQ ID NO 1202
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1202

gagcatctcc ctaggtgtgt 20

<210> SEQ ID NO 1203
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1203

aaacgggagc atctccctag 20

<210> SEQ ID NO 1204
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1204

tcccagaaac gggagcatct 20

<210> SEQ ID NO 1205
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1205

caaggttccc agaaacggga 20

<210> SEQ ID NO 1206
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1206

cgaagtttgc aggagtcggg 20

<210> SEQ ID NO 1207
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 1207

atttaccgaa gtttgcagga 20

<210> SEQ ID NO 1208
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1208

ttacacattt accgaagttt 20

<210> SEQ ID NO 1209
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1209

gtcgagttac acatttaccg 20

<210> SEQ ID NO 1210
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1210

tgcagggtcg agttacacat 20

<210> SEQ ID NO 1211
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1211

agccggtgca gggtcgagtt 20

<210> SEQ ID NO 1212
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1212

agagtgagcc ggtgcagggt 20

<210> SEQ ID NO 1213
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1213

ctgaacagag tgagccggtg 20

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<210> SEQ ID NO 1214
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1214

tcactgctga acagagtgag 20

<210> SEQ ID NO 1215
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1215

agagtttcac tgctgaacag 20

<210> SEQ ID NO 1216
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1216

cgatgcagag tttcactgct 20

<210> SEQ ID NO 1217
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1217

agtgatcgat gcagagtttc 20

<210> SEQ ID NO 1218
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1218

agtcttagtg atcgatgcag 20

<210> SEQ ID NO 1219
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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ccaggaagtc ttagtgatcg 20

<210> SEQ ID NO 1220
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1220

cctcttccag gaagtcttag 20

<210> SEQ ID NO 1221
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1221

ctgggacctc ttccaggaag 20

<210> SEQ ID NO 1222
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1222

ctcacgctgg gacctcttcc 20

<210> SEQ ID NO 1223
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1223

gcgacactca cgctgggacc 20

<210> SEQ ID NO 1224
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1224

ccagaagcga cactcacgct 20

<210> SEQ ID NO 1225
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1225

cagatgccag aagcgacact 20

<210> SEQ ID NO 1226
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1226

gaaggacaga tgccagaagc 20

<210> SEQ ID NO 1227
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1227

tggccagaag gacagatgcc 20

<210> SEQ ID NO 1228
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1228

acaggctggc cagaaggaca 20

<210> SEQ ID NO 1229
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1229

cagaccacag gctggccaga 20

<210> SEQ ID NO 1230
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1230

cttggccaga ccacaggctg 20

<210> SEQ ID NO 1231
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1231

acatcacttg gccagaccac 20

<210> SEQ ID NO 1232
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1232

agggttacat cacttggcca 20

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<210> SEQ ID NO 1233
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1233

gagaggaggg ttacatcact 20

<210> SEQ ID NO 1234
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1234

aggctggaga ggagggttac 20

<210> SEQ ID NO 1235
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1235

gtgcacaggc tggagaggag 20

<210> SEQ ID NO 1236
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1236

ctgcctgtgc acaggctgga 20

<210> SEQ ID NO 1237
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1237

cccaggctgc ctgtgcacag 20

<210> SEQ ID NO 1238
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1238

gctgttccca ggctgcctgt 20

<210> SEQ ID NO 1239
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1239

gatggagctg ttcccaggct 20

<210> SEQ ID NO 1240
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1240

gccctattta tagctgaggg 20

<210> SEQ ID NO 1241
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1241

cacgatgccc tatttatagc 20

<210> SEQ ID NO 1242
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1242

ccgggtcacg atgccctatt 20

<210> SEQ ID NO 1243
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1243

ccccggccgg gtcacgatgc 20

<210> SEQ ID NO 1244
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1244

ttcccccggc cgggtcacga 20

<210> SEQ ID NO 1245
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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ttcttccccc ggccgggtca 20

<210> SEQ ID NO 1246
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1246

agcttcttcc cccggccggg 20

<210> SEQ ID NO 1247
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1247

ggcagcttct tcccccggcc 20

<210> SEQ ID NO 1248
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1248

aacggcagct tcttcccccg 20

<210> SEQ ID NO 1249
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1249

aacaacggca gcttcttccc 20

<210> SEQ ID NO 1250
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1250

cagaacaacg gcagcttctt 20

<210> SEQ ID NO 1251
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1251

acccagaaca acggcagctt 20

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<210> SEQ ID NO 1252
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1252

agtacccaga acaacggcag 20

<210> SEQ ID NO 1253
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1253

tgtagtaccc agaacaacgg 20

<210> SEQ ID NO 1254
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1254

tgctgtagta cccagaacaa 20

<210> SEQ ID NO 1255
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1255

ttctgctgta gtacccagaa 20

<210> SEQ ID NO 1256
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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cccttctgct gtagtaccca 20

<210> SEQ ID NO 1257
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1257

ataccttct gctgtagtac 20

<210> SEQ ID NO 1258
<211> LENGTH: 20

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<212> TYPE: DNA
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<400> SEQUENCE: 1258

cgcataccct tctgctgtag 20

<210> SEQ ID NO 1259
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1259

ttccgcatac ccttctgctg 20

<210> SEQ ID NO 1260
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1260

cgcttccgca taccttctg 20

<210> SEQ ID NO 1261
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1261

gctcgcttcc gcataccctt 20

<210> SEQ ID NO 1262
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1262

ggtgctcgct tccgcatacc 20

<210> SEQ ID NO 1263
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1263

aggagccatc tcagactggg 20

<210> SEQ ID NO 1264
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1264

ggcaggagcc atctcagact 20

<210> SEQ ID NO 1265
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1265

accggcagga gccatctcag 20

<210> SEQ ID NO 1266
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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cacaccggca ggagccatct 20

<210> SEQ ID NO 1267
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1267

gctcacaccg gcaggagcca 20

<210> SEQ ID NO 1268
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1268

caggctcaca ccggcaggag 20

<210> SEQ ID NO 1269
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1269

cctcaggctc acaccggcag 20

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1270

ggccctcagg ctcacaccgg 20

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<210> SEQ ID NO 1271
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1271

ggtggccctc aggctcacac 20

<210> SEQ ID NO 1272
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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gatggtggcc ctcaggctca 20

<210> SEQ ID NO 1273
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1273

gaggatggtg gccctcaggc 20

<210> SEQ ID NO 1274
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1274

gcagaggatg gtggccctca 20

<210> SEQ ID NO 1275
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1275

gaggcagagg atggtggccc 20

<210> SEQ ID NO 1276
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1276

caggaggcag aggatggtgg 20

<210> SEQ ID NO 1277
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1277

gcccaggcca ggaggcagag 20

<210> SEQ ID NO 1278
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1278

ccagcccagg ccaggaggca 20

<210> SEQ ID NO 1279
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1279

aggccagccc aggccaggag 20

<210> SEQ ID NO 1280
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1280

gccaggccag cccaggccag 20

<210> SEQ ID NO 1281
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1281

gcagccaggc cagcccaggc 20

<210> SEQ ID NO 1282
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1282

cctgcagcca ggccagccca 20

<210> SEQ ID NO 1283
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1283

tcacctgcag ccaggccagc 20

<210> SEQ ID NO 1284
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1284

cggtcacctg cagccaggcc 20

<210> SEQ ID NO 1285
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1285

acccggtcac ctgcagccag 20

<210> SEQ ID NO 1286
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1286

tacacccggt cacctgcagc 20

<210> SEQ ID NO 1287
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1287

atgtacaccc ggtcacctgc 20

<210> SEQ ID NO 1288
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1288

tgtatgtaca cccggtcacc 20

<210> SEQ ID NO 1289
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1289

gggtgtatgt acacccggtc 20

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<210> SEQ ID NO 1290
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1290

tggatgacga ggtggaaggg 20

<210> SEQ ID NO 1291
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1291

ttgtggatga cgaggtggaa 20

<210> SEQ ID NO 1292
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1292

tcattgtgga tgacgaggtg 20

<210> SEQ ID NO 1293
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1293

ctctcattgt ggatgacgag 20

<210> SEQ ID NO 1294
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1294

gtactctcat tgtggatgac 20

<210> SEQ ID NO 1295
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1295

caggtactct cattgtggat 20

<210> SEQ ID NO 1296
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1296

tcacaggtac tctcattgtg 20

<210> SEQ ID NO 1297
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1297

tgctcacagg tactctcatt 20

<210> SEQ ID NO 1298
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1298

agctgctcac aggtactctc 20

<210> SEQ ID NO 1299
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1299

gccagctgct cacaggtact 20

<210> SEQ ID NO 1300
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1300

tttgccagct gctcacaggt 20

<210> SEQ ID NO 1301
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1301

gcctttgcca gctgctcaca 20

<210> SEQ ID NO 1302
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1302

ttggcctttg ccagctgctc 20

<210> SEQ ID NO 1303
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1303

gcattggcct ttgccagctg 20

<210> SEQ ID NO 1304
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1304

ccggcattgg cctttgccag 20

<210> SEQ ID NO 1305
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1305

ttcccggcat tggcctttgc 20

<210> SEQ ID NO 1306
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1306

ggcttcccgg cattggcctt 20

<210> SEQ ID NO 1307
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1307

ttgggcttcc cggcattggc 20

<210> SEQ ID NO 1308
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1308

tctttgggct tcccggcatt 20

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<210> SEQ ID NO 1309
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1309

ggagcaggta tgaaggtggg 20

<210> SEQ ID NO 1310
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1310

attggagcag gtatgaaggt 20

<210> SEQ ID NO 1311
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1311

tgaattggag caggtatgaa 20

<210> SEQ ID NO 1312
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1312

gcctgaattg gagcaggtat 20

<210> SEQ ID NO 1313
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1313

ttggcctgaa ttggagcagg 20

<210> SEQ ID NO 1314
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1314

gtcttggcct gaattggagc 20

<210> SEQ ID NO 1315
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1315

gatgtcttgg cctgaattgg 20

<210> SEQ ID NO 1316
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1316

agggcctttt catccacagg 20

<210> SEQ ID NO 1317
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1317

tgtagggcct tttcatccac 20

<210> SEQ ID NO 1318
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1318

tcctgtaggg ccttttcatc 20

<210> SEQ ID NO 1319
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1319

tggtcctgta gggccttttc 20

<210> SEQ ID NO 1320
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1320

agctggtcct gtagggcctt 20

<210> SEQ ID NO 1321
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1321

accagctggt cctgtagggc 20

<210> SEQ ID NO 1322
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1322

agcaccagct ggtcctgtag 20

<210> SEQ ID NO 1323
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1323

actagcacca gctggtcctg 20

<210> SEQ ID NO 1324
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1324

gcgactagca ccagctggtc 20

<210> SEQ ID NO 1325
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1325

gcagcgacta gcaccagctg 20

<210> SEQ ID NO 1326
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1326

tttgcagcga ctagcaccag 20

<210> SEQ ID NO 1327
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1327

agttttgcag cgactagcac 20

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<210> SEQ ID NO 1328
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1328

tcaagttttg cagcgactag 20

<210> SEQ ID NO 1329
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1329

gtgtcaagtt ttgcagcgac 20

<210> SEQ ID NO 1330
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1330

tcggtgtcaa gttttgcagc 20

<210> SEQ ID NO 1331
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1331

tcttcggtgt caagttttgc 20

<210> SEQ ID NO 1332
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1332

ttgtcttcgg tgtcaagttt 20

<210> SEQ ID NO 1333
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1333

aacttgtctt cggtgtcaag 20

<210> SEQ ID NO 1334
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1334

ctcaacttgt cttcggtgtc 20

<210> SEQ ID NO 1335
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1335

gccctcaact tgtcttcggt 20

<210> SEQ ID NO 1336
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1336

gcggccctca acttgtcttc 20

<210> SEQ ID NO 1337
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1337

attgcggccc tcaacttgtc 20

<210> SEQ ID NO 1338
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1338

accattgcgg ccctcaactt 20

<210> SEQ ID NO 1339
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1339

ccgaccattg cggccctcaa 20

<210> SEQ ID NO 1340
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1340

atcccgacca ttgcggccct 20

<210> SEQ ID NO 1341
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1341

agcatcccga ccattgcggc 20

<210> SEQ ID NO 1342
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1342

gccagcatcc cgaccattgc 20

<210> SEQ ID NO 1343
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1343

ttggccagca tcccgaccat 20

<210> SEQ ID NO 1344
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1344

aagttggcca gcatcccgac 20

<210> SEQ ID NO 1345
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1345

aagaagttgg ccagcatccc 20

<210> SEQ ID NO 1346
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1346

cccaagaagt tggccagcat 20

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<210> SEQ ID NO 1347
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<400> SEQUENCE: 1347

aagcccaaga agttggccag 20

<210> SEQ ID NO 1348
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cggaagccca agaagttggc 20

<210> SEQ ID NO 1349
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atacggaagc ccaagaagtt 20

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tatatacgga agcccaagaa 20

<210> SEQ ID NO 1351
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<212> TYPE: DNA
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ccatatatac ggaagcccaa 20

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atgccatata tacggaagcc 20

<210> SEQ ID NO 1353
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tgcatgccat atatacggaa 20

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ctgtgcatgc catatatacg 20

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tcactgtgca tgccatatat 20

<210> SEQ ID NO 1356
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<212> TYPE: DNA
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agctcactgt gcatgccata 20

<210> SEQ ID NO 1357
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catagctcac tgtgcatgcc 20

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ccccatagct cactgtgcat 20

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acgccccata gctcactgtg 20

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1360

accacgcccc atagctcact 20

<210> SEQ ID NO 1361
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tggaccacgc cccatagctc 20

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ccatggacca cgccccatag 20

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gccccatgga ccacgcccca 20

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<220> FEATURE:
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gtggccccat ggaccacgcc 20

<210> SEQ ID NO 1365
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acggtggccc catggaccac 20

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aggacggtgg ccccatggac 20

<210> SEQ ID NO 1367
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gagaggacgg tggccccatg 20

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tgccaaagac agccgttggg 20

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gggtgccaaa gacagccgtt 20

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ccagggtgcc aaagacagcc 20

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aggccagggt gccaaagaca 20

<210> SEQ ID NO 1372
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<212> TYPE: DNA
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gagaggccag ggtgccaaag 20

<210> SEQ ID NO 1373
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agagagaggc cagggtgcca 20

<210> SEQ ID NO 1374
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1374

gatagagaga ggccagggtg 20

<210> SEQ ID NO 1375
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ccagatagag agaggccagg 20

<210> SEQ ID NO 1376
<211> LENGTH: 20
<212> TYPE: DNA
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ctcccagata gagagaggcc 20

<210> SEQ ID NO 1377
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<213> ORGANISM: Artificial sequence
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aggctcccag atagagagag 20

<210> SEQ ID NO 1378
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<400> SEQUENCE: 1378

ccaaggctcc cagatagaga 20

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ggtccaaggc tcccagatag 20

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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tgtggtccaa ggctcccaga 20

<210> SEQ ID NO 1381
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ctgtgtggtc caaggctccc 20

<210> SEQ ID NO 1382
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1382

cagctgtgtg gtccaaggct 20

<210> SEQ ID NO 1383
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1383

tgtcagctgt gtggtccaag 20

<210> SEQ ID NO 1384
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 1384

gcctgtcagc tgtgtggtcc 20

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<210> SEQ ID NO 1385
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<213> ORGANISM: Artificial sequence
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gtagcctgtc agctgtgtgg 20

<210> SEQ ID NO 1386
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1386

cctgtagcct gtcagctgtg 20

<210> SEQ ID NO 1387
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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ttgcctgtag cctgtcagct 20

<210> SEQ ID NO 1388
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 1388

ggattgcctg tagcctgtca 20

<210> SEQ ID NO 1389
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 1389

ccaggattgc ctgtagcctg 20

<210> SEQ ID NO 1390
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<212> TYPE: DNA
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cacccaggat tgcctgtagc 20

<210> SEQ ID NO 1391
<211> LENGTH: 20

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<212> TYPE: DNA
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gaacacccag gattgcctgt 20

<210> SEQ ID NO 1392
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aaggaacacc caggattgcc 20

<210> SEQ ID NO 1393
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tccaaggaac acccaggatt 20

<210> SEQ ID NO 1394
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<400> SEQUENCE: 1394

ccttccaagg aacacccagg 20

<210> SEQ ID NO 1395
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1395

tgtccttcca aggaacaccc 20

<210> SEQ ID NO 1396
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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tcttgtcctt ccaaggaaca 20

<210> SEQ ID NO 1397
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 1397

agttcttgtc cttccaagga 20

<210> SEQ ID NO 1398
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tgcagttctt gtccttccaa 20

<210> SEQ ID NO 1399
<211> LENGTH: 20
<212> TYPE: DNA
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aggtgcagtt cttgtccttc 20

<210> SEQ ID NO 1400
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1400

gggaggtgca gttcttgtcc 20

<210> SEQ ID NO 1401
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1401

gccgggaggt gcagttcttg 20

<210> SEQ ID NO 1402
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1402

ccagccggga ggtgcagttc 20

<210> SEQ ID NO 1403
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1403

catccagccg ggaggtgcag 20

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<210> SEQ ID NO 1404
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1404

gcgcatccag ccgggaggtg 20

<210> SEQ ID NO 1405
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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tgtgcgcatc cagccgggag 20

<210> SEQ ID NO 1406
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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ccttgtgcgc atccagccgg 20

<210> SEQ ID NO 1407
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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ggaccttgtg cgcatccagc 20

<210> SEQ ID NO 1408
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1408

acaggacctt gtgcgcatcc 20

<210> SEQ ID NO 1409
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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cagacaggac cttgtgcgca 20

<210> SEQ ID NO 1410
<211> LENGTH: 20

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<212> TYPE: DNA
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gggcagacag gaccttgtgc 20

<210> SEQ ID NO 1411
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1411

gcagggcaga caggaccttg 20

<210> SEQ ID NO 1412
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1412

cctgcagggc agacaggacc 20

<210> SEQ ID NO 1413
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1413

cagcctgcag ggcagacagg 20

<210> SEQ ID NO 1414
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1414

gtacagcctg cagggcagac 20

<210> SEQ ID NO 1415
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1415

cctgtacagc ctgcagggca 20

<210> SEQ ID NO 1416
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1416

ggccctgtac agcctgcagg 20

<210> SEQ ID NO 1417
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1417

gcaggccctg tacagcctgc 20

<210> SEQ ID NO 1418
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1418

ctagcaggcc ctgtacagcc 20

<210> SEQ ID NO 1419
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1419

ccactagcag gccctgtaca 20

<210> SEQ ID NO 1420
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1420

gggccactag caggccctgt 20

<210> SEQ ID NO 1421
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1421

cctgggccac tagcaggccc 20

<210> SEQ ID NO 1422
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1422

tgccctgggc cactagcagg 20

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<210> SEQ ID NO 1423
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1423

ccctgccctg ggccactagc 20

<210> SEQ ID NO 1424
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1424

cagccctgcc ctgggccact 20

<210> SEQ ID NO 1425
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1425

tatcagccct gccctgggcc 20

<210> SEQ ID NO 1426
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1426

ggctatcagc cctgccctgg 20

<210> SEQ ID NO 1427
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1427

cctggctatc agccctgccc 20

<210> SEQ ID NO 1428
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1428

gggcctggct atcagccctg 20

<210> SEQ ID NO 1429
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1429

gctgggcctg gctatcagcc 20

<210> SEQ ID NO 1430
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1430

gcagctgggc ctggctatca 20

<210> SEQ ID NO 1431
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1431

gcagcagctg ggcctggcta 20

<210> SEQ ID NO 1432
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1432

acagcagcag ctgggcctgg 20

<210> SEQ ID NO 1433
<211> LENGTH: 20
<212> TYPE: DNA
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tggacagcag cagctgggcc 20

<210> SEQ ID NO 1434
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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ccgtggacag cagcagctgg 20

<210> SEQ ID NO 1435
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ccaccgtgga cagcagcagc 20

<210> SEQ ID NO 1436
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1436

ccaccaccgt ggacagcagc 20

<210> SEQ ID NO 1437
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1437

cgcccaccac cgtggacagc 20

<210> SEQ ID NO 1438
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 1438

acacgcccac caccgtggac 20

<210> SEQ ID NO 1439
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 1439

tgaacacgcc caccaccgtg 20

<210> SEQ ID NO 1440
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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ctgtgaacac gcccaccacc 20

<210> SEQ ID NO 1441
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1441

gggctgtgaa cacgcccacc 20

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<210> SEQ ID NO 1442
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gcttcaggtg caggcctggg 20

<210> SEQ ID NO 1443
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<212> TYPE: DNA
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gctgcttcag gtgcaggcct 20

<210> SEQ ID NO 1444
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acggctgctt caggtgcagg 20

<210> SEQ ID NO 1445
<211> LENGTH: 20
<212> TYPE: DNA
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caaacggctg cttcaggtgc 20

<210> SEQ ID NO 1446
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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gcacaaacgg ctgcttcagg 20

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cctgcacaaa cggctgcttc 20

<210> SEQ ID NO 1448
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<212> TYPE: DNA
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ggccctgcac aaacggctgc 20

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<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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ccaggccctg cacaaacggc 20

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1450

gagccaggcc ctgcacaaac 20

<210> SEQ ID NO 1451
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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agagagccag gccctgcaca 20

<210> SEQ ID NO 1452
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1452

tatagagagc caggccctgc 20

<210> SEQ ID NO 1453
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1453

gggtatagag agccaggccc 20

<210> SEQ ID NO 1454
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1454

tctgtgaagt ccagagagcg 20

<210> SEQ ID NO 1455
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1455

agttctgtga agtccagaga 20

<210> SEQ ID NO 1456
<211> LENGTH: 20
<212> TYPE: DNA
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tccagttctg tgaagtccag 20

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acatccagtt ctgtgaagtc 20

<210> SEQ ID NO 1458
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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gcaacatcca gttctgtgaa 20

<210> SEQ ID NO 1459
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1459

gcagcaacat ccagttctgt 20

<210> SEQ ID NO 1460
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<212> TYPE: DNA
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<400> SEQUENCE: 1460

tcagcagcaa catccagttc 20

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<210> SEQ ID NO 1461
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ttctcagcag caacatccag 20

<210> SEQ ID NO 1462
<211> LENGTH: 20
<212> TYPE: DNA
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atcttctcag cagcaacatc 20

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tcaatcttct cagcagcaac 20

<210> SEQ ID NO 1464
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 1464

ctgtcaatct tctcagcagc 20

<210> SEQ ID NO 1465
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 1465

aacctgtcaa tcttctcagc 20

<210> SEQ ID NO 1466
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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atgaacctgt caatcttctc 20

<210> SEQ ID NO 1467
<211> LENGTH: 20

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1467

tgcatgaacc tgtcaatctt 20

<210> SEQ ID NO 1468
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1468

gcctgcatga acctgtcaat 20

<210> SEQ ID NO 1469
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1469

acagcctgca tgaacctgtc 20

<210> SEQ ID NO 1470
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1470

gtcacagcct gcatgaacct 20

<210> SEQ ID NO 1471
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1471

cctgtcacag cctgcatgaa 20

<210> SEQ ID NO 1472
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1472

catcctgtca agcctgcat 20

<210> SEQ ID NO 1473
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1473

ttccatcctg tcacagcctg 20

<210> SEQ ID NO 1474
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1474

gtcttccatc ctgtcacagc 20

<210> SEQ ID NO 1475
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1475

ccagtcttcc atcctgtcac 20

<210> SEQ ID NO 1476
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1476

cagccagtct tccatcctgt 20

<210> SEQ ID NO 1477
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1477

gagcagccag tcttccatcc 20

<210> SEQ ID NO 1478
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1478

agggagcagc cagtcttcca 20

<210> SEQ ID NO 1479
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1479

atcagggagc agccagtctt 20

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<210> SEQ ID NO 1480
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1480

cccatcaggg agcagccagt 20

<210> SEQ ID NO 1481
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1481

gctcccatca gggagcagcc 20

<210> SEQ ID NO 1482
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1482

ctggctccca tcagggagca 20

<210> SEQ ID NO 1483
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1483

acactggctc ccatcaggga 20

<210> SEQ ID NO 1484
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1484

tccacactgg ctcccatcag 20

<210> SEQ ID NO 1485
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1485

ctgtccacac tggctcccat 20

<210> SEQ ID NO 1486
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1486

gtgctgtcca cactggctcc 20

<210> SEQ ID NO 1487
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1487

agggtgctgt ccacactggc 20

<210> SEQ ID NO 1488
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1488

gccagggtgc tgtccacact 20

<210> SEQ ID NO 1489
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1489

aaagccaggg tgctgtccac 20

<210> SEQ ID NO 1490
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1490

ttgaaagcca gggtgctgtc 20

<210> SEQ ID NO 1491
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1491

gtgttgaaag ccagggtgct 20

<210> SEQ ID NO 1492
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1492

taggtgttga aagccagggt 20

<210> SEQ ID NO 1493
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1493

tcttcccttg gaagtggacg 20

<210> SEQ ID NO 1494
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1494

tcatcttccc ttggaagtgg 20

<210> SEQ ID NO 1495
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1495

ccttcatctt cccttggaag 20

<210> SEQ ID NO 1496
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1496

agcccttcat cttcccttgg 20

<210> SEQ ID NO 1497
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1497

agaagccctt catcttccct 20

<210> SEQ ID NO 1498
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1498

gggagaagcc cttcatcttc 20

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<210> SEQ ID NO 1499
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1499

gcagggagaa gcccttcatc 20

<210> SEQ ID NO 1500
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1500

ccagcaggga gaagcccttc 20

<210> SEQ ID NO 1501
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1501

cggccagcag ggagaagccc 20

<210> SEQ ID NO 1502
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1502

gctcggccag cagggagaag 20

<210> SEQ ID NO 1503
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1503

gtccacccag aactcctggg 20

<210> SEQ ID NO 1504
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1504

gttgtccacc cagaactcct 20

<210> SEQ ID NO 1505
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1505

gctgttgtcc acccagaact 20

<210> SEQ ID NO 1506
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1506

ggtgctgttg tccacccaga 20

<210> SEQ ID NO 1507
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1507

tgaggtgctg ttgtccaccc 20

<210> SEQ ID NO 1508
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1508

cactgaggtg ctgttgtcca 20

<210> SEQ ID NO 1509
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1509

agacactgag gtgctgttgt 20

<210> SEQ ID NO 1510
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1510

aacagacact gaggtgctgt 20

<210> SEQ ID NO 1511
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1511

gggaacagac actgaggtgc 20

<210> SEQ ID NO 1512
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1512

catgggaaca gacactgagg 20

<210> SEQ ID NO 1513
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1513

gagcatggga acagacactg 20

<210> SEQ ID NO 1514
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1514

agagagcatg ggaacagaca 20

<210> SEQ ID NO 1515
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1515

gccagagagc atgggaacag 20

<210> SEQ ID NO 1516
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1516

catgccagag agcatgggaa 20

<210> SEQ ID NO 1517
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1517

gcccatgcca gagagcatgg 20

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<210> SEQ ID NO 1518
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1518

ggtgcccatg ccagagagca 20

<210> SEQ ID NO 1519
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1519

gaaggtgccc atgccagaga 20

<210> SEQ ID NO 1520
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1520

ctggaaggtg cccatgccag 20

<210> SEQ ID NO 1521
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1521

gtgctggaag gtgcccatgc 20

<210> SEQ ID NO 1522
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1522

ccagtgctgg aaggtgccca 20

<210> SEQ ID NO 1523
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1523

actccagtgc tggaaggtgc 20

<210> SEQ ID NO 1524
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1524

gtcactccag tgctggaagg 20

<210> SEQ ID NO 1525
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1525

gatgtcactc cagtgctgga 20

<210> SEQ ID NO 1526
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1526

ctggatgtca ctccagtgct 20

<210> SEQ ID NO 1527
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1527

gtcctggatg tcactccagt 20

<210> SEQ ID NO 1528
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1528

gttgtcctgg atgtcactcc 20

<210> SEQ ID NO 1529
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1529

gaagttgtcc tggatgtcac 20

<210> SEQ ID NO 1530
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1530

cgagaagttg tcctggatgt 20

<210> SEQ ID NO 1531
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1531

caccgagaag ttgtcctgga 20

<210> SEQ ID NO 1532
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1532

agtcaccgag aagttgtcct 20

<210> SEQ ID NO 1533
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1533

ttgagtcacc gagaagttgt 20

<210> SEQ ID NO 1534
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1534

cacttgagtc accgagaagt 20

<210> SEQ ID NO 1535
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1535

gggcacttga gtcaccgaga 20

<210> SEQ ID NO 1536
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1536

gaagggcact tgagtcaccg 20

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<210> SEQ ID NO 1537
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<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1537

agtgaagggc acttgagtca 20

<210> SEQ ID NO 1538
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1538

ctcagtgaag ggcacttgag 20

<210> SEQ ID NO 1539
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1539

gctctcagtg aagggcactt 20

<210> SEQ ID NO 1540
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1540

gatcagcagc aggcaggcgc 20

<210> SEQ ID NO 1541
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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ctggatcagc agcaggcagg 20

<210> SEQ ID NO 1542
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<212> TYPE: DNA
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aggctggatc agcagcaggc 20

<210> SEQ ID NO 1543
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<212> TYPE: DNA
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gtgaggctgg atcagcagca 20

<210> SEQ ID NO 1544
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atagtgaggc tggatcagca 20

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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ggcatagtga ggctggatca 20

<210> SEQ ID NO 1546
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1546

agaggcatag tgaggctgga 20

<210> SEQ ID NO 1547
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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gtcagaggca tagtgaggct 20

<210> SEQ ID NO 1548
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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caggtcagag gcatagtgag 20

<210> SEQ ID NO 1549
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1549

gtccaggtca gaggcatagt 20

<210> SEQ ID NO 1550
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1550

cttgtccagg tcagaggcat 20

<210> SEQ ID NO 1551
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<212> TYPE: DNA
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<220> FEATURE:
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caccttgtcc aggtcagagg 20

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ctccaccttg tccaggtcag 20

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accctccacc ttgtccaggt 20

<210> SEQ ID NO 1554
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gagaccctcc accttgtcca 20

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<220> FEATURE:
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agtgagaccc tccaccttgt 20

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<210> SEQ ID NO 1556
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gaaagtgaga ccctccacct 20

<210> SEQ ID NO 1557
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ctggaaagtg agaccctcca 20

<210> SEQ ID NO 1558
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1558

ttgctggaaa gtgagaccct 20

<210> SEQ ID NO 1559
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1559

gttttgctgg aaagtgagac 20

<210> SEQ ID NO 1560
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1560

ggagttttgc tggaaagtga 20

<210> SEQ ID NO 1561
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1561

gagggagttt tgctggaaag 20

<210> SEQ ID NO 1562
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1562

gttgagggag ttttgctgga 20

<210> SEQ ID NO 1563
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1563

ccagttgagg gagttttgct 20

<210> SEQ ID NO 1564
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1564

catccagttg agggagtttt 20

<210> SEQ ID NO 1565
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1565

cttcatccag ttgagggagt 20

<210> SEQ ID NO 1566
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1566

tttcttcatc cagttgaggg 20

<210> SEQ ID NO 1567
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1567

tagtttcttc atccagttga 20

<210> SEQ ID NO 1568
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1568

agatagtttc ttcatccagt 20

<210> SEQ ID NO 1569
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1569

gggagatagt ttcttcatcc 20

<210> SEQ ID NO 1570
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1570

ggtcaggtgg atggtccggg 20

<210> SEQ ID NO 1571
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1571

catggtcagg tggatggtcc 20

<210> SEQ ID NO 1572
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1572

gggcatggtc aggtggatgg 20

<210> SEQ ID NO 1573
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1573

tccttgcagc accagttggg 20

<210> SEQ ID NO 1574
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1574

agatccttgc agcaccagtt 20

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<210> SEQ ID NO 1575
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1575

ataagatcct tgcagcacca 20

<210> SEQ ID NO 1576
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1576

gtcataagat ccttgcagca 20

<210> SEQ ID NO 1577
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1577

caggtcataa gatccttgca 20

<210> SEQ ID NO 1578
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1578

ctgcaggtca taagatcctt 20

<210> SEQ ID NO 1579
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1579

gtcctgcagg tcataagatc 20

<210> SEQ ID NO 1580
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1580

caggtcctgc aggtcataag 20

<210> SEQ ID NO 1581
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1581

gagcaggtcc tgcaggtcat 20

<210> SEQ ID NO 1582
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1582

ggcgagcagg tcctgcaggt 20

<210> SEQ ID NO 1583
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1583

ctgggcgagc aggtcctgca 20

<210> SEQ ID NO 1584
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1584

agcctgggcg agcaggtcct 20

<210> SEQ ID NO 1585
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1585

ctcagcctgg gcgagcaggt 20

<210> SEQ ID NO 1586
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1586

cagctcagcc tgggcgagca 20

<210> SEQ ID NO 1587
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1587

tggcgggcag ctcagcctgg 20

<210> SEQ ID NO 1588
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1588

gaatggcggg cagctcagcc 20

<210> SEQ ID NO 1589
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1589

gcagaatggc gggcagctca 20

<210> SEQ ID NO 1590
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1590

tgtgcagaat ggcgggcagc 20

<210> SEQ ID NO 1591
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1591

cggtgtgcag aatggcgggc 20

<210> SEQ ID NO 1592
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1592

gctcggtgtg cagaatggcg 20

<210> SEQ ID NO 1593
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1593

tcagctcggt gtgcagaatg 20

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<210> SEQ ID NO 1594
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1594

ggttcagctc ggtgtgcaga 20

<210> SEQ ID NO 1595
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1595

gcaggttcag ctcggtgtgc 20

<210> SEQ ID NO 1596
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1596

tttgcaggtt cagctcggtg 20

<210> SEQ ID NO 1597
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1597

atttttgcag gttcagctcg 20

<210> SEQ ID NO 1598
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1598

tcaatttttg caggttcagc 20

<210> SEQ ID NO 1599
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1599

tgctcaattt ttgcaggttc 20

<210> SEQ ID NO 1600
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1600

cattgctcaa tttttgcagg 20

<210> SEQ ID NO 1601
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1601

ggtcattgct caatttttgc 20

<210> SEQ ID NO 1602
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1602

tgcggtcatt gctcaatttt 20

<210> SEQ ID NO 1603
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1603

tgatgcggtc attgctcaat 20

<210> SEQ ID NO 1604
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1604

ccctgatgcg gtcattgctc 20

<210> SEQ ID NO 1605
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1605

ccaccctgat gcggtcattg 20

<210> SEQ ID NO 1606
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1606

ccccaccct gatgcggtca 20

<210> SEQ ID NO 1607
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1607

cctcccccac cctgatgcgg 20

<210> SEQ ID NO 1608
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1608

gcacctcccc caccctgatg 20

<210> SEQ ID NO 1609
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1609

tcagcacctc ccccaccctg 20

<210> SEQ ID NO 1610
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1610

tgttcagcac ctcccccacc 20

<210> SEQ ID NO 1611
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1611

tgctgttcag cacctccccc 20

<210> SEQ ID NO 1612
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1612

aaatgctgtt cagcacctcc 20

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<210> SEQ ID NO 1613
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1613

aaaaaatgct gttcagcacc 20

<210> SEQ ID NO 1614
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1614

caaaaaaaat gctgttcagc 20

<210> SEQ ID NO 1615
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1615

gctcaaaaaa aatgctgttc 20

<210> SEQ ID NO 1616
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1616

caagctcaaa aaaaatgctg 20

<210> SEQ ID NO 1617
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1617

cttcaagctc aaaaaaaatg 20

<210> SEQ ID NO 1618
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1618

ccgcttcaag ctcaaaaaaa 20

<210> SEQ ID NO 1619
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1619

catccgcttc aagctcaaaa 20

<210> SEQ ID NO 1620
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1620

tctcatccgc ttcaagctca 20

<210> SEQ ID NO 1621
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1621

ctctctcatc cgcttcaagc 20

<210> SEQ ID NO 1622
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1622

gctctctctc atccgcttca 20

<210> SEQ ID NO 1623
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1623

tgggctctct ctcatccgct 20

<210> SEQ ID NO 1624
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1624

ctgtgggctc tctctcatcc 20

<210> SEQ ID NO 1625
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1625

actctgtggg ctctctctca 20

<210> SEQ ID NO 1626
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1626

tagactctgt gggctctctc 20

<210> SEQ ID NO 1627
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1627

ctgttgggta gactctgtgg 20

<210> SEQ ID NO 1628
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1628

aagctgttgg gtagactctg 20

<210> SEQ ID NO 1629
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1629

gttaagctgt tgggtagact 20

<210> SEQ ID NO 1630
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1630

cttgttaagc tgttgggtag 20

<210> SEQ ID NO 1631
<211> LENGTH: 20
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<210> SEQ ID NO 1632
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<400> SEQUENCE: 1632

ctcaggcttg ttaagctgtt 20

<210> SEQ ID NO 1633
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<212> TYPE: DNA
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<400> SEQUENCE: 1633

gacctcaggc ttgttaagct 20

<210> SEQ ID NO 1634
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<212> TYPE: DNA
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<400> SEQUENCE: 1634

caagacctca ggcttgttaa 20

<210> SEQ ID NO 1635
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1635

ctccaagacc tcaggcttgt 20

<210> SEQ ID NO 1636
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1636

cacctccaag acctcaggct 20

<210> SEQ ID NO 1637
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ggtcacctcc aagacctcag 20

<210> SEQ ID NO 1638
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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cagggtcacc tccaagacct 20

<210> SEQ ID NO 1639
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1639

gttcagggtc acctccaaga 20

<210> SEQ ID NO 1640
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1640

gcggttcagg gtcacctcca 20

<210> SEQ ID NO 1641
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1641

acaggaatgg gcggttcagg 20

<210> SEQ ID NO 1642
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1642

caaacaggaa tgggcggttc 20

<210> SEQ ID NO 1643
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1643

cagcaaacag gaatgggcgg 20

<210> SEQ ID NO 1644
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<212> TYPE: DNA
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<400> SEQUENCE: 1644

acacagcaaa caggaatggg 20

<210> SEQ ID NO 1645
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1645

catacacagc aaacaggaat 20

<210> SEQ ID NO 1646
<211> LENGTH: 20
<212> TYPE: DNA
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gatcatacac agcaaacagg 20

<210> SEQ ID NO 1647
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tttgatcata cacagcaaac 20

<210> SEQ ID NO 1648
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cgctttgatc atacacagca 20

<210> SEQ ID NO 1649
<211> LENGTH: 20
<212> TYPE: DNA
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gaagtgcagg gcagtggcgc 20

<210> SEQ ID NO 1650
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<212> TYPE: DNA
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caggaagtgc agggcagtgg 20

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<210> SEQ ID NO 1651
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gcccaggaag tgcagggcag 20

<210> SEQ ID NO 1652
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gcggcccagg aagtgcaggg 20

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cacgcggccc aggaagtgca 20

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<400> SEQUENCE: 1654

ggccacgcgg cccaggaagt 20

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<212> TYPE: DNA
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<400> SEQUENCE: 1655

gttggccacg cggcccagga 20

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cgggttggcc acgcggccca 20

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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cagcgggttg gccacgcggc 20

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<400> SEQUENCE: 1658

gctcagcggg ttggccacgc 20

<210> SEQ ID NO 1659
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1659

tgtgctcagc gggttggcca 20

<210> SEQ ID NO 1660
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1660

tgctgtgctc agcgggttgg 20

<210> SEQ ID NO 1661
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1661

tcatgctgtg ctcagcgggt 20

<210> SEQ ID NO 1662
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1662

gcctcatgct gtgctcagcg 20

<210> SEQ ID NO 1663
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1663

ctggcctcat gctgtgctca 20

<210> SEQ ID NO 1664
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1664

gccctggcct catgctgtgc 20

<210> SEQ ID NO 1665
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1665

gccaggcact gtgttctggg 20

<210> SEQ ID NO 1666
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1666

cttgccaggc actgtgttct 20

<210> SEQ ID NO 1667
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1667

ggccttgcca ggcactgtgt 20

<210> SEQ ID NO 1668
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1668

aggagaaacg gctgctttcc 20

<210> SEQ ID NO 1669
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1669

ccaaggagaa acggctgctt 20

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<210> SEQ ID NO 1670
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1670

agaccaagga gaaacggctg 20

<210> SEQ ID NO 1671
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1671

cttagaccaa ggagaaacgg 20

<210> SEQ ID NO 1672
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1672

acacttagac caaggagaaa 20

<210> SEQ ID NO 1673
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1673

agcacactta gaccaaggag 20

<210> SEQ ID NO 1674
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1674

tgcagcacac ttagaccaag 20

<210> SEQ ID NO 1675
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1675

ccatgcagca cacttagacc 20

<210> SEQ ID NO 1676
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1676

actccatgca gcacacttag 20

<210> SEQ ID NO 1677
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1677

ctcactccat gcagcacact 20

<210> SEQ ID NO 1678
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1678

cgctgcaggc ttctactgct 20

<210> SEQ ID NO 1679
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1679

tgccgctgca ggcttctact 20

<210> SEQ ID NO 1680
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1680

ttgtgccgct gcaggcttct 20

<210> SEQ ID NO 1681
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1681

catttgtgcc gctgcaggct 20

<210> SEQ ID NO 1682
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1682

gtgcatttgt gccgctgcag 20

<210> SEQ ID NO 1683
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1683

gaggtgcatt tgtgccgctg 20

<210> SEQ ID NO 1684
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1684

tgggaggtgc atttgtgccg 20

<210> SEQ ID NO 1685
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1685

aactgggagg tgcatttgtg 20

<210> SEQ ID NO 1686
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1686

gcaaactggg aggtgcattt 20

<210> SEQ ID NO 1687
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1687

ccagcaaact gggaggtgca 20

<210> SEQ ID NO 1688
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1688

aacccagcaa actgggaggt 20

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<210> SEQ ID NO 1689
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1689

ataaacccag caaactggga 20

<210> SEQ ID NO 1690
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1690

aaaataaacc cagcaaactg 20

<210> SEQ ID NO 1691
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1691

tctaaaataa acccagcaaa 20

<210> SEQ ID NO 1692
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1692

ttctctaaaa taaacccagc 20

<210> SEQ ID NO 1693
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1693

ccattctcta aaataaaccc 20

<210> SEQ ID NO 1694
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1694

cccccattct ctaaaataaa 20

<210> SEQ ID NO 1695
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1695

ccacccccat tctctaaaat 20

<210> SEQ ID NO 1696
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1696

tccccacccc cattctctaa 20

<210> SEQ ID NO 1697
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1697

gcctccccac ccccattctc 20

<210> SEQ ID NO 1698
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1698

cttgcctccc cacccccatt 20

<210> SEQ ID NO 1699
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1699

gttcttgcct ccccaccccc 20

<210> SEQ ID NO 1700
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1700

ctggttcttg cctccccacc 20

<210> SEQ ID NO 1701
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1701

acactggttc ttgcctcccc 20

<210> SEQ ID NO 1702
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1702

taaacactgg ttcttgcctc 20

<210> SEQ ID NO 1703
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1703

cgctaaacac tggttcttgc 20

<210> SEQ ID NO 1704
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1704

ccgcgctaaa cactggttct 20

<210> SEQ ID NO 1705
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1705

gtcccgcgct aaacactggt 20

<210> SEQ ID NO 1706
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1706

gtagtcccgc gctaaacact 20

<210> SEQ ID NO 1707
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1707

acagtagtcc cgcgctaaac 20

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<210> SEQ ID NO 1708
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1708

ggaacagtag tcccgcgcta 20

<210> SEQ ID NO 1709
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1709

tttggaacag tagtcccgcg 20

<210> SEQ ID NO 1710
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1710

ctttttggaa cagtagtccc 20

<210> SEQ ID NO 1711
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1711

attcttttttg gaacagtagt 20

<210> SEQ ID NO 1712
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1712

ggaattcttt ttggaacagt 20

<210> SEQ ID NO 1713
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1713

gttggaattc tttttggaac 20

<210> SEQ ID NO 1714
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1714

tcggttggaa ttctttttgg 20

<210> SEQ ID NO 1715
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1715

tggtcggttg gaattctttt 20

<210> SEQ ID NO 1716
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1716

agctggtcgg ttggaattct 20

<210> SEQ ID NO 1717
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1717

acaagctggt cggttggaat 20

<210> SEQ ID NO 1718
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1718

caaacaagct ggtcggttgg 20

<210> SEQ ID NO 1719
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1719

tcacaaacaa gctggtcggt 20

<210> SEQ ID NO 1720
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1720

gtttcacaaa caagctggtc 20

<210> SEQ ID NO 1721
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1721

tttgtttcac aaacaagctg 20

<210> SEQ ID NO 1722
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1722

gaaaagggaa cactttttg 20

<210> SEQ ID NO 1723
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1723

cttgaaaagg gaacactttt 20

<210> SEQ ID NO 1724
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1724

caacttgaaa agggaacact 20

<210> SEQ ID NO 1725
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1725

tctcaacttg aaaagggaac 20

<210> SEQ ID NO 1726
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tgttctcaac ttgaaaaggg 20

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ttttgttctc aacttgaaaa 20

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aatttttgtt ctcaacttga 20

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cccaattttt gttctcaact 20

<210> SEQ ID NO 1730
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aaacccaatt tttgttctca 20

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ttaaaaccca atttttgttc 20

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attttaaaac ccaatttttg 20

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ttaattttaa aacccaattt 20

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tgcaaaaatg tatactttaa 20

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caatgcaaaa atgtatactt 20

<210> SEQ ID NO 1736
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aggcaatgca aaaatgtata 20

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cgaaggcaat gcaaaaatgt 20

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aaccgaaggc aatgcaaaaa 20

<210> SEQ ID NO 1739
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acaaaccgaa ggcaatgcaa 20

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aatacaaacc gaaggcaatg 20

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ctaaatacaa accgaaggca 20

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acactaaata caaaccgaag 20

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aagacactaa atacaaaccg 20

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ttcaagacac taaatacaaa 20

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acattcaaga cactaaatac 20

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<210> SEQ ID NO 1746
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cttacattca agacactaaa 20

<210> SEQ ID NO 1747
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gttcttacat tcaagacact 20

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catgttctta cattcaagac 20

<210> SEQ ID NO 1749
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ggtcatgttc ttacattcaa 20

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ggaggtcatg ttcttacatt 20

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cacggaggtc atgttcttac 20

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<212> TYPE: DNA
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ctacacggag gtcatgttct 20

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acactacacg gaggtcatgt 20

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<400> SEQUENCE: 1754

cagacactac acggaggtca 20

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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1755

ttacagacac tacacggagg 20

<210> SEQ ID NO 1756
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1756

gtattacaga cactacacgg 20

<210> SEQ ID NO 1757
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1757

aaggtattac agacactaca 20

<210> SEQ ID NO 1758
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1758

actaaggtat tacagacact 20

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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1759

aaaactaagg tattacagac 20

<210> SEQ ID NO 1760
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1760

gaaaaaacta aggtattaca 20

<210> SEQ ID NO 1761
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<212> TYPE: DNA
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<400> SEQUENCE: 1761

gtggaaaaaa ctaaggtatt 20

<210> SEQ ID NO 1762
<211> LENGTH: 20
<212> TYPE: DNA
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<400> SEQUENCE: 1762

tctgtggaaa aaactaaggt 20

<210> SEQ ID NO 1763
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1763

gcatctgtgg aaaaaactaa 20

<210> SEQ ID NO 1764
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1764

caagcatctg tggaaaaaac 20

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<210> SEQ ID NO 1765
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1765

tcacaagcat ctgtggaaaa 20

<210> SEQ ID NO 1766
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1766

aaatcacaag catctgtgga 20

<210> SEQ ID NO 1767
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1767

caaaaatcac aagcatctgt 20

<210> SEQ ID NO 1768
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1768

gttcaaaaat cacaagcatc 20

<210> SEQ ID NO 1769
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1769

attgttcaaa aatcacaagc 20

<210> SEQ ID NO 1770
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1770

cgtattgttc aaaaatcaca 20

<210> SEQ ID NO 1771
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1771

aggtgcttgc atctttcacg 20

<210> SEQ ID NO 1772
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1772

ttcaggtgct tgcatctttc 20

<210> SEQ ID NO 1773
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1773

aaattcaggt gcttgcatct 20

<210> SEQ ID NO 1774
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1774

cagaaattca ggtgcttgca 20

<210> SEQ ID NO 1775
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1775

aaacagaaat tcaggtgctt 20

<210> SEQ ID NO 1776
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1776

ttcaaacaga attcaggtg 20

<210> SEQ ID NO 1777
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1777

gcattcaaac agaaattcag 20

<210> SEQ ID NO 1778
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1778

tccgcattca aacagaaatt 20

<210> SEQ ID NO 1779
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1779

ggttccgcat tcaaacagaa 20

<210> SEQ ID NO 1780
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1780

tatggttccg cattcaaaca 20

<210> SEQ ID NO 1781
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1781

agctatggtt ccgcattcaa 20

<210> SEQ ID NO 1782
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1782

accagctatg gttccgcatt 20

<210> SEQ ID NO 1783
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1783

ataaccagct atggttccgc 20

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<210> SEQ ID NO 1784
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1784

gaaataacca gctatggttc 20

<210> SEQ ID NO 1785
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1785

ggagaaataa ccagctatgg 20

<210> SEQ ID NO 1786
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1786

aagggagaaa taaccagcta 20

<210> SEQ ID NO 1787
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1787

cacaagggag aaataaccag 20

<210> SEQ ID NO 1788
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1788

taacacaagg gagaaataac 20

<210> SEQ ID NO 1789
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1789

tactaacaca agggagaaat 20

<210> SEQ ID NO 1790
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1790

tattactaac acaagggaga 20

<210> SEQ ID NO 1791
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1791

gtttattact aacacaaggg 20

<210> SEQ ID NO 1792
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1792

aggcttattg tggcaagacg 20

<210> SEQ ID NO 1793
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1793

tggaggctta ttgtggcaag 20

<210> SEQ ID NO 1794
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1794

ttttggaggc ttattgtggc 20

<210> SEQ ID NO 1795
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1795

tttttttgga ggcttattgt 20

<210> SEQ ID NO 1796
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1796

actctcattg tggatgacga 20

<210> SEQ ID NO 1797
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1797

aggtactctc attgtggatg 20

<210> SEQ ID NO 1798
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1798

acaggtactc tcattgtgga 20

<210> SEQ ID NO 1799
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1799

gctcacaggt actctcattg 20

<210> SEQ ID NO 1800
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1800

gcaccagctg gtcctgtagg 20

<210> SEQ ID NO 1801
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1801

tagcaccagc tggtcctgta 20

<210> SEQ ID NO 1802
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1802

ctagcaccag ctggtcctgt 20

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<210> SEQ ID NO 1803
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1803

cgactagcac cagctggtcc 20

<210> SEQ ID NO 1804
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1804

agcgactagc accagctggt 20

<210> SEQ ID NO 1805
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1805

ttgcagcgac tagcaccagc 20

<210> SEQ ID NO 1806
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1806

ttttgcagcg actagcacca 20

<210> SEQ ID NO 1807
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1807

caagttttgc agcgactagc 20

<210> SEQ ID NO 1808
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1808

ctgtcacagc ctgcatgaac 20

<210> SEQ ID NO 1809
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1809

tcctgtcaca gcctgcatga 20

<210> SEQ ID NO 1810
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1810

atcctgtcac agcctgcatg 20

<210> SEQ ID NO 1811
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1811

tccatcctgt cacagcctgc 20

<210> SEQ ID NO 1812
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1812

cttccatcct gtcacagcct 20

<210> SEQ ID NO 1813
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1813

tcactccagt gctggaaggt 20

<210> SEQ ID NO 1814
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1814

tgtcactcca gtgctggaag 20

<210> SEQ ID NO 1815
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1815

atgtcactcc agtgctggaa 20

<210> SEQ ID NO 1816
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1816

tggatgtcac tccagtgctg 20

<210> SEQ ID NO 1817
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1817

cctggatgtc actccagtgc 20

<210> SEQ ID NO 1818
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1818

aaggagaaac ggctgctttc 20

<210> SEQ ID NO 1819
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1819

caaggagaaa cggctgcttt 20

<210> SEQ ID NO 1820
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1820

accaaggaga aacggctgct 20

<210> SEQ ID NO 1821
<211> LENGTH: 20
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<400> SEQUENCE: 1821

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<210> SEQ ID NO 1822
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tagaccaagg agaaacggct 20

<210> SEQ ID NO 1823
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<212> TYPE: DNA
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ttagaccaag gagaaacggc 20

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<212> TYPE: DNA
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acttagacca aggagaaacg 20

<210> SEQ ID NO 1825
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1825

cacttagacc aaggagaaac 20

<210> SEQ ID NO 1826
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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cacacttaga ccaaggagaa 20

<210> SEQ ID NO 1827
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<212> TYPE: DNA
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gcacacttag accaaggaga 20

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<212> TYPE: DNA
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cagcacactt agaccaagga 20

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<223> OTHER INFORMATION: Synthetic oligonucleotide

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gcagcacact tagaccaagg 20

<210> SEQ ID NO 1830
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atgcagcaca cttagaccaa 20

<210> SEQ ID NO 1831
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1831

catgcagcac acttagacca 20

<210> SEQ ID NO 1832
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1832

tccatgcagc acacttagac 20

<210> SEQ ID NO 1833
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1833

ctccatgcag cacacttaga 20

<210> SEQ ID NO 1834
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cactccatgc agcacactta 20

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<223> OTHER INFORMATION: Synthetic oligonucleotide

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tcactccatg cagcacactt 20

<210> SEQ ID NO 1836
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<212> TYPE: DNA
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gctcactcca tgcagcacac 20

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<400> SEQUENCE: 1837

ccgctgcagg cttctactgc 20

<210> SEQ ID NO 1838
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gccgctgcag gcttctactg 20

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<400> SEQUENCE: 1839

gtgccgctgc aggcttctac 20

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 1840

tgtgccgctg caggcttcta 20

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<210> SEQ ID NO 1841
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<400> SEQUENCE: 1841

tttgtgccgc tgcaggcttc 20

<210> SEQ ID NO 1842
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1842

atttgtgccg ctgcaggctt 20

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<211> LENGTH: 20
<212> TYPE: DNA
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gcatttgtgc cgctgcaggc 20

<210> SEQ ID NO 1844
<211> LENGTH: 20
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1844

tgcatttgtg ccgctgcagg 20

<210> SEQ ID NO 1845
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ggtgcatttg tgccgctgca 20

<210> SEQ ID NO 1846
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1846

aggtgcattt gtgccgctgc 20

<210> SEQ ID NO 1847
<211> LENGTH: 20

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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ggaggtgcat ttgtgccgct 20

<210> SEQ ID NO 1848
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<400> SEQUENCE: 1848

gggaggtgca tttgtgccgc 20

<210> SEQ ID NO 1849
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1849

ctgggaggtg catttgtgcc 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1850

actgggaggt gcatttgtgc 20

<210> SEQ ID NO 1851
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1851

aaactgggag gtgcatttgt 20

<210> SEQ ID NO 1852
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1852

caaactggga ggtgcatttg 20

<210> SEQ ID NO 1853
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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agcaaactgg gaggtgcatt 20

<210> SEQ ID NO 1854
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1854

cagcaaactg ggaggtgcat 20

<210> SEQ ID NO 1855
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1855

cccagcaaac tgggaggtgc 20

<210> SEQ ID NO 1856
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1856

acccagcaaa ctgggaggtg 20

<210> SEQ ID NO 1857
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1857

aataaaccca gcaaactggg 20

<210> SEQ ID NO 1858
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1858

aaataaaccc agcaaactgg 20

<210> SEQ ID NO 1859
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1859

taaaataaac ccagcaaact 20

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<210> SEQ ID NO 1860
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1860

ctaaaataaa cccagcaaac 20

<210> SEQ ID NO 1861
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1861

ctctaaaata aacccagcaa 20

<210> SEQ ID NO 1862
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1862

tctctaaaat aaacccagca 20

<210> SEQ ID NO 1863
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1863

attctctaaa ataaacccag 20

<210> SEQ ID NO 1864
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1864

cattctctaa aataaaccca 20

<210> SEQ ID NO 1865
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1865

cccattctct aaaataaacc 20

<210> SEQ ID NO 1866
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1866

ccccattctc taaaataaac 20

<210> SEQ ID NO 1867
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1867

acccccattc tctaaaataa 20

<210> SEQ ID NO 1868
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1868

caccccccatt ctctaaaata 20

<210> SEQ ID NO 1869
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1869

cccacccccа ttctctaaaa 20

<210> SEQ ID NO 1870
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1870

ccccaccccc attctctaaa 20

<210> SEQ ID NO 1871
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1871

ctccccaccc ccattctcta 20

<210> SEQ ID NO 1872
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1872

tgcctcccca cccccattct 20

<210> SEQ ID NO 1873
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1873

ttgcctcccc acccccattc 20

<210> SEQ ID NO 1874
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1874

tcttgcctcc ccacccccat 20

<210> SEQ ID NO 1875
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1875

ggttcttgcc tccccacccc 20

<210> SEQ ID NO 1876
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1876

tggttcttgc ctccccaccc 20

<210> SEQ ID NO 1877
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1877

actggttctt gcctccccac 20

<210> SEQ ID NO 1878
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1878

cactggttct tgcctcccca 20

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<210> SEQ ID NO 1879
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1879

aacactggtt cttgcctccc 20

<210> SEQ ID NO 1880
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
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<400> SEQUENCE: 1880

aaacactggt tcttgcctcc 20

<210> SEQ ID NO 1881
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1881

ctaaacactg gttcttgcct 20

<210> SEQ ID NO 1882
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1882

gctaaacact ggttcttgcc 20

<210> SEQ ID NO 1883
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1883

gcgctaaaca ctggttcttg 20

<210> SEQ ID NO 1884
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1884

cgcgctaaac actggttctt 20

<210> SEQ ID NO 1885
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1885

cccgcgctaa acactggttc 20

<210> SEQ ID NO 1886
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1886

tcccgcgcta aacactggtt 20

<210> SEQ ID NO 1887
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1887

agtcccgcgc taaacactgg 20

<210> SEQ ID NO 1888
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1888

tagtcccgcg ctaaacactg 20

<210> SEQ ID NO 1889
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1889

agtagtcccg cgctaaacac 20

<210> SEQ ID NO 1890
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1890

cagtagtccc gcgctaaaca 20

<210> SEQ ID NO 1891
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1891

aacagtagtc ccgcgctaaa 20

<210> SEQ ID NO 1892
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1892

gaacagtagt cccgcgctaa 20

<210> SEQ ID NO 1893
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1893

tggaacagta gtcccgcgct 20

<210> SEQ ID NO 1894
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1894

ttggaacagt agtcccgcgc 20

<210> SEQ ID NO 1895
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1895

ttttggaaca gtagtcccgc 20

<210> SEQ ID NO 1896
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1896

tttttggaac agtagtcccg 20

<210> SEQ ID NO 1897
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1897

tctttttgga acagtagtcc 20

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<210> SEQ ID NO 1898
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1898

ttctttttgg aacagtagtc 20

<210> SEQ ID NO 1899
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1899

aattcttttt ggaacagtag 20

<210> SEQ ID NO 1900
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1900

gaattctttt tggaacagta 20

<210> SEQ ID NO 1901
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1901

tggaattctt tttggaacag 20

<210> SEQ ID NO 1902
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1902

ttggaattct ttttggaaca 20

<210> SEQ ID NO 1903
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1903

ggttggaatt ctttttggaa 20

<210> SEQ ID NO 1904
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1904

cggttggaat tctttttgga 20

<210> SEQ ID NO 1905
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1905

gtcggttgga attcttttttg 20

<210> SEQ ID NO 1906
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1906

ggtcggttgg aattcttttt 20

<210> SEQ ID NO 1907
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1907

ctggtcggtt ggaattcttt 20

<210> SEQ ID NO 1908
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1908

gctggtcggt tggaattctt 20

<210> SEQ ID NO 1909
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1909

aagctggtcg gttggaattc 20

<210> SEQ ID NO 1910
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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caagctggtc ggttggaatt 20

<210> SEQ ID NO 1911
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1911

aacaagctgg tcggttggaa 20

<210> SEQ ID NO 1912
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1912

aaacaagctg gtcggttgga 20

<210> SEQ ID NO 1913
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1913

acaaacaagc tggtcggttg 20

<210> SEQ ID NO 1914
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1914

cacaaacaag ctggtcggtt 20

<210> SEQ ID NO 1915
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1915

ttcacaaaca agctggtcgg 20

<210> SEQ ID NO 1916
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1916

tttcacaaac aagctggtcg 20

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<210> SEQ ID NO 1917
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1917

tgtttcacaa acaagctggt 20

<210> SEQ ID NO 1918
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1918

ttgtttcaca aacaagctgg 20

<210> SEQ ID NO 1919
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1919

ttttgtttca caaacaagct 20

<210> SEQ ID NO 1920
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1920

tttttgtttc acaaacaagc 20

<210> SEQ ID NO 1921
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1921

aacttgaaaa gggaacactt 20

<210> SEQ ID NO 1922
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1922

tcaacttgaa aagggaacac 20

<210> SEQ ID NO 1923
<211> LENGTH: 20

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1923

ctcaacttga aaagggaaca 20

<210> SEQ ID NO 1924
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1924

ttctcaactt gaaaagggaa 20

<210> SEQ ID NO 1925
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1925

gttctcaact tgaaaaggga 20

<210> SEQ ID NO 1926
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1926

ttgttctcaa cttgaaaagg 20

<210> SEQ ID NO 1927
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1927

tttgttctca acttgaaaag 20

<210> SEQ ID NO 1928
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1928

tttttgttct caacttgaaa 20

<210> SEQ ID NO 1929
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1929

atttttgttc tcaacttgaa 20

<210> SEQ ID NO 1930
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1930

caatttttgt tctcaacttg 20

<210> SEQ ID NO 1931
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1931

ccaatttttg ttctcaactt 20

<210> SEQ ID NO 1932
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1932

acccaatttt tgttctcaac 20

<210> SEQ ID NO 1933
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1933

aacccaattt ttgttctcaa 20

<210> SEQ ID NO 1934
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1934

taaaacccaa tttttgttct 20

<210> SEQ ID NO 1935
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1935

tttaaaaccc aatttttgtt 20

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<210> SEQ ID NO 1936
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1936

aatgcaaaaa tgtatacttt 20

<210> SEQ ID NO 1937
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1937

gcaatgcaaa aatgtatact 20

<210> SEQ ID NO 1938
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1938

aaggcaatgc aaaaatgtat 20

<210> SEQ ID NO 1939
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1939

gaaggcaatg caaaaatgta 20

<210> SEQ ID NO 1940
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1940

ccgaaggcaa tgcaaaaatg 20

<210> SEQ ID NO 1941
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1941

catacccttc tgctgta 17

<210> SEQ ID NO 1942
<211> LENGTH: 17

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1942

ccgcataccc ttctgct 17

<210> SEQ ID NO 1943
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1943

tcgcttccgc ataccct 17

<210> SEQ ID NO 1944
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1944

tgctcgcttc cgcatac 17

<210> SEQ ID NO 1945
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1945

ctcattgtgg atgacga 17

<210> SEQ ID NO 1946
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1946

actctcattg tggatga 17

<210> SEQ ID NO 1947
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1947

cagctgctca caggtac 17

<210> SEQ ID NO 1948
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1948

agcgactagc accagct 17

<210> SEQ ID NO 1949
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1949

cacagcctgc atgaacc 17

<210> SEQ ID NO 1950
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1950

tcctgtcaca gcctgca 17

<210> SEQ ID NO 1951
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1951

ccatcctgtc acagcct 17

<210> SEQ ID NO 1952
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1952

agtgctggaa ggtgccc 17

<210> SEQ ID NO 1953
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1953

gctggatcag cagcagg 17

<210> SEQ ID NO 1954
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1954

ctgatgcggt cattgct 17

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<210> SEQ ID NO 1955
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1955

ggctctctct catccgc 17

<210> SEQ ID NO 1956
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1956

gtgggctctc tctcatc 17

<210> SEQ ID NO 1957
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1957

ccttgccagg cactgtg 17

<210> SEQ ID NO 1958
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1958

gccttgccag gcactgt 17

<210> SEQ ID NO 1959
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1959

aggccttgcc aggcact 17

<210> SEQ ID NO 1960
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1960

gaggccttgc caggcac 17

<210> SEQ ID NO 1961
<211> LENGTH: 17

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1961

gctgctggcc tttgcct 17

<210> SEQ ID NO 1962
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1962

tatctgctgc tggcctt 17

<210> SEQ ID NO 1963
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1963

ttatctgctg ctggcct 17

<210> SEQ ID NO 1964
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1964

tgttatctgc tgctggc 17

<210> SEQ ID NO 1965
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1965

ttgttatctg ctgctgg 17

<210> SEQ ID NO 1966
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1966

gttgttatct gctgctg 17

<210> SEQ ID NO 1967
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1967

atcgctgatt tgtccgg 17

<210> SEQ ID NO 1968
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1968

catcgctgat ttgtccg 17

<210> SEQ ID NO 1969
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1969

acatcgctga tttgtcc 17

<210> SEQ ID NO 1970
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1970

cacatcgctg atttgtc 17

<210> SEQ ID NO 1971
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1971

acacatcgct gatttgt 17

<210> SEQ ID NO 1972
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1972

tgacacatcg ctgattt 17

<210> SEQ ID NO 1973
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1973

gtgacacatc gctgatt 17

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<210> SEQ ID NO 1974
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1974

cattagaaga aaaggtg 17

<210> SEQ ID NO 1975
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1975

tcattagaag aaaaggt 17

<210> SEQ ID NO 1976
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1976

cgactcatta gaagaaa 17

<210> SEQ ID NO 1977
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1977

gctcaaagtc gactcat 17

<210> SEQ ID NO 1978
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1978

agctcaaagt cgactca 17

<210> SEQ ID NO 1979
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1979

cagctcaaag tcgactc 17

<210> SEQ ID NO 1980
<211> LENGTH: 17

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1980

ccagctcaaa gtcgact 17

<210> SEQ ID NO 1981
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1981

tccagctcaa agtcgac 17

<210> SEQ ID NO 1982
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1982

ctttccagct caaagtc 17

<210> SEQ ID NO 1983
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1983

agaaacggct gctttcc 17

<210> SEQ ID NO 1984
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1984

accaaggaga aacggct 17

<210> SEQ ID NO 1985
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1985

gcatttgtgc cgctgca 17

<210> SEQ ID NO 1986
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 1986

ggtgcatttg tgccgct 17

<210> SEQ ID NO 1987
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1987

agcaaactgg gaggtgc 17

<210> SEQ ID NO 1988
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1988

ggttcttgcc tccccac 17

<210> SEQ ID NO 1989
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1989

ctaaacactg gttcttg 17

<210> SEQ ID NO 1990
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1990

gcgctaaaca ctggttc 17

<210> SEQ ID NO 1991
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1991

aacagtagtc ccgcgct 17

<210> SEQ ID NO 1992
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1992

ggttggaatt ctttttg 17

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<210> SEQ ID NO 1993
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1993

ctggtcggtt ggaattc 17

<210> SEQ ID NO 1994
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1994

acaaacaagc tggtcgg 17

<210> SEQ ID NO 1995
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1995

ttcacaaaca agctggt 17

<210> SEQ ID NO 1996
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1996

acttgaaaag ggaacac 17

<210> SEQ ID NO 1997
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1997

tcaacttgaa aagggaa 17

<210> SEQ ID NO 1998
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1998

ttctcaactt gaaaagg 17

<210> SEQ ID NO 1999
<211> LENGTH: 17

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 1999

caatttttgt tctcaac 17

<210> SEQ ID NO 2000
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2000

acccaatttt tgttctc 17

<210> SEQ ID NO 2001
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2001

atctgctgct ggccttt 17

<210> SEQ ID NO 2002
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2002

gttatctgct gctggcc 17

<210> SEQ ID NO 2003
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2003

tcgctgattt gtccggg 17

<210> SEQ ID NO 2004
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2004

acagtagtcc cgcgcta 17

<210> SEQ ID NO 2005
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

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<400> SEQUENCE: 2005

gaacagtagt cccgcgc 17

<210> SEQ ID NO 2006
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2006

gttggaattc tttttgg 17

<210> SEQ ID NO 2007
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2007

cggttggaat tcttttt 17

<210> SEQ ID NO 2008
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2008

tggtcggttg gaattct 17

<210> SEQ ID NO 2009
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2009

gctggtcggt tggaatt 17

<210> SEQ ID NO 2010
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2010

caaacaagct ggtcggt 17

<210> SEQ ID NO 2011
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2011

cacaaacaag ctggtcg 17

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<210> SEQ ID NO 2012
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2012

ttatctgctg ctggcctt 18

<210> SEQ ID NO 2013
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2013

ttgttatctg ctgctggc 18

<210> SEQ ID NO 2014
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2014

gttgttatct gctgctgg 18

<210> SEQ ID NO 2015
<211> LENGTH: 18
<212> TYPE: DNA
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catcgctgat ttgtccgg 18

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acatcgctga tttgtccg 18

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cacatcgctg atttgtcc 18

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acacatcgct gatttgtc 18

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cagctcaaag tcgactca 18

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gaacagtagt cccgcgct 18

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cggttggaat tcttttttg 18

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gctggtcggt tggaattc 18

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acaaacaagc tggtcggt 18

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cacaaacaag ctggtcgg 18

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ccagctcaaa gtcgactc 18

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ggaacagtag tcccgcgc 18

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tatctgctgc tggccttt 18

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tcggttggaa ttcttttt 18

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ttatctgctg ctggccttt 19

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ttgttatctg ctgctggcc 19

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gttgttatct gctgctggc 19

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acatcgctga tttgtccgg 19

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cacatcgctg atttgtccg 19

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acacatcgct gatttgtcc 19

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cagctcaaag tcgactcat 19

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aacagtagtc ccgcgctaa 19

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<400> SEQUENCE: 2041

gaacagtagt cccgcgcta 19

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cggttggaat tctttttgg 19

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gctggtcggt tggaattct 19

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cacaaacaag ctggtcggt 19

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tcacaaacaa gctggtcgg 19

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tatctgctgc tggcctttg 19

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tgttatctgc tgctggcct 19

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agctcaaagt cgactcatt 19

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<223> OTHER INFORMATION: Synthetic oligonucleotide

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ggttggaatt ctttttgga 19

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<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2050

acaaacaagc tggtcggtt 19

<210> SEQ ID NO 2051
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<223> OTHER INFORMATION: Synthetic oligonucleotide

<400> SEQUENCE: 2051

ccagctcaaa gtcgactca 19

1. (canceled)

2. A compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides wherein the modified oligonucleotide has a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases fully complementary to an equal length portion of nucleobases 2281 to 2300 of SEQ ID NO: 1, wherein the nucleobase sequence of the modified oligonucleotide is at least 80% complementary to SEQ ID NO: 1.

3-9. (canceled)

10. The compound of claim 2, wherein the compound is single-stranded.

11. The compound of claim 2, wherein the compound is double-stranded.

12. The compound of claim 2, wherein at least one internucleoside linkage is a modified internucleoside linkage.

13. The compound of claim 12, wherein at least one modified internucleoside linkage is a phosphorothioate internucleoside linkage.

14. The compound of claim 12, wherein each modified internucleoside linkage is a phosphorothioate internucleoside linkage.

15. The compound of claim 2, wherein the modified oligonucleotide comprises at least one modified sugar.

16. The compound of claim 15, wherein at least one modified sugar is a bicyclic sugar.

17. The compound of claim 15, wherein at least one modified sugar comprises a 2'-O-methoxyethyl, a constrained ethyl (cEt), a 3'-fluoro-HNA or a 4'-$(CH_2)_n$—O-2' bridge, wherein n is 1 or 2.

18. The compound of claim 2, wherein at least one nucleoside comprises a modified nucleobase.

19. The compound of claim 18, wherein the modified nucleobase is a 5-methylcytosine.

20. The compound of claim 2, wherein the modified oligonucleotide comprises:

a. a gap segment consisting of linked deoxynucleosides;

b. a 5' wing segment consisting of linked nucleosides; and

c. a 3' wing segment consisting of linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

21. (canceled)

22. (canceled)

23. (canceled)

24. A compound comprising a single-stranded modified oligonucleotide consisting of 20 linked nucleosides having the nucleobase sequence consisting of SEQ ID NO: 1914, wherein the modified oligonucleotide comprises:

a. a gap segment consisting often linked 2'-deoxynucleosides;

b. a 5' wing segment consisting of five linked nucleosides; and

c. a 3' wing segment consisting of five linked nucleosides;

wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a 2'-O-methoxyethyl sugar, wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine residue is a 5-methylcytosine.

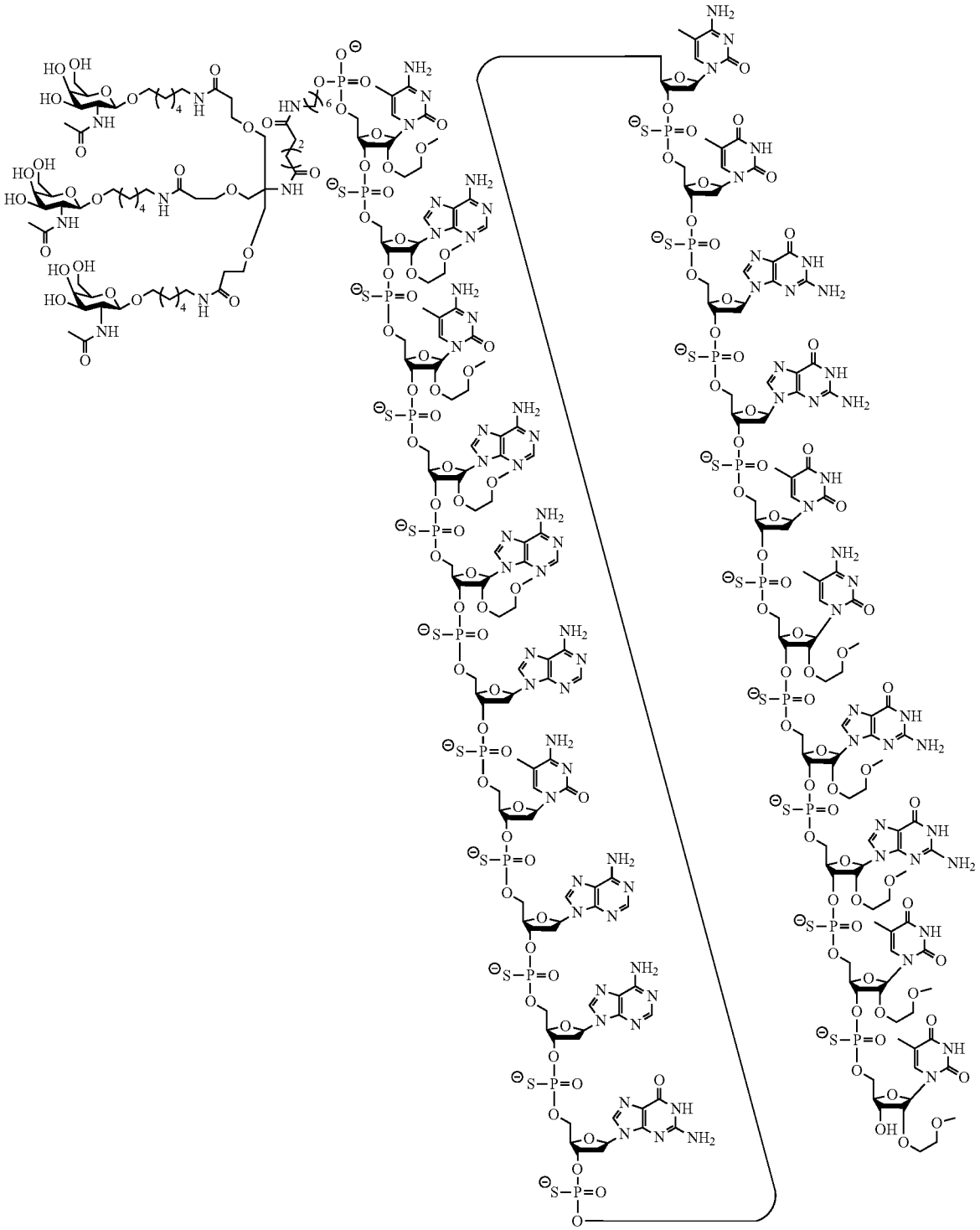
25. (canceled)

26. The compound of claim 2, comprising a conjugate group.

27. The compound of 26, wherein the conjugate group comprises a GalNAc moiety.

28. (canceled)

29. A modified oligonucleotide according to the following formula:



30. A composition comprising the compound of claim **2**, or a salt thereof, and a pharmaceutically acceptable carrier or diluent.

31-35. (canceled)

36. A method of treating a disease associated with excess angiotensinogen in a human comprising administering to the human a therapeutically effective amount of the compound of claim **2**, thereby treating the disease associated with excess angiotensinogen.

37. The method of claim **36**, wherein the disease is hypertension.

38. The method of claim **36**, wherein the disease is resistant hypertension.

* * * * *